# OpenText™ Service Manager

Get Started

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# Get Started

The following topics help you get started with Service Manager:

Service Manager modes: Classic, Codeless, and Hybrid

Accessing Service Manager

Clients

Searching records

Navigating the UI

Editing records

Accessibility

Privileges in Windows systems

Personas

Use case

Glossary

## 1. Service Manager modes: Classic, Codeless, and Hybrid

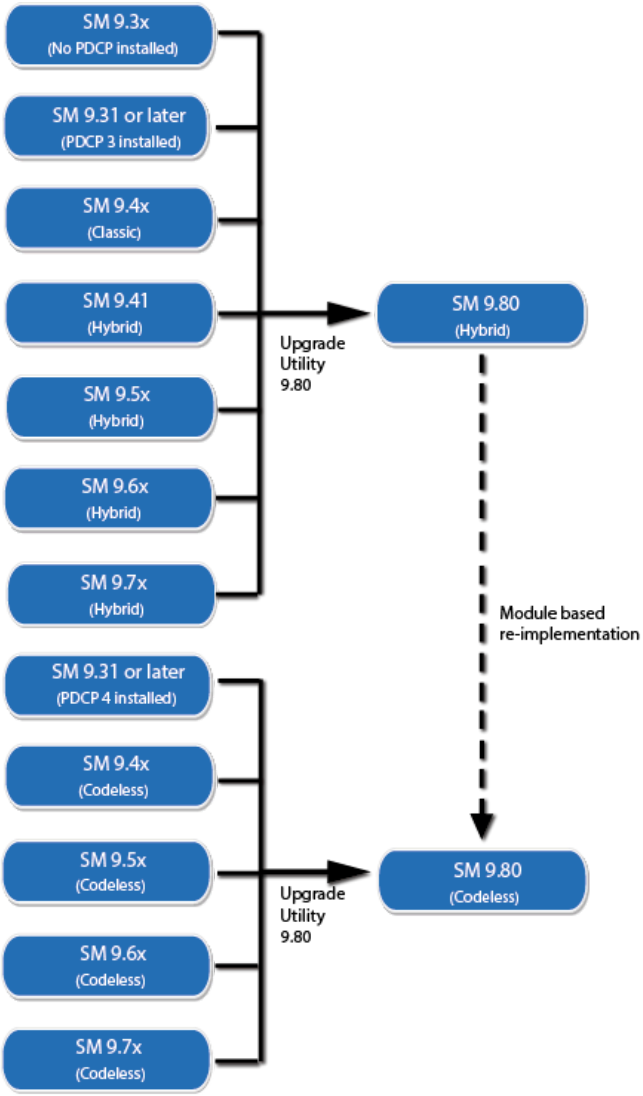
Service Manager 9.5*x* or later is available in two modes: Hybrid and Codeless. 9.4*x* versions of Service Manager are additionally available in Classic mode. The main difference between the three modes is the extent to which Process Designer technology is implemented.

- **Service Manager Classic** uses legacy workflows and functionality, with the exception of the Service Level Management and Knowledge Management modules, which are re-factored to use Process Designer-based workflows. Classic mode is unavailable in Service Manager 9.50 or later.
- In **Service Manager Codeless**, Process Designer workflows and functionality are fully implemented in the Service Desk, Incident Management, Problem Management, Change Management, Request Fulfillment, Knowledge Management, Service Level Management, and Service Catalog modules.
- **Service Manager Hybrid** is designed to ease the transition between Service Manager Classic and Service Manager Codeless. In Hybrid mode, Process Designer technology is fully implemented, however you can continue to use legacy technology such as Format Control. This enables you to continue to use tailoring that you implemented using legacy technology until the time that you choose to migrate Service Manager to Codeless mode.

### Upgrade paths

Codeless mode is directly available to new customers and to existing customers who are already running Service Manager 9.7*x* Codeless, 9.6*x* Codeless, 9.5*x* Codeless, 9.4*x* Codeless, or Service Manager 9.3*x* with Process Designer Content Pack (PDCP) 4 installed. Otherwise, you must migrate your system to Codeless mode from a Hybrid mode version (for specific upgrade paths, see the following illustration).

Hybrid mode is available to customers who are upgrading from Service Manager 9.3*x* without PDCP 4 installed, Service Manager 9.4*x* (either Classic or Hybrid), Service Manager 9.5*x* (Hybrid), Service Manager 9.6*x* (Hybrid), or Service Manager 9.7*x* (Hybrid). If you are upgrading from 9.3*x* or from 9.4*x* Classic, the Applications Upgrade Utility automatically migrates much of your data and business logic to Process Designer. For more information, refer to the Service Manager Applications upgrade documentation.




Note:
This diagram illustrates the recommended applications upgrade paths only. All paths and arrows symbolize direct upgrades.

The following tables provide an overview of the functional implementation of freshly-upgraded Service Manager Codeless and Service Manager Hybrid systems (that is, this table doesn't include the possibilities presented by additional manual tailoring).

**Service Desk module**

| | Functional implementation (Codeless) | Functional implementation (Hybrid) |
|---|---|---|
| Workflows | OOB Process Designer | OOB Process Designer workflow and migrated Classic workflow |
| Object | OOB Process Designer | Process Designer (migrated from Classic object) |
| Security | OOB Process Designer | Process Designer (migrated from Classic profiles) |
| Category records | OOB Process Designer | OOB Process Designer records and migrated Classic records |

**Incident Management module**

| | Functional implementation (Codeless) | Functional implementation (Hybrid) |
|---|---|---|
| Workflows | OOB Process Designer | OOB Process Designer workflow and migrated Classic workflow |
| Object | OOB Process Designer | Process Designer (migrated from Classic object) |
| Security | OOB Process Designer | Process Designer (migrated from Classic profiles) |
| Category records | OOB Process Designer | OOB Process Designer records and migrated Classic records |
| Alerts | OOB Process Designer | OOB Process Designer alerts and migrated Classic alerts |
| Solution matching | OOB Process Designer | Process Designer (migrated from Classic solution matching) |

**Problem Management module**

| | Functional implementation (Codeless) | Functional implementation (Hybrid) |
|---|---|---|
| Workflows | OOB Process Designer | OOB Process Designer |
| Object | OOB Process Designer | OOB Process Designer |
| Security | OOB Process Designer | Process Designer (migrated from Classic profiles) |
| Category records | OOB Process Designer | OOB Process Designer records and migrated Classic records |
| Known Errors | OOB Process Designer | Process Designer (migrated from Classic Known Errors) |

**Change Management module**

| | Functional implementation (Codeless) | Functional implementation (Hybrid) |
|---|---|---|
| Workflows | OOB Process Designer | OOB Process Designer workflow and migrated Classic workflow, or OOB Process Designer if upgrading from a PDCP3 system |
| Objects | OOB Process Designer | Process Designer (migrated from Classic objects), or OOB Process Designer if upgrading from a PDCP3 system |
| Security | OOB Process Designer | Process Designer (migrated from Classic profiles), or OOB Process Designer if upgrading from a PDCP3 system |
| Category records | OOB Process Designer | OOB Process Designer records |

**Request Management module**

| | Functional implementation (Codeless) | Functional implementation (Hybrid) |
|---|---|---|
| Workflows | N/A | N/A (unless manually configured) |
| Object | N/A | N/A (unless manually configured) |
| Security | N/A | N/A (unless manually configured) |
| Category records | N/A | N/A (unless manually configured) |

**Request Fulfillment module**

| | Functional implementation (Codeless) | Functional implementation (Hybrid) |
|---|---|---|
| Workflows | OOB Process Designer | OOB Process Designer |
| Object | OOB Process Designer | OOB Process Designer |
| Security | OOB Process Designer | OOB Process Designer |
| Category records | OOB Process Designer | OOB Process Designer |

**Service Level Management module**

| | Functional implementation (Codeless) | Functional implementation (Hybrid) |
|---|---|---|
| Workflows | OOB Process Designer | OOB Process Designer |
| Object | OOB Process Designer | OOB Process Designer |
| Security | OOB Process Designer | OOB Process Designer |
| Category records | OOB Process Designer | OOB Process Designer |

**Knowledge Management module**

| | Functional implementation (Codeless) | Functional implementation (Hybrid) |
|---|---|---|
| Workflows | OOB Process Designer | OOB Process Designer |
| Object | OOB Process Designer | OOB Process Designer |
| Security | OOB Process Designer | OOB Process Designer |
| Category records | OOB Process Designer | OOB Process Designer |

**Service Catalog module**

| | Functional implementation (Codeless) | Functional implementation (Hybrid) |
|---|---|---|
| Workflows | OOB Process Designer | OOB Process Designer |
| Object | OOB Process Designer | OOB Process Designer |
| Security | OOB Process Designer | OOB Process Designer |
| Category records | OOB Process Designer | OOB Process Designer |

**Configuration Management module**

| | Functional implementation (Codeless) | Functional implementation (Hybrid) |
|---|---|---|
| Workflows | OOB Process Designer | OOB Process Designer |
| Object | OOB Process Designer | OOB Process Designer |
| Security | OOB Process Designer | Process Designer (migrated from Classic profiles) |
| Category records | OOB Process Designer | OOB Process Designer records and migrated Classic records |
| Contract records | Classic | Classic |

## 2. Accessing Service Manager

The way that you access Service Manager depends on which of the following types of client you want to use:

- Sign in to the web client
- Sign in to the Windows client

### Related topics

Clients

### 2.1. Sign in to the web client

The Service Manager web client has four views (modes):

- Power User view (The view name is `index.do`)
- Employee Self-Service User view (The view name is `ess.do`)
- Accessible view (The view name is `accessible.do`)

- Self-Service Accessible view (The view name is `accessible_ess.do`)

These views are intended for different users. For details, see Web client.

### Prerequisite

Before signing in to the web client, make sure you have obtained the following information:

1. Contact your system administrator to obtain the following information about the Service Manager server to which you want to connect:
   - The web server host name or IP address
   - The web server port number
   - The name of the deployed Service Manager application
   - Your user name and password
2. Verify that your web browser version is supported.
3. Verify that your web browser is configured as follows:
   - Cookies are enabled.
   - JavaScript is enabled.
   - Pop-ups are enabled. You may need to add the URL to the browser exception list.

### Sign in to the web client

To sign in to the Service Manager web client, follow these steps:

1. Type the following URL in the address field of your web browser:
   `http://<servername>:<portnumber>/<appname>/<viewname>`
   This URL contains the following placeholders:
   - <servername> refers to the host name or IP address of the web server.
   - <portnumber> refers to the port number. For example, leave the port blank to default to port 80.
   - <appname> refers to the name of the deployed Service Manager application. By default, the application name is the name of the WAR file. For example, `webtier-9.80.`
   - <viewname> refers to the name of the view.

   **Important**

   To align with Microsoft's announcement, Service Manager no longer supports Internet Explorer 11. For more information about web browser support, see the Support Matrix.

   **Note**

   - If your web client requires a secure connection, Service Manager automatically redirects the URL to `https://<servername>:<HTTPSportnumber>/<appname>/<viewname>`.
   - You can append parameters to the URL to modify the web client appearance and behavior. For example, you can specify the help server location so that the web client users can easily access the Service Manager Help Center. See Setting web client preferences.
2. On the login page, type your username and password.
3. Select the language for the web client. The language defaults to the language setting in the web browser environment.
4. Click **Log In**.

### Web browser session management

Service Manager manages web browser sessions in the following two power user scenarios: a new tab or window (depending on your browser's setting for opening external links) opens always in power mode without a warning, and the original working tab is no longer expired:

- Scenario 1: You sign in through power mode first, and then click an email link in power mode as well.

- Scenario 2: You sign in through power mode first, and then click an email link in self-service mode.

**Note**

The following self-service user scenarios are based on self-service user persona analysis and the design that only one application thread is allowed in one user session for self- service mode access to avoid potential data loss:

- Scenario 3: You sign in through self-service mode first, and then click an email link in power mode.
- Scenario 4: You sign in through self-service mode first, and then click an email link in self-service mode.

The following table summarizes the web browser session management behaviors in different scenarios:

| Scenario | Does a warning display? | Is the original working tab expired? | In which mode does the new browser tab or window open? |
|---|---|---|---|
| 1 | No | No | Power mode<br><br>**Note**<br>The new tab or window directly opens up to the linked record, and retrieves only saved data of the pages that are open in the first browser tab. Either continue to work in the first tab, or save your changes in the first tab and then work in the second one. When you first sign in through accessible mode, the behavior is similar. |

| | | | |
|---|---|---|---|
| 2 | No | No | Power mode<br><br>**Note**<br>The new tab or window directly opens up to the linked record, and retrieves only saved data of the pages that are open in the first browser tab. Either continue to work in the first tab, or save your changes in the first tab and then work in the second one.<br><br>When you first sign in through accessible mode, the behavior is similar. |
| 3 | Yes | Yes | You are prompted to select one of the following modes:<br>• Self-service mode<br>• Power mode (which requires sign in again) |
| 4 | Yes | Yes | Self-service mode |

**Note**

This functionality also works when your system has Lightweight Single Sign-On (LW-SSO) enabled or is horizontally scaled.

### Limitations

- Process Designer workflow tab pages (**Tailoring** > **Process Designer** > **Workflows**) that are open in the original browser tab can't be recovered in the new browser tab.
- When you refresh the web client in the new browser tab by using the browser's Refresh feature, you may see a browser warning. Ignore the warning, and click **Retry** (in Google Chrome) or **Resend** (in Mozilla Firefox) to continue.
- A warning page is displayed when you download files from the browser (for example, when you export a record list to a text file). There are several workarounds to this issue. For example, you can hold down the Ctrl key and click the **OK** button on the Export to Text screen, until the file download dialog appears.
- In a trusted sign-on (TSO) environment, rapidly opening multiple links may cause unexpected errors or a session timeout. This is because Service Manager creates a temporary session during sign in. If you open multiple links too fast, the following issues may occur:
    - The temporary session terminates on the server after a successful sign-in, and therefore all requests with the temporary session may result in a sign-out. This sign-out will direct the web client to a goodbye page, which will terminate your session on the server.
      If you have signed in to the web client, all subsequent requests will open up a warning page. Only the most recent page is kept active, and the others expire.
    - Subsequent requests are sent before the first request is returned with a temporary session. Therefore, the server doesn't know if these requests are from the same browser session.
    - If you sign in to the web client, all subsequent requests will result in a warning page. Only the most recent page is kept active, and the others expire.

### Related topics

Windows client
Sign out of Service Manager

## 2.2. Sign in to the Windows client

To sign in to the Service Manager by using the Windows client, follow these steps:

1. Contact your system administrator to obtain the following for the Service Manager server to which you want to connect:
   • The server hostname or IP address
   • The server port number
   • Your username and password
2. From the Windows **Start** menu, click **All Programs > OpenText > Service Manager x.xx > Client**. This is the default path for the client installation. Your path may be different.
3. Click **File** > **Connect** to display the Connections dialog box. From this window, you can choose an existing connection or create a new connection.
   • To choose an existing connection, select the connection name, and then click **Connect**.
   • To create a new connection, click **New** on the Connections dialog. Specify the details for the connection, and then click **Connect**.

### Create a connection

**Applies to User roles:** All roles

To create a connection, follow these steps:

1. Click **File** > **Connect** > **Connections**.
2. Click **New** to create a new configuration.
3. Name your new configuration. Choose a name that reminds you what the configuration is for. If you use multiple servers, you may want to include the server name in the configuration name. If you use multiple login profiles, you may want to include the profile name in the configuration name.

For example, name the connection "servername admin" or "servername helpdesk."

1. Type your username and password.

2. Click **Remember my password** if you want to save your encrypted password.
3. Click **Automatically login** if you want to sign in to this server each time that you start the client. Omit this step if you want to choose a different server each time you connect to the server.
4. Type the server host name and the server port number. Contact your system administrator for this information.
5. Click the drop-down list to choose your language. The server populates the list with its supported languages. The default value is "English."
6. If you want the configuration to be identified by a background color in the Service Manager client window, click **Connection identified by color** and then choose a color.
7. Click **Apply**.
8. Click **Connect** to connect to Service Manager.

### Connections dialog

When you launch the Service Manager Windows client, the Connections dialog box appears. You can use the dialog box to set up new connection shortcuts to the server and to manage existing connections shortcuts.

The Connections dialog box is important if you are an administrator using the Client Packaging Utility to create a single branded installation.

Click the filter button to filter server connections so that only a few connections are displayed or a specific connection is displayed. By typing the beginning letters or complete name of the connection in the filter text box, so only those connections that match what you type appears in the list of server connections.



**Note**

Note

Some options may not be visible on a repackaged client.

The Connection tab on the Connections dialog box has the following options.

| Option | Default value | Definition |
| --- | --- | --- |
| Name | New_configuration | Type the name of this connection. |
| Use Login/Password | Enabled | When this option is enabled, Service Manager uses the sign in name and password information defined in the Connection dialog to create a client/server session. |
| Use Trusted Sign-on | Disabled | When this option is enabled, Service Manager uses the sign in name and password information defined in a trusted authentication source, such as Windows Integrated Authentication, to connect to the server. |
| User name | Windows user name | Type your user name to sign in to the named server. |
| Password | None | Type your password to sign in to the named server. |
| Remember my password | Disabled | Select this option to sign in automatically without retyping your user information each time. |
| Automatically login | Disabled | Select this option to sign in automatically when you start the Service Manager client. |
| Server host name | localhost | Type the name of the Service Manager server. |
| Server port number | 13080 | Type the port number assigned to the Service Manager server. The default Service Manager port number is 13080. |
| Language | None | Choose the language to use for this session. You may want to create session connections in multiple languages if you work with international customers. |
| Connection identified by a color | Disabled | This option enables you to assign a different color to each connection. By default, all client connections have the same background color as your Windows text editor. |

| | | |
|---|---|---|
| Connection color | None | Choose a standard or custom color from the color palette.<br><br>**Tip**<br>Tip<br>A custom color can improve the contrast or identify multiple connections. Lighter colors improve the readability of the forms. |

**Advanced tab**

The Advanced tab on the Connection dialog has the following settings.

| Option | Default value | Definition |
|---|---|---|
| Compress SOAP messages | Enabled | When this option is enabled, SOAP messages are compressed by using GNU zip (gzip) encoding. Gzip works similarly to WinZip™, a compression utility for Windows.<br><br>**Note**<br>Note<br>You can disable this option globally in the sm.ini file. |
| Use SSL Encryption | Disabled | When this option is enabled, server's demonstration certificate is used to encrypt your data when transmitting it.<br><br>**Note**<br>Note<br>You can enable more robust SSL encryption options from the sm.ini file. |
| Trace SOAP traffic | Disabled | When this option is enabled, the client logs SOAP messages. |
| Connect to External Load Balancer | Disabled | When this option is enabled, the Windows client supports an external hardware load balancer. |

#### Multiple sessions

Service Manager supports the following sign in scenarios.

##### Windows client

- After you sign in to Service Manager by using the Windows client, you can sign in again to the same server with the same (or a different) user name and run multiple concurrent sessions.
- After you sign in to Service Manager by using the Windows client, you can sign in again to different servers with the same (or a different) account and run multiple concurrent sessions.

Each session runs in a separate window. If you disconnect one client/server session, the associated window is closed, but all others remain open until you disconnect the relevant session. When you create a connection profile, you can specify a custom color to easily differentiate one connection from another.

##### Web client

Service Manager doesn't support multiple web client sessions.

#### Related topics

Windows client
Sign out of Service Manager

### 2.3. Sign out of Service Manager

**Applies to User roles:** All roles

It is important that you first disconnect or log off Service Manager before you exit the Windows or web clients.

#### Windows client

To sign out of the Service Manager Windows client, follow these steps:

1. Click **File** > **Disconnect from <*connection name*>**.
2. Click **File** > **Exit**.

#### Web client

To sign out of the Service Manager web client, follow these steps:

1. click **User Information** on the top-right corner of the Service Manager UI to show your User Basic Information Card.
2. Click **Logout**, and then click **OK** to confirm that you want to sign out.
3. Close the web browser.

#### Related topics

Clients
Accessing Service Manager
Sign in to the Windows client
Sign in to the web client

### 2.4. Passwords

Passwords play a major part in controlling system security within Service Manager. To protect your Service Manager system from unauthorized access, each user has a password.

A system administrator can update or reset any password. Users can also change their own passwords, unless the system administrator denies that right. In case users forgot their passwords, they can reset the passwords through the Service Manager web client login page on their own after the system administrator enable the forgot password function.

**Note:** Operators who don't have the **My Security Questions** menu in the System Navigator can't set up the security questions or reset the passwords.

### Set up my security questions

**Applies to User roles:** All roles

If your Service Manager system administrator has enabled the forgot password function, the system displays a warning message and prompts you to set up your security questions when you sign in to the web client. In case you have forgotten your password, you can answer these security questions as you selected and hence reset your password on your own.

To set up security questions, follow these steps:

1. Sign in to the web client.
2. From the System Navigator, click **My Security Questions**. The system prompts you to provide your password.
3. Type your password in the **Password** field, and then press Enter.
4. Select the system security questions and then type your answers correspondingly.

   **Note**

   Note
   Do not start your security answers with these special characters: hash symbol(#), less-than symbol(<), greater-than symbol(>), equal sign(=), and tilde(~). Otherwise, the system truncates these special characters from your security answers.

5. Save your changes.

In case you need to re-select the security questions and update the answers, follow these steps:

1. Sign in to the web client.
2. From the System Navigator, click **My Security Questions**. The system prompts you to provide your password.
3. Type your password in the **Password** field, and then press Enter.
4. Click **Clear Answers** to clear all security questions and answers. Click **Yes** in the system prompt to confirm your operation.
5. Re-select the security questions, and then type your answers accordingly.
6. Save your changes.

**Tip:** If the system administrator updates the number of security questions, the system informs and prompts you to update the security questions accordingly when you sign in to the Service Manager web client.

### Change my password

**Applies to User roles:** All roles

In case you still remember your password and hope to change it, you can use the command line (if this feature is enabled for you) or the Change Password option in the ESS client.

To change your password by using the command line, follow these steps:

1. On the web client or the Windows client, type password in the command line field. The Change Password form opens, and Service Manager displays the password form with your user name.

   **Note**

   Note This option is available only if your administrator has given you access to the Service Manager command line.

2. Type your old password in the **Old password** text box.
3. Type your new password in the **New password** text box.
4. Re-type your new password in the **Confirm new password** text box.
5. Click **OK**.

To change your password by using the ESS client, follow these steps:

1. Sign in to the ESS client.
2. Click **Change Password** on the navigator.

3. Enter your old and new passwords, and save the change.

The new password is required the next time you sign in. If you start another session while the current session exists, you must use the new password. If you have a connection defined in the Connections dialog box, you must change the sign-in password in that dialog box and click **Apply** to save the connection information.

**Reset my password**

**Applies to User roles:** All roles

After you set up your security questions, you can reset the password on your own if you forgot the password. Follow these steps:

1. Access the web client login page, and then click the **Forgot Password** link.
2. Type your user name or email address, and then click **Continue**. The system directs you to the password reset URL and displays your security questions.
3. Enter your answers to the security questions, and then click **Continue**. The system sends a verification code to the email address as you specified.
   **Note:** If your answers are incorrect and you have exceeded the maximum retry attempts, the system sends an email notification and informs you that the reset password function is temporarily locked. Wait for some time till the system automatically unlocks this function. You may need to consult the system administrator for this time duration.
4. Enter the verification code on the login page, and then click **Continue**.
   **Note:** Make sure to enter the verification code before it expires.
5. Enter and then confirm your new password, and then click **Continue**.
6. The system displays that your password has been successfully reset. Click **Done** and the system directs you to the login page again. You can then sign in immediately using the new password.

**Tip:** If the system administrator updates the number of security questions, you can still reset the password based on your security questions settings before. Later, when you sign in to Service Manager using a new password, the system informs and prompts you to update the security questions accordingly.

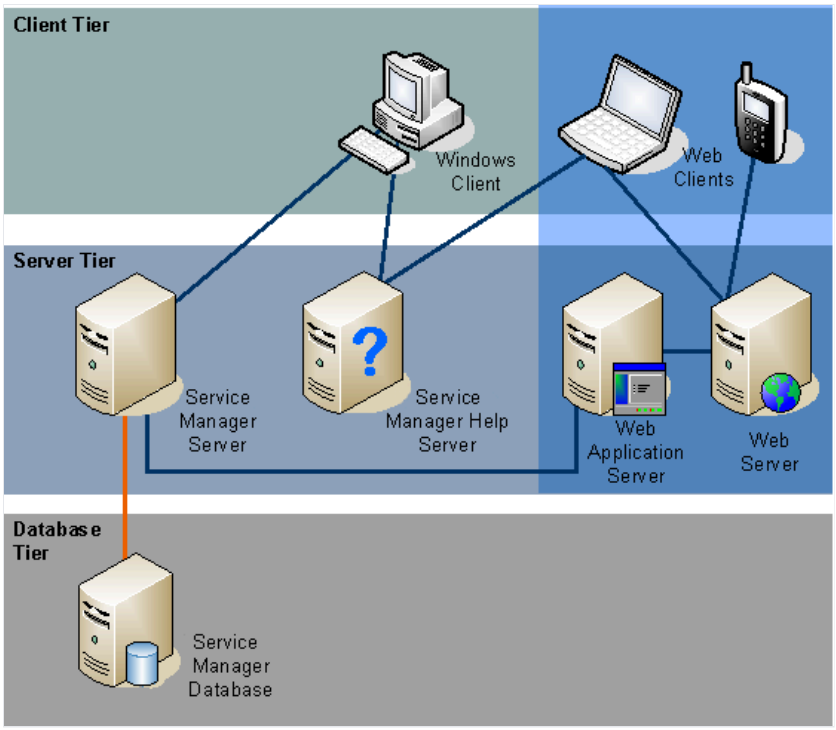
**Note:** If you have not set up your security questions before, you can't start the reset process. In this case, contact the system administrator to reset your password.

## 3. Clients

Service Manager includes the following two client interfaces:

- A web client that runs in a web browser and accesses the Service Manager server through a URL.
- A Windows client that runs as an application on a Windows host and accesses the Service Manager server through a client/server connection.

The following diagram illustrates how each type of client connects to the Service Manager server.




Both clients provide end users access to the Service Manager applications. The Windows client, however, offers additional functionality, such as the ability to create forms and access the database dictionary. The Windows client is intended for administrators and implementers.

### Related topics

Windows client
Web client
Accessing Service Manager

## 3.1. Web clients

The web client offers the same basic features as the Windows client, but provides a different look and feel, tree view navigation, and other features.

The web client is the only Service Manager client that runs on a non-Windows platform. The web client supports accessibility and meets Section 508 requirements. For more information, see Accessibility.

#### Web client setting requirements

Verify the following before using your web client.

- Enable cookies.
- Enable JavaScript.
- Enable pop-ups in all browser versions. You may need to add the Service Manager URL to the browser exception list.

Verify that your browser version is compatible by checking the Service Manager Support matrix.

**Note**

Note

- The web client doesn't support multiple logins. Therefore, you should ensure that you only log on to Service Manager once in your browser, as browser sessions aren't independent and multiple logins result in unpredictable system responses.
- Do not use the buttons (such as Refresh, Back, and Forward) on your browser toolbar or their keyboard shortcuts (such as Ctrl+R, Ctrl+Left, and Ctrl+Right) to perform Service Manager actions. Instead, use the buttons on the Service Manager interface, as described in List and detail toolbars.

### Web client views

The web client offers several different interface views. The following views are included in out-of-box installations of Service Manager:

| Web client view | Description |
|---|---|
| Power user | This web client view is the standard interface. It most closely matches the Windows client and is intended for Service Desk technicians, managers, and administrators. |
| Accessible | The accessible view is intended for Service Desk technicians, managers, and administrators who need accessibility features. The accessible view includes the following features:<br>• Larger default fonts<br>• The ability to increase the browser font size<br>• Larger form spacing<br>• Simplified navigation<br>The following differences exist between the accessible and standard web clients views:<br>• The accessible web client doesn't support the graphical workflow feature.<br>• Thread navigation links (the tabs that identify open forms) aren't available in the accessible web client.<br>For more information about web client accessibility, see Accessibility.<br>Be aware that Service Manager accessibility has some limitations. See Limitations. |
| Employee self-service (ESS) | The ESS view provides a simplified Service Desk interface for users to create, view, and update service requests. Intended for end-users as an entry point to Service Desk.<br>**Note**<br>Note The ESS interface is designed so that users can easily understand what they need to do; therefore, no Service Manager help icon is available. If users want to see the online help, their System Administrator can provide them with a URL that they can enter into their web browser. |
| Accessible self-service | This web client view is a modified version of the self-service user view with high contrast colors and larger fonts. This view is intended for end-users who need accessibility features. |

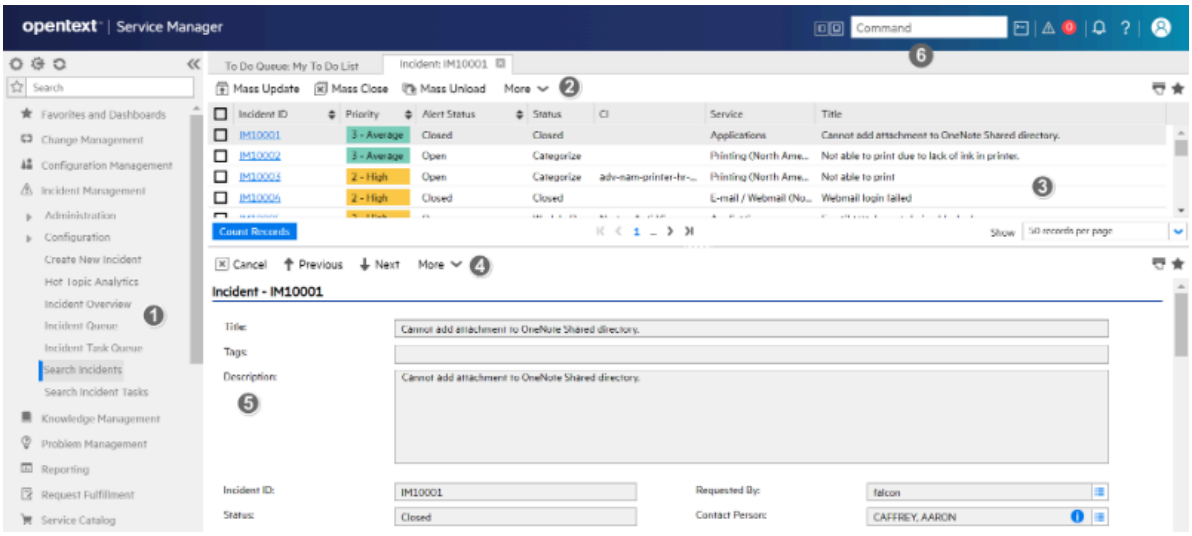
Each of these views has default behavior and appearance. Administrators can customize the views by setting web client preferences in the web.xml file. In addition, administrators can customize the web client interface by creating custom style sheets and images.

Each view is accessed by a different kind of URL. For more information, see Accessing Service Manager.

> **Note**
>
> Note Service Manager also has a web interface for mobile clients. The mobile client interface runs on smartphones and supports key tasks of the Change Management and Incident Management applications. This interface requires setup and configuration. For more information about how to do this, see Service Manager Mobile Applications.

### Parts of the web client interface

The following image illustrates where you can find the main components of the Service Manager web client interface:




The numbers in the image above represent the following components.

| Component | Description |
| --- | --- |
| 1 | **System Navigator**. Provides navigation through the Service Manager applications and their forms. |
| 2 | **Record List toolbar**. Contains commands that affect the record list portion of the editor. Click the **More** button to access additional commands. |
| 3 | **Record List area**. The record list is populated after you execute a **Search** command. The currently-selected record that in the record list is displayed for editing directly below the record list, in the record detail area. |
| 4 | **Detail List toolbar.** Contains commands that affect the record that's opened for editing, in the record detail area. |
| 5 | **Record Details area.** Displays the record that's selected in the record list area. |
| 6 | **Command/Search line**. Allows you to enter shortcut commands or record's prefix and IDs. Click the **Command/Search line** toggle button to switch between the Command line and the Search line.<br>**Note**<br>Note The **Command/Search line** toggle button is available in the web client only and is disabled by default. For more information about how to activate this button, see the "Activate the command/search line toggle button" topic. |

### Horizontal and vertical layouts

The Service Manager web client provides the following two layouts for a list-detail page:

- Horizontal layout: In the horizontal layout, the record list pane is above the record detail pane. This is the default layout of the list-detail page.
- Vertical layout: In the vertical layout, the record list pane is on the left side of the record detail pane.

You can use the  switch to toggle between the two layouts.

Your web browser stores the following layout information of your current login session:

- Horizontal layout or vertical layout
- Columns and the width of each column

In a newly opened page or in your next login, the web client can keep the same layout. If you clear the cache of the web browser, the web client uses the horizontal layout.

#### Hover information in the vertical and horizontal layouts

In both layouts, the content in the first column appears as links. When the mouse hovers over one of these record links, a hover window appears and displays the values of all columns. Hidden columns are displayed below a separation line.

To customize the fields that the hover window can display, follow the instructions in Record list.

**Related topics**

Clients
Windows client
Differences between the Windows and web clients
Accessing Service Manager
Web client keyboard interactions and shortcuts

## 3.2. Windows client

Service Manager includes both Windows and web clients. The Windows client is intended for administrators and implementers, while the web client is suitable for most end users. The Windows client enables you to design forms and work with the dbdict; capabilities that aren't available to web client users.

The Windows client includes the following features:

**Interface**

- A dynamic user interface that you can customize (by using preferences) to change the way you interact with perspectives and views
- Windows client update capability

**Client/server connection**

- A Connection dialog that enables you to define multiple connections to different servers and saves them from one session to another
- Automatic connection for connection profiles with the automatic logon option enabled

**System Navigator**

- Favorites folder to save shortcuts to frequently used applications, forms, and queries
- Simplified access to forms, files, and records through the System Navigator

**Applications**

- Application-specific user interface elements
- Charts and dashboards to show data graphically

**Help**

- Field help
- Help for end users and administrators

**Other features**

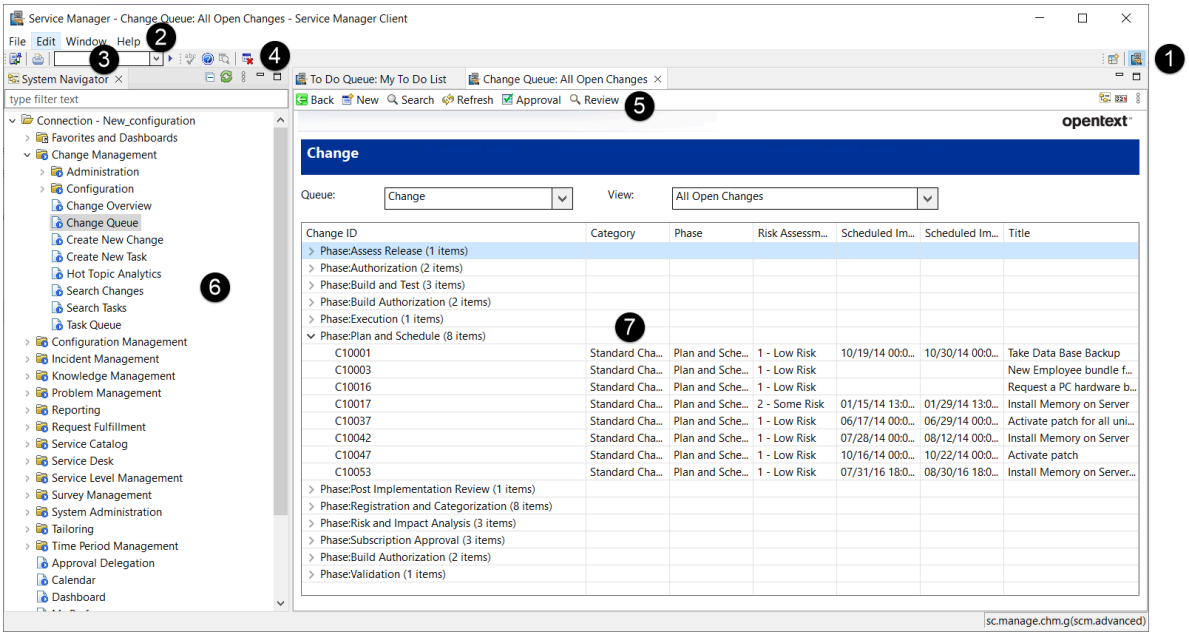
- UTF-8 (Unicode) support to display data in any language
- SOAP-based client/server communication
- Automatic reconnection of client/server sessions after brief network connection disruptions

**Windows client features**

| Feature | Functionality |
| --- | --- |
| Window design | You can drag and drop client components to arrange them in the following ways:<br>• Drag the title bar (or tab) of any view to a different position within the client window.<br>• Drag any view or editor outside the client window.<br>• Drag the shortcut bar to dock on the right, left, or bottom of the client window.<br>• Resize any view or editor by dragging the split bar up, down, or sideways.<br>• Drag any view to a vertical or horizontal position in the client window. |
| Views | Views have a minimize and maximize button. Click **Maximize** to fill the client window. Click **Minimize** for the view to appear as a tab in the lower right section of the editor. |
| Views for all users | Dashboard<br>Messages<br>Secondary System Navigator<br>System Navigator |
| Views for administrators | Console<br>Detail Data Detail Form<br>List Data<br>List Form<br>Request<br>Response<br>Stopwatches |
| Editor tabs | If you drag multiple views or editors to the same location, they arrange themselves in a tabbed notebook format. |
| Perspectives | The Administration perspective contains views for administrators only. You must have the Admin plug-in installed to access this perspective. |

### Parts of the Windows client interface

The following image illustrates where you can find the main components of the Service Manager Windows client interface:




The numbers in the image above represent the following components.

| Component | Description |
|---|---|
| 1 | **Shortcut bar**. Shortcuts to different perspectives and views. For more information, see *Shortcut bar*. |
| 2 | **Menu bar**. Menus of commands that apply to or affect the Windows client. |
| 3 | **Command line**. Quick access to areas of Service Manager brought up by entering the command for them. For more information, see the Enable an operator to see the command line topic. |
| 4 | **Toolbar**. Commands and operations that aren't specific to the current editor. |
| 5 | **Editor Toolbar**. Commands and operations that are specific to the current editor. The editor is determined by the type of entry selected in the System Navigator tab. |
| 6 | **System Navigator**. The forms available to the user, grouped by their applications. |
| 7 | **Form open for editing**. The current form being edited. |

The Windows client interface also has *views* and *perspectives*. See the related topics for more information.

All of the views, perspectives, and editors are collectively called the *workbench*.

#### Workbench

> **Note**
>
> The workbenches is a Windows client-only feature.

The workbench is the window that contains all the Service Manager interface. Its primary panes are views (such as System Navigator or Properties) and the editor.

The workbench uses perspectives to organize the views and the editor into task-oriented configurations. It changes the title bar according to your perspective and what form you use. You can also configure the workbench in the preferences. The form name appears at the bottom right corner of the workbench.

When you open a connection to a different server or to the same server, you open a new workbench. There is no limit to the number of workbenches you can have open at one time, although you can only run one client at a time.

#### System Navigator

The System Navigator is a visual hierarchical structure that enables you to navigate through Service Manager applications and to drill down into the system to access forms and other information. You can drag and drop favorites (such as queries, lists, and forms) into the System Navigator view, store charts and dashboards, access Service Manager applications, and complete database tasks.

Administrators can configure user roles with permissions that enable or disable viewing of System Navigator components. For example, some users should not have direct access to database information or be able to create public folders.

Some available actions appear on the System Navigator contextual (right-click) menu. The list of actions varies, depending on user roles and the current task. The following table describes two of the commands found in the System Navigator contextual menu.

| Command | Definition |
| --- | --- |
| New (favorites or dashboards) | Creates a new folder or dashboard. |
| **Refresh**  | Refreshes the display.<br>**Note**<br>Do not refresh the System Navigator by refreshing your browser. Instead, use the Service Manager System Navigator Refresh button. |

#### Client differences: System Navigator

The following differences between the System Navigator of the Windows client and of the web client occur because the browser environment is different from the Windows client environment.

| Feature | Windows client? | Web client? | Comment |
| --- | --- | --- | --- |
| Create favorites | Yes | Yes | Favorites are restricted to lists, detail, and menu screens.<br><br>**Windows client**<br>From a record list, application, or form, drag the System Navigator icon to the Favorites folder, or to a subordinate folder.<br>Web client<br>From a record list, click **Save as View**. |
| Create subordinate nodes | Yes | Yes | You can customize the System Navigator by adding a folder.<br><br>**Windows client**<br>Right-click the parent node to make a new folder.<br>**Web client**<br>1. Click **Manage Favorites** .<br>2. Expand the Favorites directory to locate the parent node. Click **New Folder** .<br>3. Type the name of the new folder and click **OK**. |

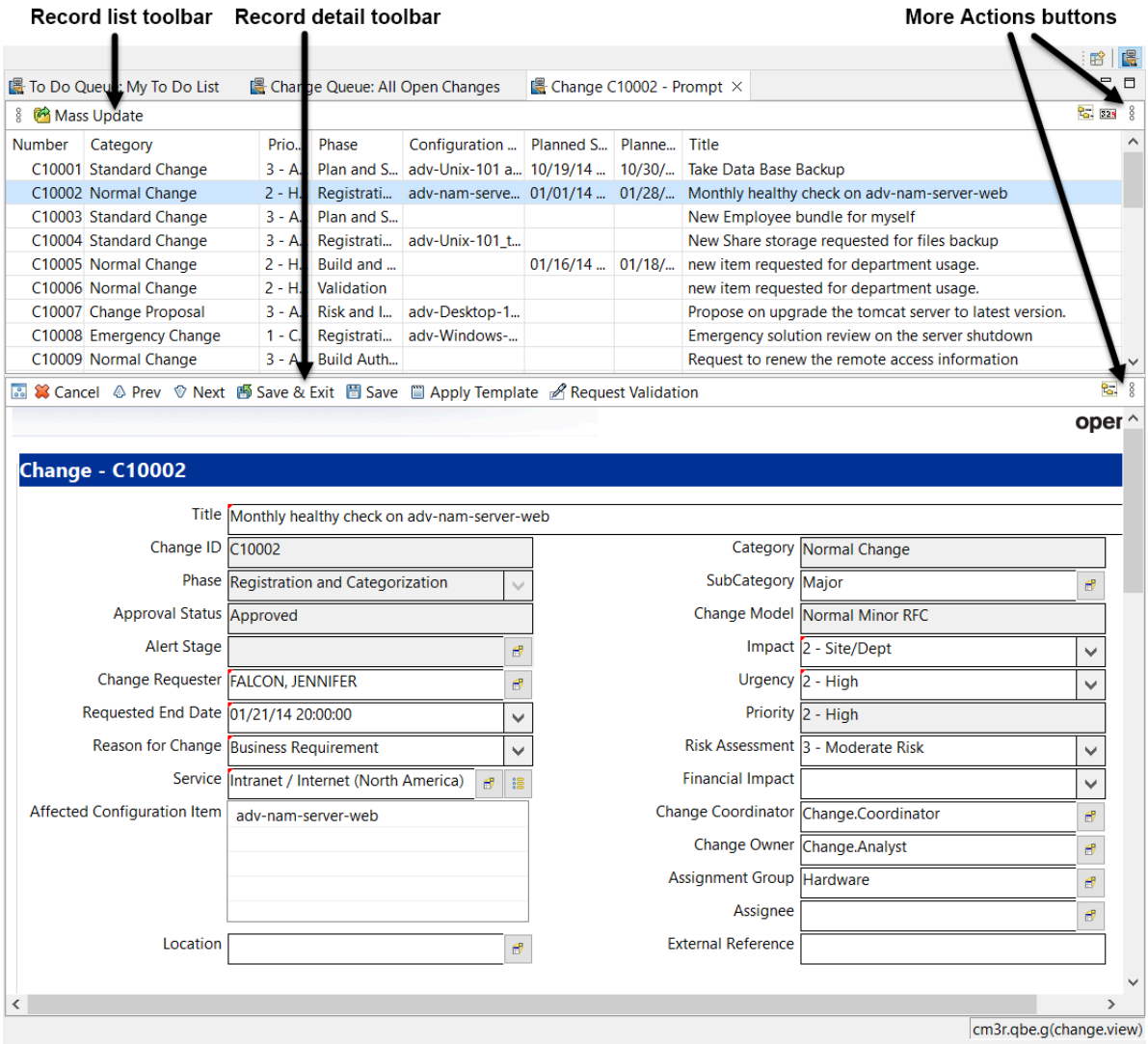
| | | | |
|---|---|---|---|
| Delete nodes | Yes | Yes | You can customize the System Navigator by removing nodes.<br><br>**Windows client**<br>Right-click to delete nodes from the System Navigator.<br><br>**Web client**<br>1. Click **Manage Favorites** .<br>2. Expand the Favorites directory, select the node to delete, and then click **Delete**. |
| Collapse All | Yes | No | You can collapse subordinate nodes when you select a parent node.<br>**Windows client**<br>Right-click the parent node, and then click **Collapse All** to close all subordinate child nodes.<br>**Web client**<br>**Collapse All** is unavailable for the web client. |

| | | | |
|---|---|---|---|
| Search field | Yes | Yes | You can perform searches in system menus and favorites.<br>**Windows client**<br>• The search line has a Clear icon **x**. Click this icon to clear the keyword.<br>• If you select a child node in the search result, the selection remains unchanged after clearing the keyword with the Clear icon.<br>• If there is no selection in the search result, the system restores the original system menus and favorites after clearing the keyword with the Clear icon. If you clear the filter text by pressing Backspace, the system restores the original system menus and favorites after clearing the keyword even if you have selected a child node.<br>• The search result lists the menu entries that begin with the search string.<br>• The system highlights the entire menu entry that consists of the keyword.<br>**Web client**<br>• The search line has no Clear icon. Press Backspace to clear the keyword.<br>• When the search result locates a |

| | | | |
|---|---|---|---|
| | | | parent node (such as Change Management), click this parent node to expand.<br>• The search result lists the menu entries that consist of the search string.<br>• The system only highlights the keyword in the search result. |

**Editor**

The following image illustrates different parts of the editor:




The editor is the primary pane in which you can edit forms and work in Service Manager applications. Service Manager opens new editors as needed. You can drag and drop editors to display them side–by–side or stacked in a tabbed notebook. Click any tab to move that editor to the top of the stack.

#### Editor toolbar

The editor toolbar contains the tools that are used by associated applications. The tools that are available change depending on the Service Manager application and the current task.

The editor toolbar also contains the following icons:

- The **Favorites** icon ( ). Drag this icon to the Favorites folder to create a shortcut to forms or applications.
- The **More Actions** icon ( ). Click this icon to display a list of related Service Manager tasks.
- The **Menu** icon ( ). Click this icon to rearrange the editor space. For example, you can relocate or hide the record list.
- Application icons. Click these icons to complete standard Service Manager actions.

The record list toolbar contains the following icons:

- The **Favorites** icon ( ). Drag this icon to the Favorites folder to create a shortcut to forms or applications.
- The **Print** icon ( ). Click this icon to send the current form to a networked printer.
- The **Count** icon ( ). Click this icon to display the number of records in a record list.
- The **More Actions** icon ( ). Click this icon to display a list of related Service Manager tasks.

#### Service Manager toolbar

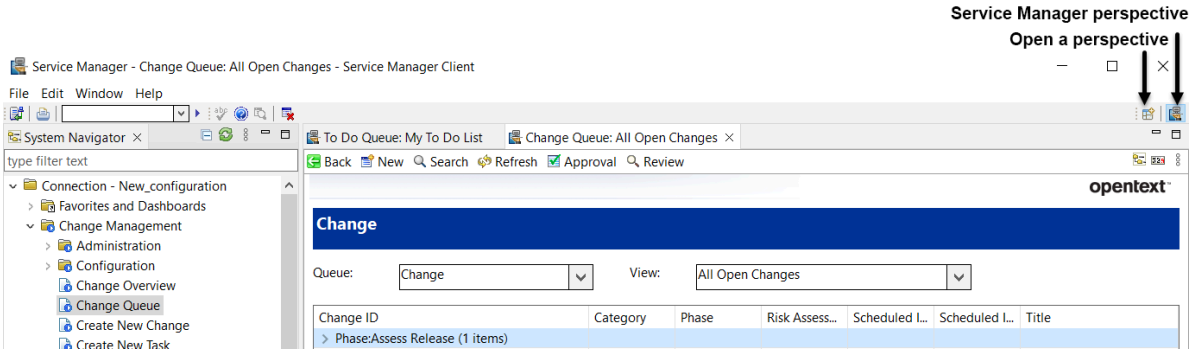
The Service Manager toolbar in the Windows client is located just below the main Menu bar. It contains buttons that launch the Connections dialog, print, check spelling, open the field help, and close Service Manager tools to use with Service Manager applications.

The toolbar also displays the Service Manager command line, if that privilege is granted on the Startup tab of your operator record.

#### Shortcut bar

> **Note**
>
> The shortcut bar is a Windows client feature.

When you create perspectives and collapsed views, the icons for these views are displayed in the shortcut bar. Storing the System Navigator, on the shortcut bar frees up more space for the editor.




To minimize a view that you opened from the shortcut bar, click the editor.

| Icon | Option | Definition |
|---|---|---|
| – | Minimize option | Minimizes the pane into the shortcut bar |
| × | Close option | Closes the view and remove it from the shortcut bar |

Each view has additional options and menus, according to its purpose.

#### Fonts

Using font styles (for example, bold or italics) affects the spacing of the characters on a form. If labels in forms are truncated, select a different font, style, or size. You can also increase the width of the form to accommodate a different font, style, or size. To do this, specify an increased width percentage in the Service Manager preferences.

#### Help menu

The Help menu has six selections.

| Menu item | Description |
|---|---|
| Help Contents | Open the Help window. |
| Help on Field | View pop-up help for the current field. Press Ctrl+H as a shortcut, or click the **Help on Field** icon on the toolbar. This is disabled on the out-of-the-box system. |
| Help on Form | View pop-up help for the current form. This is disabled on the out-of-the-box system. |
| About Service Manager Client | Access version and copyright information; link to the company website or Eclipse.org. You can also link to details about features, plug-ins, and configuration. |
| Service Manager Server Information | Access information about the version, network address, host, port, hardware, and operating system. |

#### Forms

Service Manager displays forms in the Service Manager workspace. Each form has an associated More Actions menu, which contains additional options for that form. The available options vary, depending on the current Service Manager application and user profile rights.

To access menu options in a Windows client, right-click the form, and then choose an option from the Context menu. Or, click the pull-down menu icon in the upper right corner of the workspace toolbar.

To access menu options in a web client, click **More**, and then choose an option from the More Actions menu.

#### Link label

A link label provides a caption and an icon that links to another object, such as a wizard. Link labels are enabled or disabled depending upon your actions in the associated object. For example, to add a new filter to filter a long list of incidents, you create a filter in the Conditions table. If you specify no other conditions, only the "Add New Filter Criteria" and "Add New Compound" links are enabled. After you add new filter criteria, additional links (Edit Selection and Remove Selection) are enabled.

When a link is enabled, it is displayed in blue type and with a bright icon. When a link is disabled, the link's icon and text are gray or dull.

#### Drag–and–drop feature

The drag–and–drop feature moves or copies objects within the System Navigator in the Windows client only. Your user rights determine what objects you can move and copy.

| Option | Definition |
| --- | --- |
| **Copy**  | When you copy an object in the System Navigator, the mouse pointer icon changes to include the pointer arrow, a small rectangle, and a plus sign. |
| **Move**  | When you move an object in the System Navigator, the mouse pointer icon changes to the pointer arrow and a small rectangle. |
| **Forbidden**  | When you don't have the user rights to move or copy, the mouse pointer icon changes to a circle with a slash. |



**Note**

In the web client, you can use **Manage Favorites** to move or copy nodes.

**Windows client keyboard shortcuts**

The following are the Windows client keyboard shortcuts.

| Keyboard shortcut | Action |
| --- | --- |
| Ctrl+F6 | Cycle through multiple Editor panes.<br>This is similar to the Alt+Tab functionality in Windows. |
| Ctrl+F7 | Cycle through multiple views.<br>This is similar to the Alt+Tab functionality in Windows. |
| Ctrl+F8 | Cycle through multiple perspectives.<br>This is similar to the Alt+Tab functionality in Windows. |
| Ctrl+H | View help on field. |
| Ctrl+S | Spellcheck. |
| Ctrl+Shift+M | Expand a scrollable text field by opening a separate window that magnifies the text for easier viewing and editing. |
| Ctrl+Shift+Tab | When the current focus is on a notebook page, this keyboard shortcut allows you to move back to the previous notebook page. |
| Ctrl+Tab | When the current focus is on a notebook page, this keyboard shortcut allows you to select the next notebook page. |
| Ctrl+U | Expand or collapse the status of the list frame in a list detail page. |
| F12 | Transfer focus from the left navigation pane to the main viewing pane. |

**Related topics**

Clients
Web client
Differences between the Windows and web clients
Accessing Service Manager
Sign in from the Windows client

### 3.3. Differences between the Windows and web clients

The web client offers the same basic features as the Windows client. Each client also has unique features.

| Feature | Windows client? | Web client? | Comment |
| --- | --- | --- | --- |
| Accessibility support | No | Yes | |
| Charts | Yes | Yes | You can't change chart colors in the web client. The Context (right-click) menu is unavailable. You can use favorite properties to edit chart properties if you make the chart a favorite. |
| Collapsible Groups | No | Yes | In the web client, groups can be displayed as collapsible sections on forms. In the Windows clients, groups are expanded and can't be collapsed by the user. |
| Computer Telephony Integration (CTI) | Yes | Yes | |
| Count records | Yes | Yes | **Windows client**: Click **Count** on the right side of the record list toolbar.  The Windows client displays the total number of records to the right of the Count icon.<br>**Web client**: There are two ways to count records in the web client.<br>1. Click **More**, and then **Count**. The web client displays a message with the total number of records<br>2. From the paging control on the bottom of the record list frame, click the **Count Records** button. The web client displays the total number of records to the right of the Count Records button in the paging control. |
| Database administration | Yes | No | |

| | | | |
|---|---|---|---|
| Debugger | Yes | No | |
| Dynamic Data Exchange (DDE) support | Yes | Yes | **Note**<br>Note DDE is available in the web Client only in the CTI interface. |
| Employee self-service mode | No | Yes | A simplified interface to submit requests to Service Desk. |
| Export lists to Excel | Yes | No | **Note**<br>Note The web client can only export lists to text files, which you can manually import to Excel. |
| Forms designer | Yes | No | |
| Multi-platform client support | No | Yes | Use a supported web browser listed in the Service Manager Support Matrix. |
| More Actions menu | Yes | Yes | **Windows client**: Click the down-arrow icon in the upper right corner. ▾<br>**Web client**: Click the **More** button on the detail or record list toolbar. |

| | | | |
|---|---|---|---|
| Right-clicking (to open the More Actions menu) | Yes | No | **Windows client**: Right-clicking in a form displays the More Actions menu for that form, and right-clicking in a record list displays the More Actions menu for that record list.<br>**Web client**: Right-clicking a form or record list displayed in the web client doesn't open the More Actions menu. Instead, the web browser's context menu opens. |
| System tailoring | Yes | No | |
| System Definition utility | Yes | No | You must enable the Administration plug-in. |
| URL navigation | No | Yes | |
| Mandatory fields | Yes | Yes | **Windows client**: Indicated by a small red triangle in the upper left hand corner of the field<br>**Web client**: Indicated by a red asterisk to the left of the field |
| Process Designer workflow | No | Yes | |

**Related topics**

Clients
Web client
Windows client

## 4. Searching records

Service Manager enables you to search for records and content in a number of ways.

**Advanced Search**

Advanced Search enables you to search for a record or group of records in a table. The Advanced Search form enables you to specify a range of search criteria to narrow a search.



**Note**

Note The tables in which you can perform an Advanced Search depend upon the access rights that you are assigned. For example, if you have limited rights, you may be limited to searches in interaction records or incident records only. However, a system administrator can search most, if not all, tables that are available for record searches.

**Quick Search**

Quick Search enables you to search for individual records directly from the Service Manager UI. To access the Quick Search search line, click the Command icon on the top right of the Service Manager UI, and then click the Command/Search line toggle button.

**Note**

Note

- The Command/Search line toggle button is available in the web client only and is disabled by default.
- After you execute the command/universal search action in the Edge browser, the whitespace characters in the search string will not disappear if you have pressed the spacebar and typed whitespace characters.

**Smart Search**

You can search for content from a range of sources inside and outside of Service Manager by using Smart Search. To access the Smart Search search box, click the Smart Search icon on the top right of the Service Manager UI.

**Note**

Note The Smart Search icon is displayed only to customers who have Smart Analytics enabled.

### Related topics

Search for a record by using Advanced Search
Search for a record by using Quick Search
Search for content by using Smart Search

## 4.1. Search for a record by using quick search

**Applies to User roles:** All roles

The Quick Search feature in Service Manager provides a search line from which you can access single records. This enables you to enter the record ID and open the associated record directly, without having to navigate through menus.

To search for a record using the Quick Search search line, follow these steps:

1. If you have access to the Service Manager command line, click the **Command** icon, and then click the **Command/Search line** toggle button to switch to the search line. If you have no command line access, a search line box is already available.

   **Note**

   Note The **Command/Search line** toggle button is available in the web client only and Service Manager disables this button by default. To activate this button, see Activate the command/search line toggle button.

2. Type the record's prefix and ID in the search Line, and then press Enter.
   For example, type `IM10005` in the search line, and then press Enter. Record number `IM10005` opens. However, if you type `IM` in the search line and then press Enter, Service Manager displays no records because you specified no ID.

**Note**

Note

- The default record prefixes that you can search for include SD (interactions), C (change), IM (incidents), and CIs (configuration items). For more information about how to add a customized prefix for a record type, see Add a prefix for a record type.
- You can enable more record types that already have a prefix (such as Problem) searchable from the search line.
- If you try to search a record with the **Command/Search line** toggle button set to the command line, the system displays the following message: `Unrecognized command. Reenter`.
- If you try to search for a table or a record that doesn't exist in Service Manager with the **Command/Search line** toggle button set to the search line, the system displays the following message: `No records found`.

### Configuring more record types for quick search

Out-of-the-box, the search line only supports searches by record ID (with ID prefix) for the following record types:

- SD (interactions)
- C (change)
- IM (incidents)
- CI (configuration items)

You can enable more record types (for example, Problem records) for quick search.

To configure a record type for quick search, follow these steps:

1. Sign in to Service Manager as a system administrator.
2. Type `gl` in the command line, and press Enter.
3. In the List Name field, enter `UniSearch Types`.
4. Click **Search**.
5. Add the table of the record type and the record type prefix to the global list. For example, for Problem records, do the following:
    1. In the Value List field, add **rootcause**.
    2. In the Display List field, add **PM**.

    **Note**

    Note Make sure the types and prefixes are within double quotation marks and separated by commas with no spaces between them. For example: `{"incidents","cm3r","probsummary","request","rootcause"}` and `{"SD","C","IM","RF","PM"}`.
6. Click **Save**, and then click **More** > **Rebuild Global List**.
7. Click **OK**.
8. Sign out and then sign in to the web client again.
9. Click the toggle icon to switch the command line to the search line, and enter a problem record (such as `PM10001`), and then click the **Search** icon. Service Manager displays the problem record.

### Related topics

Search for a record by using Advanced Search
Search for content by using Smart Search

## 4.2. Search for a record by using advanced search

With Advanced Search, you can select from several types of search criteria to limit a search to help you locate a specific record or group of records when there is a large number of records in a table. The search provides two entries where users

can specify a table to search (Look For) and a view to search (View). In addition to these two entries, the following tabs are available to further define a search.

> **Note**
>
> Note Not all these tabs are available for all tables, as not all tables include all the search criteria available on each tab.

- The **Table** tab identifies the table being searched (for example, Change or Incident). This tab includes the main fields for the table being searched. Smart Search is used so that the search automatically includes the ID prefix ( IM for incident records) as part of a record number search. For example, when you search for record number 1003, the search will locate IM1003. This tab also includes more choices (checkboxes) that provide additional search criteria to further limit a search.
- The **Advanced Filter** tab enables users to create and save filters for advanced searches that locate a specific set of records, such as all interaction records closed after a certain date. A wizard helps users construct a complex query that combines various search criteria and compounds to create an advanced search. The selections include all available fields for the table you are searching with comparison criteria based on the data type of the field you choose. The **Advanced Filter** tab also includes a **Not** checkbox, which gives you the ability to negate any search criteria or whole compounds.
- Where applicable, there is also a **Text Search** tab that enables you to search for text in IR Key fields defined for the table in the dbdict. For example, an IR Key that has a description defined could be searched for a text string.

**Example: Search for a record**

**Applies to User roles:** All roles

This topic demonstrates how to search for an incident record as an example of how to search for any type of record or set of records in Service Manager. Most of the actions that are described below are common to any record search. However, the search criteria that are available will depend on the fields in the table that you are searching.

To search for an incident record, follow these steps:

1. In Incident Management, click **Search Incidents**.
2. Use the search form to specify additional search criteria to make your search more efficient. For example, in the View field, select **Open incidents assigned to my group**.
3. For more search options, click the following tabs:

| Tab | Action |
|---|---|
| **Incident** | You can leave any or all fields on this tab blank. For the fastest searching, use the fields identified as key (indexed) fields at your site. If these fields aren't clearly identified on the form, ask your Service Manager administrator for a list of key fields. You can also select some options that provide additional search criteria. |
| **Advanced Filter** | Add new filter criteria to your search.<br>• To write your own search query to sort the results by keyed fields, click **More** or the **More Actions** icon, and then select **Expert Search**.<br>• To specify the field, comparison, and value for the filter, click **Add New Filter Criteria**. |

4. Click **Search** to display the list of records that matches your search criteria. Select a record in the list to view it.

**Note**

Note If Service Manager generates a message that states no records were found, you can broaden the search by eliminating some of the search parameters. You can sort a record list by clicking the header of the column that you want to sort. For example, to sort the list by status, click the **Status** column header.

5. To modify or end your search, choose one of the following options:
   • To clear the search criteria you selected and return to a blank search form, click **Clear** or press Alt+F7.
   • To start the Add a New Device wizard and add a new business service or device to your configuration items, click **New** or press Alt+F1.
   • To restore the search criteria you have cleared, click **Restore** or press Alt+F11.
   • To end your search and return to the previous screen, click **Cancel** or press Alt+F3.

You can choose to create a new view, so that you have the query readily available the next time you want to search for these records.

- In the web client, click **More**, and then select **Save List as View**.
- In the Windows client, click the **More Actions** menu (the down-arrow button on the far right of the toolbar), and then select **Save List as View**.

### Advanced filter

Service Manager provides a standard search screen for the main tables of each module. The search screens provide the main fields to perform the search on the first tab as well as the capability to perform more complex queries using the Advanced Filter tab. The Advanced Filter tab maintains search functionality when migrating from Service Desk to Service Manager.

The Advanced Filter tab enables you to specify queries against any other field in the table in addition to the fields that appear on the initial search screen. You can use the Advanced Filter wizard to combine the fields with other expressions using either "and" or "or" operators as well as parentheses to create complex queries. The comparison operators will vary depending on the field type that you select. On array fields, the values will also change depending on whether you are searching on an array field that has been configured to be used as an IR key or as a regular array. The "Is Null" and "Not Equal to" operators allow you to search for records where a field is equal to a specified value or is null. A "Not" checkbox appears as part of the wizard to help you create negated queries on any field except arrays that are configured as IR keys.

When you create a filter, you should consider that the execution time for the query may vary, depending on the following factors:

- The number of records in the table
- The complexity of the query
- The use of non-keyed fields

We recommend that you try to avoid queries against non-keyed fields and avoid using multiple negations that would cancel each other, such as not(not <field> <operator> <value>).

### Create an advanced filter

**Applies to User roles:** All roles

The following example demonstrates how to use an advanced filter to search for a record or set of records in Service Manager. In this example, a new advanced filter is created to search for a set of records that satisfy the specified search criteria for incident records.

To create an advanced filter, follow these steps:

1. Click **Incident Management** > **Search Incidents**.
2. Click the **Advanced Filter** tab, and then click **Add New Filter Criteria**.

3. Select a field from **Field in Incident** to be part of the search criteria. For example, select **Assignment Group**. If you don't find the fields that are appropriate for your search criteria (for example, you want **Manager Name** as a field ), follow these steps:
   a. From **Field in Incident**, select a field that's linked to a table that contains the field that you want to add as search criteria. In this example, select **Assignment Group** from **Field in Incident**, as the assignment table contains both the **Manager Name** and **Assignment Group** fields.
   b. Click **Use Multi Level Chooser**, select **Manager Name** from **Incident Assignment Records**, and then click **Next**.

   **Note**
   Note **Incident Assignment Records** may not appear if the field you selected in step a is not linked to any other table.

4. Select a comparison from the drop-down list (for example, **Equals**). You can also negate the operator field you select by selecting the Not checkbox, except for an IR field, which can't be negated.
5. Type or use Fill to select a value for the **Value** field. For example, enter the manager's name, and then click **Next**.

The new advanced filter is created, and the system now uses the filter to search for and display a list of records that satisfy the search criteria you provided.

Related topics

Searching records
IR keys and non-IR keys
Enable IR search for a file
Regenerate IR keys

### 4.3. Search for content by using Smart Search

**Applies to User roles:** All roles

**Important**
Important Smart Search is available only if you have Smart Analytics enabled in Service Manager.

The Smart Search search box in Service Manager enables you to use the Smart Search feature of Smart Analytics to search for content directly, without having to navigate through menus.

To use Smart Search, follow these steps:

1. Click the **Smart Search** icon on the top-right corner of the Service Manager UI to display the search box.
2. Enter the desired search terms in the Smart Search box, and then press Enter. For example, type IM10005 in the Smart Search box, and then press Enter.
   The Smart Search window opens and a list of documents that match your search terms is displayed. If no result is displayed, type a new search string for your search or use Advanced Search to specify additional search criteria.

   **Note**
   Note You need to set up and verify the server connectivity for multiple servers and connectors. For detailed information to configure and monitor the connectors and servers, see Configure connectors.

3. Optionally, configure the following additional search conditions to narrow the Smart Search results, and then click **Search**.

| Field | Description |
| --- | --- |
| Search within results | This option is only activated when there are results returned. Click this option button to add it as a filter to the filter panel on the right. You can click the remove button on the filter to remove it.<br>For example, type Service in the search box and then click **Enter**. After the results are returned, you click this option and a **Previous Search: Service** filter is added to the search results panel. Then, type **Server** in the search box and then click **Enter**. The results that are returned are the same as those returned if you search for **Service AND Server**. |
| General filter | Click the **Filter** button to cascade the general filter to filter the content by the last modified time. The option you select is added to the filter panel on the right. You can click **Clear All** to remove all the filters. |
| Library | Specifies the knowledgebases for your search repository. |
| All | Click the checkbox to select all the knowledgebases as your search repository. |
| *<Knowledgebase name>* | Click the checkbox to include this knowledgebase to your search repository.<br>**Note**<br>Note Your library options in Smart Search are remembered for your next login. |

4. Click a document title or identifier to open it.

In the search results, if you click the button to vote up or vote down an article, this will affect the ranking of the search result next time. This is not configurable. Also, if the document is re-used for a solution, the ranking for this document is also affected. For detailed information, see Use adaptive learning in Smart Search.

**Related topics**

Search for a record by using Advanced Search
Search for a record by using Quick Search
Use Smart Search as a general search tool

## 5. Navigating the UI

### The System Navigator

The System Navigator is a navigational structure to view menus, tables, fields, forms, and other integral components. You can also create new nodes that give you quick access to triggers, links, incidents, and personal favorites. The System Navigator simplifies switching forms, applications, and tools and enables administrators to view forms, files, and links quickly.

### Favorites

Favorites are frequently-visited forms, queries, business services, or links that you want to preserve for quick navigation. If you bookmark a favorite query, that query runs each time that you click that link. You can create, rename, copy, or delete the favorite. If you add a query to your Favorites list, Service Manager re-runs the query each time that you select it. Service Manager supports both public and private Favorites lists.

### Personalization

You can define your own business view of records, incidents, contacts, and so on. By customizing the System Navigator and adding folders for different activities or watch lists, you can easily review all your high priority information.



**Note**

Note Do not use the browser buttons (such as Refresh, Back, and Forward) on your browser toolbar or their keyboard shortcuts (such as Ctrl+R, Ctrl+Left, and Ctrl+Right) to navigate the Service Manager UI in the web client. Instead, use the buttons on the Service Manager interface, as described in the "List and detail toolbars" topic.

### Related topics

Accessing an application
Menus
List and detail toolbars
Add a folder to the System Navigator
Collapse open folders in the System Navigator
Delete a node in the System Navigator

## 5.1. Menus

The Service Manager Services, Support, Utilities, and Toolkit menus appear after you log on to Service Manager. The main menu provides access to all Service Manager applications. The number of menu items that are displayed depends on your user role and on the capability words that you are assigned. For example, if you have administrator rights, all menu items are displayed.

| Menu | Purpose |
|---|---|
| Applications | Accesses the main Service Manager applications. |
| System Administration | Accesses utilities that administer Service Manager. |
| Tailoring | Accesses applications that tailor Service Manager. |

### Related topics

Navigating the UI
Accessing an application
List and detail toolbars
Add a folder to the System Navigator
Collapse open folders in the System Navigator
Delete a node in the System Navigator

## 5.2. More Actions menu

The More Actions menu contains various actions and functions that apply to the specific record list or detail form that you have open. The menu items available for each list or detail form vary based on your user profile, your permissions, and how Service Manager was implemented at your location.

To access the More Actions menu on a web client, click the **More** button.

To access the More Actions menu on a Windows client, click the down-arrow button on the far right of the toolbar.

**Related topics**

Windows client
Web client
Differences between the Windows and web clients

### 5.3. List and detail toolbars

Toolbars are displayed above the record list and record details sections of Service Manager applications. These toolbars use standard buttons to trigger common actions. Some of the actions are listed below.



**Note**

Note

If you are using a web client, don't use the browser buttons (such as Refresh, Back, and Forward) on your browser toolbar or their keyboard shortcuts (such as Ctrl+R, Ctrl+Left, and Ctrl+Right) to perform Service Manager actions. Instead, use the buttons on the Service Manager interface.

| Windows Client | Web Client | Icon name | Meaning |
|---|---|---|---|
|  |  | Add | Add a new record to the database. |
|  |  | Back | Return to the last form. |
|  |  | Cancel | Reverse your last action. |
|  |  | Clear | Erase the data that you type on a form. |
|  |  | Close | Change the status of the current record. |
|  | N/A | Count | Count the number of records in a record list. If you are using a web client, you can access this information by clicking the Count Records below the record list. |
|  |  | Fill | Populate the current field with data. |
|  |  | Find | Locate related data. |
|  | N/A | Menu | Refresh the record list and option list. |
|  |  | New | Create a new record. |
|  |  | Next | Advance to the next record. |
|  |  | Save and Exit | Update the current record and return to the previous form. |
|  | N/A | More Actions menu | Access the additional actions available for the current form. This icon is located on the far right side of the toolbar in the Windows client. |
| N/A | MORE | More Actions menu | Access the additional actions available for the current form. This button is displayed in the web client only. |
|  |  | Previous | Return to the previous record. |
|  |  | Print | Send the current form to the printer. |

| | | | |
|---|---|---|---|
|  |  | Qopen (Quick Open) | Create a service desk interaction or incident report. |
|  |  | Refresh | Update to display changes. |
|  |  | Save | Update the current record and display the current form. |
|  |  | Search | Generate a record list. |

**Related topics**

Navigating the UI
Accessing an application
Menus
Add a folder to the System Navigator
Collapse open folders in the System Navigator
Delete a node in the System Navigator

### 5.4. To Do list

The To Do list is the first screen all Service Manager users see when they sign in to the Windows client. Users who sign in to the web client will also see a To Do list, unless they're using an operator with the configuration to see a dashboard home page instead.

The To Do list displays a list of records in the operator's To Do queue. The operator can use the Queue and View fields to display different record lists, depending on the selections made in each of these fields. The operator's user role and profile determine the selections available in the Queue and View fields.

For example, an operator might select the Interaction queue in the Queue field and Self-Service Interactions in the View field to display a list of all open self-service interactions to which the operator has access. Once a list of records appears, the operator double-clicks a record to display the details of the record.

The To Do list includes active links on the left side of the form that allow an operator to perform tasks, such as searching for or creating a new record, depending on the queue and view selected and operator access privileges.

### 5.5. Views and favorites

Views and favorites enable you to define and save a query. For example, the following out–of–box views or favorites are queries that generate record lists:

- Calls I opened
- High priority problems
- All my approvals
- My tasks
- Active contracts
- All my open requests

When you display the contents of a view or favorite, the query that's associated with it automatically runs and produces the list of records that meet the query criteria. Favorites appear in the System Navigator under **Favorites and Dashboards**. Views appear as queues within most Service Manager applications. Whether you access the record list from Favorites and Dashboards, or a view from an application, the list of records in each is identical.

You can create favorites if you have the user.favorites capability word in your profile. You can share favorites with other users if you have the public.favorites capability word in your profile. Service Manager stores favorites as unique records in the inbox table.

### Public and private favorites

Any user can create private favorites (such as common queries and frequently used forms) for their own use if their profile contains the user.favorites capability word. A named user who has the public.favorites capability word in their profile can share their private favorites and dashboards with other users. Only the owner of a favorite or a system administrator can modify the favorite properties.

The public.favorites capability word is only for administrators to create and modify shared favorites. An administrator can create a favorite for an individual user, user groups, or the entire user community, and then specify one or more content owners. A designated content owner can't remove the favorite folder, but they can define or modify subordinate folders or dashboards.

You can click the **Show Private Favorite Only** button to display the private favorites only. In addition, you can configure the *sortNavFavorites* parameter to display all favorites in ascending, alphabetical order. See Web parameter: sortNavFavorites for more information.

### Add a view or favorite

**Applies to User roles:** All roles except Self-Service User

You can use Add Favorite to define and save a query. When you save a record list as a favorite, you are saving a dynamic query that extracts information from the database. When the database changes, the resulting output changes.

#### Windows client

To add a view or favorite in the Windows client, follow these steps:

1. Navigate to the Service Manager form or application that you want to save as a view or favorite. Or, enter search criteria to find the record or list of records that you want to save, and then click **Search**.
2. Click the System Navigator icon and drag it to Favorites and Dashboards or to a subordinate favorites folder.
3. Type a name for the new favorite.
4. Click **OK**. Service Manager displays the new favorite at the bottom of the Favorites and Dashboards folder.

#### Web client

To add a view or favorite in the web client, follow these steps:

1. Navigate to the Service Manager form or application that you want to save as a view or favorite. You can also enter search criteria and then click **Search** to find the record or list of records that you want to save.
2. Click the **Add Favorite** button in the Record List toolbar or the Detail List toolbar to save the record list or the single record.
3. Type a name for the new favorite.
4. Click **OK**. Service Manager displays the new favorite in the Favorites and Dashboards folder.

> **Note**
> Note You can't add a favorite in a workflow page.

**Private favorites**: You can create favorites only if you have the user.favorites capability word in your operator record.

**Public favorites**: You can share favorites with other users only if you have the public.favorites capability word in your profile. If you create a public favorite and store it in the Favorites and Dashboards parent node, the favorite is available to all users. If you create a public favorite, and store it in a node that's subordinate to the Favorites and Dashboards parent node, the availability of the favorite depends on the properties of the parent node.

Contact your administrator if you can't create favorites.

**Tip**

Tip All out-of-the-box view names are localized. When you create a new view (inbox) and there are active languages other than English, you need to search for the scmessage record and then translate the content of Text for the language:

Language Code: <*language code*>

Class: local:inbox

Message Number: <*inbox id*>

### Edit a view or favorite

**Applies to User roles:** All roles except Self-Service User

You can edit a view or favorite to change the title, the query, or the view definitions. Additionally, you can move or copy views and favorites to other folders.

#### Windows client

To edit a view or favorite in the Windows client, follow these steps:

1. Click **Favorites and Dashboards**.
2. Right-click the favorite that you want to edit, and then select **Properties**. The View Definition form opens.
3. Update any fields that require revision, click **Save**, and then click **OK**.

To move or copy the view or favorite, click the view or favorite and drag it to the new destination.

#### Web client

To edit a view or favorite in the web client, follow these steps:

1. Click **Manage Favorites** in the Navigator pane.
2. Expand a favorite folder to display all the favorite links in the folder.
3. Click the associated checkbox to edit an individual favorite.
4. To edit the view or favorite:
    1. Click the favorite text link. The View Definition form opens.
    2. Update any fields that require revision.
    3. Click **Save**.
    4. Click **OK**.
5. To move the view or favorite:
    1. Select the view or favorite you want to move.
    2. Click **Move To** in the toolbar and select the destination folder.
    3. Click **Yes**.
6. To copy the view or favorite:
    1. Select the view or favorite you want to copy.
    2. Click **Copy To** in the toolbar and select the destination folder from the drop-down list.
    3. Click **Yes**.

**Private favorites**: You can create favorites only if you have the user.favorites capability word in your operator record.

**Public favorites**: You can share favorites with other users only if you have the public.favorites capability word in your profile. If you create a public favorite and store it in the Favorites and Dashboards parent node, the favorite is available to all users. If you create a public favorite, and store it in a node that's subordinate to the Favorites and Dashboards parent node, the favorite availability depends on the properties of the parent node.

Contact your administrator if you can't create favorites.

### Delete a view or favorite

**Applies to User roles:** All roles except Self-Service User

You can delete a view or favorite if it is no longer needed.

#### Windows client

To delete a view or favorite in the Windows client, follow these steps:

1. Click **Favorites and Dashboards**.
2. Right-click a favorite, and then select **Delete**.
3. Click **Yes** to confirm the deletion.

#### Web client

To delete a view or favorite in the web client, follow these steps:

1. Click **Manage Favorites** in the Navigator pane.
2. Expand a favorite folder to display all favorite links in the folder.
3. Select the associated view or favorite that you want to delete.
4. Click **Delete** in the toolbar, and then click **Yes** to confirm the deletion.

### Enable the auto-refresh feature for views and favorites

**Applies to User roles:** All roles with the `Queue.AutoRefresh` execute capability word

You can configure views and favorites in the web client to be automatically refreshed.



**Note**

Note You can't enable this feature for record lists.

#### Windows client

To enable auto-refresh by using the Windows client, follow these steps.



**Note**

Note Although views don't auto-refresh in the Windows client, you can still enable the feature from the Windows client.

1. Click **Favorites and Dashboards**.
2. Right-click the view or favorite that you want to edit, and then select **Properties**. The View Definition form opens.
3. On the Auto refresh tab, select the **Enable Auto Refresh** option. This option is disabled by default.
4. Select a refresh interval from the **Refresh Interval** drop-down list, and then save the view. When you open this view or favorite in the web client, it will be automatically refreshed at the selected interval.

#### Web client

To enable auto-refresh in the web client, follow these steps:

1. In the Navigator pane, click the **Manage Favorites** icon.
2. Click a view to open the View Definition form.
3. On the Auto refresh tab, select the **Enable Auto Refresh** option. This option is disabled by default.
4. Select a refresh interval from the **Refresh Interval** drop-down list, and then save the view.

Alternatively, if you are a system administrator, you can follow these steps:

1. In the Navigator pane, click **System Administration** > **Base System Configuration** > **Miscellaneous** > **Views/Favorites**.
2. Use search or advanced search to find a view or favorite.
3. Open the view or favorite, and then select the **Enable Auto Refresh** option on the Auto refresh tab. This option is disabled by default.
4. Select a refresh interval from the **Refresh Interval** drop-down list, and then save the view.

#### Set the auto-refresh interval for a view or favorite

**Applies to User roles:** All roles with the `Queue.AutoRefresh` execute capability word

You can set the interval at which views and favorites in the web client are automatically refreshed.

> **Note**
>
> Note
>
> - When you configure your own auto-refresh interval for a specific view or favorite, you overwrite the values set in the view record.
> - The auto-refresh feature must be enabled for the view or favorite in order for you to set the refresh interval.

To set your own auto-refresh interval for a view or favorite, follow these steps:

1. Open the view or favorite for which you want to set a refresh interval. For example, click **Service Desk** > **Interaction queue** in the Navigator pane.

   > **Note**
   >
   > Note You can't set an auto-refresh interval for record lists.

2. Click **Set Auto Refresh Interval** in the Interaction pane menu.
3. Select a refresh interval from the **Refresh Interval** drop-down list, and then click **Finish**.

The interval starts as soon as you save it.

#### Related topic

Web parameter: sortNavFavorites

### 5.6. Switching the System Navigator

By default, the System Navigator is displayed in the left-hand panel of the Service Manager interface.

#### Web client

In the web client, you can switch the System Navigator between the following two modes:

- Sidebar mode
  In the sidebar mode, the System Navigator only displays the icons of the top-level menu items. When the mouse hovers over an icon, a pop-up menu immediately appears on the right side of this icon and displays the submenu. In this mode, the System Navigator is only about one-third as wide as in the accordion mode. The sidebar mode provides more working space and offers better efficiency for you to locate a menu item.

  > **Note**
  >
  > Note
  >
  > - The sidebar mode only works in Power User view (index.do) and Employee Self-Service view (ess.do), but doesn't work in Accessible view (accessible.do) and Self-Service Accessible view (accessible_ess.do).
  > - The sidebar mode is designed for users who have normal vision. Therefore, when the focus is on an icon in the sidebar mode, JAWS doesn't read the name of the item, but notifies the user to switch the System Navigator to the accordion mode.

- Accordion mode
  The accordion mode uses both icons and text. In addition, the accordion mode allows you to manage favorites.

You can switch the System Navigator to the sidebar mode by clicking the **Collapse Navigator** (**<<**) button. Click the **>>** button to switch to the accordion mode. Alternatively, you can use the Alt+N keyboard shortcut to toggle between the two

modes.

Your web browser can remember the current mode of the System Navigator when you log out, and the web client keeps the same mode in your next login. If you clear the cache of the web browser, the System Navigator is in the accordion mode in your next login.

#### Windows client

In the Windows client, you can collapse or hide the System Navigator to increase space on the screen for regular activities. To do this, click the **Minimize** button on the right side of the panel toolbar. To display the panel again, click the **Restore** button on the far left of the screen.

#### Related topics

Collapse open folders in the System Navigator
Delete a node in the System Navigator

### 5.7. Menu item order in the System Navigator

By default, menu items in the Service Manager System Navigator are displayed in ascending alphabetical order. You can disable this option so that menu items are displayed in the order in which they arrive from the server. Additionally, system administrators can group menu items to make it easier for end users to navigate within the system.

In the web client, the alphabetical sorting of System Navigator menu items is defined by the **sortNavItems** parameter in the web.xml file. The default is `true` (enabled).

In the Windows client, the alphabetical sorting of System Navigator menu items is defined by the **Sort menu items alphabetically in the System Navigator** option. To configure this option, click **Window** > **Preferences** > **Service Manager** > **Appearance**.

#### Related topics

Web parameter: sortNavItems

### 5.8. Change the System Navigator menu item order option

**Applies to User roles:** System Administrator

#### Web client

To change the System Navigator menu item order option in the web client, set the following parameter in the web.xml file:

```
<init-param>
  <param-name>sortNavItems</param-name>
  <param-value>true</param-value>
</init-param>
```

The default value of the parameter is "true" (enabled).

#### Windows client

To change the System Navigator menu item order option in the Windows client, follow these steps:

1. Click **Window** > **Preferences > Service Manager > Appearance**.
2. Clear the **Sort menu items alphabetically in the System Navigator** option to disable alphabetical sorting. Or, select the **Sort menu items alphabetically in the System Navigator** option to enable alphabetical sorting.

#### Related topics

Menu item order in the System Navigator
Enter a parameter in the web.xml file
Web parameter: sortNavItems

### 5.9. Add a folder to the System Navigator

**Applies to User roles:** All roles

#### Windows client

To add a folder to the System Navigator in the Windows client, follow these steps:

1. Right-click the parent node and choose **New Folder**.
2. Type the name of the new folder.
3. Click **OK**.

#### Web client

To add a folder to the System Navigator in the web client, follow these steps:

1. Click **Manage Favorites**.
2. Expand the Favorites directory to locate the parent node.
3. Click **Add folder or dashboard**, and then select **New Folder**.
4. Type the name of the new folder.
5. Click **OK**.

#### Related topics

Navigating the UI
Accessing an application
Menus
List and detail toolbars
Collapse open folders in the System Navigator
Delete a node in the System Navigator

### 5.10. Collapse open folders in the System Navigator

**Applies to User roles:** All roles

To collapse the open folders in the System Navigator in the Windows client, right-click the parent node, and then click **Collapse All** to close all the subordinate child nodes.



**Note**

Note This feature is unavailable in the web client.

#### Related topics

Navigating the UI
Accessing an application
Menus
List and detail toolbars
Add a folder to the System Navigator
Delete a node in the System Navigator

### 5.11. Delete a node in the System Navigator

**Applies to User roles:** All roles

#### Windows client

To delete a System Navigator node in the Windows client, right-click the nodes in the System Navigator, and then click **Delete**.

**Web client**

To delete a System Navigator node in the web client, follow these steps:

1. Click **Manage Favorites**.
2. Expand the Favorites directory and select the node that you want to delete.
3. Click **Delete**.

> **Note**
>
> Note You can permanently delete nodes that you create only. If you delete a node that's common to all users, it reappears when you refresh the connection.

**Related topics**

Navigating the UI
Accessing an application
Menus
List and detail toolbars
Add a folder to the System Navigator
Collapse open folders in the System Navigator

### 5.12. Accessing an application

You can access applications from the System Navigator. You can expand nodes in the System Navigator to see all available applications and tasks at the same time. You can create favorites that make any part of Service Manager, or a frequently used query, accessible with a single click. The System Navigator can remain visible while you work in any application, which enables quick transitions to other tasks and forms.

To search for an application, type a keyword in the search line embedded in the System Navigator. The system searches for the application automatically.

To use a particular application, click the name of the application. For example, click **Service Desk** > **Create New Interaction** to open the New Interaction form and add information.

**Related topics**

Navigating the UI

## 6. Editing records

Service Manager applications enable users to access and edit records. Multiple users can view the same record concurrently. If one of those users begins to modify the record, the server locks the record for updates only from that user. When the user saves the record, the server unlocks it. One of the following scenarios occurs when two users access the same record:

- If another other user is updating a record, Service Manager generates a message when you try to change it. Service Manager locks the record  until the other user completes the update.
- If the record isn't in use, but you don't have the current version of the record, Service Manager prompts you to refresh the record.

  > **Note**
  >
  > Note Do not refresh the record by refreshing your browser. Instead, use the Service Manager Refresh button.

  To disable the pop-up window that appears when you reload a record, follow these steps:

  1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
  2. Click the **General** tab.

3. Select the **Disable the Pop-up Window to Reload Records** option.
4. Click **Save**.

- If the record isn't locked and you have the most current version, you may continue typing without interruption. Other users can't update that record until you complete the update.

Service Manager applications also enable background process to access and edit records. When Service Manager is integrated with other products, the background process updates the records automatically in many scenarios. One of the following scenarios occurs when a user and a background process access the same record:

- If you and the background process are updating different fields in the record simultaneously, Service Manager generates a message after you save your update. The system merges the updated fields automatically and integrates these fields into the record.
- If you and the background process are updating the same field in the record concurrently, Service Manager prompts you to merge the conflicted updates when you try to save your update.
  To merge the conflicted updates manually, follow these steps:
  1. Click **Merge** in the toolbar. The "Conflicts" section opens.
  2. In the "Conflicts" section, select the value in one of the following fields:
     - **Original Value in DB**: The original value of the field in the database.
     - **Current Value in DB**: The current value in the database as updated by the background process.
     - **Unsaved Update**: The unsaved value as updated by you.
  3. Click **OK** to save the conflicted value with your selection. Or, click **Cancel** to return to the Details page of the record with unsaved updates.
  4. Click **Save** or **Save & Exit** to save the record to the database.



**Note**

Note

- The **Merge** button is available only when there are conflicted updates.
- The Merge Conflicted Updates feature is available for business data modules only, including Change Management, Configuration Management, Incident Management, Problem Management, and Service Desk.
- The merge function is available only when there are no conflicts on fields of the system usage type. For more information about how to define the field usage type, refer to Data Policy.

### Related topics

Fields
Fill function
View Detailed Information function

## 6.1. Record list

The Service Manager database is a set of files that contains records. You can search, modify, create, and delete these records. Records may contain information about the following items:

- Users
- Configuration items
- Interaction reports
- Incidents
- Changes
- Requests

You can locate records by using a query, which is a search performed on the Service Manager database. A record list query is a search of the database using the information you know about a specific record. If Service Manager finds multiple

records that match your search parameters, it displays a record list. You can double-click any record from the list to view it in a form.

When you access a record, you can browse (view only) or update the record. Your ability to update the record depends on your user profile. For example, a Service Desk user may not have permission to approve a change record.

### Alternate forms

System Administrators can modify Service Manager so that alternate forms are available to some users. In this situation, the Alternate Forms option in the More Actions menu is available. User can select the form to use to display the information. Administrators do this to display different views such as financial or security information.

Administrators can specify alternate forms to display a list of records or to display a single record.

### Enable or disable a record list

**Applies to User roles:** All roles

To enable a record list in the Windows client, follow these steps:

1. Click **Window** > **Preferences**.
2. In the **Preferences** dialog box, expand **Service Manager**.
3. Click **Appearance**, and then select **Show detail pane with record list**.
4. Click **Apply**, and then click **OK**.

To disable a record list in the Windows client, follow these steps:

1. Click **Window** > **Preferences**.
2. From the Preferences dialog, expand **Service Manager**.
3. Click **Appearance**, and then clear the **Show detail pane with record list** option.
4. Click **Apply**, and then click **OK**.



**Note**

Note This method is only available in the Windows client and doesn't work in the web client.

### Create customized column views for record lists

**Applies to User roles:** All roles

To create customized column views for record lists, follow these steps:

1. Click **More** or the **More Actions** icon.
2. Click **Modify Columns** to display a form that enables you to select the columns (fields) that the record list displays.
3. Modify the columns as desired. For more information about how to do this, see Working with record lists.
4. Click **More** or the **More Actions** icon.
5. Click **Save as Named Grid**, type a name for your new grid, and then click **OK**.



**Note**

Note You can repeat this process for each view that you want to create.

### Create customized column views for record lists

**Applies to User roles:** All roles

To create customized column views for record lists, follow these steps:

1. Click **More** or the **More Actions** icon.
2. Click **Modify Columns** to display a form that enables you to select the columns (fields) that the record list displays.
3. Modify the columns as desired. For more information about how to do this, see Working with record lists.
4. Click **More** or the **More Actions** icon.

5. Click **Save as Named Grid**, type a name for your new grid, and then click **OK**.

> **Note**
>
> Note You can repeat this process for each view that you want to create.

#### Invoke a saved column view

**Applies to User roles:** All roles

To invoke a saved column view, follow these steps:

1. Display any record list.
2. Click **More** or the **More Actions** icon.
3. Click **Modify Columns**, click **More** or the **More Actions** icon again, and then select **Load Named Grid**.
4. Click the drop-down list to choose the name of a saved column view (grid), and then click **OK**.

#### Delete all saved column views

**Applies to User roles:** All roles

To delete all saved column views, follow these steps:

1. Display any record list.
2. Click **More** or the **More Actions** icon.
3. Click **Modify Columns**.
4. Click **More** or the **More Actions** icon again, and then select **Clear Saved Grids**.

#### Related topics

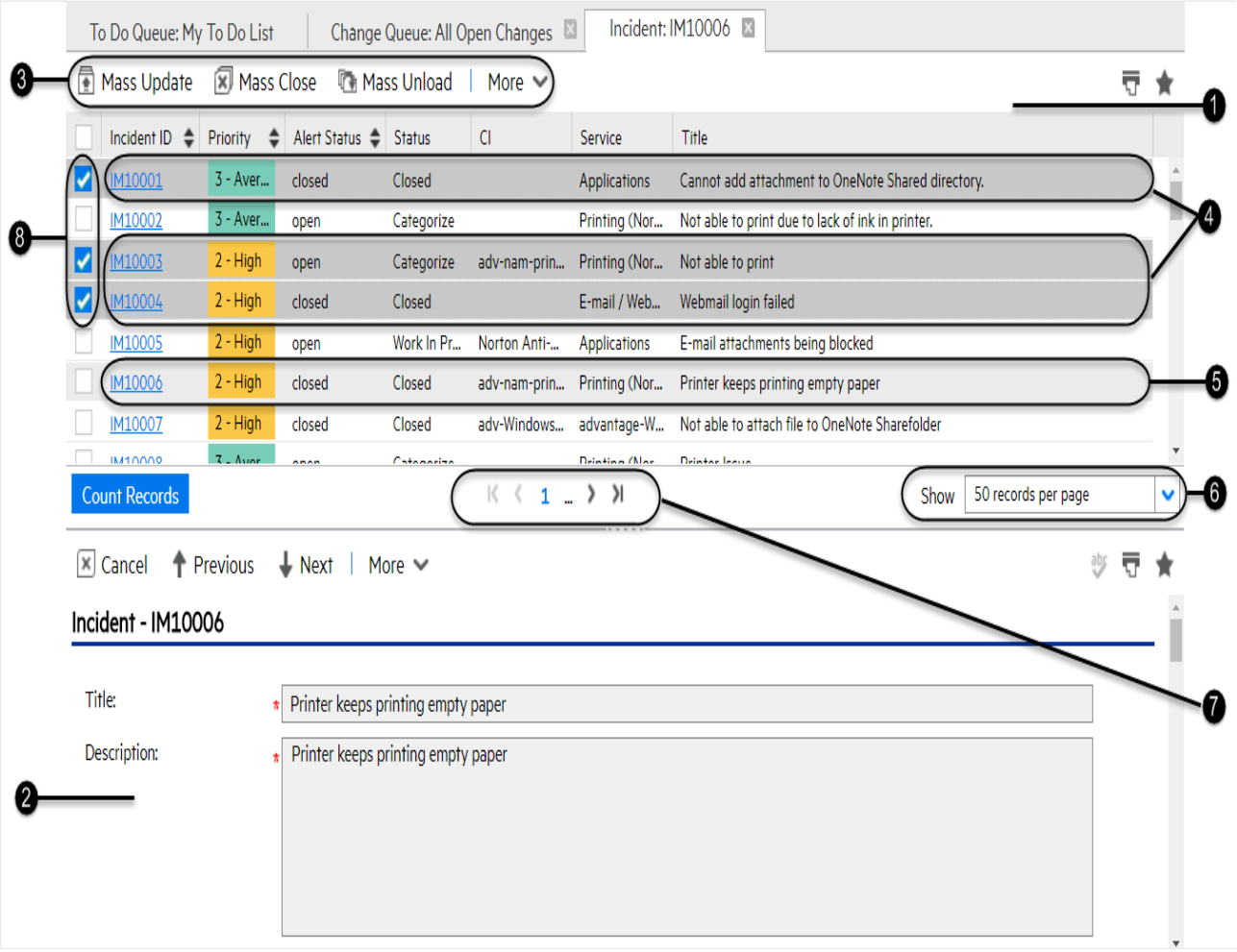
More Actions menu
Format Control processes
Working with record lists
View multiple records using an alternate form

## 6.2. Work with record lists

The RecordList functionality in the web client has been improved to achieve better performance and usability. To learn about the differences between the old and new RecordList functionality, see RecordList improvements (web client). Unless noted otherwise, record lists function similarly in the web and Windows clients.

The following figures depict the components of the list/detail and list-only record list pages in the web client.

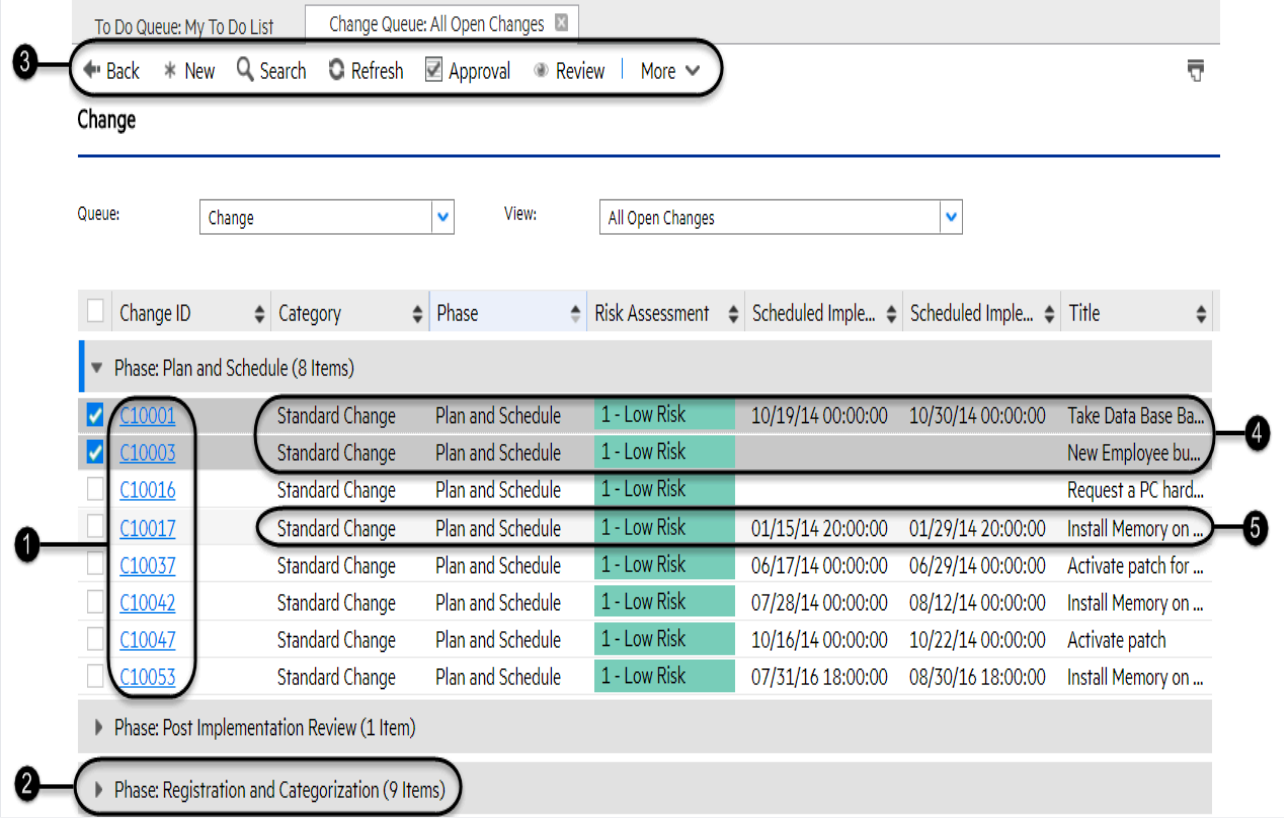
**List/detail page (web client)**




1. List pane: The record list is displayed on this pane.
2. Detail pane: The details of a record are displayed on this pane.
3. List toolbar: Contains options that take effect on the entire list or selected rows in the list.
4. Selected rows: Selected rows are highlighted but without a black border.
5. Focused row (in the unselected state in this figure): The row that's currently in focus. The focused row has a black border and a blue (when selected) or dark gray (when unselected) background. When selected, the focused row is also a selected row. Keyboard operations take effect only on the focused row (not on selected rows). For example, pressing Enter drills down to the focused row.
6. Page size selection box: Defines how many records are displayed on each page.
7. Paging bar: Allows users to navigate through the pages of the record list.
8. Checkbox: Indicates whether the row is selected.

9. Truncated records:

**List-only page (web client)**




1. Drill-down link: Each cell in the first column is a drill-down link. Clicking a link opens the details of the record in the detail pane.
2. Group: A group in the record list (in the collapsed state in this figure).
3. List toolbar
4. Selected rows
5. Focused row

**Select multiple records**

> **Note**
>
> Note You can't use only the Shift key to select a range of records that span multiple groups or pages. You can press Ctrl+Shift to select records that span multiple pages, but you can't do so for records that span multiple groups.

To select multiple records, use one of the following methods:

- Hold down the Ctrl key and click multiple rows. To deselect a row, click the row again.
- Hold down the Shift key, click one row, and then click another row to select a range of rows. To deselect a row, hold down the Ctrl key and click the row again.

To select multiple records in the web client, use one of the following methods:

- Select the checkboxes of multiple rows. Clear the checkbox to deselect a row.
- Hold down the Ctrl key, and click multiple rows. Click a row again to deselect it.
- Hold down the Ctrl key, press the Up or Down key, and then press the Space key to select multiple rows one by one. Press the Space key again to deselect a row.

- Hold down the Shift key, click one row, and then click another tow to select a range of rows. To deselect a row, hold down the Ctrl key and click the row.
- Hold down the Shift key, and press the Up or Down arrow key to select a range of rows. To deselect a range of rows, hold down Shift and press the Up or Down key again. To deselect a row, hold down Ctrl and press Up or Down to move the focus to the row and press the Space key.

### Expand/collapse a group

In the web client, you can expand only one group in a record list at one time. The grouping behavior varies depending on the viewrecordlist setting (no matter whether it is set in the server's sm.ini file, in the web tier configuration file (web.xml), or in the web client login URL).

- **If the viewrecordlist setting is set to true**:
  Service Manager always expands the group in which the current row is located. By default, when you initially open a record list, the first row is the current row (therefore, the first group is expanded). In list/detail pages, the current row is the row currently displayed in the detail pane. In list-only pages, the current row is the last selected row.
  Whenever a record list page is reloaded (for example, when you sort the record list), Service Manager expands the group in which the current row is located.
  The following table describes the different ways in which you can change the current row in the list/detail and list-only pages in the web client.

| In list/detail pages | In list-only pages |
|---|---|
| You can change the current row in either of the following ways:<br>○ Click a row (not the checkbox of a row, which allows multi-select).<br>○ Use the Up or Down key to move the focus to a row (without holding the Ctrl or Shift key) | Select a row to change the current row. The last selected row is the current row.<br>**Note**<br>Note It doesn't matter how you select the row (for example, by clicking the row, by selecting the checkbox, or by using the Up or Down key or the Ctrl+Up/Down+Space keys). |

- **If the viewrecordlist setting is set to false**:
  Whenever a list-only record list is reloaded (for example, when you refresh or sort the list), the first row in the list becomes the current row and is expanded.
  To expand or collapse a group in the web client, use one of the following methods:
  - Click the plus/minus (+/-) symbol of a group.
  - Use the Tab key to move the focus to a group and then press the Right or Left arrow key.
  - Use the Tab key to move the focus to a group and then press the Space key.

  To expand or collapse a group in the Windows client, use one of the following methods:
  - Click the plus/minus (+/-) symbol of a group.
  - Use the Tab key to move the focus to a group, and then press the Right or Left arrow key.

### Sort a record list

To sort a record list in the Windows client, use one of the following methods:

- Click a column header to resort the list using that column's information. For example, to sort the list by status, click the **Status** column header.
- Click again to reverse the order (for example, from ascending to descending).

To sort a record list in the web client, use one of the following methods:

- Click a column header.
- Use the Tab key to move the focus to a column header and then press Enter.

### Drill down to a record

When you drill down to a record, the details of a record are displayed in the detail pane.

To drill down to a record in the web client, use one of the following methods:

- In the list-only view: Click the link in the first column of a row, or double-click a row.
- In the list/detail view: Click a row.
- In both views: Press the Up or Down arrow key to select a row and then press Enter.

> **Note**
> Note If the focus moves to a collapsed group, press the Space key or the Right arrow key to expand the group and then continue.

To drill down to a record in the Windows client, use one of the following methods:

- In the list-only view, double-click a record.
- In the list/detail view, click a record.

### Count the records in a list

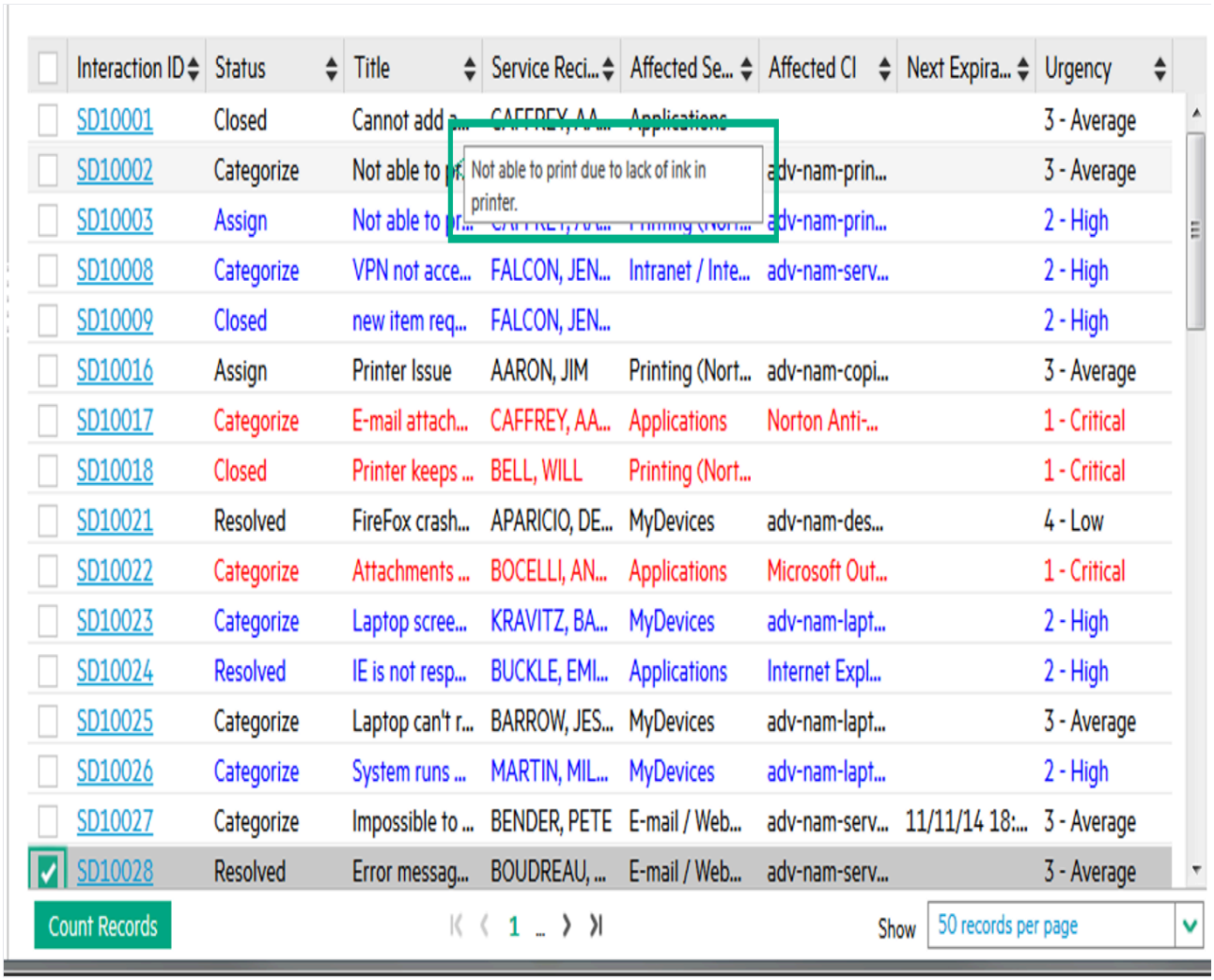
To count the records in a list in the web client, use one of the following methods:

- Click the **Count Records** button located below the record list.
- From the list toolbar, click **More**, and then select **Count**.
- Click the last page number on the paging bar or press the Tab key until the focus moves to the last page number, and then press Enter.

To count the records in a list in the Windows client, click the **Count Records** button located on the right side of the record list Toolbar.

### View content that has been truncated to fit in a column

If the content of a column can't be fully displayed within the configured column width, the content is truncated and appended by an ellipsis (...). To view the truncated content, hover the mouse over it. After a short time, the content appears in a tooltip, as displayed in the following image.




**Change the page size (web client only)**

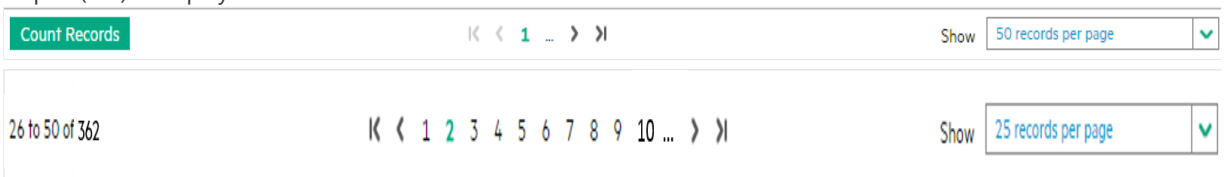
Use the page size selection box located below the record list to select the number of records that are displayed on each page.

**Navigate through pages (web client only)**

You can use the controls on the paging bar below the record list to view the records, page by page.

The following figures summarize the paging bar behavior.

- There are up to 10 page links on the paging bar. If there are more pages after or before the last or first page link, an ellipsis ("...") is displayed.




- If you don't click the **Count Records** button, the paging bar remembers the largest page number that you have visited.




- When you click the **Count Records** button or reach the last record in the list, the **Count Records** button is replaced by the record count of the current page. For example, "1 to 25 of 302".

### Move the focus using the keyboard

Use the Tab key to move from a detail record back to the record list. Press Tab to move forward or Shift+Tab to move backward through the fields of a detail record until the focus returns to the record list pane. Once the focus is positioned in the record list, you can scroll through the list of records by using the up and down arrow keys. When you move the focus to a record you want to view, press Enter.

### Set the focus on a record for keyboard actions (web client only)

Keyboard actions take effect only on the focused row. For example, pressing Enter selects and opens the focused record, instead of a selected record.

To set the focus on a record for keyboard actions in the web client, use one of the following methods:

- Click a row.
- Press the Up or Down arrow key to select a row.

### Show or hide columns (web client only)

To show or hide columns, follow these steps:

1. Move the mouse to any column header.
2. Click the  icon. A drop-down menu appears.

   **Note**

   Note If you want to sort the record list by a column, click the column name.

3. Select or clear the checkbox before a column name to show or hide the column.

**Note**

Note

- By default, the list pane only displays the first four columns in the vertical layout.
- You can't hide the first column.
- The columns that appears in the drop-down menu are configurable. For more information, see Record list.
- The content in the drop-down menu doesn't support JAWS because this menu is provided for users who have normal vision.

### Switch on or off the color indicator for a column

In the record list, you can quickly switch on or off the color indicator of a column. To do this, use the **Color Indicate** option in the drop-down menu of that column.

### Change the width of a column

To change the width of a column, follow these steps:

1. Position the mouse pointer in the column heading near the cell border until it appears as a double-headed arrow.
2. Perform one of the following actions
   - Drag the column boundary to the left or right until the column is the desired width.
   - Double-click the mouse to restore the default width of the column.

### Change column definition and order

There is a default order to the columns that appear when you display a record list. Your administrator can determine the order, or you can create a personal view that orders columns according to your preference. Each view that you define has a unique name.

### Change the columns in a record list

**Applies to User Roles:** All roles

To change the columns in a record list, follow these steps:

1. Display any record list.
2. Click **More** or the **More Actions** icon.
3. Click **Modify Columns** to display a form that enables you to modify the columns (fields) that are displayed in the record list. Each field appears in a table row that you can edit.
4. Click the column name row to select a new field from the drop-down list.
5. To add a column between two existing column, insert the cursor in an existing row, and then click **Insert Line**. A new row appears above the current row. This creates a new column in the record list.
6. To specify the field to display in a column, select it from the drop-down list in the row.
7. To delete an existing column, insert the cursor by the name of the column that you want to delete, and then click **Delete Line**.
8. To cancel your changes, click **Back**.
9. To return to the default column settings, click **Use Default**.
10. When you finish changing the columns, click **Proceed** on the Toolbar. The record list displays the new column order.

Service Manager saves the column changes as a logon preference. Your logon preferences don't affect other users.

### Zoom in or out record lists (web client only)

To zoom in or out record list pages in the web client (including the text and images), press the Ctrl key together with the plus or minus key (+/–) or together with the mouse wheel.

### Related topics

Record list
More Actions menu
RecordList improvements (web client)

## 6.3. RecordList improvements (web client)

The RecordList functionality in the web client has been improved to achieve better performance and usability. For more information about working with record lists, see Working with record lists.

**Usability improvements in the new RecordList**

- On a list-only page (for example, a To-Do queue), the first visible column in each row is a drill-down link. If you click this link, the record opens.
- On a list-only page, a single click on a row no longer opens the record (it selects the row instead).
- The focused row now has a black border and an orange or gray background (when selected and not selected, respectively), while selected rows only have an orange background.
- More keyboard operations are supported.

**Differences between the old and new RecordList functionality**

The following table summarizes the major differences between the old and new RecordList behaviors.

| Concept | Description | Before version 9.31 | In version 9.31 or later |
| --- | --- | --- | --- |
| Focused row | This is the row where the focus is set for keyboard actions (unique). | To set the focus on a row, use one of the following methods:<br>• Hold down the Ctrl key, and click a row.<br>• Press the Up or Down arrow key to move the focus to the row. | To set the focus on a row, use one of the following methods:<br>• Click a row.<br>• Press the Up or Down arrow key to move the focus to the row (This will clear previous selections).<br>• Hold down the Ctrl key, and press the Up or Down arrow key to move the focus to the row (This will keep previous selections). |
| Single- select | Select a single row for list toolbar options (unique). | Not supported | To select a single row, use one of the following methods:<br>• Click the row (not the checkbox).<br>• Press the Up or Down key to move the focus to the row. |

| | | | |
|---|---|---|---|
| Multi- select | Select multiple rows for list toolbar options. | Select the checkboxes of the rows. | To select multiple rows, use one of the following methods:<br>• Select the checkboxes of the rows. To deselect a row, clear the checkbox.<br>• Hold down the Ctrl key, and click the rows. To deselect the focused row, press the Space key; to deselect a selected row, click the row again.<br>• Hold down the Shift key, and click the first and last rows. To deselect the focused row, press the Space key; to deselect a selected row, click the row again. |
| Drill down | On a list-only page: Open the details of a row (record).<br>On a list/detail page: Refresh the detail pane to display the details of a row (record). | To drill down, use one of the following methods:<br>• Click the row.<br>• Press Enter to drill down to the focused row. | To drill down, use one of the following methods:<br>• **List-only**: Click the link in the first column of a row, or double-click a row.<br>• **List/detail**: Click a row.<br>• **Both views**: Press the Up /Down arrow key to select a row and then press Enter. |

**Mouse/keyboard action improvements**

The following table provides a comparison between the mouse and keyboard actions of the old and new RecordList, for list-only and list/detail pages. The actions are identical for both views unless otherwise specified.

| User's Mouse/Keyboard Operation | Old Action | New Action |
| --- | --- | --- |
| Click a row | Drills down to the record (updates the detail pane). | **List-only**: Selects the row.<br>**List/detail**: Selects the row, and drills down to the row (updates the detail pane). |
| Press the Ctrl key+ click on a row | Sets the focus on the row. | Selects or deselects the row. |
| Press the Shift key + click on multiple rows | Not supported | Selects or deselects a range of rows. |
| Select or clear a checkbox | Select or deselects the row (Multi-select is allowed). | Selects or deselects the row. |
| Press the Up or Down arrow key | Moves the focus up/down. | Deselects existing selected rows, and moves both the focus and selected row up/down. |
| Press the Ctrl key + Up or Down arrow key | Moves the focus up/down. | Keeps existing selected rows, and moves the focus up/down. |
| Press the Shift key + Up or Down arrow key | Not supported | Selects a range of rows. |
| Press the Space key on the focused row | Selects or deselects the row. (Multi-select is allowed) | Selects or deselects the row. |
| Press Enter on the focused row | Drills down to the focused row. | Drills down to the focused row. |
| Click the drill-down link | Not supported | **List-only**: Drills down to the row.<br>**List/detail**: Not supported |
| Double-click a row | Not supported | Drills down to the record. |

| | | |
|---|---|---|
| Refresh or sort a list-only record list | Collapses all groups, regardless of the viewrecordlist setting. | **If viewrecordlist is set to true**: Expands the group that contains the current row (that is, the last selected row).<br>**If viewrecordlist is set to false**: Expands the first group in the list (the first row becomes the current row).<br>**Note**<br>Note When a list-only page (a ToDo list or Fill list) is initially opened or reloaded (refreshed or resorted), the current row (either the first row or the last selected row) is only focused but not selected, regardless of the viewrecordlist setting. |

**Related topics**

Record list
Working with record lists
Web parameter: recordListArrayDisplayStyle
Web parameter: recordListArrayMaxChars

### 6.4. Work with tags

Service Manager provides out-of-the-box tag fields to Interaction, Incident, and Knowledge Management forms and enables you to add one or more keywords or terms to a record. These tags are searchable and you can use these tags to highlight important information.

**Tip:** For detailed instructions about how to support tags in more Service Manager modules, see Supporting tags in Service Manager.

To add a tag, type a word and then press Enter or the comma key. Do not use angle brackets, exclamation marks, slashes, backslashes, and double quotation marks in the tags.

**Note:** To separate multiple tags, you can press a comma in the English input method only. If you are working with another input method such as Chinese, press Enter instead.

When you are typing, the system automatically opens the auto-suggestion list as long as your input matches one or more tags that are commonly used. The system also filters the list and displays the tags that match your input. By default, this list accommodates up to 100 tags. You can select a tag from the list by using either of the following methods:

- Press Up/Down Arrows to highlight a tag, and then press Enter.
- Use the mouse to click the tag directly.

**Tip:** You can also press Down Arrow to open the complete auto-suggestion list that displays the most commonly used tags for this type of record.

To remove a tag, click **x** on the tag. To remove the last tag, click **x** on the tag or press Backspace.

#### Tag assignment group support

As of version 9.70, the taghints table has the Assignment property in the database. After an IT operator adds, edits, or selects a tag and then saves the record, the system automatically sets the Assignment property of this tag to the assignment group that the IT operator belongs to. If an IT operator is a member of multiple assignment groups, the system creates or updates multiple tags of the same word or phrase with different Assignment properties.

The tag assignment group support feature empowers an IT operator to firstly view the most commonly used tags based on the operator's assignment group settings. When this feature is enabled and an IT operator is typing tags, the auto-suggestion list sorts the tags used by the operator's assignment groups by frequency of usage and then displays these tags in descending order of counts. Tags with specific Assignment properties take precedence over the ones with null Assignment properties in the auto-suggestion list. If there are less than 100 tags with specific Assignment properties, the system appends tags with null Assignment properties to the bottom of the auto-suggestion list in descending order of counts.

IT operators can view and select the tags used by their assignment groups. IT operators who don't belong to any assignment groups can't view the tags used by various assignment groups. Nevertheless, tags with null Assignment properties are visible to all IT operators.

The tag assignment group support is disabled by default. To enable this feature, click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record > General** tab **> More**, and then select the **Tag Based on Assignment Groups** checkbox.

**Note:** If you have worked with the auto-suggestion list in earlier applications versions from 9.50 to 9.64 and then upgrade the applications to version 9.7x, the Assignment properties of the existing tags stored in the taghints table are actually null.

**Note:** All tags for Knowledge Management have null Assignment properties. The tag assignment group support feature is inapplicable to Knowledge Management because Knowledge Management uses different group mechanisms other than Assignment Groups.

#### Related topics

Web client keyboard interactions and shortcuts
Supporting tags in Service Manager
Tag control

### 6.5. Adding prefix for a record type

The default records prefixes that you can search for include SD (interactions), C (change), IM (incidents), and CIs (configuration items). To search for a record type that doesn't have an out-of-the-box prefix (for example, approvals), you need to add a prefix for this record type.

To add a customized prefix for a record type, follow these steps:

1. Sign in to Service Manager as a System Administrator.
2. Type `gl` on the command line, and then press Enter.
3. Type Universal Search Customize List in the **List Name** field, and then click **Search**.
4. Add a record type with the corresponding prefix.
   For example, to add an approval record, type "Approval" in the **Value List** field and then type `A` in the **Display List** field as the prefix.
   When you add multiple record types and prefixes, make sure the types and prefixes are within double quotation marks and separated by commas with no spaces between them. For example: `{"svcCartItem","Approval"}` and `{"sci","A"}`.
5. Click **Save**, and then click **OK**.

When you search for `A10232` by using the Service Manager search line, Service Manager displays the approval record with the ID of 10232.

#### Related topics

Searching records
IR keys and non-IR keys

Enable IR search for a file
Regenerate IR keys

### 6.6. Printing lists and records

You can print record lists and record details from both the Service Manager Windows client and the web client.

To print the record list or record details, follow these steps:

1. Use search or advanced search to display a list of records or the details of a specific record.
2. (Windows client) Click **File** > **Print**. Or, click the **Print** icon and select **List View** to print the list of records or **Detail View** to print a single record.
   (Web client) Click the **Print** button, and then right-click to use the print function of your browser.

#### Related topics

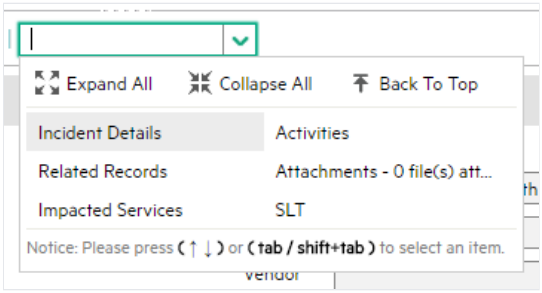
Web client
Windows client

### 6.7. Using Quick Jump in record detail

When the record detail is displayed in groups, you can use the Quick Jump feature to navigate to the desired section in record detail.

**Note**

Note For more information about how to configure the display style of the record detail, see **Preferred Notebook Style** in Set home page preferences.




You can expand the drop-down list and then click one of the section names to jump to the section.

Alternatively, if you type the section name in the text box, the matching section names then appear in the drop-down list. You can use the up or down arrow to select a section name and then press Enter to jump to the section.

You can also use the three buttons on the top of the drop-down list:

- **Expand All**: Expand all first-level sections in the record detail.
- **Collapse All**: Collapse all first-level sections in the record detail.
- **Back To Top**: Go back to the top of the record detail.

#### Related topics

Set home page preferences

### 6.8. Fields

Fields are areas of a form in which information for a record is displayed. For example, that information might be an asset name or the description of an incident. You can type information into some of these fields. Service Manager fills the other fields automatically.

When you type new data in a field and click Save, Service Manager stores the data in the appropriate Service Manager records.

### Required fields

Required fields contain the information that's necessary to complete a form. For example, if you create an incident record, you must specify a category. A red asterisk is displayed to the left of a required field. If you don't type data in a required field before you save the record, Service Manager generates an error message. In some forms, the message includes a list of possible values.

### Protected fields

You can't edit protected fields. For example, you can't change the incident number when you view an existing incident record. The fields that are protected depends on the form and your user rights.

### Tabbing through fields

To tab through fields, press Ctrl+Tab. If you use the Tab key, Service Manager processes the Tab character as the content of the field and treats it as invalid data for multi-line text fields. However, you can use the Tab key to tab through regular input fields, such as fill fields or comfill fields.

### View Field Help

**Applies to User roles:** All roles

Field Help can help you understand the use and function of each field or control on a form.



**Note**

Note In the web client, make sure the **disableKeyHelp** parameter is set to false before you view Field Help.

To view Field Help, follow these steps:

1. Insert the cursor in a field or click a control (such as a checkbox) on the form.
2. Perform one of the following actions:
    - In the Windows client, press Ctrl+H.
    - In the web client, press F1 or Shift + F1.

Service Manager displays the help information for the current field or control. "No description" is displayed if no help text is available.



**Note**

Note Service Manager also displays the file name, format name, and field name of the current field or control, if one of the following conditions is fulfilled:

- In the Windows client, the “Show context-sensitive help debug information” preference is enabled.
- In the Web client, “viewcontexthelp=true” is appended to the URL before you log in. The “viewcontexthelp” is only supported in Power User view (index.do).

You can use such information to add or edit a field help record, or add a stored query using certain field names.

### Related topics

Editing records
Fill function
View Detailed Information function
Field help editor
Stored queries
Add or edit help records

## 6.9. Fill function

The Fill function allows you to create or update a record quickly. When you click the **Fill** icon next to an empty field or a field that contains partially-keyed leading information, Service Manager identifies the applicable related record and adds the related information to the field. For example, if you type Br in a name field and click the **Fill** icon, Service Manager identifies records with last names that begin with "Br,"such as the record for Nicholas Brown, along with any assets that are linked to this contact record.

**Note**

Note If you attempt to locate information for a field that depends on one or more fields that contain invalid or unrelated information, you receive an error message that resembles the following:

`Cannot find related information in dept using query: company#"advantage" and dept.full.name#"advantage/edd"`

In this example, the value of the department name field depends upon the values in the company and dept.full.name fields. You must provide valid values in the company field and dept.full.name field.

### Use the Fill functionality

**Applies to User roles:** All roles

To use the Fill functionality, follow these steps:

1. Insert the cursor in any field that has a **Fill** icon adjacent to the text field.
2. Perform one of the following actions:
   - Click the **Fill** icon adjacent to the field.
   - Press F9.
3. A record list that contains possible entries for the selected field is displayed. Double-click a record to fill the selected field and any related fields.

### Use the Fill Selected functionality with multiple records

**Applies to User roles:** All roles

The Fill Select functionality is available for array fields only. An array field is a field that accepts a list of entries rather than a single entry. This topic describes how to use the Fill Selected functionality to add Downstream CIs to a Configuration Item Relationship as an example of how to use the Fill Selected functionality with multiple records. This example was selected because the Downstream CIs field is an example of an array field. This example uses out-of-box data for Applications CI.

**Note**

Note The multi-select functionality must be enabled by a System Administrator.

#### Windows client

To use the Fill Selected functionality to add multiple items to an array field, follow these steps:

1. Click **Configuration Management** > **Resources** > **Configuration Item Relationships**.
2. Click **Search** to display a list of existing Configuration Item Relationships, and then select **Applications**.
3. Select the next available blank line in Downstream CIs, and then click the **Fill** icon for the entry.
4. Click **Search** or type search criteria, and then click **Search**.
5. To select multiple items from the list, follow these steps:
   1. Click an item in the list.
   2. Press and hold down the **Ctrl** key.
   3. Select another item in the list.
   4. Continue to hold down the Ctrl key while you select other items in the list, until you have selected several items.
6. Click **Fill Selected**. The items that you selected are added to the list of Downstream CIs.

Web client

> **Note**
>
> Note In the web client, you can select multiple records to fill on one page at a time only.

To use the Fill Selected functionality to add multiple items to an array field, follow these steps:

1. Click **Configuration Management** > **Resources** > **Configuration Item Relationships**.
2. Click **Search** to display a list of existing Configuration Item Relationships, and then select **Applications**.
3. Select the next available blank line in Downstream CIs, and then click the **Fill** icon for the entry.
4. Click **Search** or type search criteria, and then click **Search**.
5. To select multiple items from the list, follow these steps:
    1. Select the checkbox next to the records that you need from one page.
    2. Click **Fill Selected**.

Related topics

Editing records
Fields
View Detailed Information function

## 6.10. View Detailed Information function

Certain fields in a record are linked to related information in different files. The Find function locates associated records for fields such as asset, logical or contact names, locations, vendors, or users. For example, when you use the Find function, you can populate the Reported By field in an incident record.

The function behaves differently for reference fields and normal fields (which don't have a referenced table defined in the data policy).

Behavior on normal fields

For fields that aren't a reference field, the **View Detailed Information** function uses the line in the link record for its query.

- In the web client, the **View Detailed Information** button becomes active immediately after the user starts typing in the field.
- In the Windows client, the button is disabled when the field is empty.

You can view or modify a linked record within the record's application. Depending on your user profile, you may be able to view Asset Management records, but not modify those records.

> **Note**
>
> Note If you attempt to locate information for a field that depends on one or more fields that contain invalid or unrelated information, you receive an error message that resembles the following:
>
> `Cannot find related information in dept using query: company#"advantage" and dept.full.name#"advantage/edd"`
>
> In this example, the value of the department name field depends upon the values in the company and dept.full.name fields. You must provide valid values in the company field and dept.full.name field.

Behavior on reference fields

The **View Detailed Information** function no longer uses the query defined in the link line; instead, the system performs an exact match query based on "id".

- In the web client, only when the reference field contains a value that exactly matches a value in the referenced table, the button is enabled (otherwise it is disabled).
- in the Windows client, the button is disabled when the field has no reference value (either you enter a value without clicking Fill or the field is empty).

#### Use the Find function

**Applies to User roles:** All roles

To perform a find on data in tables, double-click the first item in the table. (If you are using the web client, you only need to click once.)

To perform a find in other locations, follow these steps:

1. Insert the cursor in a field where you want to view the related Service Manager records. You can enter partial data in a field to narrow the search.
2. Perform one of the following actions:
    - Click the **Find** icon at the end of the field.
    - Press F8.

Service Manager displays a record that contains information related to that field. If no records are found or the selected field is not linked to the Find function, Service Manager generates an error message.

To exit the related record, click **Cancel**.

> **Note**
>
> Note Find doesn't insert the information from the related record. Use the Fill function to enter data automatically.

#### Related topics

Editing records
Fields
Fill function

### 6.11. Smart Indicator function

The Smart Indicator provides information on related issues, and enables you to drill down to this data to troubleshoot a user's issue. You can configure Smart Indicators to display the following information:

- Any other open or closed interactions and incidents that belong to the user
- Incidents associated with the same service or configuration item (CI)
- Known errors for a service (this enables you to immediately communicate information related to the service to the user)

#### Use the Smart Indicator function

**Applies to User roles:** All roles

You can configure Smart Indicators to display the following information:

- Any other interactions and incidents that were open and closed for the same service user
- Incidents associated with the same service
- Incidents associated with the same configuration item (CI)

You can also configure Smart Indicators to show known errors for a service, which enables you to immediately communicate to a caller information that's related to the service.

The following steps offer an example of the Smart Indicator function in use. In this example, a user contacts Service Desk to report an issue with their printer. The issue is first created by a Service Desk Analyst as an interaction, escalated to an incident, and then analyzed by an Incident Analyst. During this process, users use the Smart Indicator to browse through other interactions and incidents to look for issues that are associated with the service and CI.

**Service Desk Analyst**

1. Sign in to Service Manager and open an Interaction for the employee `Aaron Caffrey`.
2. Click the **Smart Indicator** icon to the right of Aaron's name to check whether Aaron is calling about an existing interaction.

3. To identify this issue with the Printer service, click **Fill**.
4. To see if this issue is already reported, click the **Smart Indicator** icon next to the service (image).
   A wizard opens to gather more information about this service. This wizard enables you to perform the following actions:
   - Double-click a line to browse through and drill down to a detailed overview of the open incidents, closed incidents, or possible workarounds for this service.
     To exit the related record, click **Cancel**, **OK**, or **Close**.
   - Click **Next** to return to the interaction record.
5. Click **Fill** to identify the primary CI.
6. Click the **Smart Indicator** for the affected CI.
7. Type the title and a description of the incident.
8. Categorize the interaction as: **Incident**, **Failure**, and **Function or Feature not working**.
9. Fill out the **Impact**, **Urgency**, and then click **Escalate**.
10. Select the **Location**, which helps decide how to create and route a new incident for assignment. Service Manager creates an incident in the background and you are returned to your original interaction record. From here, you can let Aaron know what the Service Level Agreement (SLA) target time will be and that someone will be working to resolve this incident.
11. Sign out of Service Manager. An Incident Coordinator now assigns the new incident record to an Incident Analyst.

**Incident Analyst**

1. Sign in to Service Manager, locate the incident record in your **To Do** queue, and then accept the incident record to begin your analysis.
2. Mark the incident as a candidate for the Knowledge Base.
3. Create an activity record to describe how you reproduced the issue and how much time you spent on the issue, and then log any other information.
4. Click the **Smart Indicator** icon for the **Service** and **Affected CI** fields to display the visualization wizard that contains interaction and incident records with the same service and CI issue.
5. Double-click a line to browse through the records, and then drill down to a specific record for details. Service Manager displays the record with information related to that field.
6. To exit the related record, click **Cancel**, **OK**, or **Close**.

**Related topics**

Fill function
View Detailed Information function

## 6.12. Attachments

You can attach files to incident, change, problem, and knowledge records in Service Manager. For more information, see the below.

### Add an attachment

To add an attachment to a record, follow these steps:

1. Open the record to which you want to add an attachment. To do this, select a record from the queue or search for a specific record.

   **Note**

   Note You can also add an attachment when you create a new record.

2. Based on your needs, attach files using any of the following methods:
   • Scroll down and click the **Attachments** tab. Click **Add files**, browse to the file or files that you want to attach to the record, and then confirm your selection.

• Scroll down and click the **Attachments** tab, and then directly drag and drop the file or files from your local drive into the Attachments area.
• Press **CTRL** + **V** to attach the screenshot saved in your clipboard. If the cursor is placed in an editable field or text area of the record, the text of `<Screenshot name>.png` will be automatically added right behind the cursor.
A progress bar in the **File Name** column displays the progress of the file upload process.
The file is now uploaded. However, the file is not attached to the record until you click **Save**. To remove a file that's uploaded in error, click the **X** icon in the **Remove** column before you click **Save**.

3. Click **Save**.

**Note**
Note

- The size limit for individual attachments and the space that's available for storing attachments are displayed in the upper-right corner of the **Attachments** section.
- If you try to attach a file that exceeds the size limit for individual attachments or the total available space, you receive an error message, and the attachment is not uploaded.
- If you try to attach a type of file that's not permitted (for example, an .exe file), you receive a message that prompts you to remove the attachment. If you don't remove the attachment, it is removed automatically when you click **Save**.
- Special characters contained in a filename (such as :;/"<>*?\|) are changed to %27 once the file is uploaded.
- There is no limit to the number of files that you can attach to a record, provided that they don't exceed the size limit. However, we recommend that you don't attach more than 20 files to a single record.
- If you refresh the browser or click certain comfill buttons that refresh the browser before the file upload process is complete, the file is not uploaded.
- Whether you can attach a file with a duplicated name against the attachment list depends on the setting of the **preventDuplicatedAttachmentName** parameter.

### Open an attachment

To open a file that's attached to a record, follow these steps:

1. Open the record to which the file that you want to open is attached. To do this, select a record from the queue or search for a specific record.
2. Scroll down and click the **Attachments** tab.
3. To open a single file, click the file name or the download icon in the **Download** column.
   To open multiple files, select the files that you want to open by using the checkboxes next to the file names, and then click **Download**.

   **Note**
   Note When you download multiple attachments concurrently, Service Manager packages the files in a compressed (zipped) folder. Some third-party unzipping tools may not correctly handle file names that contain non-Roman characters. In this situation, the name of the unzipped file may change unexpectedly. We recommend that you use WinRAR to unzip the compressed folder.

4. Click **Save**.

### View the details of an attachment

To view the details of the files that are attached to a record, scroll down to and expand the **Attachments** section of the appropriate record.

**Note**

Note The number of attached files is displayed on the **Attachments** tab heading. This enables you to identify whether a record has attachments quickly without having to expand the **Attachments** section.

However, the number of attached files is not displayed if a custom dynamic view dependency is configured for the section or tab title. This is because the custom dynamic view dependency may include file count information.

If a file is attached to the record, the following information is displayed in the table in the **Attachments** section:

- The name of the attached file
- The size of the attached file (in KB)
- The login name of the person who attached the file
- The date when the file was attached to the record

Attached files are displayed in the order in which they were uploaded.

#### Delete an attachment

To delete an attachment from a record, follow these steps:

1. Open the record from which you want to delete an attachment. To do this, select a record from the queue or search for a specific record.
2. Scroll down and click the **Attachments** tab.
3. To delete a single file, click the **X** icon in the **Remove** column.
   To delete multiple files, select the files that you want to delete by using the checkboxes next to the file names, and then click **Remove**.
4. In the dialog box that appears, confirm the deletion.
5. Click **Save**.

## 7. Accessibility

Our goal is to provide products and services that meet the needs of the widest range of possible users, which includes persons with disabilities and age-related limitations. Our software products are designed and developed to conform to the applicable United States Section 508 Standards from the US Access Board and the W3C Web Content Accessibility Guidelines (WCAG) 2.1 level A and AA success criteria when and wherever possible.

The Service Manager web client contains features that are intended to provide a level of access that meets the needs of persons with disabilities. It is a fully functional client that integrates successfully with assistive technology tools. How the product conforms to the individual standards criteria can be found in the company's conformance reports.

You can find more information about the accessibility policy, the Section 508 requirements, the Voluntary Product Accessibility Template (VPAT) conformance report, and WCAG 2.1 here.

### 7.1. Product usage recommendations

The standard web client and accessible web client both meet the needs of non-administrative users performing basic Service Desk tasks. Administrators need to use the standard web client or the Windows client for administrative tasks with complex forms.

Be aware that Service Manager accessibility has some limitations. See Limitations.

#### Related topics

Web client views
Sign in to the Web client

## 7.2. Web client keyboard interactions and shortcuts

This section introduces keyboard interactions and shortcuts for the Service Manager web client.

### Keyboard shortcuts for general operations

You can customize the Service Manager web client keyboard shortcuts that have a name in the shortcut.xml configuration file. By default, most of the web client shortcuts use the Alt key plus a character to produce a combined keystroke.

The following table shows the default actions assigned to the keystrokes, but your own browser may override these actions.

| Area | Keyboard shortcut | Name in shortcut.xml | Action |
|---|---|---|---|
| Page | **Tab** | N/A | To move the focus into a field, press the Tab key until it reaches that field. To return the focus to the previous location, press Shift + Tab.<br>**Note:** On some forms, there may be fields that aren't displayed. In this case, tabbing does move focus to the invisible field until you press the Tab key again. |
| Button | **Space/Enter** | N/A | Activate the focused button.<br>**Note:** If a button has a shortcut, you can use the shortcut even if the button is not focused. |
| Check boxes | **Space** | N/A | Select or clear the checkbox. |
| Radio buttons | **Arrow keys** | N/A | Select a radio button from a group of buttons. |

Navigation through sections

| Keyboard shortcut | Name in shortcut.xml | Action |
|---|---|---|
| **Alt+Ctrl+N** | keyNavigator | Move the focus to the last focused toolbar button in the navigator |
| **Alt+Ctrl+M** | keyMainContent | Move the focus to the last focused toolbar button in the main content pane<br>**Note:** The **Alt+Ctrl+M** shortcut doesn't work as expected in Edge because this keyboard shortcut is already used as a browser shortcut. In this case, manually change the key value for this shortcut in shortcut.xml to use another key. For more information, see Editing Web client keyboard shortcuts. |
| **Alt+Ctrl+L** | keyList | (QBE view only) Focus on the last focused toolbar button on the list pane |
| **Ctrl+Alt+D** | keyDetail | (QBE view only) Focus on the last focused toolbar button on the detail pane |
| **Alt+Q** | keyTabSections | Navigate through the following panes (not including the toolbar buttons):<br>• Navigator + main content pane<br>• Navigator + list pane + detail pane (if the active tab page is a list-detail record list)<br>**Note:** When you navigate from one pane to another by pressing Alt+Q, the focused element in the original pane is remembered. When you return to the original pane by pressing Alt+Q again, the focus is set on the remembered element. If no element in the original pane was focused, the focus is set on the frame window of the pane. However, if the original pane is refreshed, the focus is set on the first element of the navigator pane or the frame window of the main content pane, list pane, or detail pane. |

| | | |
|---|---|---|
| **Alt+Ctrl+H** | keyFocusTabHeader | Move the focus to the header of the current active tab .<br>When using Ctrl + Alt + H to focus a tab header, and then press the Left/Right Arrow key, the headers of the inactive previous or next tabs are still focusable. |
| **Alt+Ctrl+T** | keyCloseActiveTab | Close the current active tab<br>**Note:** You can't use this shortcut to close the first To-Do queue tab. |
| **Space** | N/A | Activate the tab when the focus is on the header of an inactive tab in a notebook on a form |
| **Enter** | N/A | When the focus is on the header of an inactive tab, pressing **Enter** makes the tab active |
| **Left/Right Arrow** | N/A | Moves the focus to the previous or the next tab in a notebook on a form |

**Navigation through toolbars**

| Keyboard shortcut | Name in shortcut.xml | Action |
|---|---|---|
| **Left Arrow** | N/A | Moves the focus to the previous control. The focus movement wraps from the first element to the last element. |
| **Right Arrow** | N/A | Moves the focus to the next control. The focus movement wraps from the last element to the first element. |
| **Home** | N/A | Moves the focus to the first element. |
| **End** | N/A | Moves the focus to the last element. |

Navigation Menu

| Keyboard shortcut | Name in shortcut.xml | Action |
|---|---|---|
| **Left/Right Arrow** | NavMenu.LRArrows.Key | Expand or collapse an item in the navigator.<br>Right arrow:<br>• Expand a closed node that has child nodes.<br>• Move the focus from an open parent node to its first child node.<br>Left arrow:<br>• Collapse an open node.<br>• Move the focus from a child node to its parent node.<br>Note: For the right-to-left (RTL) language environment, use the opposite arrow instead. |
| **Space** | NavMenu.Space.Key | Expand or collapse an item or a panel in the navigator.<br>For a parent node, expand or collapse the node. For an end node without children, perform the action associated with the node. Usually open a new tab in the central pane. |
| **Up/Down Arrow** | NavMenu.UDArrow.Key | Move the focus to the previous or the next node without expanding or collapsing a node. |
| **Alt+1** | keyManageFav | Open the Manage Favorites pane |
| **Alt+2** | keyAddListFav | Add a favorite that links to the current list |
| **Alt+3** | keyAddDetailFav | Add a favorite that links to the current record |
| **Alt+N** | keyCollapseExpandNav | Switch the System Navigator between the expanded mode and the collapsed mode |
| **Enter** | NavMenu.Enter.Key | • For a parent node, expand or collapse the node.<br>• For an end node without children, perform the action associated with the node. Usually, open a new tab in the central pane. |

| | | |
|---|---|---|
| **Home/End** | N/A | Move the focus to the first or the last node in the tree without expanding or collapsing a node. |
| **A-Z** | N/A | If you type a letter, move the focus to the next node with a name that begins with the typed letter. |

**Note:** When the System Navigator receives the focus, if a node is focused before, the focus is set on the previously focused node. Otherwise, the focus is set on the first node.

Record list

| Keyboard shortcut | Name in shortcut.xml | Action |
|---|---|---|
| **Space** | Recordlist.Space.Key | Select or deselects a record |
| **Up/Down Arrow** | Recordlist.UDArrow.Key | Move the focus through the rows and the focused row is selected by default |
| **Left/Right Arrow** | Recordlist.LRArrow.Key | Expand or collapse a group in a list<br>**Tip:** Press the Right arrow key to open the column header context menu for a table column header |
| **Ctrl+Up/Down Arrow** | Recordlist.CtrlUDArrow.Key | (Multi-selectable record list only) Focus on the previous/next row, but keep the selection state of rows |
| **Shift+Up/Down Arrow** | Recordlist.ShiftUDArrow.Key | (Multi-selectable record list only) Select multiple adjacent rows |
| **Ctrl/Alt+Page Up/Down** | Recordlist.Ctrl+PageUpDown.Key | Navigate to the previous or next page of a ungrouped record list. The first row of the target page is selected by default. If the target page has been accessed before, the previously selected row is selected.<br>**Note:** The **Ctrl+Page Up/Down** shortcut doesn't work in Firefox and Chrome. |
| **Enter** | Recordlist.Enter.Key | Drill down to the current focused row |
| **Home/End** | Recordlist.HomeEnd.Key | Move the focus to the first/last cell in the row that contains focus. |
| **Ctrl+Home/End** | Recordlist.CtrlHomeEnd.Key | • Move the focus to the first or last cell in the current table in a ungrouped record list<br>• Move the focus to the first or last group node in a grouped record list |
| **Alt+U** | keyToggleList | (QBE view only) Collapse/expand the list pane |

Detail Form

| Keyboard shortcut | Name in shortcut.xml | Action |
|---|---|---|
| **Alt+J** | keyJumpAddress | Focus on the combo box containing links to each group on the detail form and toggle the state of the combo box. Note that the combo box is visible only when the **Preferred Notebook Style** of the operator is set to **Groups**. |
| **F1** | functionKeyHelp | View help on field |
| **F2** | functionKeyMagnify | Expand a scrollable text field by opening a separate window that magnifies the text for easier viewing and editing |
| **Enter** | N/A | If the focus is in a link, pressing Enter will also drill down to the link.<br>If the focus is not in a link, pressing Enter will select the current row and move the focus to the drill-down link of the row (Double–Click Field). The Attachment control in a detail form doesn't use Enter to select a row. |

Collapsible groups in the detail form

| Keyboard shortcut | Name in shortcut.xml | Action |
|---|---|---|
| **Enter** | N/A | When the focus is on the title of a collapsible group, press Enter to expand or collapse the selected group. Once the group is expanded, you can press the Tab key to navigate through the controls in the group. |

Table on the detail form

| Keyboard shortcut | Name in shortcut.xml | Action |
|---|---|---|
| **Home/End** | Recordlist.HomeEnd.Key | Move the focus to the first/last cell in the row that contains focus. |
| **Ctrl + Home** | Table.CtrlHome.Key | Move the focus to the first cell in the first row. However, if a table has sortable column headers, move the focus to the first column header. |
| **Ctrl + End** | Table.CtrlEnd.Key | Move the focus to the last cell in the last row |
| **Enter** | Recordlist.Enter.Key | If the focus is on a sortable column header, sort the column. |
| **Space** | Recordlist.Space.Key | If the focus is on a sortable column header, sort the column. |

Collaboration

| Keyboard shortcut | Name in shortcut.xml | Action |
|---|---|---|
| **Alt+Ctrl+C** | N/A | Open or close the conversation window in both the power user view (index.do) and the ESS view (ess.do) |
| **Alt+Ctrl+V** | keySmcNotification | Open or close the Notification panel |
| **Alt+Ctrl+B** | keySmcChatRequest | Open or close the Chat Request panel |

Smart Search

| Keyboard shortcut | Name in shortcut.xml | Action |
|---|---|---|
| **Alt+Ctrl+S** | keyOpenSmartSearch | Open the Smart Search Dialog |
| **Alt+Ctrl+U** | keyOpenQuickSearch | Open the Context Aware Search panel |
| **Alt+Ctrl+1** | keySmartSearchMainTopRegion | Move the focus to the Smart Search Editor on the Smart Search dialog |
| **Alt+Ctrl+2** | keySmartSearchMainLeftRegion | Move the focus to the first library of the library list on the Smart Search dialog |
| **Alt+Ctrl+3** | keySmartSearchMainRightRegion | Move the focus to the first search result on the Smart Search dialog |

Miscellaneous

| Keyboard shortcut | Name in shortcut.xml | Action |
|---|---|---|
| **Alt+Ctrl+A** | keyAlertWindow | Open/Close the alert window if there is a To-Do alert message at minimum |
| **Alt+Ctrl+G** | N/A | Clear the message bar on a page |
| **Alt+G** | keyMessages | Open or close the Messages and Alerts window |
| **Alt+P** | keyPrint | • Print the current page content<br>• (QBE view only) Print all content on the detail pane |
| **Alt+Ctrl+P** | keyPrintList | • Print the current page content<br>• (QBE view only) Print all content on the list pane |
| **Alt+O** | keyAbout | Open the About the Software window and display the installed Service Manager components |
| **Alt+R** | keyRefresh | Refresh the current screen<br>**Note:** Do not refresh Service Manager screens by refreshing your browser. Instead, use the Service Manager **Refresh** button or shortcut (Alt+R). |
| **Alt+L** | keyLogout | Open the logout dialogue |
| **Alt+Ctrl+/** | keyShortcut | Show or hide system keyboard shortcut list |

### Keyboard shortcuts for date picker

The Service Manager web client provides a date picker widget that assists you to input date and time. The following table shows the keyboard shortcuts for the date picker widget.

| Keyboard shortcut | Action |
| --- | --- |
| Esc | Close the date picker widget without selection |
| Enter | Choose the highlighted date |
| Shift+Enter | Choose today |
| Left Arrow | Previous day |
| Right Arrow | Next day |
| Up Arrow | Previous week |
| Down Arrow | Next week |
| Shift+Left Arrow | Previous month |
| Shift+Right Arrow | Next month |
| Shift+Up Arrow | Previous year |
| Shift+Down Arrow | Next year |
| Tab | Switch focus between date and time |
| Down Arrow (when the focus is on the time field in the date picker widget) | Open the time list |
| Enter (when the time list is open) | Choose highlighted time |

**Keyboard shortcuts for list builder**

The list builder widget allows you to construct a list of choices. The following table shows the keyboard shortcuts for the list builder widget.

| Keyboard shortcut | Action |
| --- | --- |
| Left Arrow<br>Right Arrow<br>Up Arrow<br>Down Arrow | Select items in the list builder |
| Shift+Left Arrow<br>Shift+Right Arrow | Move the selected item to the left or right column |
| Shift+Up Arrow<br>Shift+Down Arrow | Move the selected item up or down in the right column |

Keyboard shortcuts for decimal control

| Keyboard shortcut | Action |
|---|---|
| Up Arrow | Increase the value by one. |
| Down Arrow | Decrease the value by one. |
| Right Arrow | (Slider view only) Increase the value by one. |
| Left Arrow | (Slider view only) Decrease the value by one. |
| Page Up | (Slider view only) Increase the value by the granularity as specified in the Step property.<br>For example, suppose the current value is 3 and the step value is 5. When you press Page Up, the value increases to 8. When you press Page Up again, the value increases to 13. |
| Page Down | (Slider view only) Decrease the value by the granularity as specified in the Step property.<br>For example, suppose the current value is 23 and the step value is 5. When you press Page Down, the value decreases to 18. When you press Page UpDown again, the value decreases to 13. |
| Home | Set the value to the minimum if the decimal control has a minimum value. |
| End | Set the value to the maximum if the decimal control has a maximum value. |
| Number key | (Spinner view only) Set a valid value directly. |

Keyboard shortcuts for tag control

| Keyboard shortcut | Action |
|---|---|
| Down Arrow | Open the complete auto-suggestion list that displays the most commonly used tags for this type of record. When you are typing, the system automatically filters the list and displays the tags that match your input. |
| Enter<br>Comma | Separate multiple tags. |
| Backspace | Remove the last tag. |



**Note**

When JAWS is running, use JAWS keystrokes to read the list of tags.

**Keyboard shortcuts for detail tab headers and notebook tab headers**

| Keyboard shortcut | Action |
|---|---|
| Tab | When the focus is on the header of a tab, moves the focus to the first element inside the active tab panel. |
| Left Arrow | When the focus is on the header of a tab, moves the focus to the previous tab.<br>(Notebook tab only) If the focus is on the first tab, moves the focus to the last tab. |
| Right Arrow | When the focus is on the header of a tab, moves the focus to the next tab.<br>(Notebook tab only) If the focus is on the last tab, moves focus to the first tab. |
| Home | When the focus is on the header of a tab, moves the focus to the first tab. |
| End | When the focus is on the header of a tab, moves the focus to the last tab. |
| Enter/Space | When the focus is on the header of an inactive tab, you can press Enter or Space to make the tab active. |

**Keyboard shortcuts for combo box**

| Keyboard shortcut | Action |
|---|---|
| Alt+Down Arrow | Expand the drop-down list of a combo box. |
| Down Arrow | Expand the drop-down list of a combo box. |
| Up/Down Arrow | When the drop-down list of a combo box is expanded, move through items in the list box. During this process, The value in the input box is NOT updated. |
| ESC | When the drop-down list of a combo box is expanded, close the list without updating the value in the input box. |
| Enter | When the drop-down list of a combo box is expanded, update the value in the input box to the highlighted item and close the list. |
| Printable Characters | Type a text into the input box of a combo to search for a valid item.<br>• If there is no matching value, the drop-down list is not expanded;<br>• If there are matching values, the drop-down list is expanded, showing all matching items. For a select-only combo box, the text is also auto-completed in the input box; for a non-select-only combo box, the text is not auto-completed. |

**Keyboard shortcuts for HTML Editor**

| Keyboard shortcut | Action |
|---|---|
| Alt+F10 | Focus the opened popup or toolbar of the rich text editor. |
| Esc | Return focus to the previous position. |
| Ctrl + C | Copies selection from the editable are to clipboard. |
| Ctrl + X | Cuts the selection from the editor to clipboard. |
| Ctrl + V | Pastes from clipboard. |
| Ctrl + Z | Same effect as clicking on the Undo button in the editor. |
| Ctrl + Y | Same effect as clicking on the Redo button in the editor. |
| Ctrl + K | Opens the insert link popup to insert a link in the editable area. |
| Ctrl + P | Opens the insert image popup to insert an image in the Javascript rich text editor. |
| Ctrl + A | Selects all text in the editable area of the rich text editor. |
| Ctrl + B | Same effect as clicking on the Bold button in the editor. |
| Ctrl + I | Same effect as clicking on the Itallic button in the editor. |
| Ctrl + U | Same effect as clicking on the Underline button in the editor. |
| Ctrl + S | Same effect as clicking on the Strikethrough button in the editor. |
| Ctrl + ] | Indents the current paragraph. |
| Ctrl + [ | Outdents the current paragraph. |

### Keyboard shortcuts for Javascript editor

#### Editable mode

| Keyboard shortcut | Action |
| --- | --- |
| Esc | Activate the Tab Move Focus mode.<br>When the focus is on the Javascript editor, press Esc to activate the Tab Move Focus mode. And the subsequent Tab/Shift+Tab keys will move the focus out of the editor.<br>To exit from the Tab Move Focus mode, move the focus to the Javascript editor, and then press any keyother than Esc, Tab, or Shift. The subsequent Tab/Shift+Tab key will restore the original function of indenting. |
| Tab | If something is selected, indent it by one indent unit. If nothing is selected, insert a tab character. |
| Ctrl+] | Indent the current line or selection by one indent unit. |
| Ctrl+[ | Dedent the current line or selection by one indent unit. |
| Shift+Tab | Auto-indent the current line or selection. |

#### Read-only mode

| Keyboard shortcut | Action |
| --- | --- |
| Tab<br>Shift+Tab | Move the focus out of the Javascript editor. |

### Keyboard shortcuts for Calendar

#### Time Period record list panel & Object record list panel

| Keyboard shortcut | Action |
| --- | --- |
| Tab | Switch between panels |
| Up/Down Arrow | Select a record in the panel |
| Enter/Space | Drill down to the details of each record |

#### Date picker panel

| Keyboard shortcut | Action |
| --- | --- |
| Left/Right/Up/Down Arrow | Navigate between days or weeks |
| Shift+Up/Down Arrow | Switch between years |
| Shift+Left/Right Arrow | Switch between months |

**Submenu**

| Keyboard shortcut | Action |
|---|---|
| Right Arrow | Show the submenu |
| Left Arrow | Hide the submenu |

**Tree panel**

| Keyboard shortcut | Action |
|---|---|
| Up/Down Arrow | Select filters |
| Space | Select the highlighted child filter |
| Delete | Delete the selected child filter |
| Enter | Open the selected child filter |

**Advanced Selector window**

| Keyboard shortcut | Action |
|---|---|
| Up/Down Arrow | In the **Available** panel, highlight the field for your selection |
| Shift+= | Add the selected field to the **Selected** panel |
| - | In the **Selected** panel, remove the selected field |

### Keyboard shortcuts for Reports

**Reporting dashboard**

| Keyboard shortcut | Action |
|---|---|
| Tab | Switch between the dashboard toolbar buttons and the widget toolbar buttons |
| Alt+Left/Right Arrow | Select a report widget on the report dashboard |

**Standard List widget & Title List widget**

| Keyboard shortcut | Action |
|---|---|
| Tab | Switch between the records |
| Enter | Drill down to the details page of the selected record |
| Tab | Move the focus to the List All button on the bottom of the screen, and then press Enter to open the full records list page |

Number widget

| Keyboard shortcut | Action |
| --- | --- |
| Tab | Focus the total Number |
| Enter | Drill down to the full list |

### Keyboard shortcuts for Smart Analytics

Smart Search pop-up window

| Keyboard shortcut | Action |
| --- | --- |
| Alt+Ctrl+1 | Move the focus to the Smart Search editor |
| Alt+Ctrl+2 | Move the focus to the first library of library list |
| Alt+Ctrl+3 | Move the focus to the first search result record |

Suggest Solution button

| Keyboard shortcut | Action |
| --- | --- |
| Alt+Ctrl+U | Move the focus to the Suggest Solution button if it is displayed on the detail view of Incident, Request, Change, Problem, and Interaction |
| Enter | Open the Context Aware Search panel |

Other areas

| Keyboard shortcut | Action |
| --- | --- |
| Tab | Switch between the active components |
| Alt+Ctrl+S | Open the Smart Search windows |

### Quick help for keyboard shortcuts

In the web client, you can open quick help to view the keyboard shortcuts for the web client. To do this, use one of the following methods:

- Click the **Help** button
  - If the web client doesn't have online help, the shortcut list directly opens.
  - If the web client has online help, a drop-down menu appears. Click **Shortcut List** from the drop-down menu.
- Use the keyboard shortcut: Alt+Ctrl+/

You can adjust the size of the list or move the list to anywhere within the browser window. When the browser window resizes or you open the list again, the list restores to the default size and position.

The keyboard shortcuts that the quick help presents are based on the settings in the shortcut.xml file. If the configuration of the shortcut.xml file is incorrect, the quick help displays an error message.

## 7.3. Accessibility designs for Service Manager graphic features

Service Manager graphic features have the following accessibility designs:

| Service Manager graphic features | Accessibility designs |
| --- | --- |
| Service Manager color indicator | No accessibility support.<br>Color indicator doesn't support Accessible view (accessible.do) and Self-Service Accessible view (accessible_ess.do) because this feature is designed for people with normal vision. JAWS reads the value in a column instead of reading the color of the column. |
| Service Manager navigation menu | The sidebar mode only works in Power User view (index.do) and Enterprise Self-Service view (ess.do), but doesn't work in Accessible view and Self-Service Accessible view. You can use JAWS to read the accordion mode of the Service Manager navigation menu. |
| Service Manager Process Designer framework | Limited accessibility support.<br>You can use the keyboard to navigate to the Workflow Viewer panel. When the focus is on a record's detail in the Service Manager Process Designer Workflow Viewer, JAWS reads the current phase and the next phase value. When the focus is on another place in the Workflow Viewer (such as on the category detail page), JAWS reads "this is workflow diagram."<br>The Process Designer framework administration tools support keyboard navigation only. These tools include Task Planner, Condition Editor, Query Editor, Workflow List, and Workflow Editor.<br>To modify the phase property of a Process Designer framework administration tool, you need to press the Tab key to focus on a phase and then press Enter to open the phase property. You can update the property by using keyboard navigation. |
| Service Manager Calendar | Limited accessibility support.<br>You can use keyboard to navigate through the user interface elements, See the *Keyboard shortcuts for Calendar* section in Web client keyboard interactions and shortcuts. |
| Service Manager Reports | Limited accessibility support.<br>You can use keyboard to navigate through the user interface elements. See the *Keyboard shortcuts for Report* section in Web client keyboard interactions and shortcuts. |
| Service Manager Smart Analytics | You can use keyboard to navigate through the user interface elements. See the *Keyboard shortcuts for Smart Analytics* section in Web client keyboard interactions and shortcuts. |

**Related topics**

Limitations

## 7.4. Limitations

Service Manager accessibility in this software release has the following limitations.

### General accessibility limitations

Service Manager has the following general accessibility limitation:

- Background images introduced as of the Service Manager 9.40 release don't support the Windows high contrast mode.

### Table accessibility limitations

Service Manager table accessibility has the following limitations:

- Table accessibility support is not implemented for the following modules: Process Designer, and Calendar.

### To-Do alerts accessibility limitations

Service Manager To-Do alerts accessibility has the following limitations:

- The To-Do alerts supports keyboard navigation in Power User view (index.do) only.
- The To-Do alerts doesn't support Accessible view (accessible.do) and Self-Service Accessible view (accessible_ess.do).

### Reports accessibility limitations

Service Manager Reports accessibility has the following limitations:

- You can't move the focus to the Area chart, Bar chart, Line chart, and Pie chart by using keyboard navigation.
- JAWS can't read the Area chart, Bar chart, Line chart, and Pie chart meaningfully.
- JAWS can't read the Pivot table meaningfully.

### Smart Analytics accessibility limitations

Service Manager Smart Analytics accessibility has the following limitations:

- You can't move the focus to the Hot Topic Analytics chart by using keyboard navigation.
- JAWS can't read the content of the Hot Topic Analytics chart.

### Service Manager Collaboration accessibility limitations

Service Manager Collaboration accessibility has the following limitations:

- Even if a user has changed the photo, the **User Information** on the top-right corner of the Service Manager UI always displays the default image in Windows high contrast mode.

## 7.5. Client configuration recommendations

There are a number of accessibility tools available from third-party vendors. These tools vary in the disabilities they address and the technology they support. We strive to ensure that our software products don't interfere with the proper functioning of any accessibility product. We test the Service Manager accessible Web client with several screen reader products; however, it is important that users perform due diligence to ensure that any accessibility tools integrate successfully with Service Manager.

### Screen readers and the Service Manager web client

The Service Manager Web client conforms to Web Content Accessibility Guidelines (WCAG) 2.1, satisfies most of the Success Criteria in WCAG 2.1, and can work well with a screen reader such as JAWS. You can configure JAWS to read all of the tags in a custom hierarchy. For example, you can configure form fields and links with the Custom Search String to read the texts, titles, and alternative texts. Other recommendations are:

- Use text/title for all elements except images.

- Configure graphics settings to read only Tagged Graphics.
- Do not use JAWS in the Graphics Mode with the Service Manager Web client.
- Configure buttons to use both Label and Title if different.

Service Manager uses many long titles to describe the action or the result of following a link or button. Forms can be very complex with many form fields and links, resulting in redundant, but necessary, oral descriptions. We recommend that novice Service Manager users slow down the words-per-minute setting until they are familiar with the application. Some out-of-box (OOB) forms may be complex for screen readers. If this occurs, application administrators can re-design complex forms and add accessibility attributes as needed.

The Service Manager Web client supports all accessible users and is designed to integrate with screen reader assistive technology.

#### JAWS

Service Manager supports different browsers; however, Microsoft new Edge based on Chromium provides the best integration with JAWS and the Service Manager accessible web client (accessible.do and accessible_ess.do). Visually or hearing impaired users can use the Service Manager accessible web client with the out-of-box configuration. If you customize the JAWS advanced configuration settings, you can improve the usability and understanding of the application for some users. For more information, see the JAWS documentation. We don't provide third-party technical support.



**Caution**

Caution

JAWS is unsupported on the Service Manager standard web client or the Windows client.



**Note**

Note

Only the following combination of JAWS and browser is supported: JAWS 2024 and new Edge based on Chromium.

## 8. Privileges in Windows systems

### Install and uninstall

To install or uninstall the Service Manager server, a user must have administrator privileges.

### Run the Service Manager server

#### Run the Service Manager Server using Windows Command

To run the Service Manager Server Command in a command Window, the user must have administrator privileges and open an elevated command prompt.

1. Click the **Start** menu.
2. Type **cmd**, right-click the **Command Prompt** tile, and then click **Run as administrator**.
3. If prompted, click **Yes**.
4. Start the server from the Windows command prompt.

#### Run the Service Manager Server using Windows Service

You can run the Service Manager Server service as an administrator user. See Start the server from Windows services.

If you need to run the Service Manager Server service as a non-administrator user, follow these steps.

**Note**

Note  The feature to run the Service Manager Server service as a non-administrator user is available after you appy the Service Manager 9.80 patch 1.

1. Create or specify a non-admin user to run the Service Manager server.
2. Create or specify a group to run the Service Manager server. Add the user to this group.
3. Press Windows key + R to open the RUN tool, type **gpedit.msc**, and press **Enter**. The **Local Group Policy Editor** console will open.
4. In the left pane, navigate to **Computer Configuration** > **Windows Setting** > **Security Settings** > **Local Policies** > **User Rights Assignment**.
5. In the right pane, among the list of policies, double click on the **Create global objects** policy.
6. Click on **Add User or Group**, and then add the group defined in step 2 to the list.
7. In the right pane, among the list of policies, double click on the **Manage auditing and security log** policy.
8. Click on **Add User or Group**, and then add the group defined in step 2 to the list.
9. Grant **Modify** permission to the SM install folder for the group defined in step2.
   Note that you also need to grant **Modify** permission to those folders that require SM access, such as **svc_rootdir** and **dashboard_export_path**.
10. Grant the privilege to the group defined in step 2 for smmasterkey folder permission.
11. In the sm.ini file, add the group defined in step 2 to the ntsecuritygroup parameter: `ntsecuritygroup:<groupname>`
12. Press Windows key + R to open the RUN tool, type **services.msc**, and press **Enter**. The service console will open.
13. In the Windows service console, double click on the Service Manager server service, and then select the **Log On** tab. Select **This account**, and add the user defined in step 1.
14. Start the server from Windows services.

### References

OpenSCManagerA function (winsvc.h)
How to grant users rights to manage services
Standard Access Rights
Process Security and Access Rights
MapViewOfFile function (memoryapi.h)
Synchronization Object Security and Access Rights
Access Rights for Access-Token Objects
CreateFileMappingA function (winbase.h)

## 9. Personas

Service Manager has the following personas:

### System administrator

Has full access to system administration. The default user name is **falcon** whose initial password is specified after the first login. The falcon user can add additional system administrators in **System Administration** > **Ongoing Maintenance** > **Operators**.

### IT agent

Has access to the business modules, as well as the Service Portals. The IT agents can be assigned different roles with permissions that enable them to complete their daily work. For example, the Change Manager role manages changes and runs the change process activities. For more information about Change Management user roles, see Change Management user roles (Codeless) and Change Management user roles (Hybrid).

For more information about the user roles for other business modules, see the user roles topics in Use Service Manager.

### Service Portal user

Has access to the Service Portals. The Service Portal user is an end user persona.

## 10. Use case

This section walks you through the following end-to-end use cases:

- Send web tier URLs through e-mail notifications
- Creating an email notification for Change updates
- Use Feature Tracker for Service Manager version control
- Visualize SLM in the record list
- Visualize SLM on SLT View tab with a suspended ticket
- Visualize SLM on SLT View tab with altered assignment groups
- Collaborate on Teams with war room experience
- Interact with Service Manager by using agent bot
- Interact with Service Manager by using virtual agent
- Synchronize Service Manager Change to ALM Octane defect
- Synchronize Service Manager Change to ALM Octane feature
- Synchronize Service Manager Change to ALM defect
- Synchronize Service Manager Change to ALM requirement
- Synchronize Service Manager Problem to ALM defect
- Synchronize ALM defect to Service Manager Problem
- Synchronize Service Manager Incident to Jira bug
- Synchronize Service Manager Change to Jira new feature

### 10.1. Send web tier URLs through email notifications

**Applies to User roles:** System Administrator

This use case creates a notification record with the following properties.

- Creates an email message whenever someone updates a Service Desk interaction record.
- The email message contains a direct URL to the updated interaction record.
- The email message contains a URL to the self-service interface if the record was opened from self-service or a URL to the Web client interface if it was opened from any other interface.

**Note**: This use case assumes you have installed the Service Manager Web tier and email service.

Complete the following tasks in sequential order:

- Task 1: Update the Web configuration file
- Task 2: Update the system information record
- Task 3: Update the SD.incident.mail form
- Task 4: Update the SM Update notification record
- Task 5: Create and update a self-service interaction record
- Task 6: Review the output event

#### Update the Web configuration file

Follow these steps:

1. Sign in to the Service Manager web client.
2. Stop the Web tier Web application server.
3. Open the Service Manager Web tier's Web configuration file (web.xml) in a text editor.

> **Note**
> Note The web.xml file is in the WEB-INF folder of your Web tier's context root.

4. Search for the following entry:

```
<init-param>
  <param-name>serverHost</param-name>
  <param-value>localhost</param-value>
</init-param>
```

5. Change the `localhost` param-value to the fully-qualified domain name of your Service Manager server, for example, myserver.mydomain.com.
   Replace the values *myserver* and *mydomain.com* with the server hostname and domain name of the Service Manager server.

> **Important**
> Important You must specify a fully qualified domain name for the **serverHost** parameter or the server will generate invalid URLs.

6. Save the Web configuration file.

### Update the system information record

Follow these steps:

1. Restart your Web application server.
2. Sign in to the Service Manager server as a system administrator.
3. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
4. Click the **Active Integrations** tab.
5. In the **WebServer URL** field, type the fully qualified URL to your Web tier. For example:

```
http://myserver.mydomain.com:myport/SM/index.do
```

   Replace the values *myserver* and *mydomain.com* with the server hostname and domain name of the server running the Service Manager Web tier. Replace *myport* with the communications port your Service Manager Web tier Web server listens for HTTP requests.
   The server stores the value of this field in the **$L.web.url** global variable.
6. In the **ESS URL** field, type the fully qualified URL to your Web tier. For example:

```
http://myserver.mydomain.com:myport/SM/ess.do
```

   Replace the values *myserver* and *mydomain.com* with the server host name and domain name of the server running the Service Manager Web tier. Replace *myport* with the communications port your Service Manager Web tier Web server listens for HTTP requests.
   The server stores the value of this field in the **$L.ess.url** global variable.
7. Save the system information record.

### Update the SD.incident.mail form

Follow these steps:

1. Sign out and restart the Service Manager server.
2. Sign in to Service Manager again as a system administrator.
3. Click **Tailoring** > **Forms Designer**.

4. Search for the form SD.incident.mail.
5. Click **Design**.
6. Add a new text area field with the following properties.

| Property | Value |
|---|---|
| Input | $L.web.url |
| Width | 400 |

   This form will display the URL to the Web client interface.
7. Make any other changes to the form.
   You will use this form as the template for your email messages.
8. Click **OK** twice to exit Forms Designer and save the form.
9. Search for and select the **SD.incident.mail** form .
10. Click **More** or the More Actions icon, and then click **Copy/Rename**.
11. In the **New Name** field, type `SD.incident.mail.ess` .
12. Select the **Copy** option.
13. Click **OK**.
14. Click **Design**.
15. Select the text area field you created to display the Web tier URL and update it to have the following properties.

| Property | Value |
|---|---|
| Input | $L.ess.url |
| Width | 400 |

   This form will display the URL to the self-service interface.
16. Click **OK** twice to exit Forms Designer and save the form.

**Update the SM Update notification record**

Follow these steps:

1. Click **Tailoring** > **Notifications** > **Notifications**.
2. Search for the **SM Update** notification record.
3. Add the following new rows to the notification record.

| URL sent | Msg class | Msg No. | Arguments | Condition | Format | Notify Method | Recipient(s) |
|---|---|---|---|---|---|---|---|
| Self-service | sm | 10 | incident.id in $L.file | ess.entry in $L.file=true | SD.incident.mail.ess | email | contact.name in $L.file |
| Web client | sm | 10 | incident.id in $L.file | nullsub(ess.entry in $L.file,false)=false | SD.incident.mail | email | contact.name in $L.file |

4. Click **OK** to save the notification record.

#### Create and update a self-service interaction record

Follow these steps:

1. Open a Web browser and sign in to the self-service interface.
2. Create a new self-service interaction record and note the interaction record ID.
3. From a windows client, click **Service Desk > Search Interaction Records**.
4. Use the interaction record ID of the self-service interaction record to retrieve the interaction record details.
5. Add a category, a subcategory, and an update to the self-service interaction record.
6. Exit out of the self-service interaction record and return to the Service Desk search form.
7. Search for SD1001.
8. Open the **Activities** section.
9. In the **Update** field, type an update.
10. Click **Save**.

#### Review the output event

Follow these steps:

1. Click **Tailoring** > **Event Services** > **Output Events**.
2. Click **Search**.
3. Look for an output event with an **Event Code** of email and the ID number of the self-service interaction record that you created in the **External Information String** field.
4. If you are working in a Windows client, right-click in the **External Information String** field and click **Magnify** to display the field text in a new window.
5. Search the form for a URL.
   The URL should contain the path to your self-service interface, for example:

   ```
   http://myserver.mydomain.com:myport/SM/ess.do?
   ctx=docEngine&file=incidents&query=incident.id=%22SD10005%22
   &queryHash=eb12d3d8&action=&title=Interaction%20;SD10005
   ```

6. Look for an output event with an **Event Code** of email and SD1001 in the **External Information String** field.
7. If you are working in a Windows client, right-click in the **External Information String** field and click **Magnify** to display the field text in a new window..
8. Search the form for a URL.
   The URL should contain the path to your Web client interface, for example:

   ```
   http://myserver.mydomain.com:myport/SM/index.do?
   ctx=docEngine&file=incidents&query=incident.id=%22SD1001%22
   &queryHash=89472347&action=&title=Interaction%20;SD1001
   ```

### 10.2. Creating an email notification for Change updates

**Applies to User roles:** System Administrator

The following example describes how to set up a notification that meets the following requirements:

- The notification sends an email message when a Change record is updated.
- The email message displays `Change [Num] Updated by [Operator]` where `[Num]` represents the number of the updated change and `[Operator]` represents the name of the operator who updated the change.
- The email message is sent to the Change Manager.

Complete the following tasks in sequential order:

- Task 1: Create a custom message record
- Task 2: Create a custom notification definition record

- Task 3: Enable the notification in the object record
- Task 4: Test the notification

**Create a custom message record**

To create a custom message record, follow these steps:

1. Click **Tailoring** > **Notifications** > **Messages**.
2. Specify the properties as listed in the following table.

| Name | Value |
|---|---|
| Language Code | en |
| Class | ndp |
| Message Number | 9001 |
| Severity | 3 |
| Text | Change %S Updated by %S |
| Comments | Sent when a change is updated. |

3. Click **Add**.

**Create a custom notification definition record**

To create a custom notification record, follow these steps:

1. Click **Tailoring** > **Notifications** > **Notifications**.
2. Type NDP Change Update in the **Name** field, and then click **Add**. An ID is automatically created.
3. Type true in the **Condition** field.
4. On the **Message** tab, add a message entry in the list with properties listed in the following table:

| Name | Value |
|---|---|
| Msg Class | ndp |
| Msg No. | 9001 |
| Arguments | {number in $L.file,operator in $L.file} |
| Condition | true |
| Notify Method | email |
| Recipient(s) | $G.chmmanagerEmail |

   Note Set **Msg No** to an unassigned number. Use a custom message class for the **Msg Class** field.
5. (Optional) On the **Email Attachment** tab, define the conditions and query filters under which Service Manager should add attachments from the original change record to the notification.
   **Tip:** We recommend keeping the **scheduled.action.processor** background process running when working with attachments in notifications. Hence, the **Run Scheduled Action** rules work at run time and Service Manager performs scheduled automatic cleaning of the redundant attachments that accumulate in the current record.
6. Click **Save**.

#### Enable the notification in the object record

To enable the notification in the object record, follow these steps:

1. Click **Tailoring** > **Document Engine** > **Objects**.
2. Type =cm3r in the **File name** field, and click **Search**.
3. Click the **Notifications** tab, verify that the **Update** field is empty, and then type NDP Change Update in the **Update** field.
4. Click **Save**.

#### Test the notification

To test the notification, follow these steps:

1. Temporarily turn off the email agent that you are using, such as JavaMail, or switch it to debug mode. This is to ensure that newly created email messages remain in the eventout table.
2. Click **Change Management** > **Search Changes**.
3. Use the Search feature to locate a change record. For example, C10001.
4. Modify the **Description** field.
5. In the **Activities** section, select **Update** for the **New Update Time** field, and write a description in the **New Update** field.
6. Click **Save** to save the changes.
7. Click **Tailoring** > **Event Services** > **Output Events**.
8. Type email in the **Event Code** field, and click **Search**.
9. Verify that an event record is created that has an External Information String that resembles the following.
   `change.manager@advantage.com^falcon^change.manager@advantage.com^Change C10001 Updated by falcon^Change C10001 Updated by falcon`

### 10.3. Use Feature Tracker for Service Manager version control

Service Manager provides comprehensive and fully-integrated IT service management through variously customized applications. Although Service Manager already provides an in-tool revision control based on the individual record, it has always been a challenge to implement version control for customization stacks. Service Manager can integrate with Apache Subversion (SVN) or Git to provide a version control solution for your customized applications.

This use case describes a typical use case to leverage Feature Tracker for version control of Service Manager customization:

1. Prepare Git.
   1. The system administrator exports all the applications records from Service Manager and check in the exported files to a Git server to create a baseline repository. For details, see the *Prepare a code repository* section in the Prepare Git topic.
   2. Developers check out the latest version of the applications records and import them into a local Service Manager development environment. For details, see the *Prepare a development environment* section in the Prepare Git topic.
2. Prepare Service Manager.
3. Before performing configurations and using Feature Tracker, make sure an operator have the **FeatureTrackerCommit** and **FeatureTrackerReleaseManager** capability words in the user profile record.
4. Configure Feature Tracker settings. See the *Configure settings to track features* section in the Using Feature Tracker topic.
5. Create a release, and then create a feature by using Feature Tracker. For details, see the *Create a feature or release record* section in the Using Feature Tracker topic.
6. Open the feature record, and then relate the feature to a release by selecting the release in the **Related Release** drop-down list.
7. Developers customize application data records of Service Manager.
8. Open the feature record and then click the **Objects** tab to view the modified objects. Click the **Refresh Objects** button to refresh the object list. Select the objects of your interest from the list and then click **Link selected objects**

**with the feature**. For details, see the *View detailed information of a feature record* section in the Using Feature Tracker topic.

9. Committing customization changes to your source control repositories by using Feature Tracker. For details, see the *Commit changes using Feature Tracker* section in the Using Feature Tracker topic.
   Operators with commit right to Feature Tracker can check in or discard the changes.
10. Developers check the commit in the Git repository and then merge it to the main repository.
11. When the feature is ready to release, the release manager can go to the related Release record and then build a release by using Feature Tracker. For details, see the *Build releases using Feature Tracker* section in the Using Feature Tracker topic.
12. The release manager deploys the desired release on the test Service Manager system by using Feature Tracker. For details, see the *Deploy releases using Feature Tracker* section in the Using Feature Tracker topic.
13. The release manager deploys the desired release on the production Service Manager system by using Feature Tracker.

### 10.4. Visualize SLM in the record list

Both the Incident and Interaction queues in Service Manager provide the **Next SLA Target Time** and **Next OLA Target Time** columns that display the operation status of the Service Level Targets (SLTs). In the columns, the progress bars visualize the natural time and the color indicators highlight the status of the remaining time for the latest expiring SLT. An IT agent can click the column headers to sort the records by their natural remaining time and hence arrange the priority of the current work.

This use case describes how the SLM visualization works in the record list when an SLA and an OLA work on an Incident. The IT agent can view the indicators' color changes as time elapses.

#### Prepare SLA and OLA

This example uses the following OLAs, SLAs, and Process Targets (SLTs) and most of them are available out-of-the-box. Create similar data if you're not working with the out-of-the-box data and don't have the examples.

| Item | Values |
| --- | --- |
| OLA 176 | • Title: OLA Email Team<br>• Assignment Groups: E-mail / Webmail (Africa), E-mail / Webmail (Asia)<br>• Related process targets: 296, 303 |
| Process Target 296 | • Name: Response Time - Prio 1<br>• Schedule: 24x7<br>• Initial State: Open<br>• Final State: Work In Progress<br>• Duration Type: Interval<br>• Duration: 00:10:00<br>• Duration scope: Time for the current assignment group |
| Process Target 303 | • Name: Resolution Time - Prio 1<br>• Schedule: 24x7<br>• Initial State: Open<br>• Final State: Resolved<br>• Duration Type: Interval<br>• Duration: 01:00:00<br>• Duration scope: Total time for each assignment group |
| SLA 168 | • Title: Base Monitoring SLA for IT Services<br>• Related process targets: 179, 313 |
| Process Target 179 | • Name: All Levels - incident - Fix - All Priorities<br>• Schedule: Mon to Fri from 8am to 4pm<br>• Initial State: Open<br>• Final State: Resolved<br>• Duration Type: Interval<br>• Duration: 20:00:00 |
| Process Target 313 | • Name: SLA168_Incident_SLT1<br>• Schedule: 24x7<br>• Initial State: Open<br>• Final State: Assign<br>• Duration Type: Interval<br>• Duration: 00:20:00 |

**Create an Incident record**

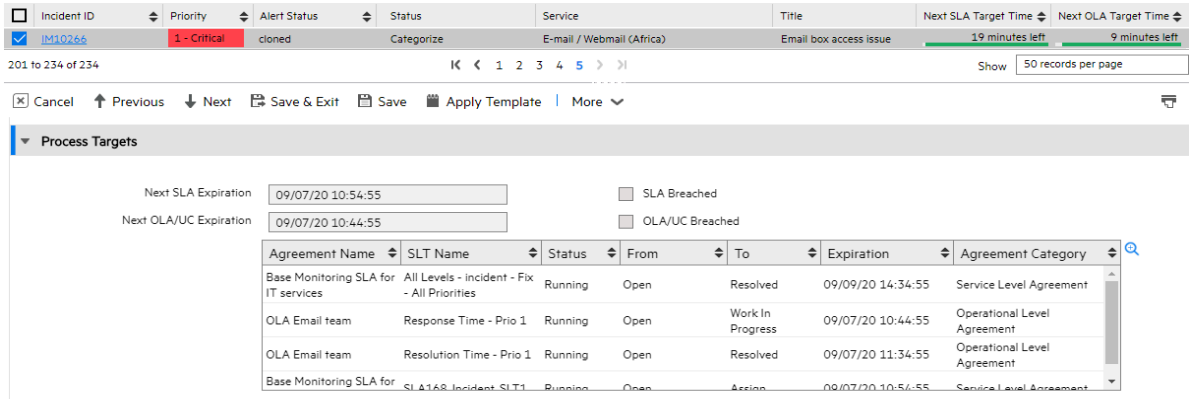
Create an Incident record that both the SLA and OLA can apply. This example uses the Incident record IM10266.

### View the indicators' status change

#### Green

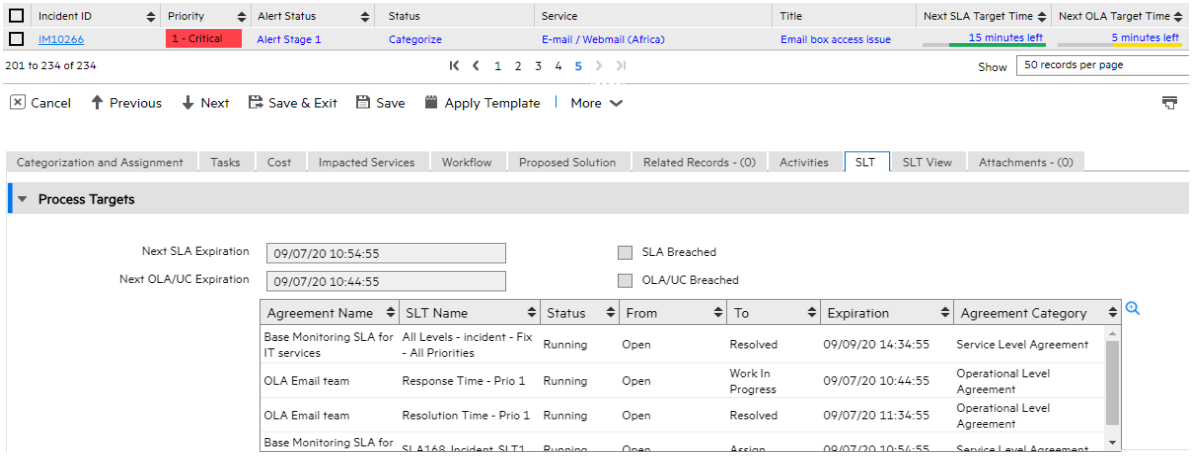
Shortly after you create IM10266, open this record and then click the **SLT** tab to view the status of the SLTs. All of them are **Running**.

In the record list, the **Next SLA Target Time** column displays the status of the next SLA target (313) and **Next OLA Target Time** column displays the status of the next OLA target (296). Both indicators are green because the remaining time is more than 50% of the next SLT.



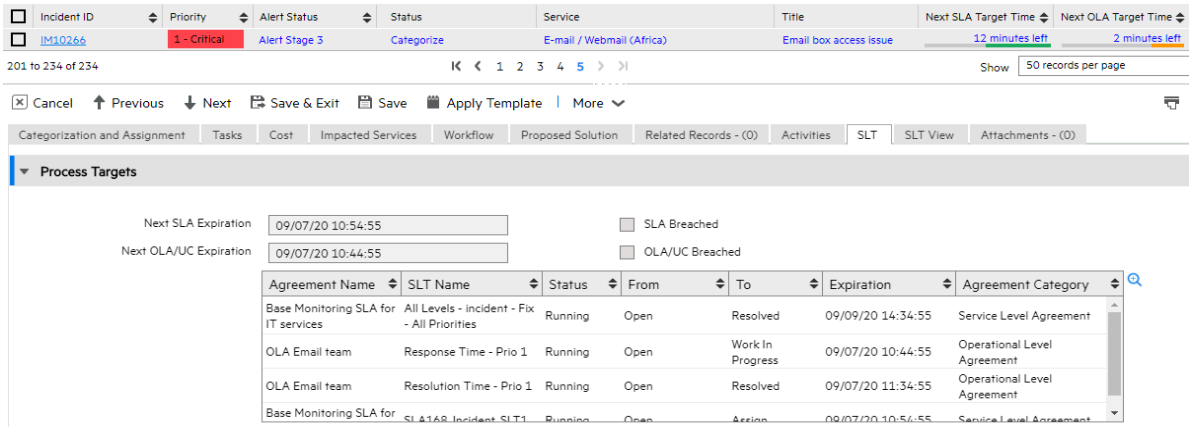

#### Yellow

Refresh the record list after a short while. Now the color indicator in the **Next OLA Target Time** column turns yellow because the remaining time is between 25% and 50% of the next SLT (296). The color indicator in the **Next SLA Target Time** column is still green because the duration of the next SLT (313) is 20 minutes.



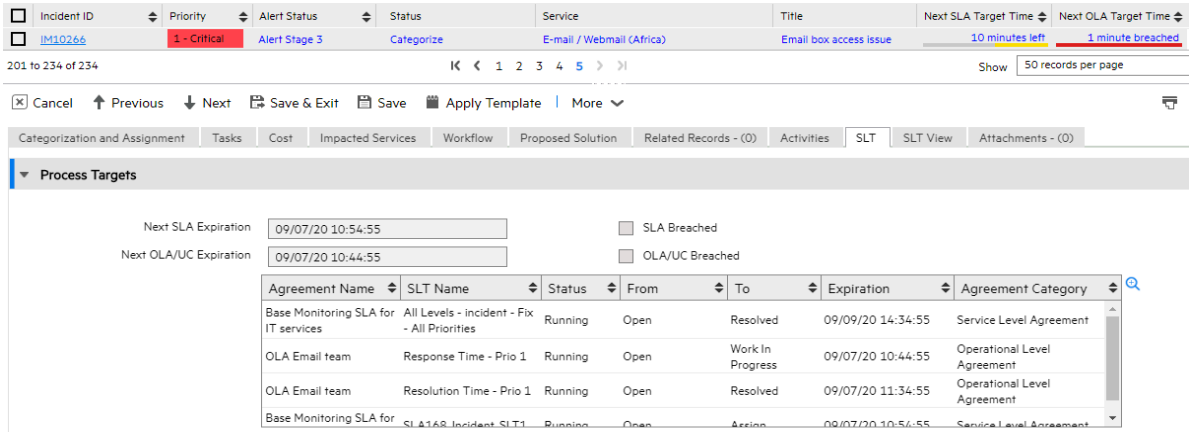

#### Orange

Refresh the record list after a short while. Now the color indicator in the **Next OLA Target Time** column turns orange because the remaining time is less than 25% of the next SLT (296).



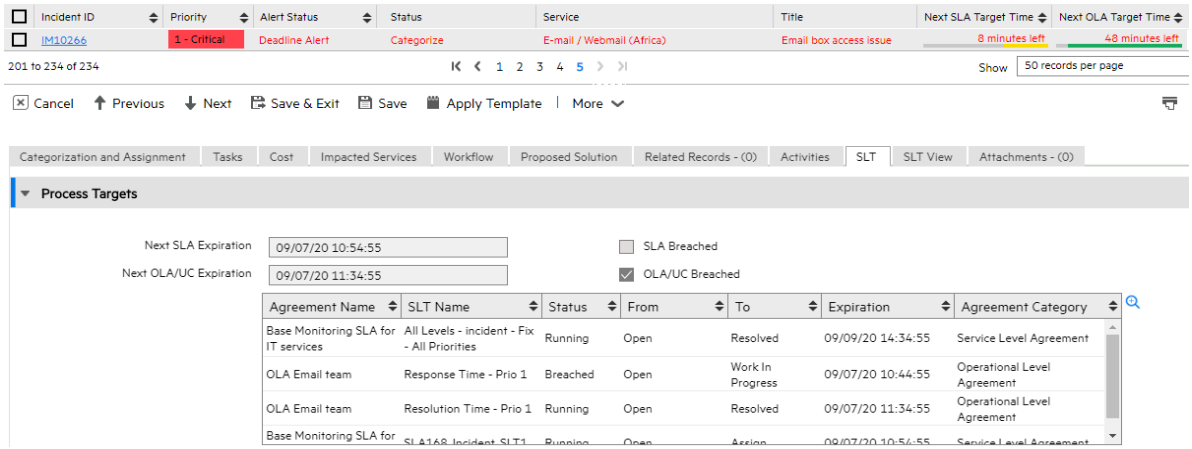

### Red

Refresh the record list after a short while. Now the color indicator in the **Next OLA Target Time** column turns red because the SLT (296) has been breached for 1 minute. The color indicator in the **Next SLA Target Time** column turns yellow because the remaining time is between 25% and 50% of the next SLT (313).

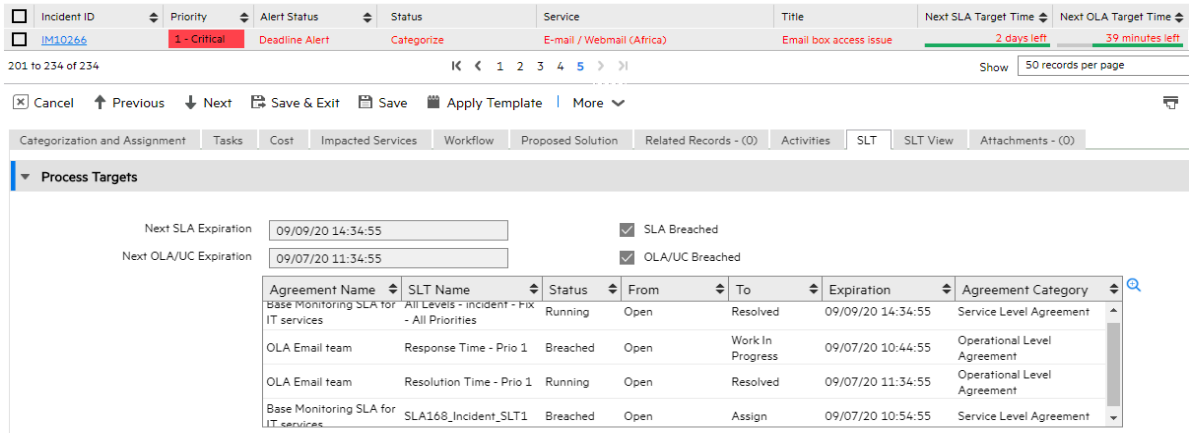



### Green again

After SLT (296) breaches, the color indicator in the **Next OLA Target Time** column is green again and displays the status of the next SLT (303).

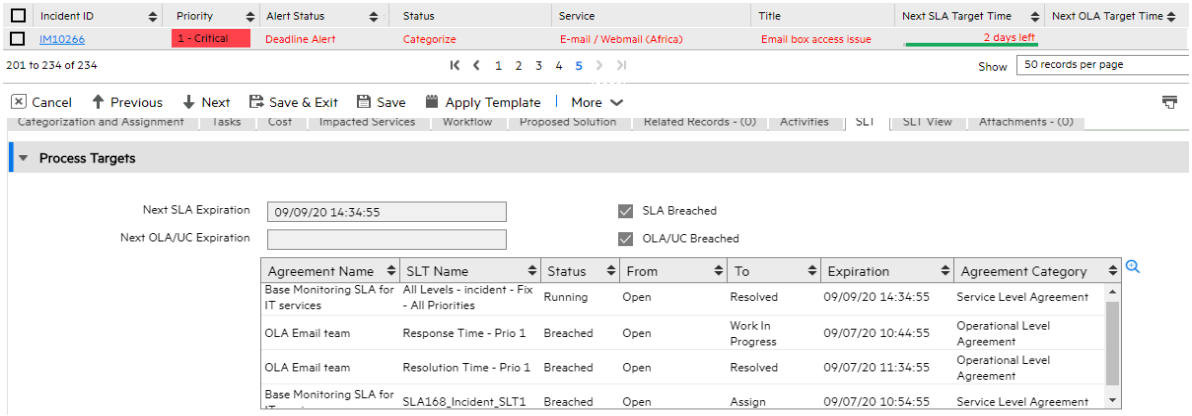



After SLT (313) breaches, the status of the next SLT (179) is displayed in the **Next SLA Target Time** column.




**Blank**

Refresh the record list after some time. Now SLT (303) is also breached and the system displays the **Next OLA Target Time** column as blank.




## 10.5. Visualize SLM on SLT View tab with a suspended ticket

The out-of-the-box Incident and Interaction forms have the **SLT View** tab that displays the operation status for all related SLA, OLA, and UC of a record. SLAs, OLAs, SLTs, and the related assignment groups are displayed in hierarchies on the left side. On the right side, the progress bars consist of various color indicators and visualize the actual working time of the process targets. An IT agent can view the definition of all indicators and duration of a color indicator by tooltips and hence arrange the priority of the current work.

This use case describes how the SLM visualization works on the SLT View tab when an SLA works on an Interaction with suspended status. An IT agent can view the indicators' color changes as time elapses. When this Interaction's status becomes **Suspended**, the indicator's color can reflect this update as well.

### Prepare SLA

This example uses the following SLA and Process Targets (SLTs) and most of them are available out-of-the-box. Create similar data if you're not working with the out-of-the-box data and don't have the sample data.

| Item | Values |
| --- | --- |
| SLA 172 | • Title: Platinum Service<br>• Related process targets: 307, 311 |
| Process Target 307 | • Name: Platinum - Initial Response<br>• Schedule: 24x7<br>• Initial State: Open<br>• Final State: Dispatched<br>• Duration Type: Interval<br>• Duration:20:00:00<br>• Suspend processing: Pending Vendor<br>• for these states: Suspended |
| Process Target 311 | • Name: Platinum - Time to Fix<br>• Schedule: 24x7<br>• Initial State: Open<br>• Final State: Closed<br>• Duration Type: Interval<br>• Duration: 20:00:00 |

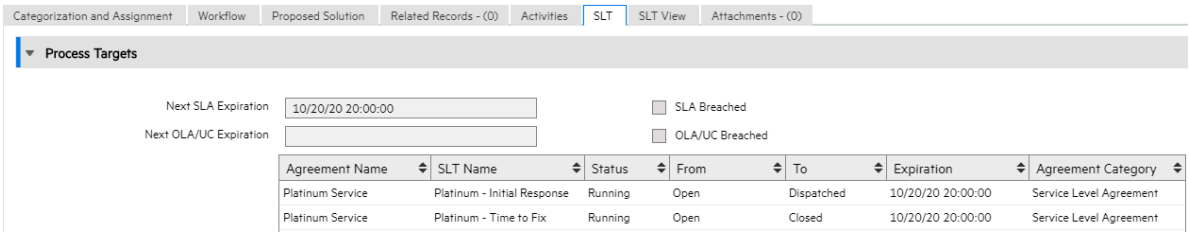
### Create a Service Desk Interaction record

Create a Service Desk Interaction record that both the SLA can apply. This example uses the Interaction record SD10222.

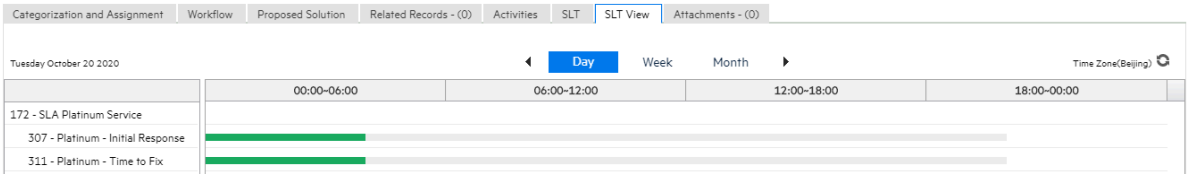
### View the indicators' status change

#### Green

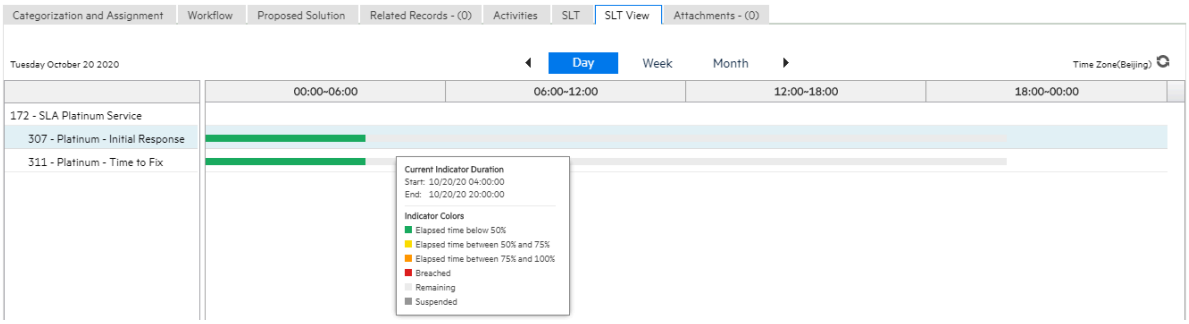
Refresh the record list and open SD10222 four hours later. Click the **SLT** tab to view that the SLTs' status is Running.




Next, click the **SLT View** tab to view the status of the SLTs.



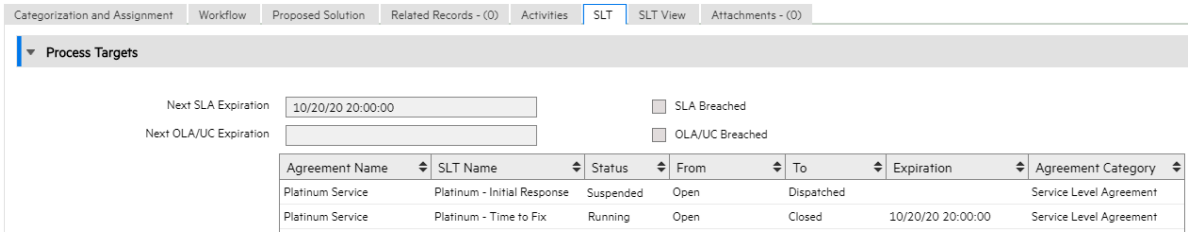

Hover your mouse over the remaining time area (the light grey indicator) of SLT 307, the time information on the tooltip may resemble the following. The Start time is the actual time when this tooltip is displayed, and the End time is the SLT breach time.



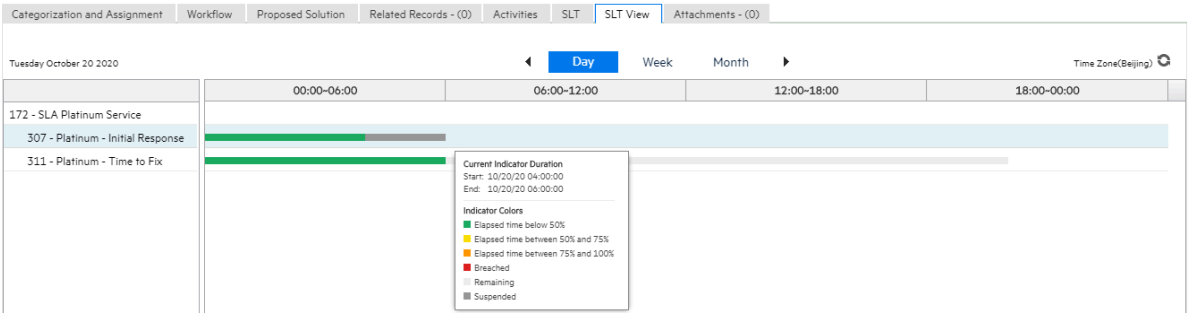

Update the ticket's status to **Suspended**, and then set the suspend time to 2 hours. Save your changes.

Dark grey

Refresh the record list and check the ticket's status after 2 hours. Click the **SLT** tab to view the SLTs' status.

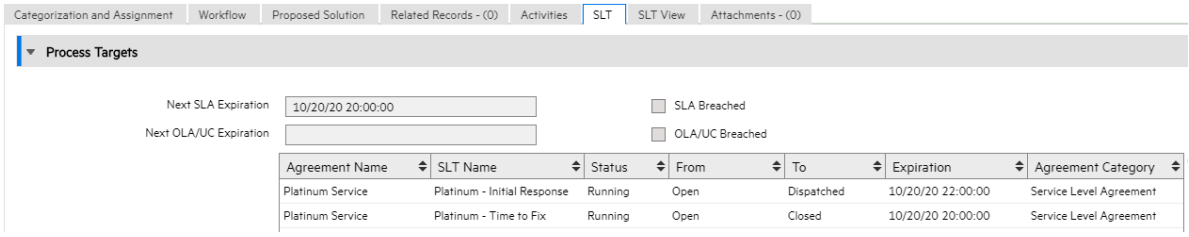



Next, click the **SLT View** tab to view the status of the SLTs. Hover your mouse over the dark grey indicator of SLT 307, the time information on the tooltip may resemble the following. The Start time is the start time when this ticket is suspended, and the End time is the actual time when this tooltip is displayed.

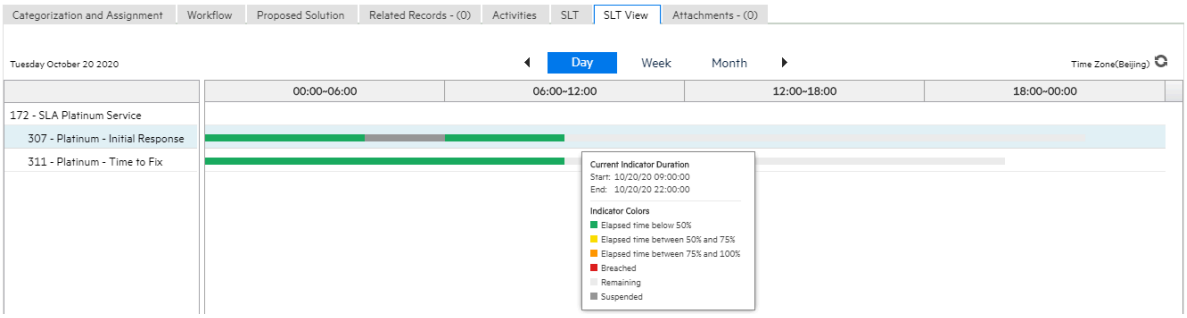



Green again

Refresh the record list and check the ticket's status after 3 hours. Now the ticket is unsuspended. Click the **SLT** tab to view the SLTs' status.

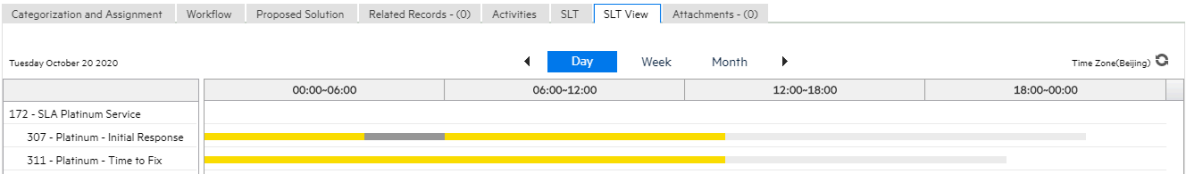



Next, click the **SLT View** tab to view the status of the SLTs. Hover your mouse over the remaining time area (the light grey indicator) of SLT 307, the time information on the tooltip may resemble the following. The Start time is the actual time when this tooltip is displayed, and the End time is the SLT breach time. Note that the SLT breach time equals to the original breach time plus the suspended time.



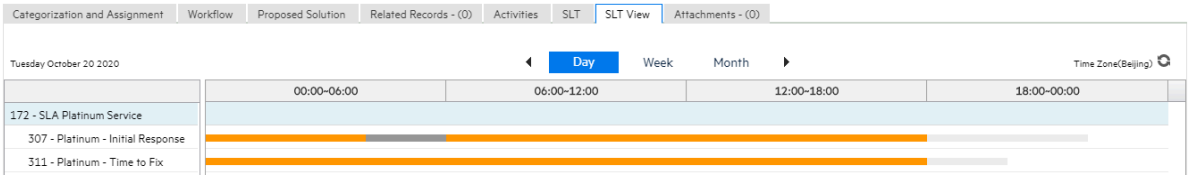

Yellow

Refresh the record list and check the ticket's status after 4 hours. Click the **SLT View** tab to view the status of the SLTs.



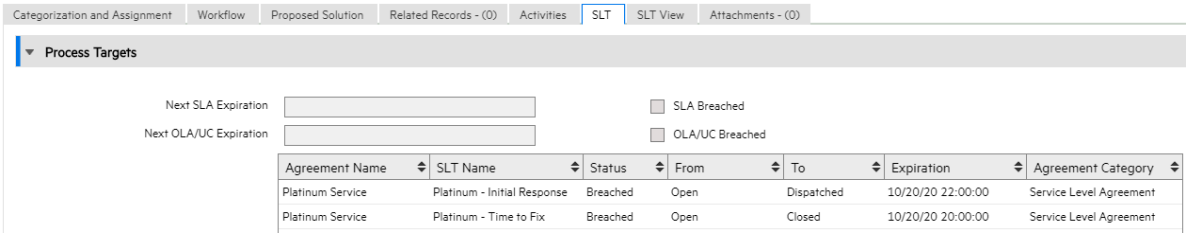

Orange

Refresh the record list and check the ticket's status after 6 hours. Click the **SLT View** tab to view the status of the SLTs.

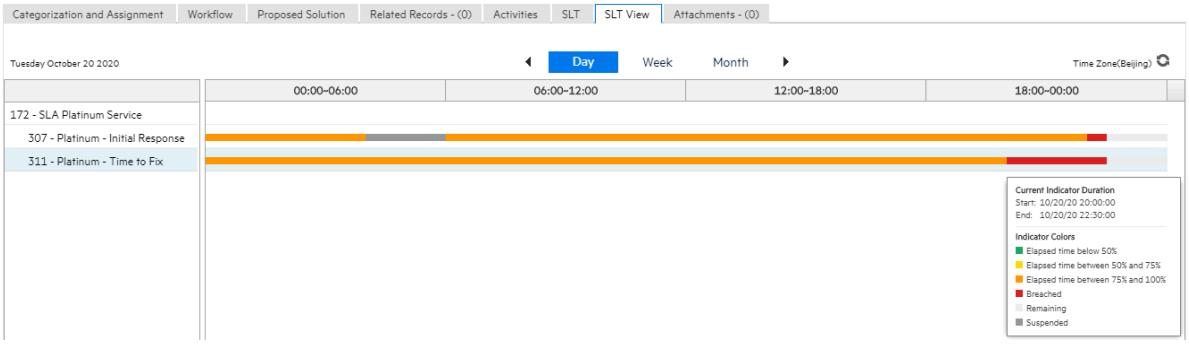



Red

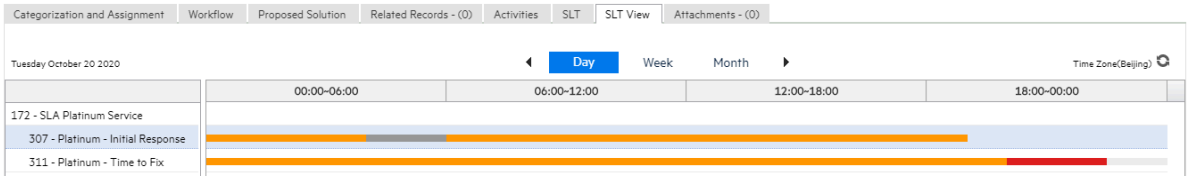
Refresh the record list and check the ticket's status after 4.5 hours. Click the **SLT** tab to view that both SLTs' status is Breached.




Next, click the **SLT View** tab to view the status of the SLTs. Hover your mouse over the remaining time area (the light grey indicator) of SLT 307, the time information on the tooltip may resemble the following. The Start time is the SLT breach time, and the End time is the actual time you refresh the SLT View tab and display this tooltip.



Note that if you escalate the Service Desk Interaction record and its status become Dispatched (the Final State as defined in SLT 307), the light grey indicator for SLT 307 disappears, whereas the light grey indicator for SLT 311 sustains because the ticket has not reached the Final State as defined in SLT 311 yet. Suppose you escalate the Service Desk Interaction record at 19:00 and then check the status of the SLTs at 22:30, the screenshot may resemble the following:




## 10.6. Visualize SLM on SLT View tab with altered assignment groups

The out-of-the-box Incident and Interaction forms have the **SLT View** tab that displays the operation status for all related SLA, OLA, and UC of a record. SLAs, OLAs, SLTs, and the related assignment groups are displayed in hierarchies on the left side. On the right side, the progress bars consist of various color indicators and visualize the actual working time of the process targets. An IT agent can view the definition of all indicators and duration of a color indicator by tooltips and hence arrange the priority of the current work.

This use case describes how the SLM visualization works on the SLT View tab when an OLA works on an Incident with altered assignment groups. You can view the indicators' color changes as time elapses. When you change this Incident's assignment group setting, the progress bars can reflect this update as well.

### Prepare OLA

This example uses the following OLAs and Process Targets (SLTs) and most of them are available out-of-the-box. Create similar data if you're not working with the out-of-the-box data and don't have the examples.

| Item | Values |
| --- | --- |
| OLA 176 | • Title: OLA Email Team<br>• Assignment Groups: E-mail / Webmail (Africa), E-mail / Webmail (Asia)<br>• Related process targets: 296, 303, 314 |
| Process Target 296 | • Name: Response Time - Prio 1<br>• Schedule: 24x7<br>• Initial State: Open<br>• Final State: Work In Progress<br>• Duration Type: Interval<br>• Duration: 12:00:00<br>• Duration scope: Time for the current assignment group |
| Process Target 303 | • Name: Resolution Time - Prio 1<br>• Schedule: 24x7<br>• Initial State: Open<br>• Final State: Resolved<br>• Duration Type: Interval<br>• Duration: 12:00:00<br>• Duration scope: Total time for each assignment group |
| Process Target 314 | • Name: Fix Time - Prio 1<br>• Schedule: 24x7<br>• Initial State: Open<br>• Final State: Work In Progress<br>• Duration Type: Interval<br>• Duration: 12:00:00<br>• Duration scope: Total time for all assignment groups |

**Create an Incident record**

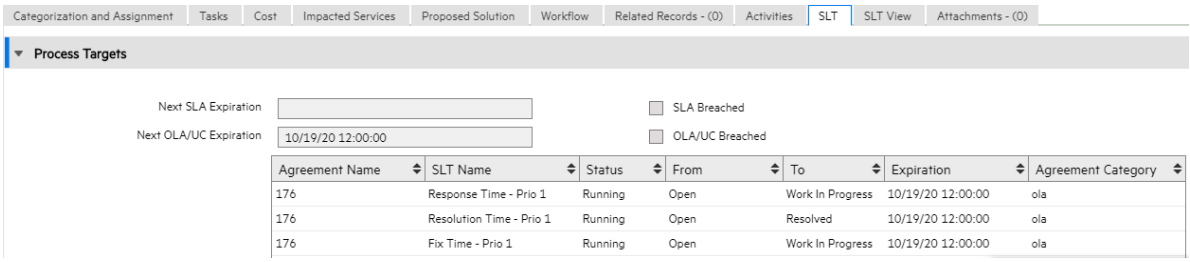
Create an Incident record that the OLA can apply. This example uses the Incident record IM10421 with the following specific values and you need to fill other fields as required.

- Assignment Group: E-mail / Webmail (Africa)
- Assignee: Incident.Coordinator

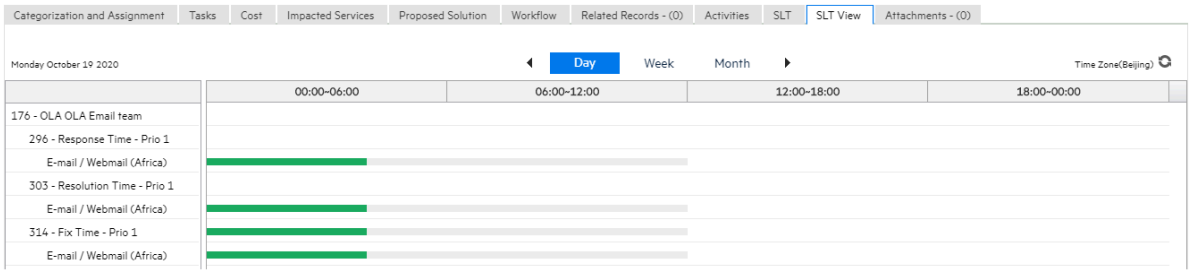
**View the indicators' status change**

**Green**

Refresh the record list and open IM10421 four hours later. Click the **SLT** tab to view the status of the SLTs.

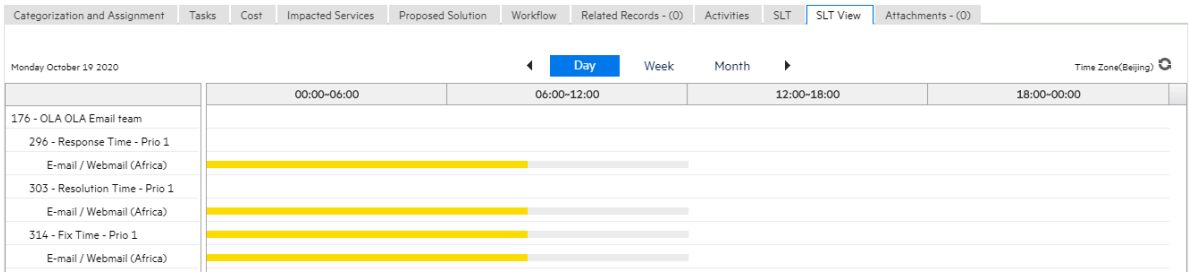



Next, click the **SLT View** tab to view the status of the SLTs. All related OLA, SLT, and assignment groups that are working on the OLA are displayed hierarchically. The color indicator for SLT 296, 303, and 314 are green. Note that SLT 314 has a progress bar because the duration scope of this SLT is **Total time for all assignment groups**.




### Yellow

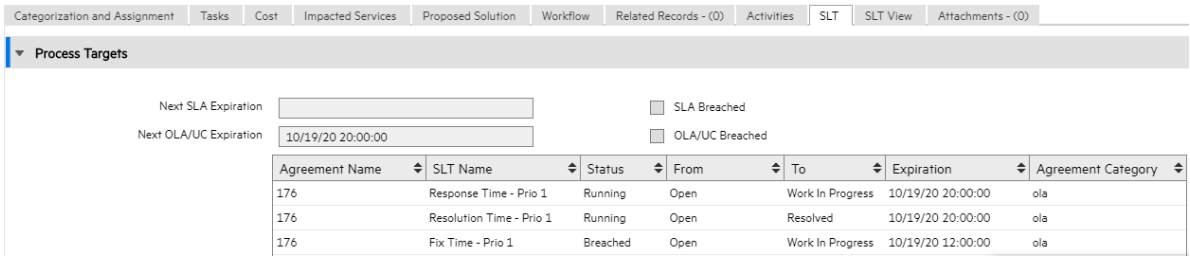
Refresh the record list and check the ticket's status after 4 hours.



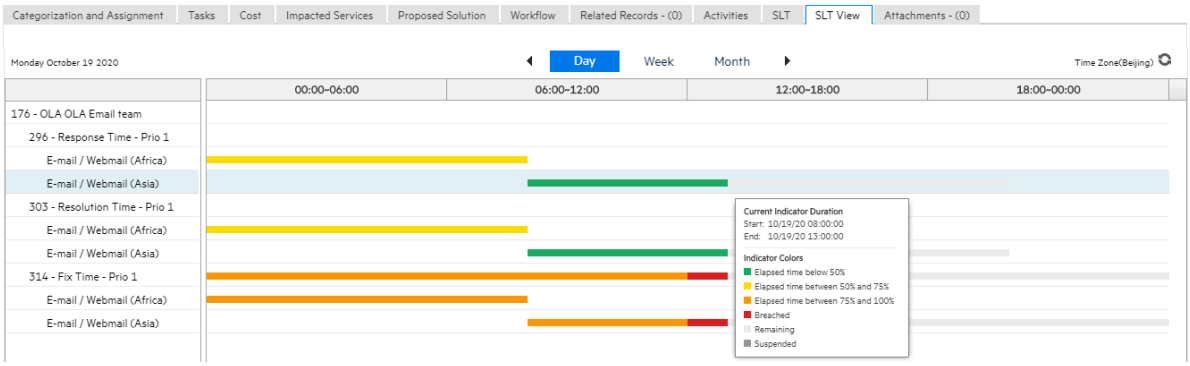

Update the ticket's assignment group to **E-mail / Webmail (Asia)**, and then save your changes.

### Green again

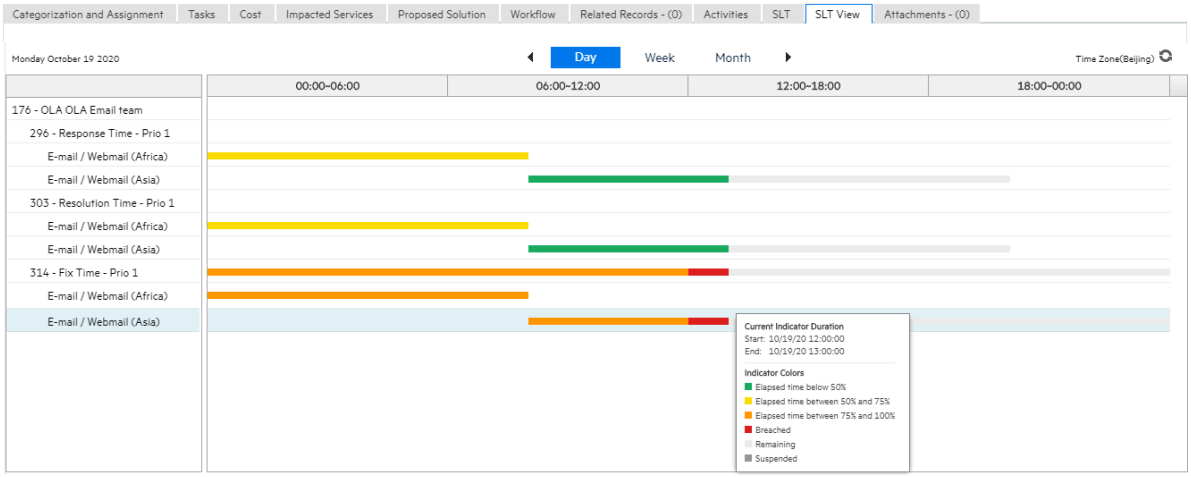
Refresh the record list and check the ticket's status after 4 hours. Click the **SLT** tab to view the status of the SLTs.




Next, click the **SLT View** tab to view the status of the SLTs. Now the new assignment group E-mail / Webmail (Asia) appears on this tab. Hover your mouse over the green indicator for the **E-mail / Webmail (Africa)** assignment group under SLT 296, the time information on the tooltip may resemble the following. The Start time is when you update the ticket's assignment group, and the End time is the actual time you refresh the SLT View tab and display this tooltip.

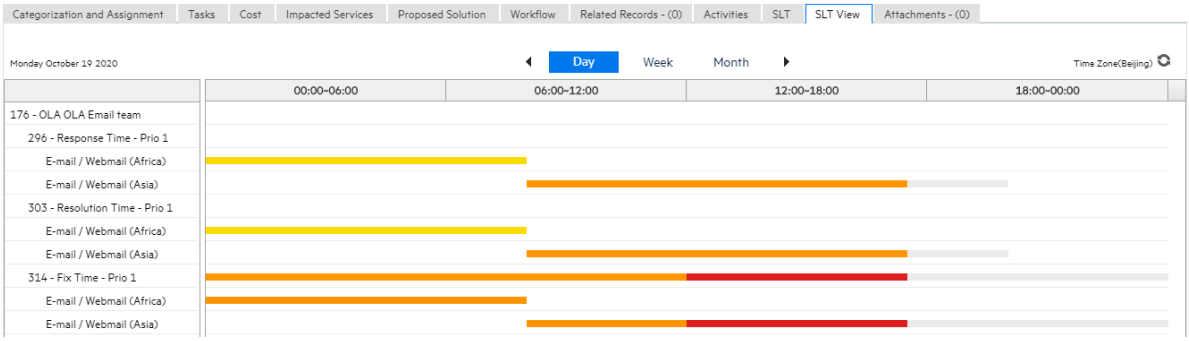



Hover your mouse over the red indicator for the **E-mail / Webmail (Africa)** assignment group under SLT 314, the time information on the tooltip may resemble the following. The Start time is the SLT breach time, and the End time is the actual time you refresh the SLT View tab and display this tooltip.




#### Orange

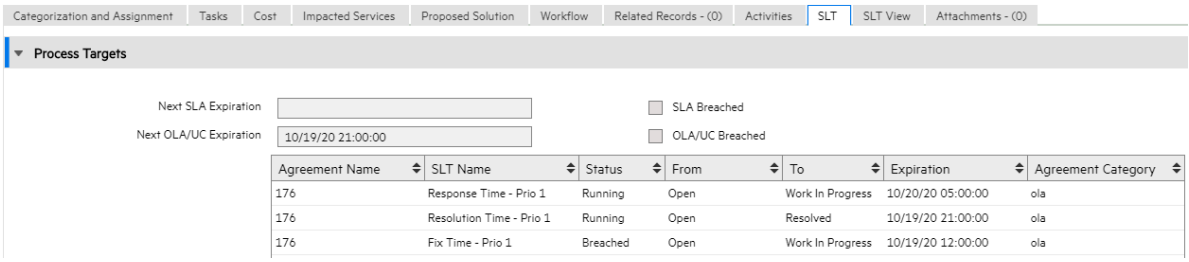
Refresh the record list and check the ticket's status after 5 hours. Click the **SLT View** tab to view the status of the SLTs.



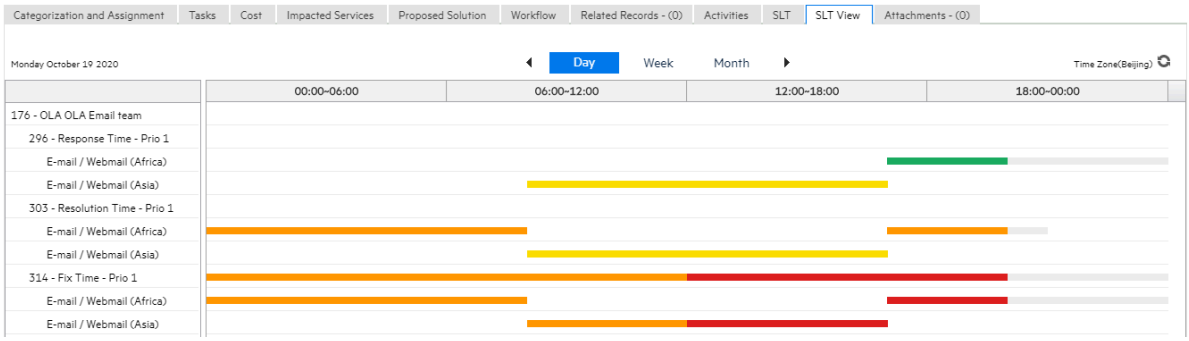

Update the ticket's assignment group to **E-mail / Webmail (Africa)** again, and then save your changes.

#### Green again

Refresh the record list and check the ticket's status after 3 hours. Click the **SLT** tab to view the status of the SLTs. The system recalculates the expiration time for both SLT 296 and SLT 303 due to assignment group changes.

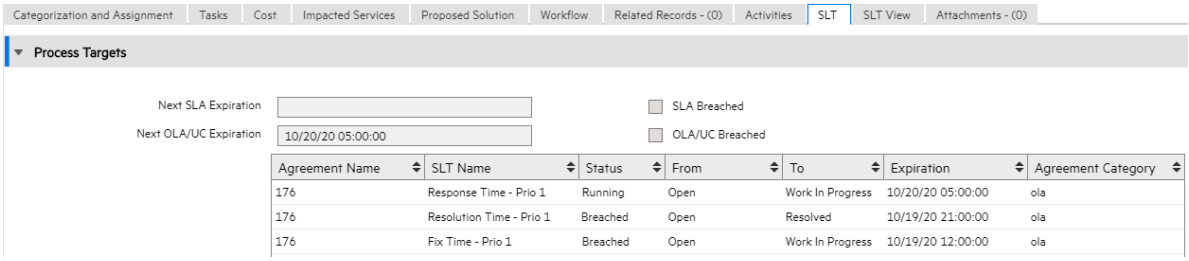



Next, click the **SLT View** tab to view the status of the SLTs. Note that the progress bar for the **E-mail / Webmail (Africa)** assignment group under SLT 303 now has a gap due to the group changes.

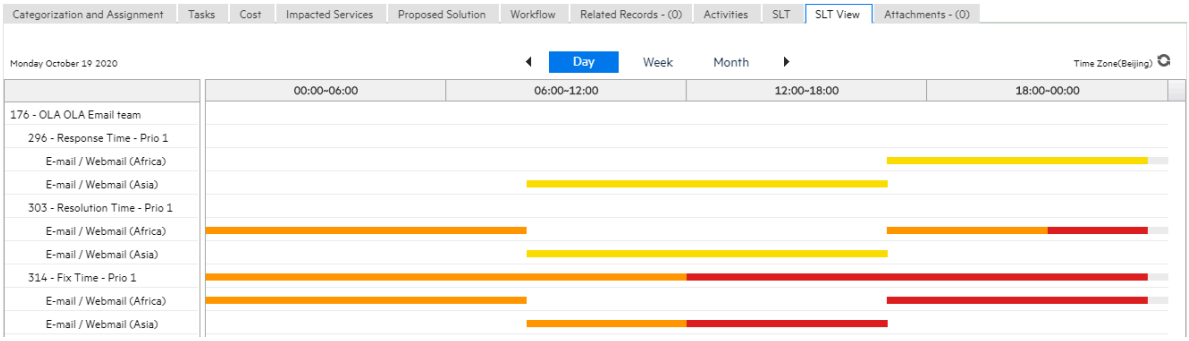



**Yellow again**

Refresh the record list and check the ticket's status after 3.5 hours.




Next, click the **SLT View** tab to view the status of the SLTs.




## 10.7. Collaborate on Teams with war room experience

This use case describes the following user scenario:

A critical email server connection failure occurs. An Incident Analyst creates a major Incident to track the issue and restore service as soon as possible. To collaborate quickly, the Incident Analyst initiates a Teams conversation. The Incident details (such as Incident ID, Incident Title, and Description) about the case are displayed in the Teams channel. Major Incident Team stakeholders including the Incident Coordinator and Incident Manager and other IT agents can collaborate in real-time in the Teams channel.

### Prerequisites

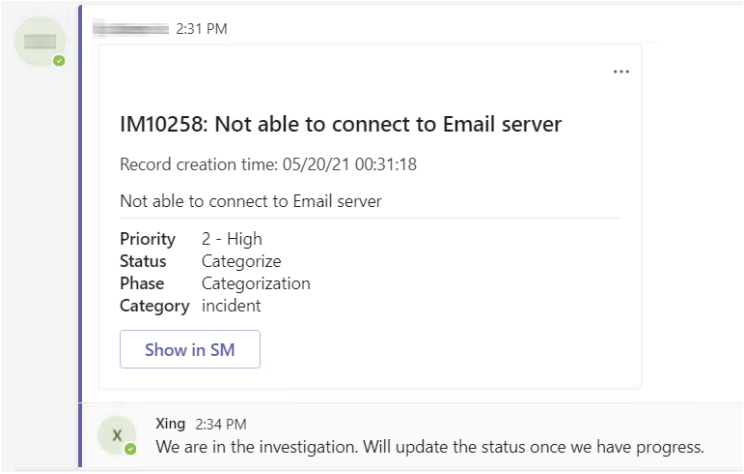
As a prerequisite of this use case, the Service Manager system administrator must complete the following tasks in advance:

1. Verify the **Major incident** checkbox is available on the Incident form. Otherwise, add the MajorIncident field to the Incident form.
2. Select both the **Synchronize Message** and the **Synchronize Attachment** checkboxes in SMIS.
3. Navigate to **Tailoring** > **Integration Manager** to start the Teams integration instance.
4. Navigate to **Tailoring** > **Allowlist** > **Attachment Allowlist** to confirm that the txt file extension exists in the Service Manager attachment allowlist.

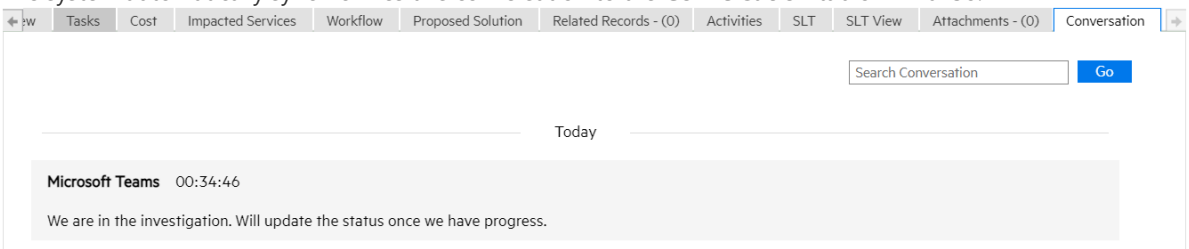
### Collaborate with IT agents on Teams

The following steps walk you through this use case:

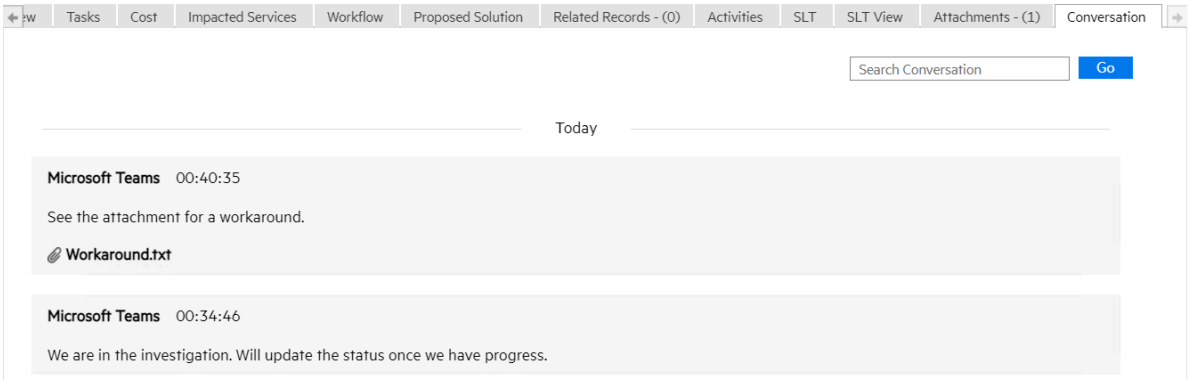
1. An Incident Analyst creates a major Incident (For example, IM10258) in Service Manager.
2. The Incident Analyst opens IM10258 and then clicks **Discuss on Teams**.
3. The Incident Analyst signs in to Teams and then checks that a message about the major Incident is sent to the channel.




   As shown in the sample screenshot above, an Incident Coordinator (Xing) in this channel has replied to the message. To access the Incident and view the details, this Incident Coordinator can click the **Show in SM** button and then log in to Service Manager web tier on a new browser tab.
   **Note:** If multiple conversations are underway in a channel, Teams displays the channel messages in chronological order of the latest replies.
4. The system automatically synchronizes this conversation to the **Conversation** tab of IM10258.




   The synchronization time depends on the System Administrator's setting to the **Interval Time (s)** and the **Synchronize Interval Time (min)** option in SMIS.
5. As the conversation continues, the system groups the incoming messages by days and displays them in reverse chronological order.
   To search for certain content in the conversation, the IT agents can type the keywords in the search field and then click **Go**. If the system administrator has enabled Smart Analytics, the IT agents can also search for the keywords in the conversation history by using Smart Search.




6. The IT agents can still post messages in the Teams channel after IM10258 is closed. However, Service Manager no longer synchronizes messages to the closed Incident.

#### Related topics

Microsoft Teams
Set up the integration for war room experience

### 10.8. Interact with Service Manager by using agent bot

This use case describes the following user scenario:

A customer reports a case in which Outlook keeps asking for user credentials. To handle this case quickly, an Incident Analyst initiates a Teams conversation with the bot and uses the bot to create an Incident in Service Manager to track the issue. Later, the Incident Analyst provides a solution and then closes this Incident through the bot as well.

#### Prerequisites

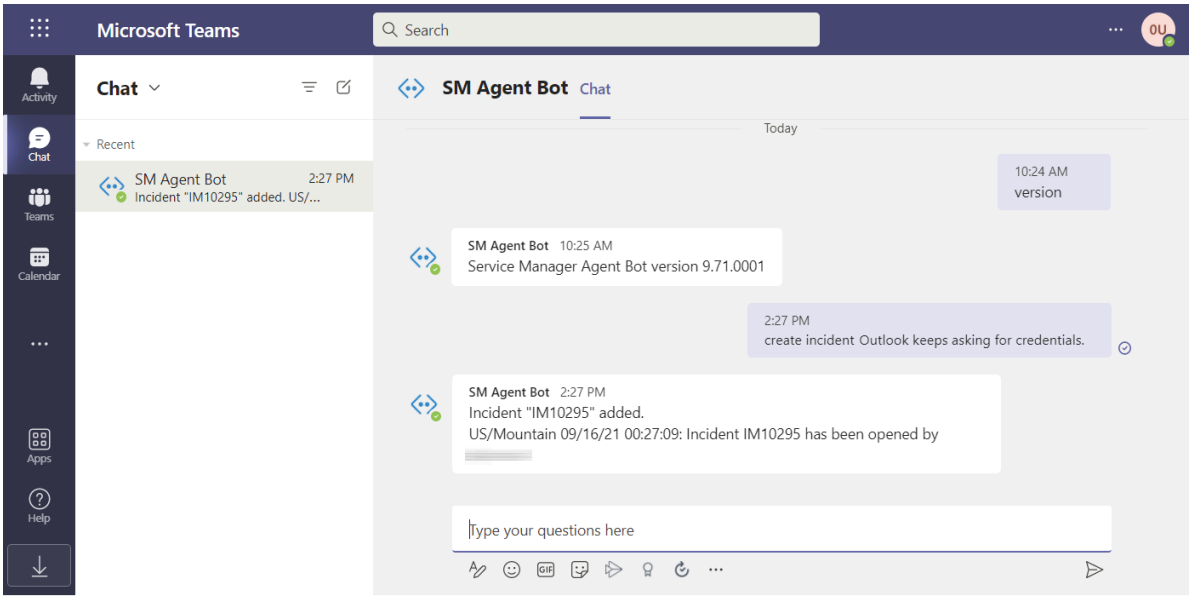
As a prerequisite of this use case, the Service Manager system administrator must complete the following tasks in advance:

1. Navigate to **Tailoring** > **Integration Manager** to start the Teams integration instance.
2. Verify that the Incident Analyst has the following permission:
    - Create Incidents
    - View, update, and close Incidents

#### Interact with Service Manager on Teams

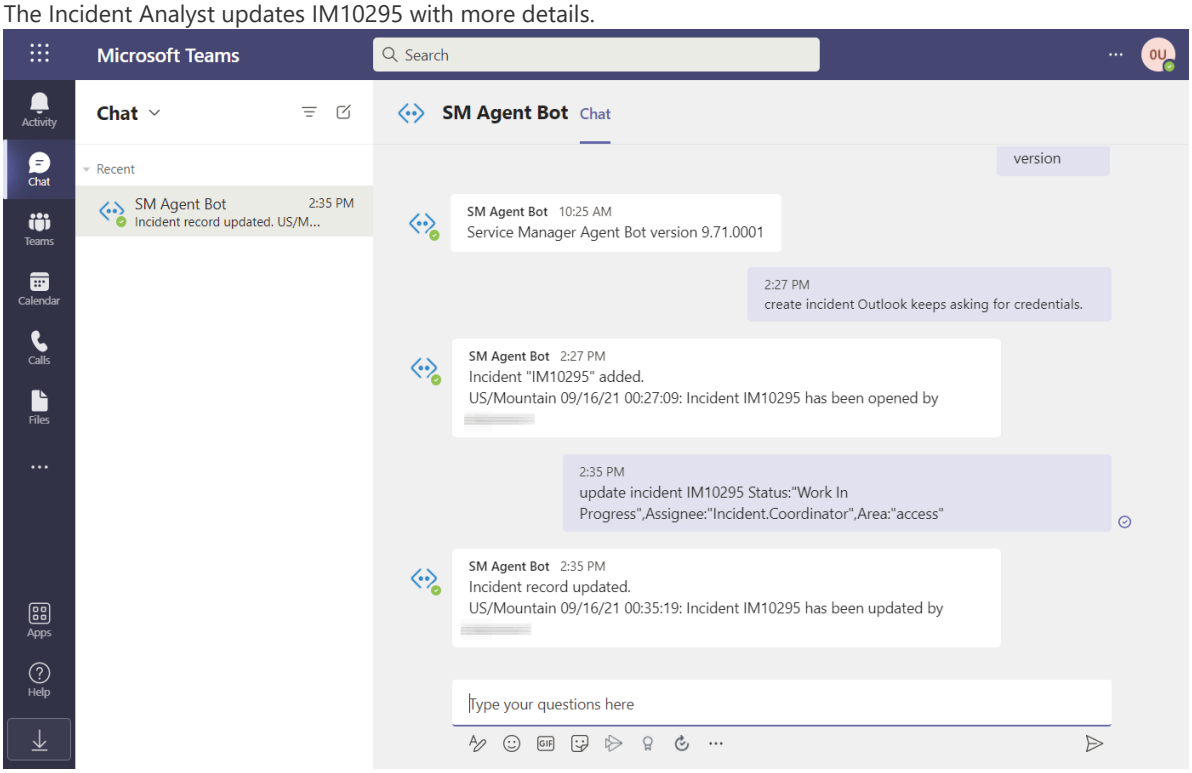
The following steps walk you through this use case:

1. An Incident Analyst initiates a Teams conversation with the bot and then enters a command to create an Incident in Service Manager.



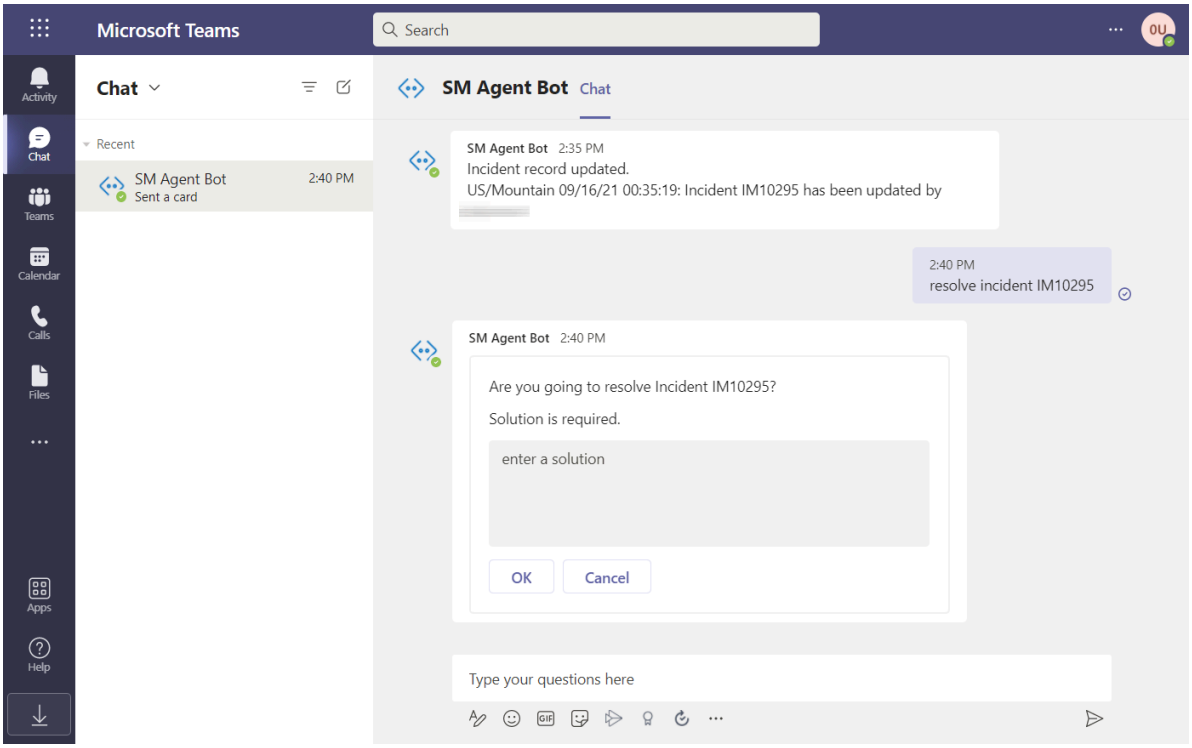

As shown in the sample screenshot above, the agent bot creates Incident IM10295 with the description of **Outlook keeps asking for credentials**.

2. The Incident Analyst updates IM10295 with more details.

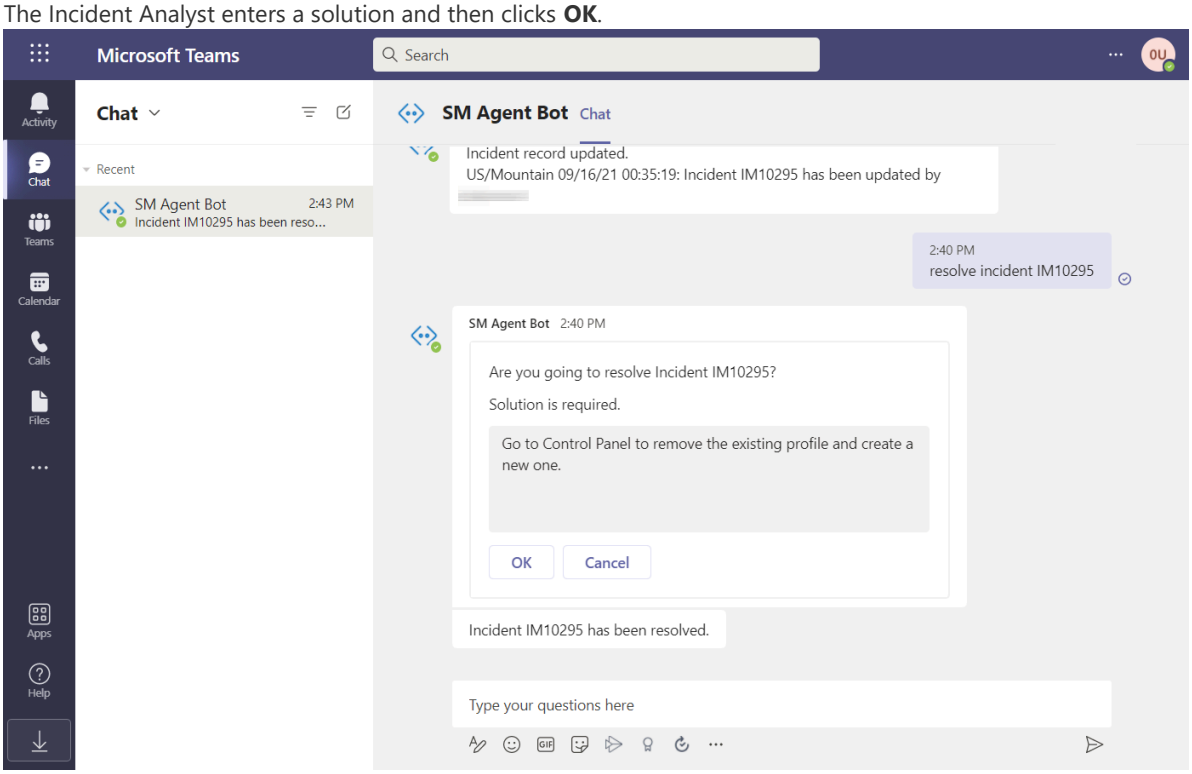



As shown in the sample screenshot above, the status of Incident IM10295 is **Work in Progress** and the Subcategory is **access**. The Incident Analyst assigns this Incident to **Incident.Coordinator**.

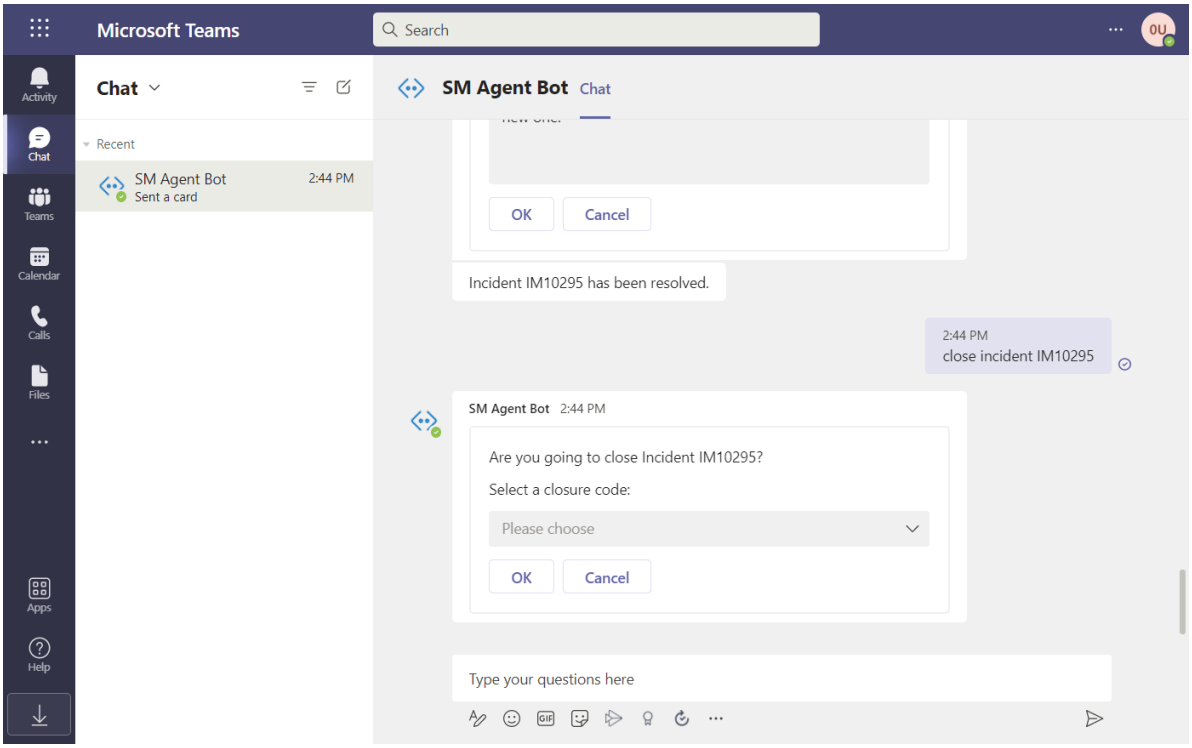
3. Later, the Incident Coordinator finds a solution and the Incident Analyst resolves this Incident.



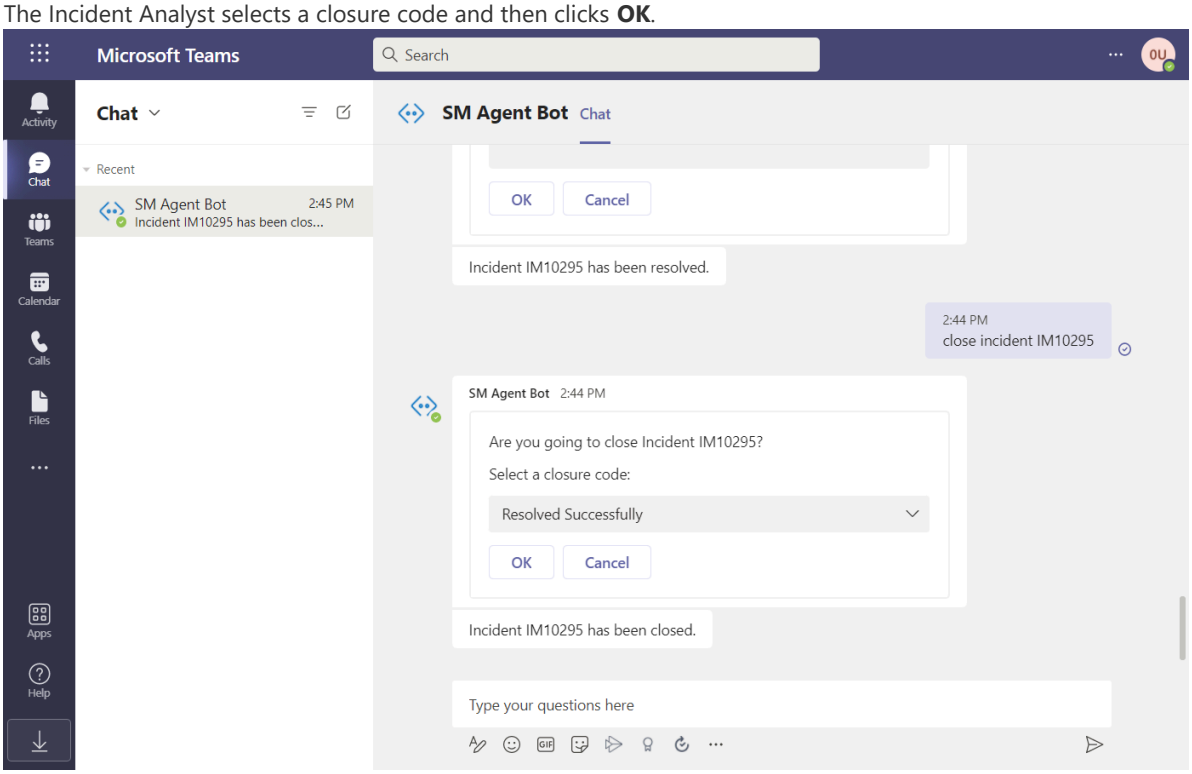

The Incident Analyst enters a solution and then clicks **OK**.




4. Finally, the Incident Analyst enters a command to close this Incident.




The Incident Analyst selects a closure code and then clicks **OK**.




Now, IM10295 is closed.

### Related topics

Microsoft Teams
Set up agent bot

## 10.9. Interact with Service Manager by using virtual agent

This use case describes the following user scenario:

A Service Manager employee self-service (ESS) user is working remotely from home as the office has been temporarily closed according to Social Distancing policies during a pandemic. Recently, this user encounters a company website access problem. To solve this problem quickly, the user initiates a Teams conversation and then searches the knowledge base for a solution by using the virtual agent. However, the knowledge search result doesn't match the user's problem. Hence, this user raises a Service Desk ticket by using the virtual agent. Later, this user can check the ticket's status with the virtual agent's assistance in Teams as well.

### Prerequisites

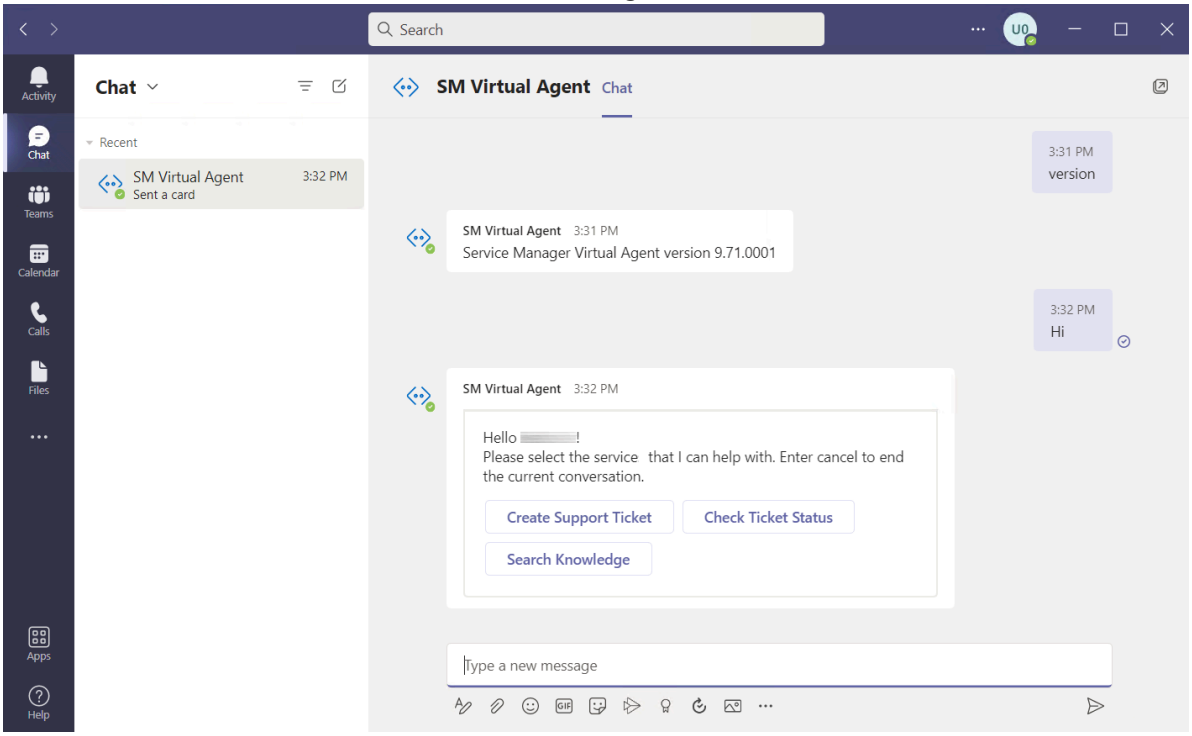
As a prerequisite of this use case, the Service Manager system administrator must complete the following tasks in advance:

1. Navigate to **Tailoring** > **Integration Manager** to start the Teams integration instance.
2. Enable IDOL.
3. Verify that the ESS user has the following permission:
    - Search for a knowledge article
    - Create and view Interactions

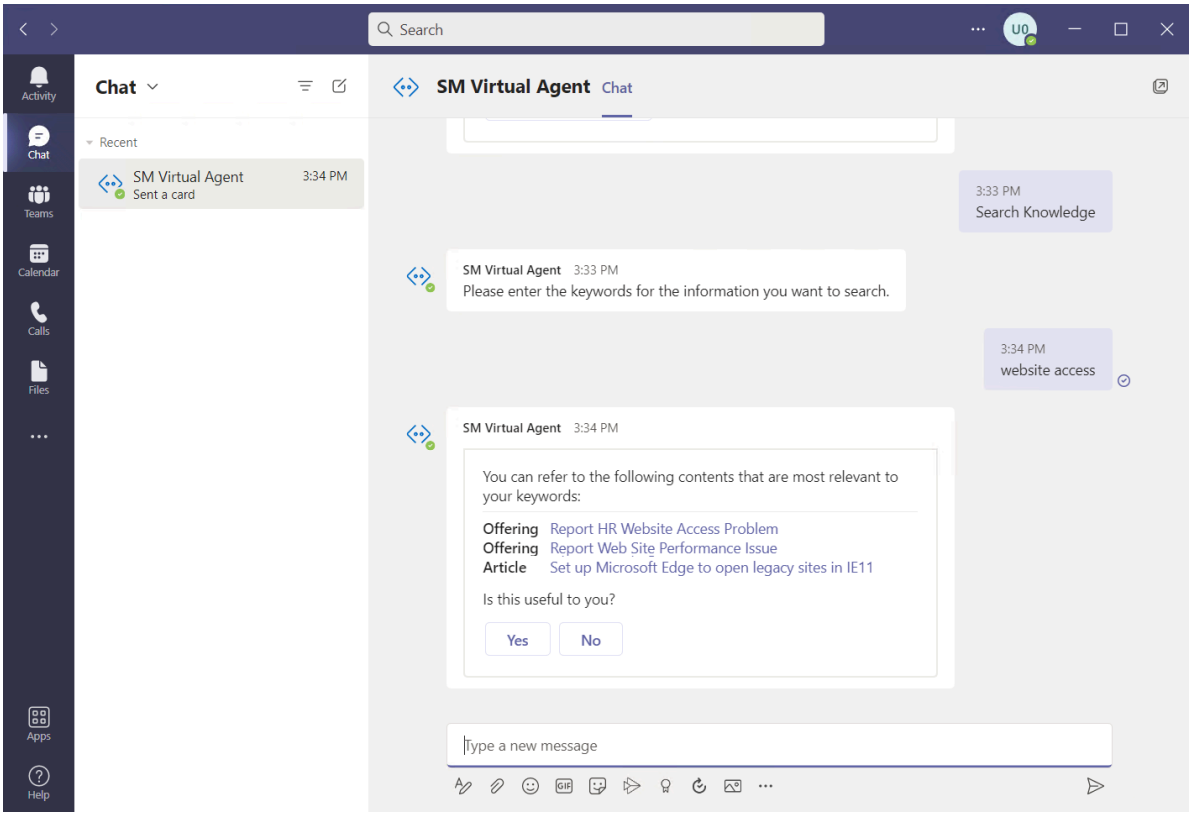
### Interact with Service Manager on Teams

The following steps walk you through this use case:

1. An ESS user initiates a Teams conversation with the virtual agent.

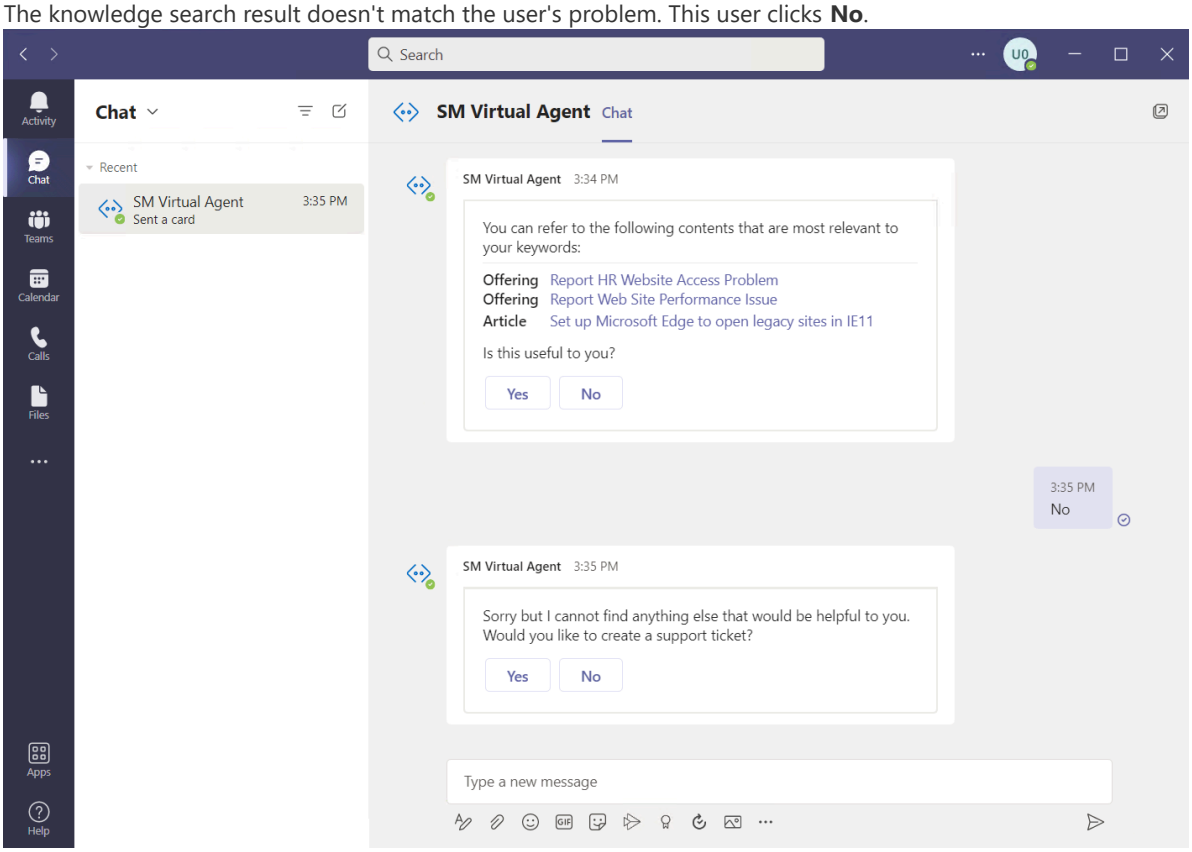



   As shown in the sample screenshot above, the virtual agent responds with several services.
2. This user clicks **Search Knowledge** and then enters some keywords to find a solution.



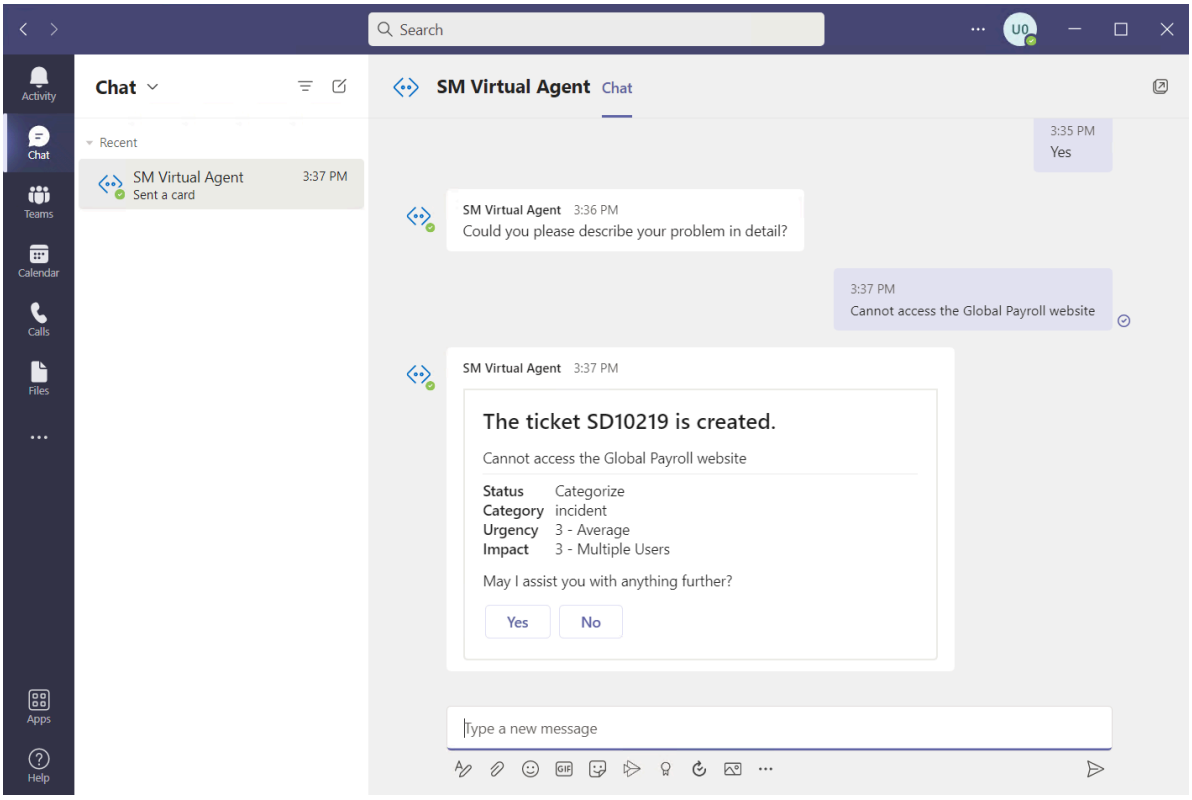

As shown in the sample screenshot above, the virtual agent responds with two offerings and one knowledge article after performing a Smart Search in Service Manager.

3. The knowledge search result doesn't match the user's problem. This user clicks **No**.



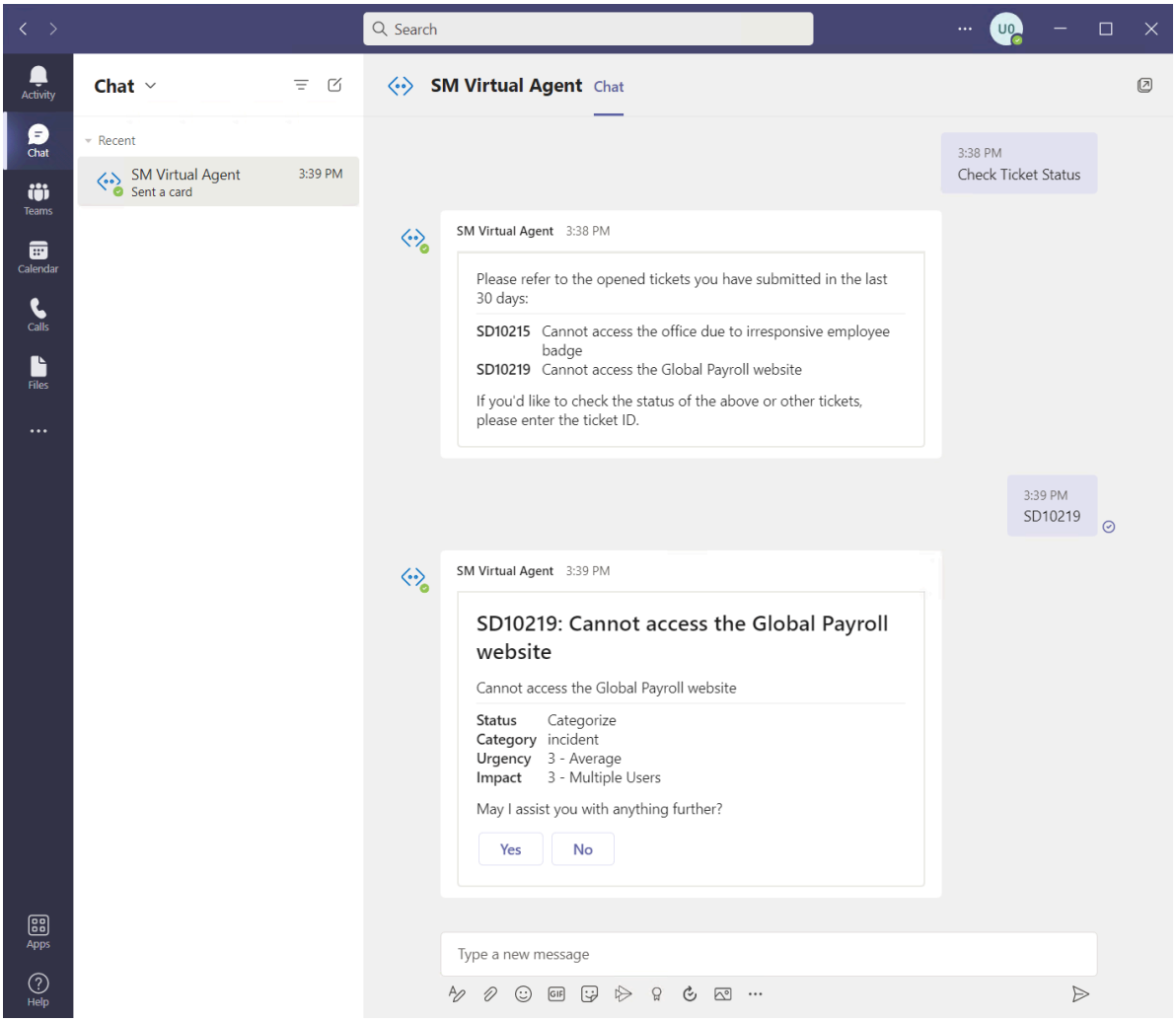

As shown in the sample screenshot above, the virtual agent suggests creating a support ticket.

4. The user clicks **Yes** and then creates a Service Desk ticket




As shown in the sample screenshot above, the virtual agent creates Interaction SD10219 with the description of `Cannot access the Global Payroll website`.

5. Later, this user can check the Service Desk ticket's status by clicking **Check Ticket Status**. The virtual agent lists the most recent Service Desk tickets that relate to the user in the last 30 days.




As shown in the sample screenshot above, the user enters the ticket's ID and then presses Enter to get the ticket's status from the virtual agent.

### Related topics

Microsoft Teams
Set up virtual agent

## 10.10. Synchronize Service Manager Change to ALM Octane defect

This use case describes the following user scenario:

Spring Boot team was recently notified that they needed to upgrade their Tomcat version in time to avoid the new vulnerability issue in Tomcat. The Product Owner proposes a Change with the aim of upgrading Tomcat to the latest version 10.x. After assessing the risks and impact, the Change is approved and now it's in the Build and Test phase. Since ALM Octane has a complete defect tracking process, the Change Manager decides to synchronize the Change to ALM Octane as a defect and hence to test it and track it thoroughly.

### Prerequisites

As a prerequisite of this use case, the Service Manager system administrator must complete the following tasks in advance:

1. Navigate to **Tailoring** > **Integration Manager** to start the ALM Octane integration instance.
2. There are two ways to select the ALM Octane synchronization action and the ALM Octane workspace: manually or automatically.
   To automatically select the ALM Octane synchronization action and the ALM Octane workspace, check the following prerequisites.

1. Navigate to **Tailoring** > **Process Designer** > **Workflows**. Open the workflow for Normal Change, and then check that the **chm.octane** rule set exists in the **Build And Test** phase.
2. Check that the **OctaneCustom** JavaScript exists in Script Library. Its content may resemble the following.

```
/**
 * If some changes need to Sync to Octane automatically
 * Update the condition and return {action, workspace}.
 */
function getOctaneConfig(change) {
    // The condition, a default one is to contain CI 'CI1000910, adv-Server-101_tomcat'
    let list = change.assets && change.assets.toArray() || [];
    if (isChangeToDefectEnabled() && list.includes('CI1000910')) {
        let workspace = vars.$G_OCT_WS_LIST_CHANGE_TO_DEFECT[0];
        if (workspace) {
            return {
                action: 'changeToDefect',
                workspace
            }
        }
    }

    // The condition for "changeToFeature", a default one is to contain CI 'CI1000914, adv-
Server-102_tomcat'
    if (isChangeToFeatureEnabled() && list.includes('CI1000914')) {
        let workspace = vars.$G_OCT_WS_LIST_CHANGE_TO_FEATURE[0];
        if (workspace) {
            return {
                action: 'changeToFeature',
                workspace
            }
        }
    }

    return null;
}

function isChangeToDefectEnabled() {
    return vars.$G_OCT_WS_LIST_CHANGE_TO_DEFECT &&
vars.$G_OCT_WS_LIST_CHANGE_TO_DEFECT.length()>0;
}

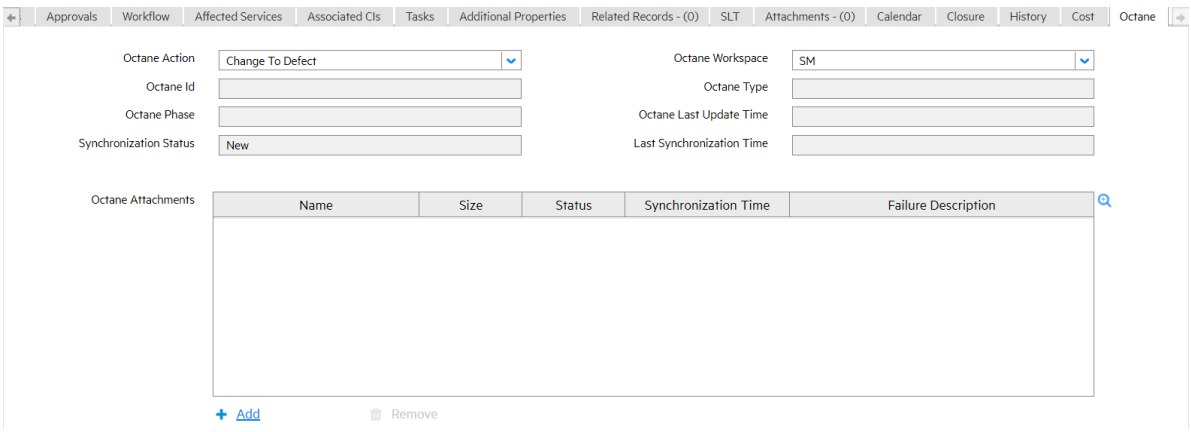
function isChangeToFeatureEnabled() {
    return vars.$G_OCT_WS_LIST_CHANGE_TO_FEATURE &&
vars.$G_OCT_WS_LIST_CHANGE_TO_FEATURE.length()>0;
}
```

In this use case, `adv-Server-101_tomcat` is the Affected Configuration Item, and its CI Identifier is `CI1000910`. Otherwise, refer to *step 2* below to manually select the ALM Octane synchronization action and the ALM Octane workspace.

**Synchronize Service Manager Change to ALM Octane defect**

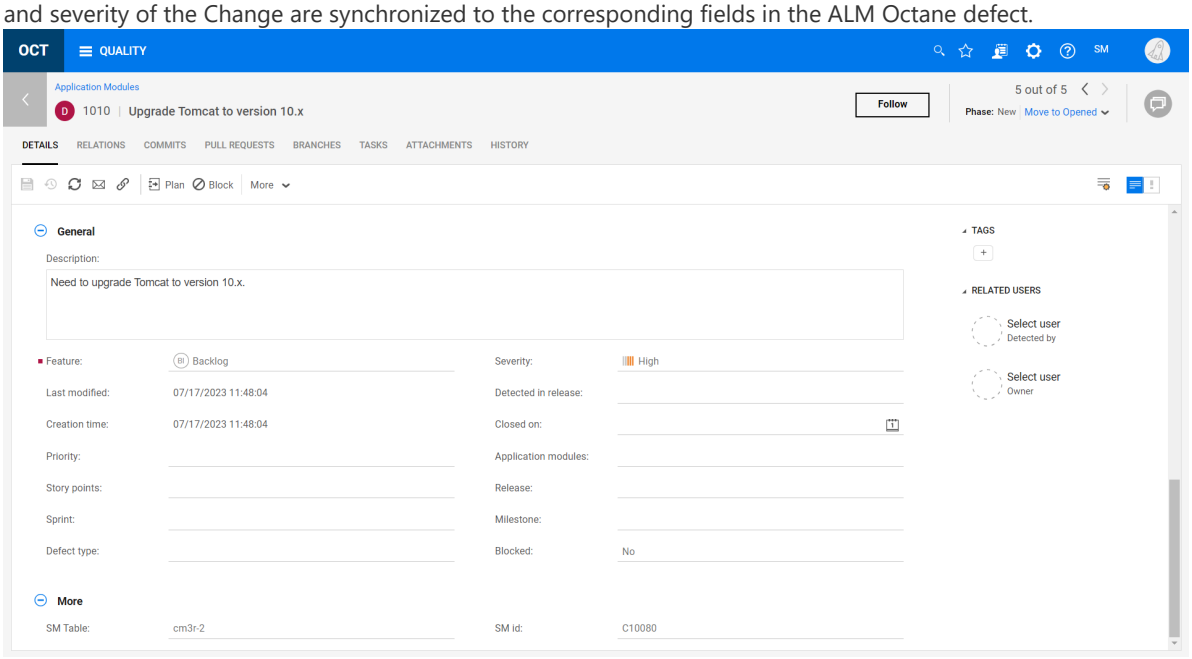
The following steps walk you through this use case:

1. The Change Manager opens a Change and then sets **Affected Configuration Item** to adv-Server-101_tomcat.
2. Suppose the **chm.octane** ruleset exists in the **Build And Test** phase and the **OctaneCustom** JavaScript exists in Script Library. When the Change moves to the **Build and Test** phase, the **Octane action** and **Octane Workspace** fields are filled automatically.

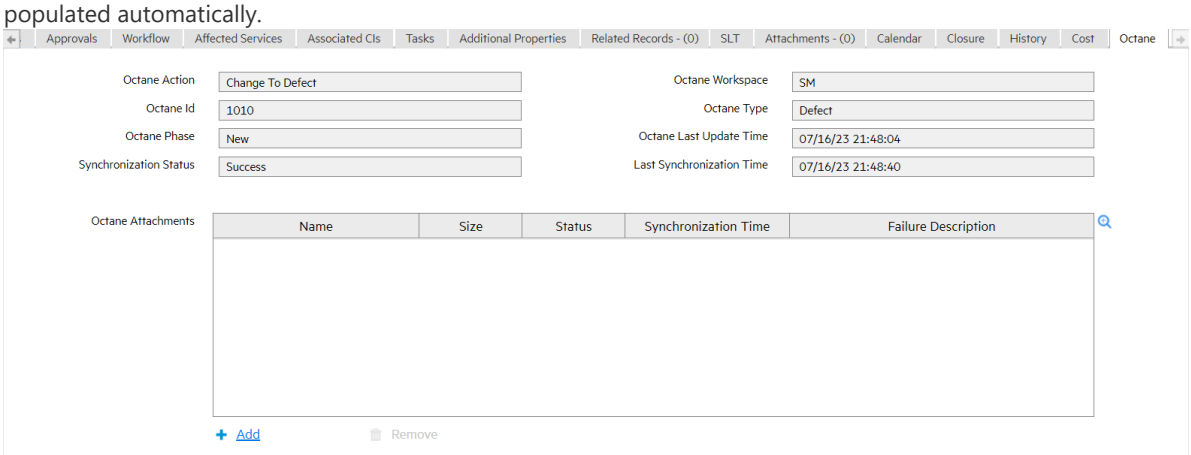



Otherwise, the Change Manager selects **Change To Defect** in the **Octane Action** drop-down list, and then selects the ALM Octane workspace in the **Octane Workspace** drop-down list.

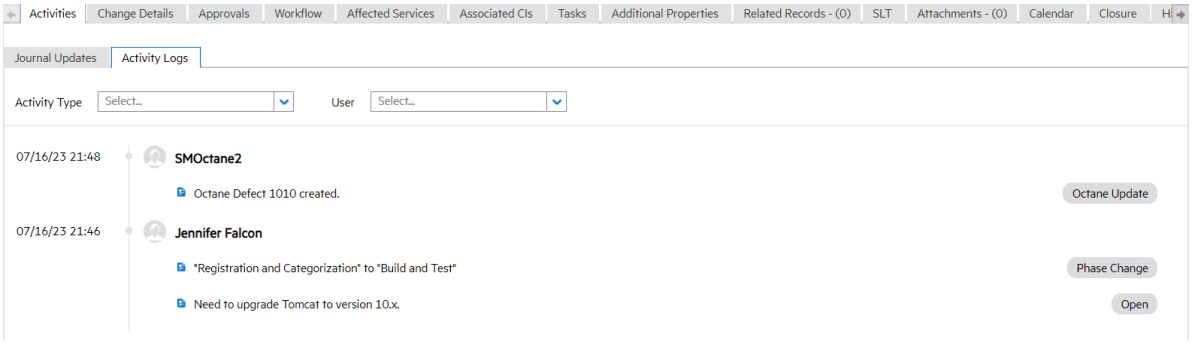
3. After Service Manager runs the integration background process, ALM Octane creates a defect. The title, description, and severity of the Change are synchronized to the corresponding fields in the ALM Octane defect.



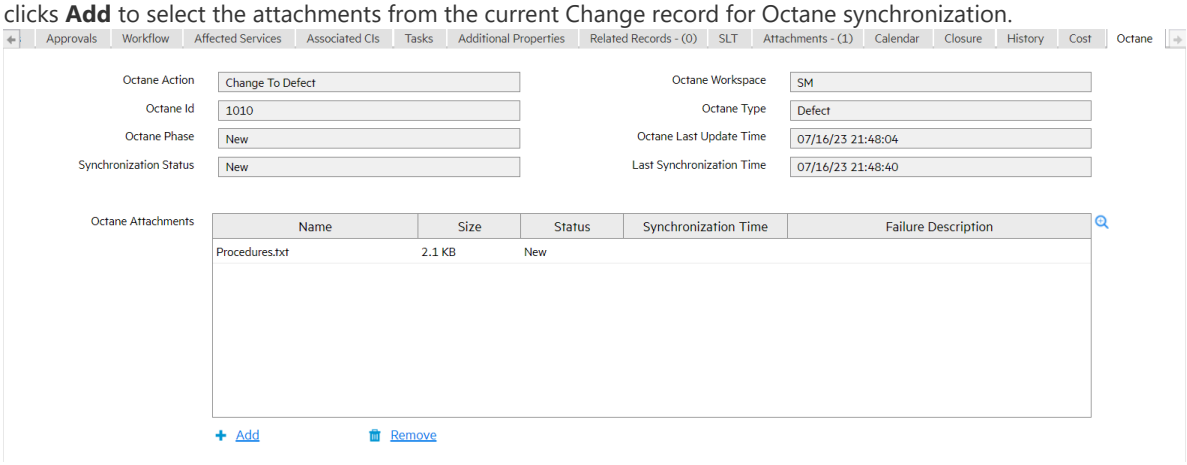

Meanwhile, the Octane Id, Octane Type, Octane Phase, Octane Last Update Time, Synchronization Status, and Last Synchronization Time fields on the **Octane** tab of the Change in Service Manager have read-only values that are populated automatically.



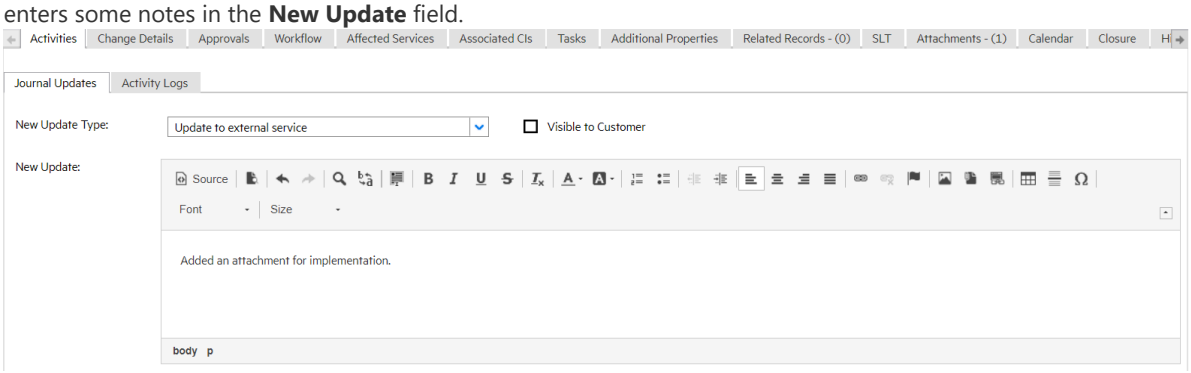

Service Manager also adds an activity to the **Activity Logs** for defect creation.




4. The Change Manager adds an attachment to the Change and then saves this update. Next, the Change Manager clicks **Add** to select the attachments from the current Change record for Octane synchronization.

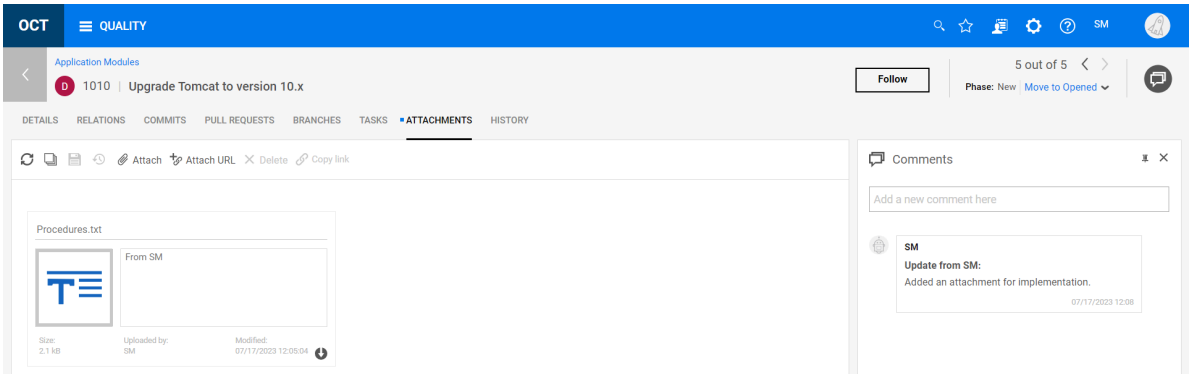



Then, the Change Manager selects **Update to external service** in the **New Update Type** drop-down list and then enters some notes in the **New Update** field.

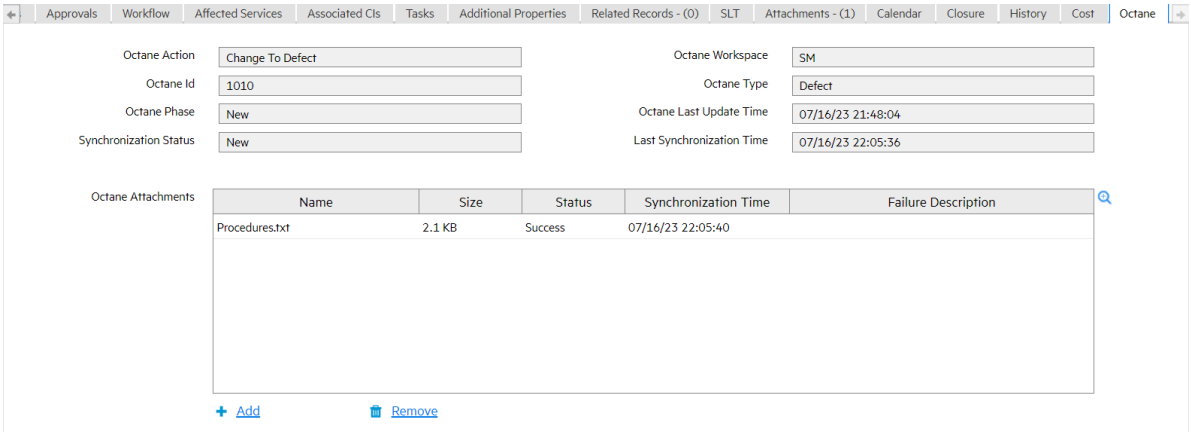



The Change Manager saves this update.

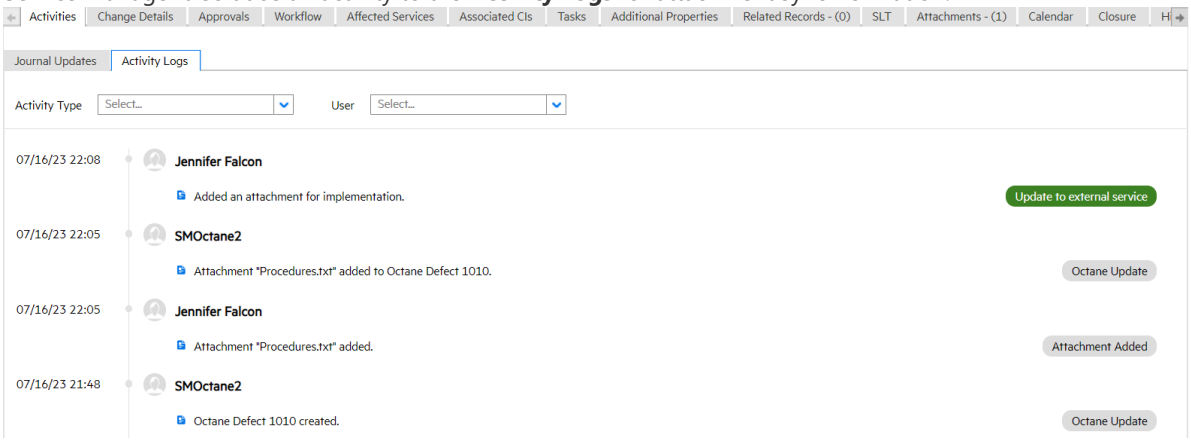
5. After synchronization, the Change Manager's notes in the **New Update** field now appear as a comment in the Defect record. The selected attachment in the Change record now appears as the ALM Octane defect's attachment as well.



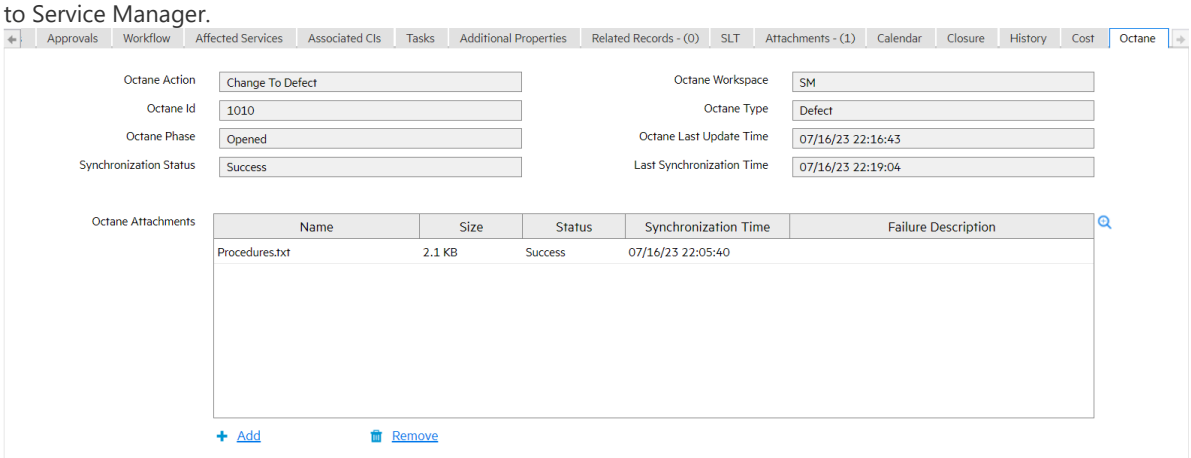

Meanwhile, the **Octane** tab displays the attachment synchronization status.



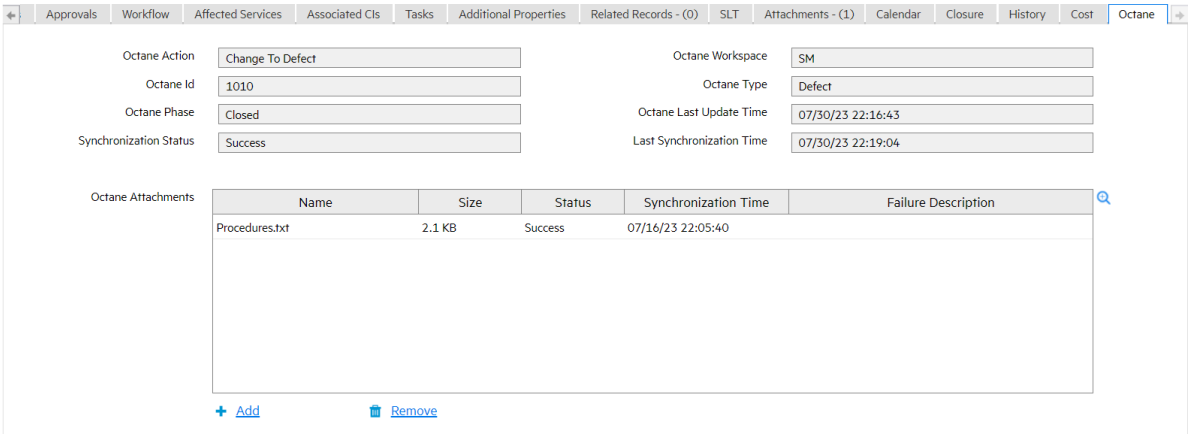

Service Manager also adds an activity to the **Activity Logs** for attachment synchronization.




6. As the defect follows the defect management workflows in ALM Octane, the phase updates are synchronized back to Service Manager.




7. Later, when an ALM Octane Team Member closes the defect, the integration synchronizes this phase information to Service Manager. The Change workflow continues.




### Related topics

ALM Octane

## 10.11. Synchronize Service Manager Change to ALM Octane feature

This use case describes the following user scenario:

An online shopping website is exploring foreign markets. To increase its competitiveness, the decision-making department decided to support Paypal as a new payment method. The product manager responsible for developing the online platform made a technical evaluation of the decision and proposed to update the payment interface of the platform to be compatible with the new payment method. After assessing the risks and impacts of this change, this proposal is finally approved. Since ALM Octane has full lifecycle management of features, the change manager synchronizes the change to ALM Octane as a feature for development, test, and delivery jobs.

### Prerequisites

As a prerequisite of this use case, the Service Manager system administrator must complete the following tasks in advance:

1. Navigate to **Tailoring** > **Integration Manager** to start the ALM Octane integration instance.
2. There are two ways to select the ALM Octane synchronization action and the ALM Octane workspace: manually or automatically.
   To automatically select the ALM Octane synchronization action and the ALM Octane workspace, check the following prerequisites.
   1. Navigate to **Tailoring** > **Process Designer** > **Workflows**. Open the workflow for Normal Change, and then check that the **chm.octane** ruleset exists in the **Build And Test** phase.
   2. Check that the **OctaneCustom** JavaScript exists in Script Library. Its content may resemble the following.

```
/**
 * If some changes need to Sync to Octane automatically
 * Update the condition and return {action, workspace}.
 */
function getOctaneConfig(change) {
    // The condition, a default one is to contain CI 'CI1000910, adv-Server-101_tomcat'
    let list = change.assets && change.assets.toArray() || [];
    if (isChangeToDefectEnabled() && list.includes('CI1000910')) {
        let workspace = vars.$G_OCT_WS_LIST_CHANGE_TO_DEFECT[0];
        if (workspace) {
            return {
                action: 'changeToDefect',
                workspace
            }
        }
    }

    // The condition for "changeToFeature", a default one is to contain CI 'CI1000914, adv-
Server-102_tomcat'
    if (isChangeToFeatureEnabled() && list.includes('CI1000914')) {
        let workspace = vars.$G_OCT_WS_LIST_CHANGE_TO_FEATURE[0];
        if (workspace) {
            return {
                action: 'changeToFeature',
                workspace
            }
        }
    }

    return null;
}

function isChangeToDefectEnabled() {
    return vars.$G_OCT_WS_LIST_CHANGE_TO_DEFECT &&
vars.$G_OCT_WS_LIST_CHANGE_TO_DEFECT.length()>0;
}

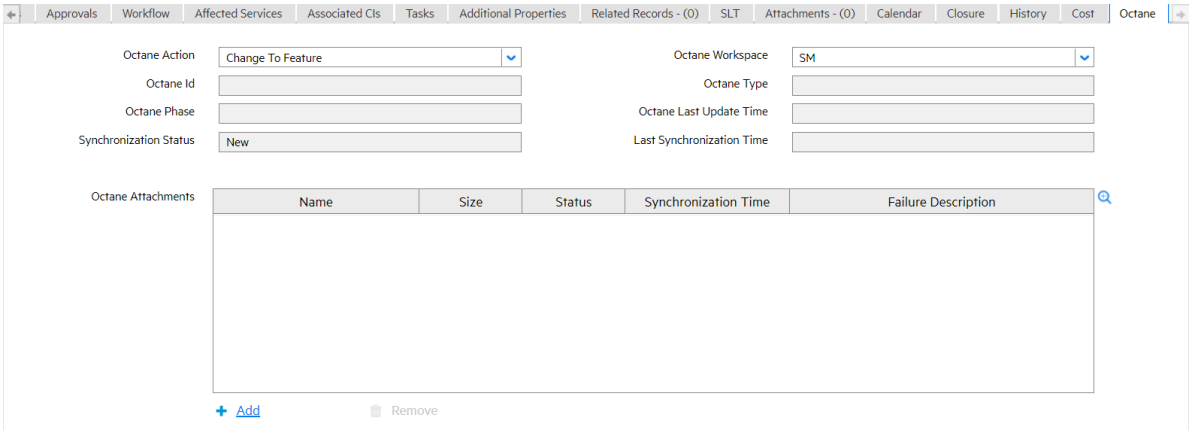
function isChangeToFeatureEnabled() {
    return vars.$G_OCT_WS_LIST_CHANGE_TO_FEATURE &&
vars.$G_OCT_WS_LIST_CHANGE_TO_FEATURE.length()>0;
}
```

In this use case, `adv-Server-102_tomcat` is the Affected Configuration Item, and its CI Identifier is `CI1000914`. Otherwise, refer to *step 2* below to manually select the ALM Octane synchronization action and the ALM Octane workspace.

**Synchronize Service Manager Change to ALM Octane feature**

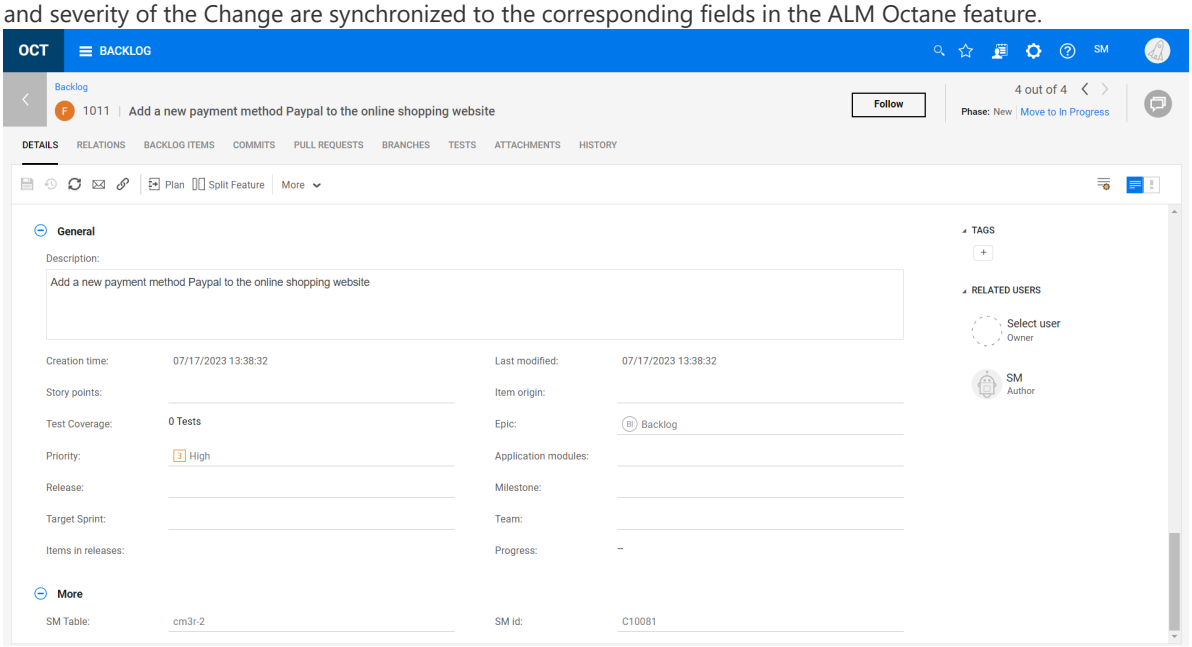
The following steps walk you through this use case:

1. The Change Manager opens a Change and then sets **Affected Configuration Item** to adv-Server-102_tomcat.
2. Suppose the **chm.octane** ruleset exists in the **Build And Test** phase and the **OctaneCustom** JavaScript exists in Script Library. When the Change moves to the **Build and Test** phase, the **Octane action** and **Octane Workspace** fields are filled automatically.

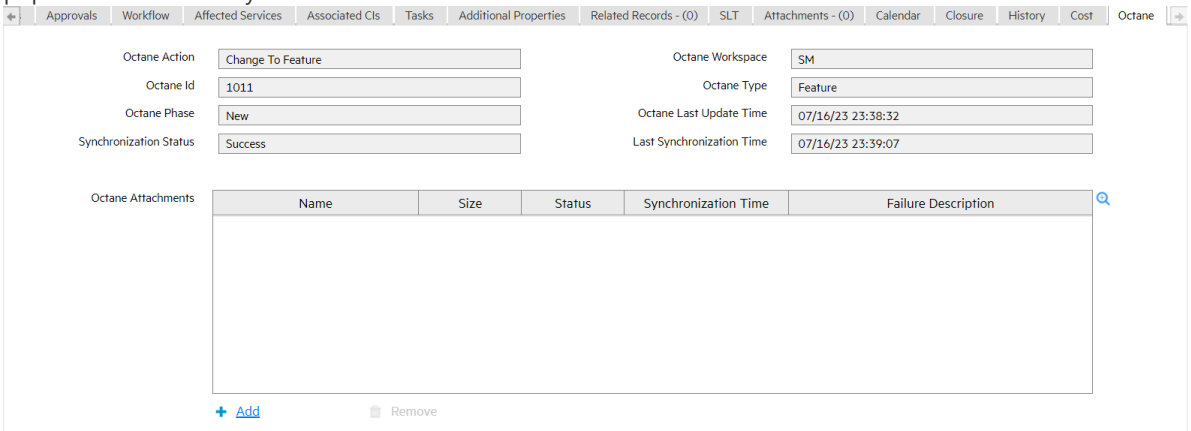



Otherwise, the Change Manager selects **Change To Feature** in the **Octane Action** drop-down list, and then selects the ALM Octane workspace in the **Octane Workspace** drop-down list.

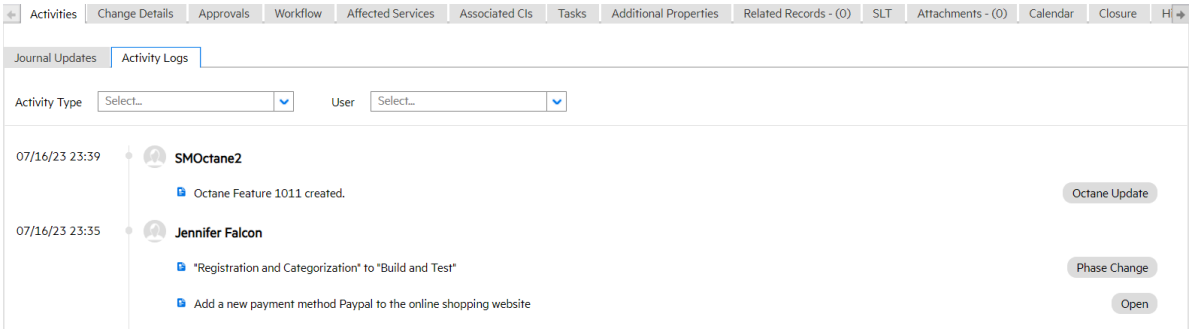
3. After Service Manager runs the integration background process, ALM Octane creates a feature. The title, description, and severity of the Change are synchronized to the corresponding fields in the ALM Octane feature.



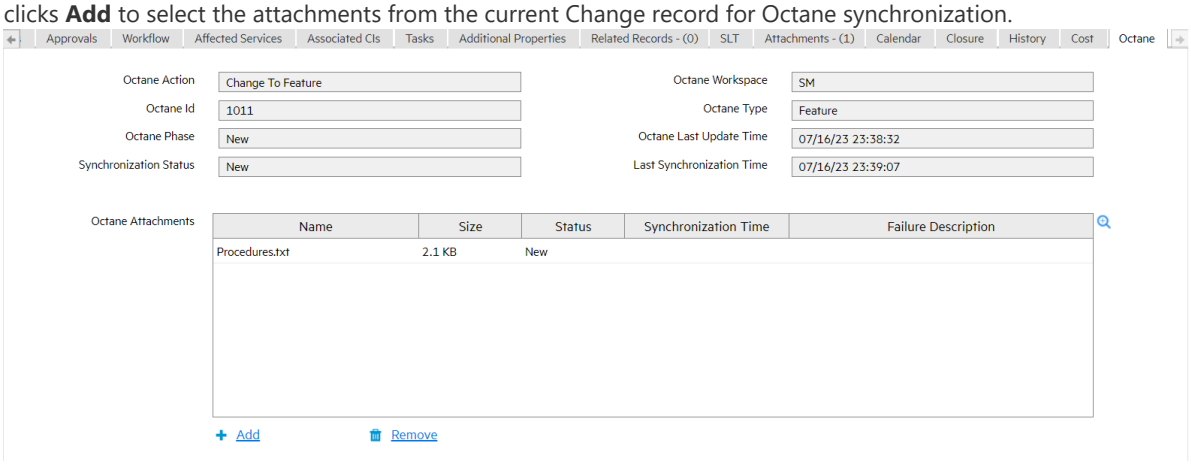

Meanwhile, the Octane Id, Octane Type, Octane Phase, Octane Last Update Time, Synchronization Status, and Last Synchronization Time fields on the **Octane** tab of the Change in Service Manager have read-only values that are populated automatically.



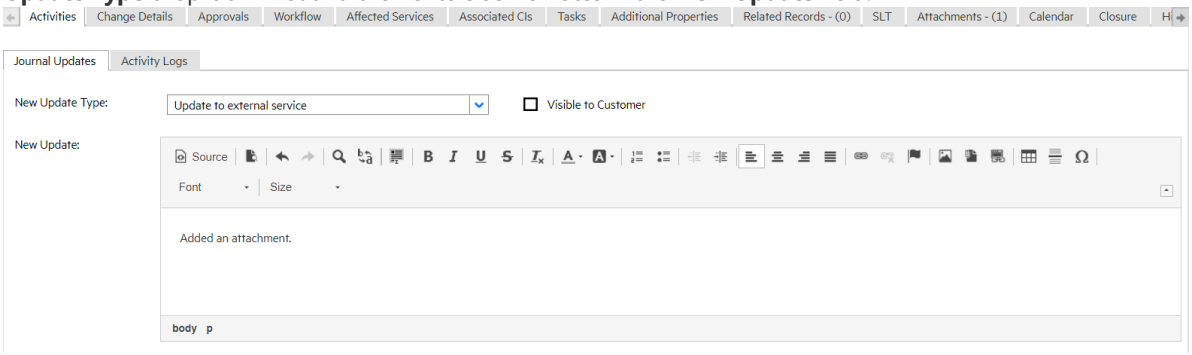

Service Manager also adds an activity to the **Activity Logs** for feature creation.




4. The Change Manager adds an attachment to the Change and then saves this update. Next, the Change Manager clicks **Add** to select the attachments from the current Change record for Octane synchronization.

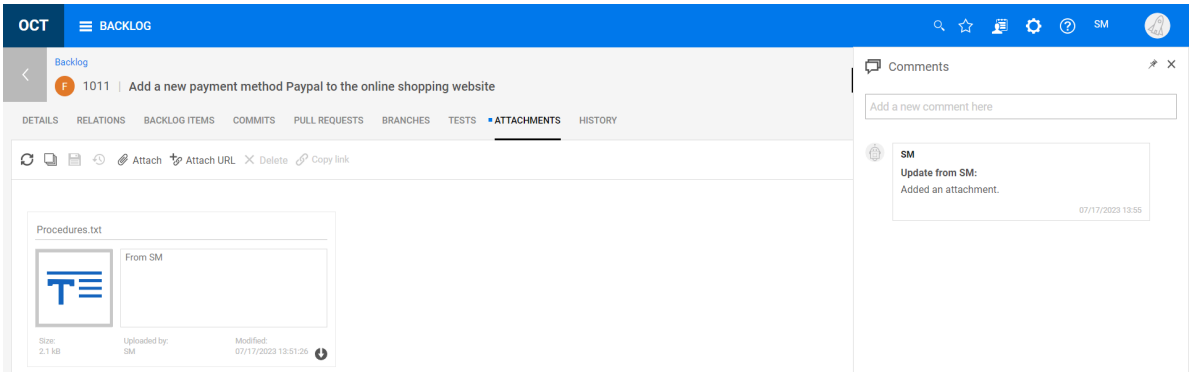



The Change Manager saves this update. Next, the Change Manager selects **Update to external service** in the **New Update Type** drop-down list and then enters some notes in the **New Update** field.

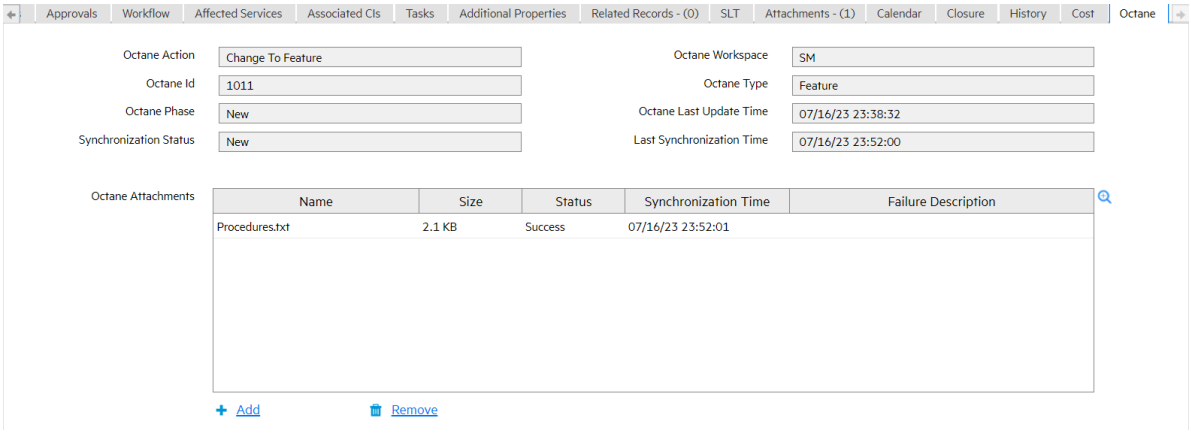



The Change Manager saves this update as well.

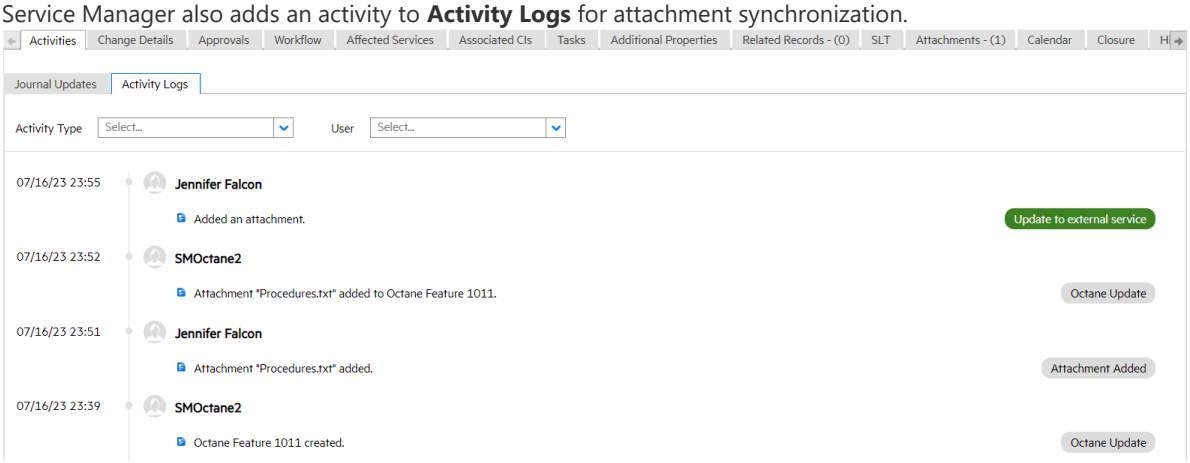
5. After synchronization, the Change Manager's notes in the **New Update** field now appear as a comment in the feature record. The selected attachment in the Change record now appears as the ALM Octane feature's attachment as well.



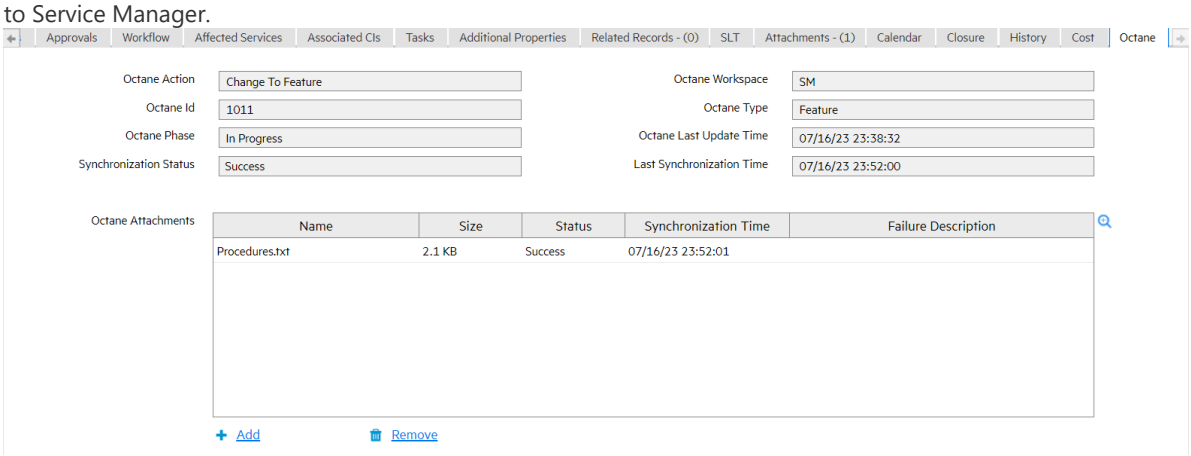

Meanwhile, the **Octane** tab displays the attachment synchronization status.



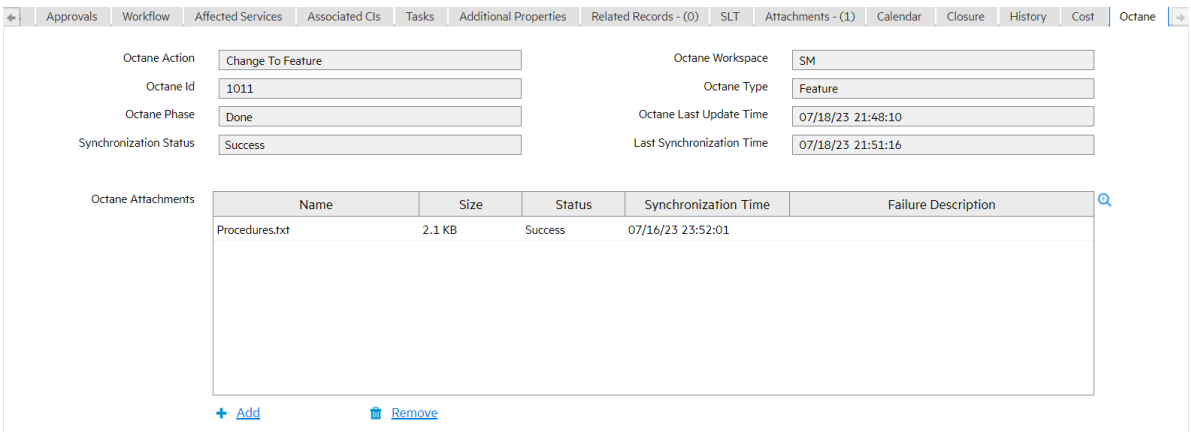

Service Manager also adds an activity to **Activity Logs** for attachment synchronization.




6. As the feature follows the feature management workflows in ALM Octane, the phase updates are synchronized back to Service Manager.




7. Later, when the feature moves to the Done phase in ALM Octane, the integration synchronizes this phase information to Service Manager. The Change workflow continues.




### Related topics

ALM Octane

## 10.12. Synchronize Service Manager Change to ALM defect

This use case describes the following user scenario:

Spring Boot team was recently notified that they needed to upgrade their Tomcat version in time to avoid the new vulnerability issue in Tomcat. The Product Owner proposes a Change with the aim of upgrading Tomcat to the latest version 10.x. After assessing the risks and impact, the Change is approved and now it's in the Build and Test phase. Since ALM has a complete defect tracking process, the Change Manager decides to synchronize the Change to ALM as a defect and hence to test it and track it thoroughly.

### Prerequisites

As a prerequisite of this use case, the Service Manager system administrator must complete the following tasks in advance:

1. Navigate to **Tailoring** > **Integration Manager** to start the ALM integration instance.
2. The background synchronization tasks start after you select the ALM synchronization action and the ALM project. There are two ways: manually or automatically.
   To automatically select the ALM synchronization action and the ALM project, check the following prerequisites.
   1. Navigate to **Tailoring** > **Process Designer** > **Workflows**. Open the workflow for Normal Change, and then check that the **chm.alm** rule set exists in the **Build And Test** phase.
   2. Check that the **AlmCustom** JavaScript exists in Script Library. Its content may resemble the following.

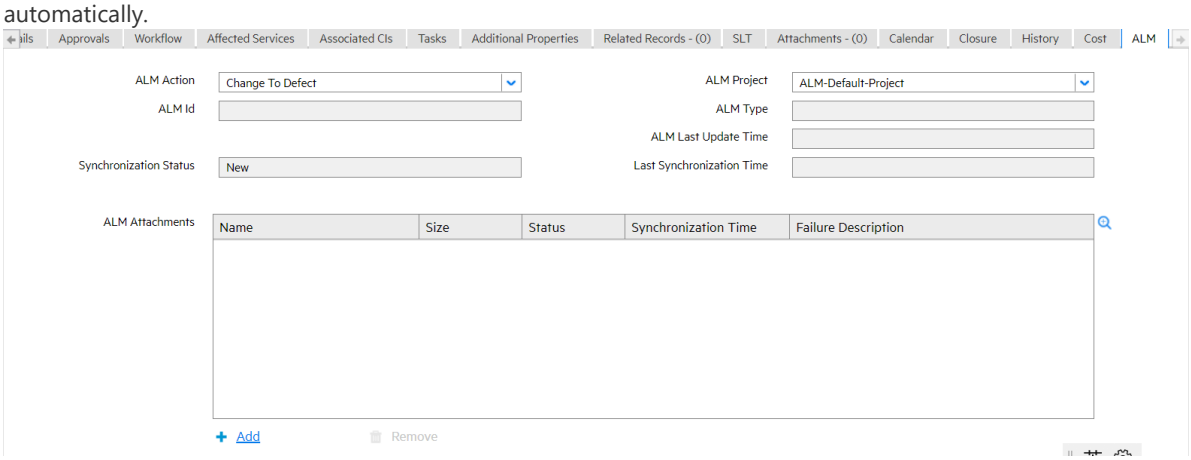
```
/**
* If some changes need to sync to ALM automatically
* Update the condition and return {action, workspace}.
*/
function getAlmConfig(file) {
    var fileName = funcs.filename(file);
    if ('cm3r' == fileName) return getAlmConfigForChange(file);
    if ('rootcause' == fileName) return getAlmConfigForProblem(file);
    return null;
} function getAlmConfigForChange(change) {
    // The condition for Change To Defect, a default one is to contain CI 'CI1000910, adv-
Server-101_tomcat'
    let list = change.assets && change.assets.toArray() || [];
    if (isChangeToDefectEnabled() && list.includes('CI1000910')) {
        let workspace = vars.$G_alm_WS_LIST_CHANGE_TO_DEFECT[0];
        if (workspace) {
            return {
                action: 'changeToDefect',
                workspace
            }
        }
    }     // The condition for Change To Requirement, a default one is to contain CI
'CI1000914, adv-Server-102_tomcat'
    if (isChangeToFeatureEnabled() && list.includes('CI1000914')) {
        let workspace = vars.$G_alm_WS_LIST_CHANGE_TO_FEATURE[0];
        if (workspace) {
            return {
                action: 'changeToFeature',
                workspace
            }
        }
    }     return null;
} function getAlmConfigForProblem(problem) {
    // The condition for Problem To Defect, a default one is to contain CI 'CI1000910, adv-
Server-101_tomcat'
    if (isProblemToDefectEnabled() && problem.logical_name == 'CI1000910') {
        let workspace = vars['$G.alm.WS.LIST.PROBLEM.TO.DEFECT'][0];
        if (workspace) {
            return ({
                action: 'problemToDefect',
                workspace
            })
        }
    }     return null;
}
function isEnabled(name) {
    var list = vars[name];
    return list && list.length() > 0;
} function isChangeToDefectEnabled() {
    return isEnabled('$G.alm.WS.LIST.CHANGE.TO.DEFECT');
} function isChangeToFeatureEnabled() {
    return isEnabled('$G.alm.WS.LIST.CHANGE.TO.FEATURE');
} function isProblemToDefectEnabled() {
    return isEnabled('$G.alm.WS.LIST.PROBLEM.TO.DEFECT');
}
```

In this use case, `adv-Server-101_tomcat` is the Affected Configuration Item, and its CI Identifier is `CI1000910`. Otherwise, refer to *step 2* below to manually select the ALM synchronization action and the ALM project.

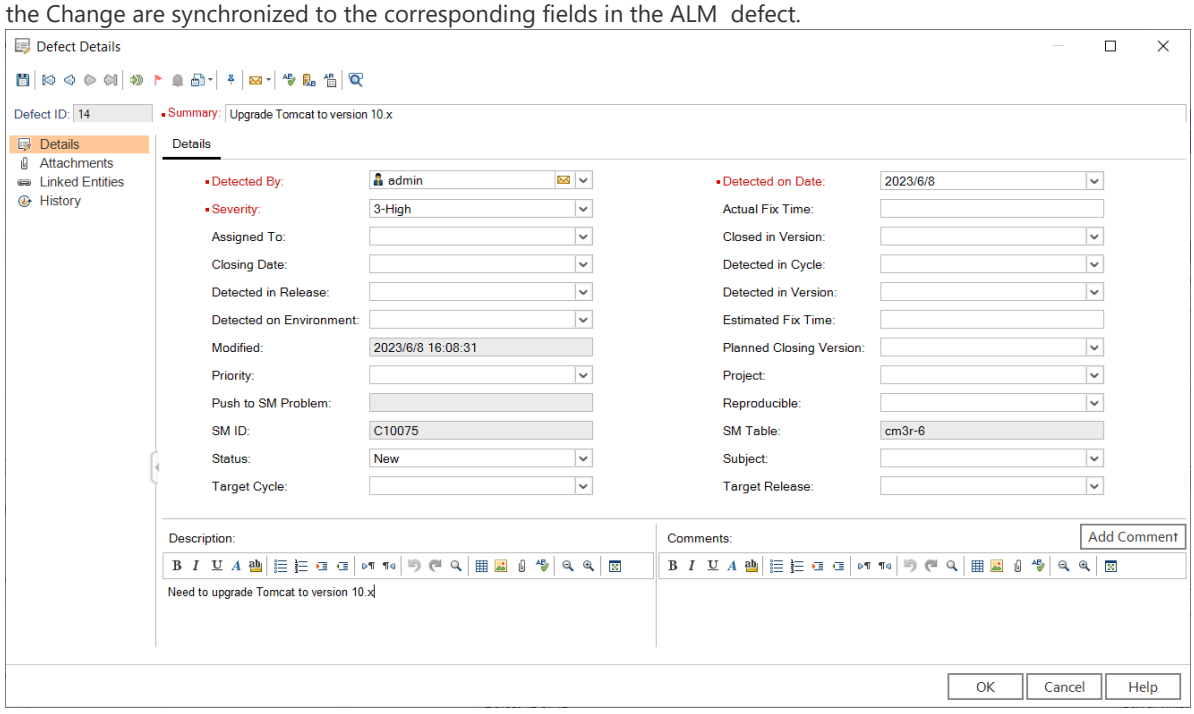
### Synchronize Service Manager Change to ALM defect

The following steps walk you through this use case:

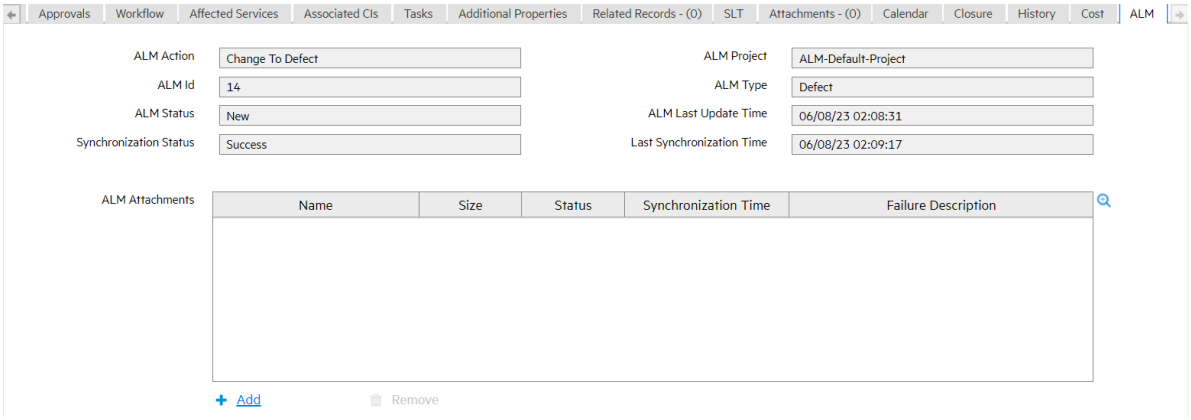
1. The Change Manager opens a Change and then sets **Affected Configuration Item** to adv-Server-101_tomcat.
2. Suppose the **chm.alm** ruleset exists in the **Build And Test** phase and the **AlmCustom** JavaScript exists in Script Library. When the Change moves to the **Build and Test** phase, the **ALM Action** and **ALM Project** fields are filled automatically.



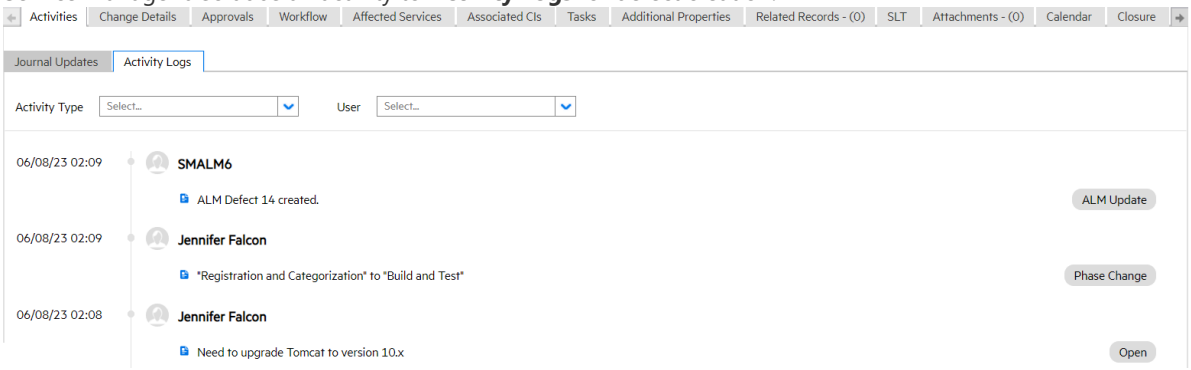

   Otherwise, the Change Manager selects **Change To Defect** in the **ALM Action** drop-down list, and then selects the ALM project (project alias) in the **ALM Project** drop-down list.
3. After Service Manager runs the integration background process, ALM creates a defect. The title and description of the Change are synchronized to the corresponding fields in the ALM  defect.



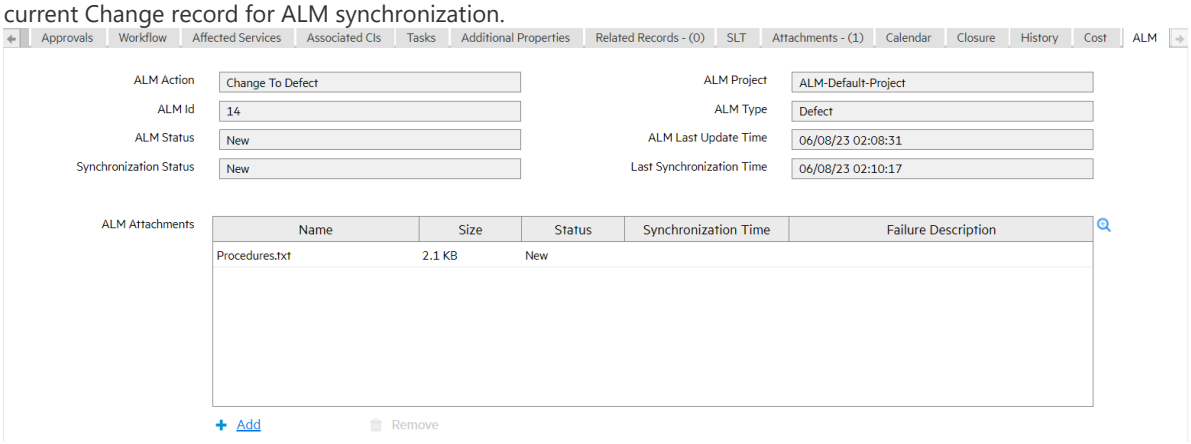

   Meanwhile, the ALM Id, ALM Type, ALM Status, ALM Last Update Time, Synchronization Status, and Last Synchronization Time fields on the **ALM** tab of the Change in Service Manager have read-only values that are populated automatically.



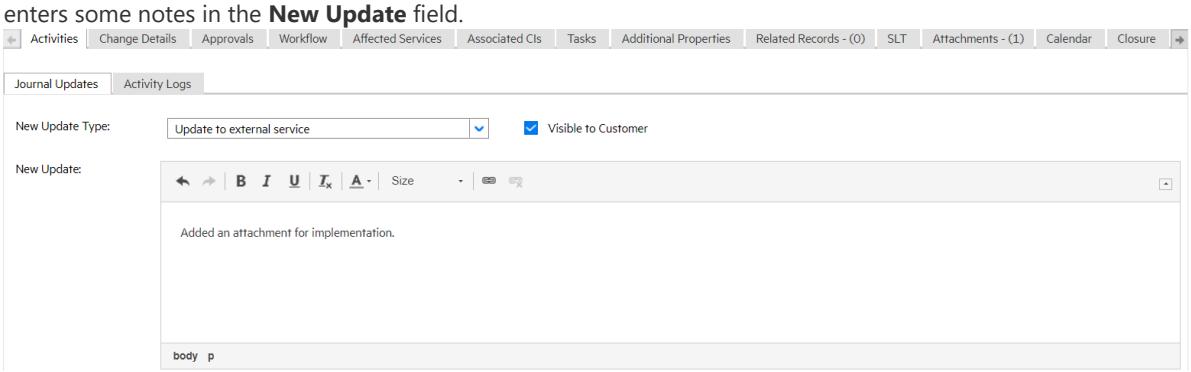

Service Manager also adds an activity to **Activity Logs** for defect creation.




4. The Change Manager adds an attachment to the Change and then clicks **Add** to select the attachments from the current Change record for ALM synchronization.

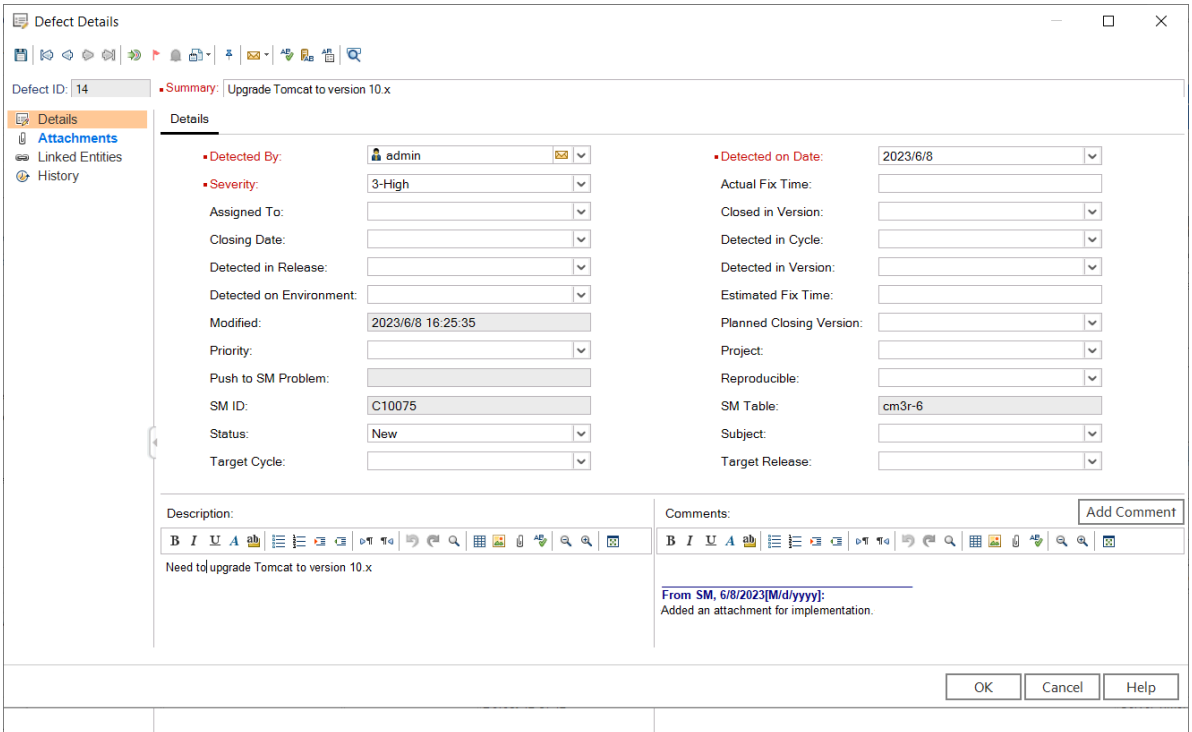


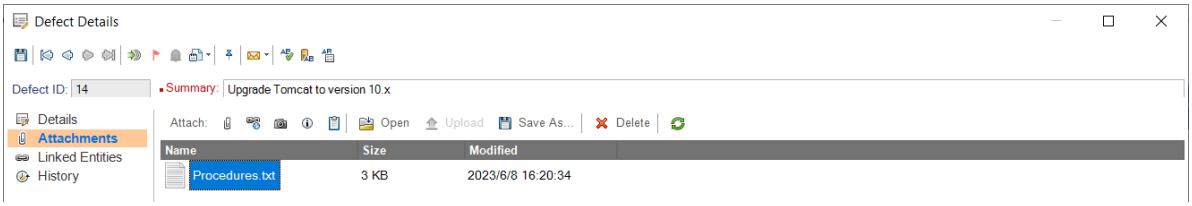

Next, the Change Manager selects **Update to external service** in the **New Update Type** drop-down list and then enters some notes in the **New Update** field.



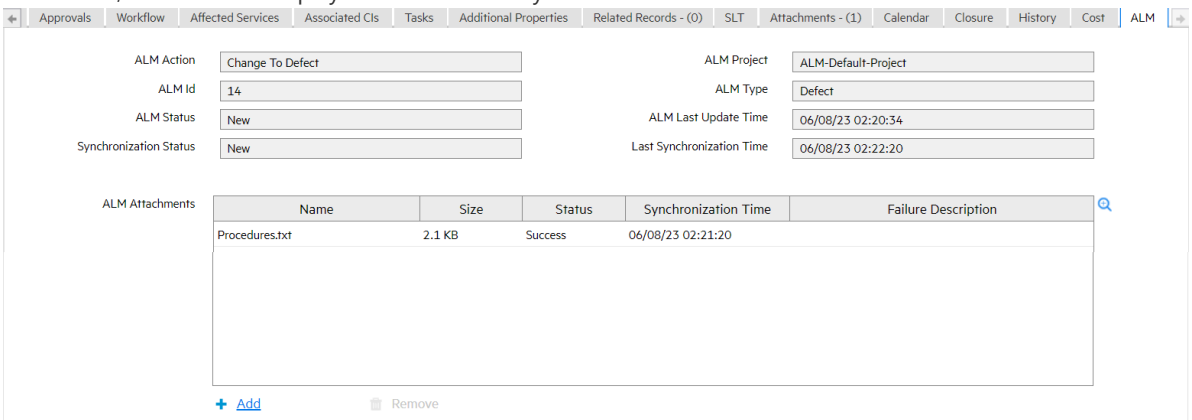

The Change Manager saves this update.

5. After synchronization, the Change Manager's notes in the **New Update** field now appear as a comment in the Defect record. The selected attachment in the Change record now appears as the ALM defect's attachment as well.

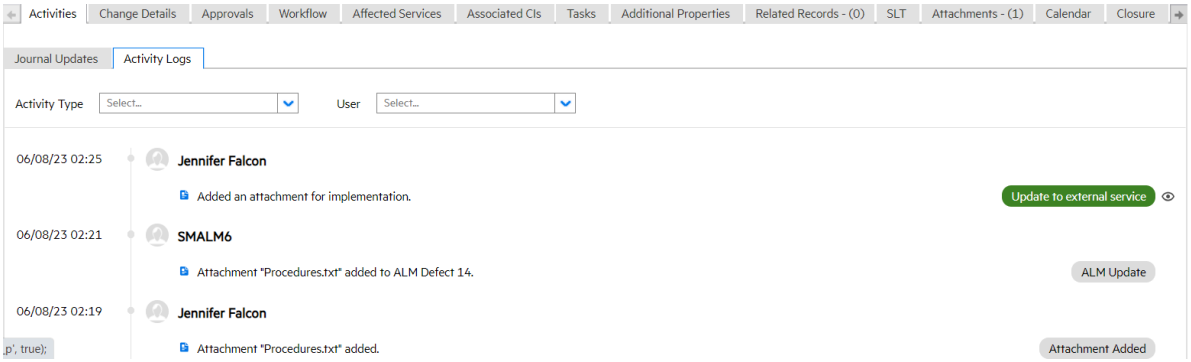


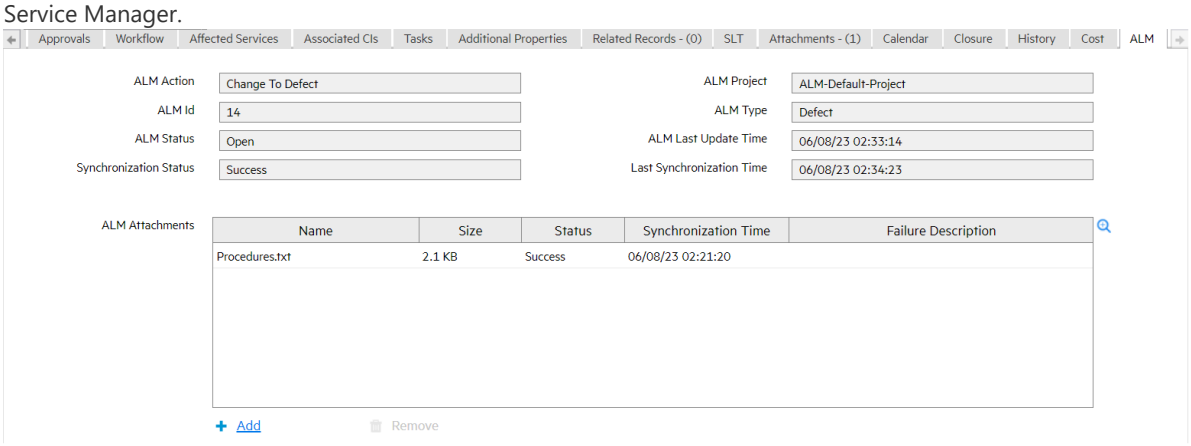

Meanwhile, the **ALM** tab displays the attachment synchronization status.



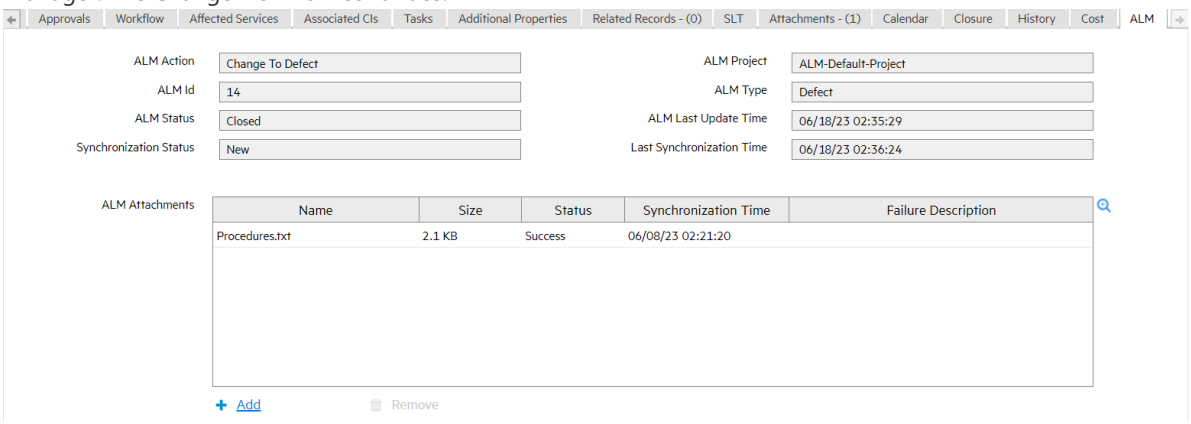

Service Manager also adds an activity to the **Activity Logs** for attachment synchronization.




6. As the defect follows the defect management workflows in ALM, the status updates are synchronized back to Service Manager.




7. Later, when an ALM Team Member closes the defect, the integration synchronizes this phase information to Service Manager. The Change workflow continues.




### Related topics

ALM

## 10.13. Synchronize Service Manager Change to ALM requirement

This use case describes the following user scenario:

An online shopping website is exploring foreign markets. To increase its competitiveness, the decision-making department decided to support Paypal as a new payment method. The Product Manager responsible for developing the online platform made a technical evaluation of the decision and proposed to update the payment interface of the platform to be compatible with the new payment method. After assessing the risks and impacts of this change, this proposal is finally approved. Since ALM has full lifecycle management of features, the Change Manager synchronizes the Change to ALM as a requirement for development, test, and delivery jobs.

**Prerequisites**

As a prerequisite of this use case, the Service Manager system administrator must complete the following tasks in advance:

1. Navigate to **Tailoring** > **Integration Manager** to start the ALM integration instance.
2. The background synchronization tasks start after you select the ALM synchronization action and the ALM project. There are two ways: manually or automatically.
   To automatically select the ALM synchronization action and the ALM project, check the following prerequisites.
   1. Navigate to **Tailoring** > **Process Designer** > **Workflows**. Open the workflow for Normal Change, and then check that the **chm.alm** ruleset exists in the **Build And Test** phase.
   2. Check that the **AlmCustom** JavaScript exists in Script Library. Its content may resemble the following.

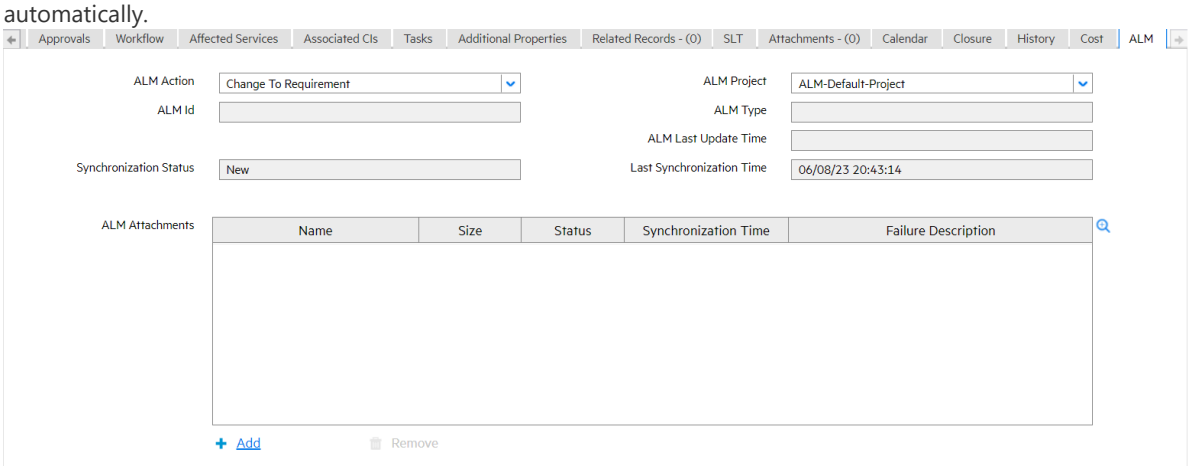
```
/**
* If some changes need to sync to ALM automatically
* Update the condition and return {action, workspace}.
*/
function getAlmConfig(file) {
    var fileName = funcs.filename(file);
    if ('cm3r' == fileName) return getAlmConfigForChange(file);
    if ('rootcause' == fileName) return getAlmConfigForProblem(file);
    return null;
} function getAlmConfigForChange(change) {
    // The condition for Change To Defect, a default one is to contain CI 'CI1000910, adv-
Server-101_tomcat'
    let list = change.assets && change.assets.toArray() || [];
    if (isChangeToDefectEnabled() && list.includes('CI1000910')) {
        let workspace = vars.$G_alm_WS_LIST_CHANGE_TO_DEFECT[0];
        if (workspace) {
            return {
                action: 'changeToDefect',
                workspace
            }
        }
    }     // The condition for Change To Requirement, a default one  Cis to contain CI
'CI1000914, adv-Server-102_tomcat'
    if (isChangeToFeatureEnabled() && list.includes('CI1000914')) {
        let workspace = vars.$G_alm_WS_LIST_CHANGE_TO_FEATURE[0];
        if (workspace) {
            return {
                action: 'changeToFeature',
                workspace
            }
        }
    }     return null;
} function getAlmConfigForProblem(problem) {
    // The condition for Problem To Defect, a default one is to contain CI 'CI1000910, adv-
Server-101_tomcat'
    if (isProblemToDefectEnabled() && problem.logical_name == 'CI1000910') {
        let workspace = vars['$G.alm.WS.LIST.PROBLEM.TO.DEFECT'][0];
        if (workspace) {
            return ({
                action: 'problemToDefect',
                workspace
            })
        }
    }     return null;
}
function isEnabled(name) {
    var list = vars[name];
    return list && list.length() > 0;
} function isChangeToDefectEnabled() {
    return isEnabled('$G.alm.WS.LIST.CHANGE.TO.DEFECT');
} function isChangeToFeatureEnabled() {
    return isEnabled('$G.alm.WS.LIST.CHANGE.TO.FEATURE');
} function isProblemToDefectEnabled() {
    return isEnabled('$G.alm.WS.LIST.PROBLEM.TO.DEFECT');
}
```

In this use case, `adv-Server-102_tomcat` is the Affected Configuration Item, and its CI Identifier is `CI1000914`. Otherwise, refer to *step 2* below to manually select the ALM synchronization action and the ALM project.

### Synchronize Service Manager Change to ALM requirement

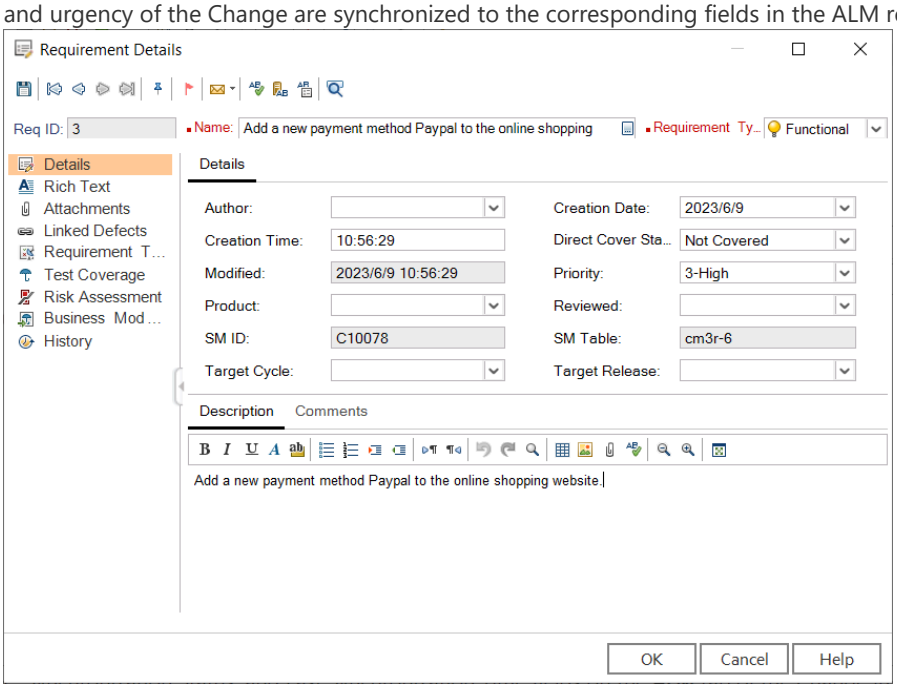
The following steps walk you through this use case:

1. The Change Manager opens a Change and then sets **Affected Configuration Item** to adv-Server-102_tomcat.
2. Suppose the **chm.alm** ruleset exists in the **Build And Test** phase and the **AlmCustom** JavaScript exists in Script Library. When the Change moves to the **Build and Test** phase, the **ALM Action** and **ALM Project** fields are filled automatically.

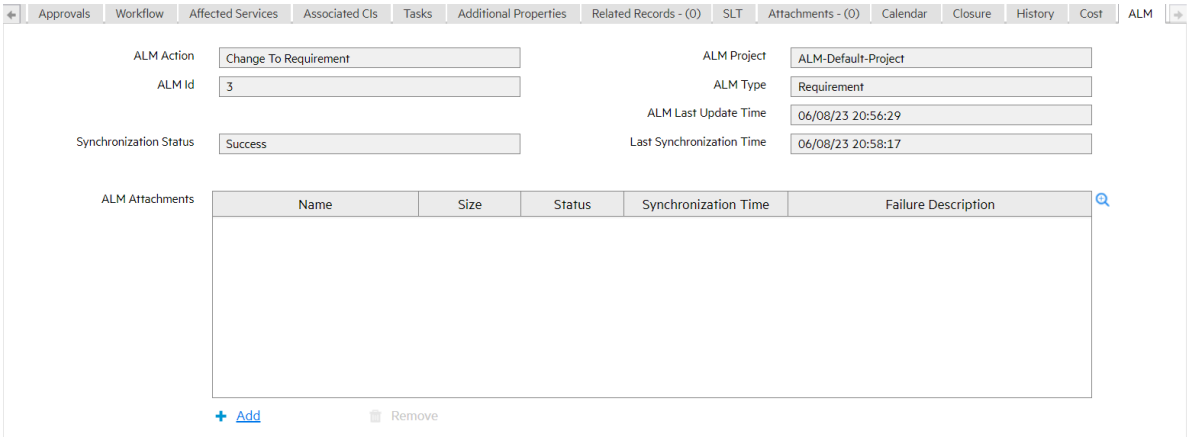



Otherwise, the Change Manager selects **Change To Requirement** in the **ALM Action** drop-down list and then selects the ALM project (project alias) in the **ALM Project** drop-down list.

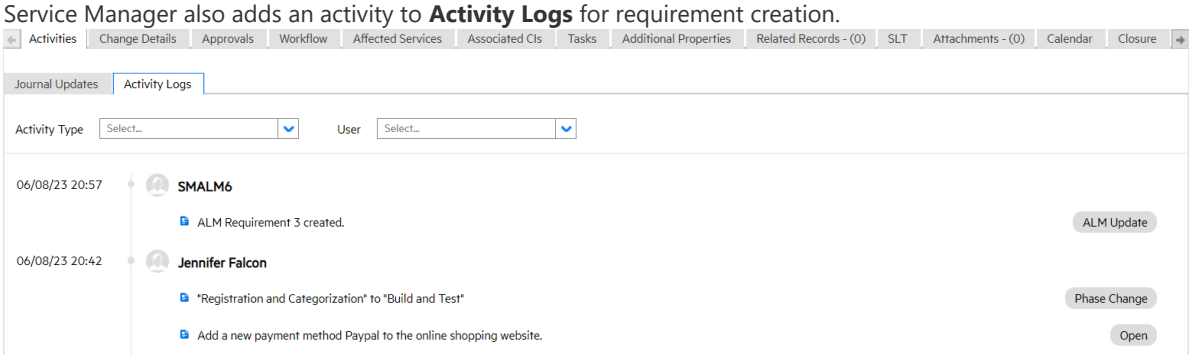
3. After Service Manager runs the integration background process, ALM creates a requirement. The title, description, and urgency of the Change are synchronized to the corresponding fields in the ALM requirement.



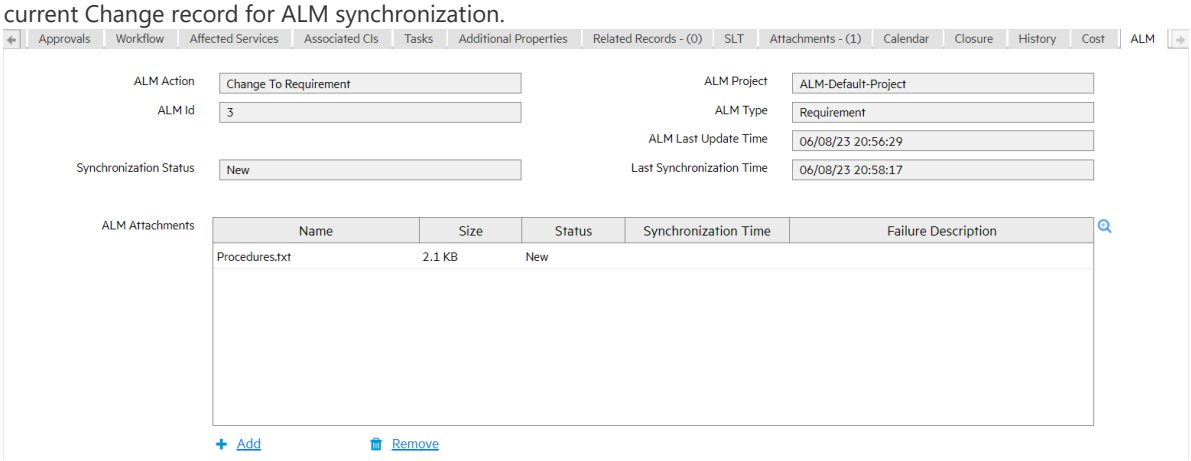

Meanwhile, the ALM Id, ALM Type, ALM Last Update Time, Synchronization Status, and Last Synchronization Time fields on the **ALM** tab of the Change in Service Manager have read-only values that are populated automatically.



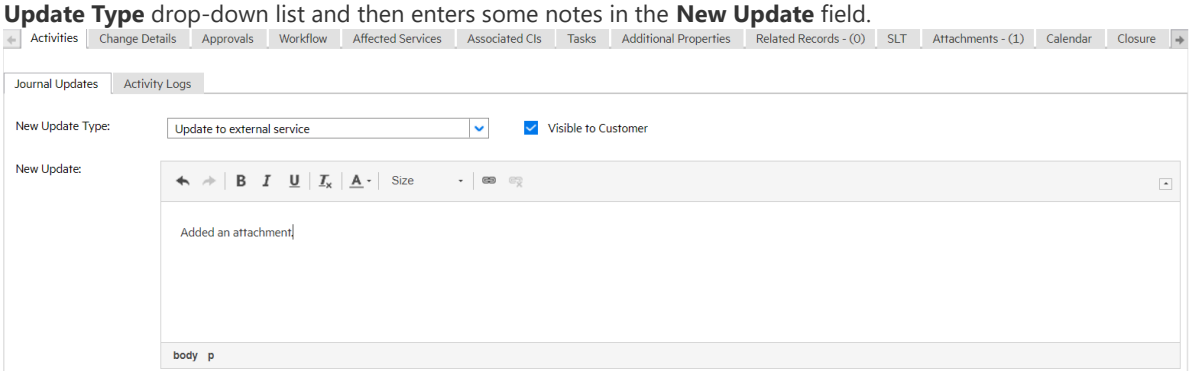

Service Manager also adds an activity to **Activity Logs** for requirement creation.




4. The Change Manager adds an attachment to the Change and then clicks **Add** to select the attachments from the current Change record for ALM synchronization.



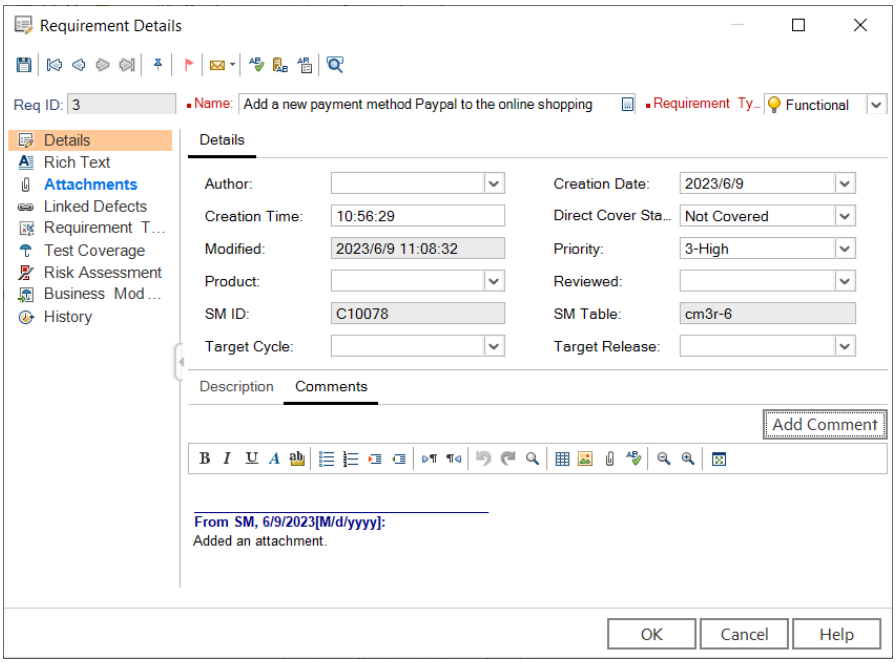

The Change Manager saves this update. Next, the Change Manager selects **Update to external service** in the **New Update Type** drop-down list and then enters some notes in the **New Update** field.

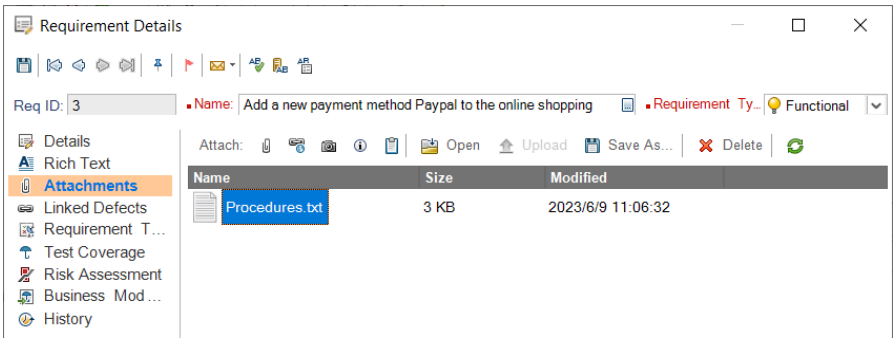



The Change Manager saves this update as well.

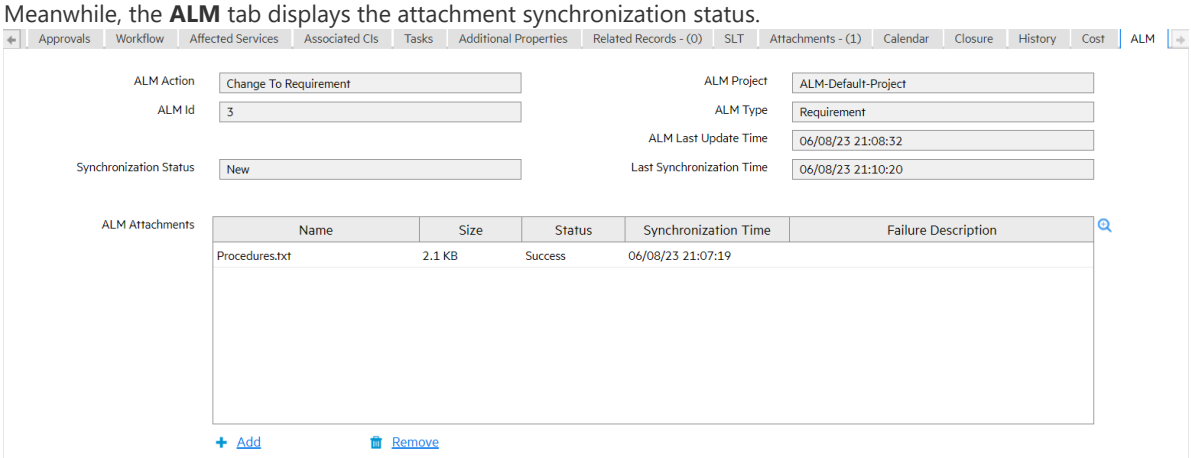
5. After synchronization, the Change Manager's notes in the **New Update** field now appear as a comment in the requirement record. The selected attachment in the Change record now appears as the ALM requirement's

attachment as well.

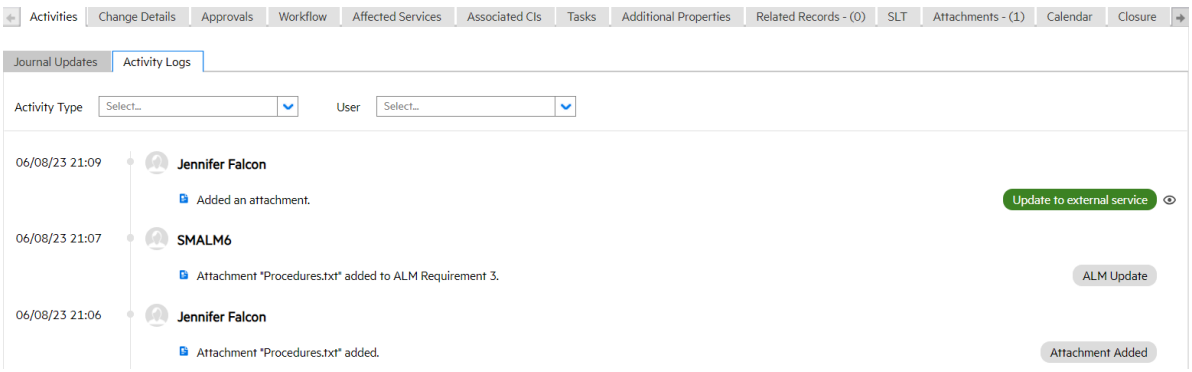



Meanwhile, the **ALM** tab displays the attachment synchronization status.




Service Manager also adds an activity to **Activity Logs** for attachment synchronization.




### Related topics

ALM

## 10.14. Synchronize Service Manager Problem to ALM defect

This use case describes the following user scenario:

Recently the Performance team receives a lot of reports about Norton antivirus program downgrading user performance. The process seems to peg the CPU at 60% or more and consumes at least 1 GB of RAM locally. This can be a potential issue. The Problem Manager decides to synchronize the Problem to ALM as a defect. Hence the vendor will supply a hotfix and use ALM to track the issue thoroughly.

### Prerequisites

As a prerequisite of this use case, the Service Manager system administrator must complete the following tasks in advance:

1. Navigate to **Tailoring** > **Integration Manager** to start the ALM integration instance.
2. The background synchronization tasks start after you select the ALM synchronization action and the ALM project. There are two ways: manually or automatically.
   To automatically select the ALM synchronization action and the ALM project, check the following prerequisites.
   1. Navigate to **Problem Management** > **Configuration** > **Workflows**. Open the workflow for Problem, and then check that the **pbm.alm** ruleset exists in the **Categorization**, **Investigation**, **Resolution**, and **Review** phase.
   2. Check that the **AlmCustom** JavaScript exists in Script Library. Its content may resemble the following.

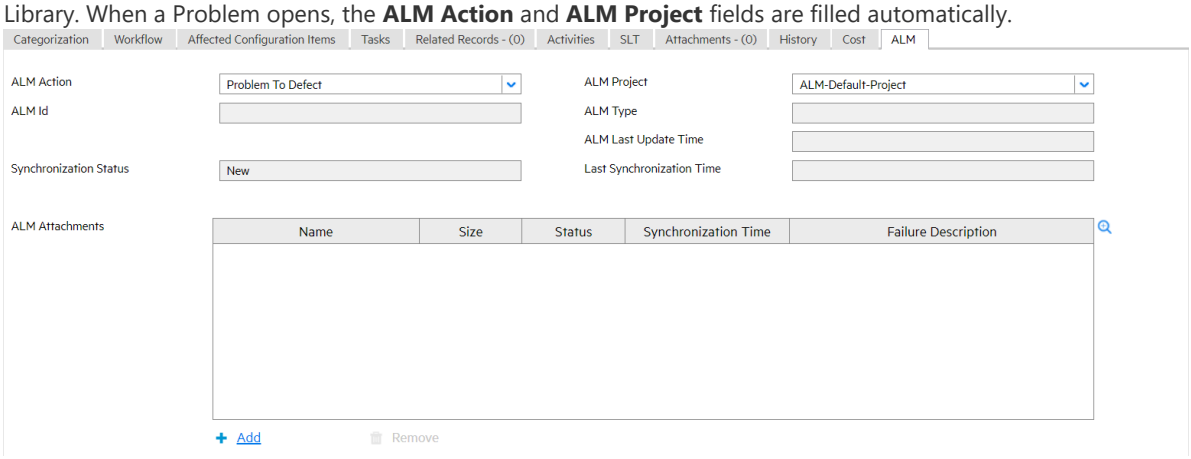
```
/**
* If some changes need to sync to ALM automatically
* Update the condition and return {action, workspace}.
*/
function getAlmConfig(file) {
    var fileName = funcs.filename(file);
    if ('cm3r' == fileName) return getAlmConfigForChange(file);
    if ('rootcause' == fileName) return getAlmConfigForProblem(file);
    return null;
} function getAlmConfigForChange(change) {
    // The condition for Change To Defect, a default one is to contain CI 'CI1000910, adv-
Server-101_tomcat'
    let list = change.assets && change.assets.toArray() || [];
    if (isChangeToDefectEnabled() && list.includes('CI1000910')) {
        let workspace = vars.$G_alm_WS_LIST_CHANGE_TO_DEFECT[0];
        if (workspace) {
            return {
                action: 'changeToDefect',
                workspace
            }
        }
    }     // The condition for Change To Requirement, a default one  Cis to contain CI
'CI1000914, adv-Server-102_tomcat'
    if (isChangeToFeatureEnabled() && list.includes('CI1000914')) {
        let workspace = vars.$G_alm_WS_LIST_CHANGE_TO_FEATURE[0];
        if (workspace) {
            return {
                action: 'changeToFeature',
                workspace
            }
        }
    }     return null;
} function getAlmConfigForProblem(problem) {
    // The condition for Problem To Defect, a default one is to contain CI 'CI1000910, adv-
Server-101_tomcat'
    if (isProblemToDefectEnabled() && problem.logical_name == 'CI1000910') {
        let workspace = vars['$G.alm.WS.LIST.PROBLEM.TO.DEFECT'][0];
        if (workspace) {
            return ({
                action: 'problemToDefect',
                workspace
            })
        }
    }     return null;
}
function isEnabled(name) {
    var list = vars[name];
    return list && list.length() > 0;
} function isChangeToDefectEnabled() {
    return isEnabled('$G.alm.WS.LIST.CHANGE.TO.DEFECT');
} function isChangeToFeatureEnabled() {
    return isEnabled('$G.alm.WS.LIST.CHANGE.TO.FEATURE');
} function isProblemToDefectEnabled() {
    return isEnabled('$G.alm.WS.LIST.PROBLEM.TO.DEFECT');
}
```

In this use case, `adv-Server-101_tomcat` is the Affected Configuration Item, and its CI Identifier is `CI1000910`. Otherwise, refer to *step 2* below to manually select the ALM synchronization action and the ALM project.

### Synchronize Service Manager Problem to ALM defect

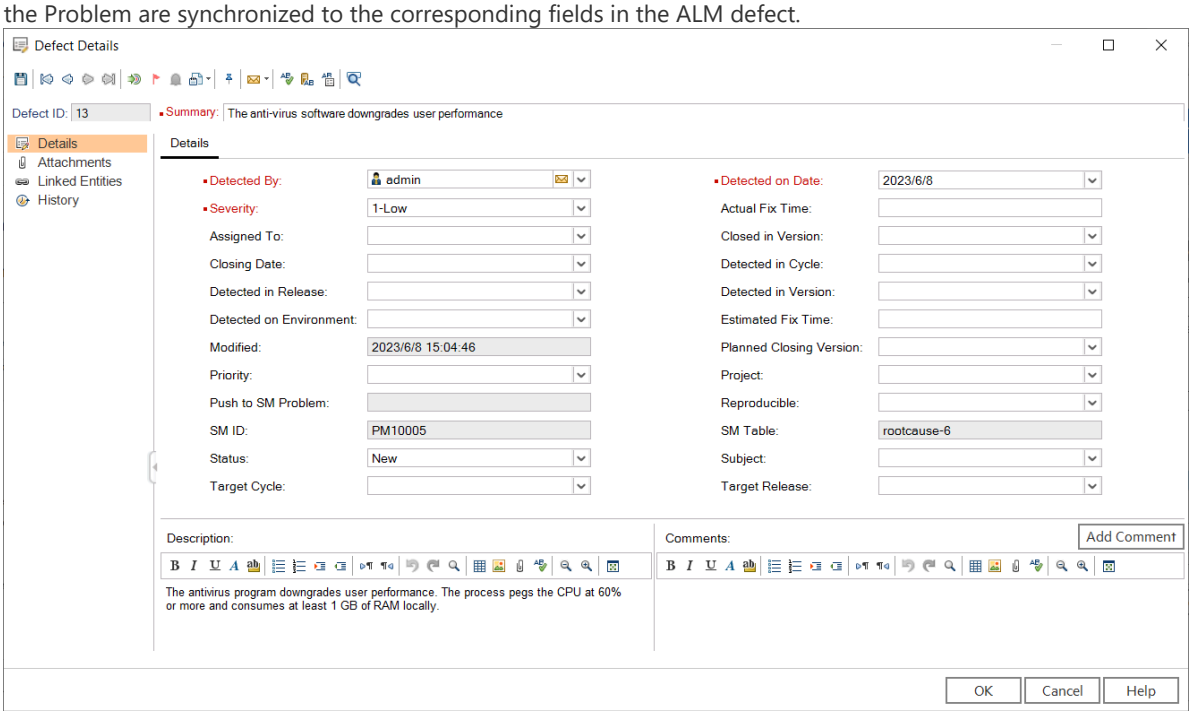
The following steps walk you through this use case:

1. The Problem Manager opens a Problem and then sets **Affected Configuration Item** to adv-Server-101_tomcat.
2. Suppose the **pbm.alm** ruleset exists in any phases of a Problem and the **AlmCustom** JavaScript exists in Script Library. When a Problem opens, the **ALM Action** and **ALM Project** fields are filled automatically.

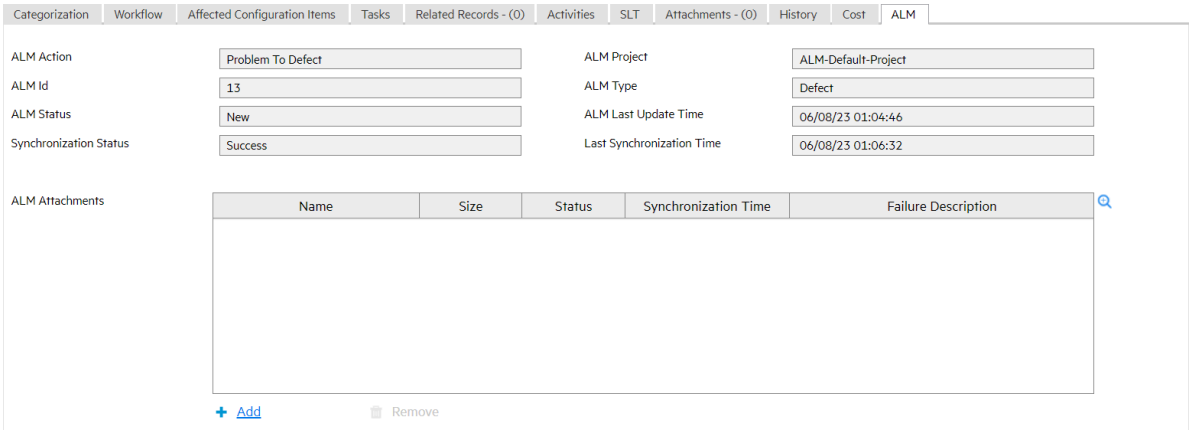



   Otherwise, the Problem Manager selects **Problem To Defect** in the **ALM Action** drop-down list, and then selects the ALM project (project alias) in the **ALM Project** drop-down list.

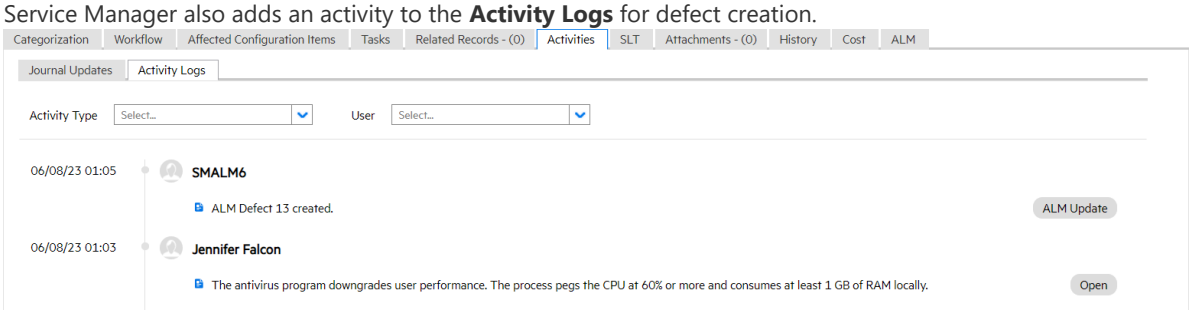
3. After Service Manager runs the integration background process, ALM creates a defect. The title and description of the Problem are synchronized to the corresponding fields in the ALM defect.



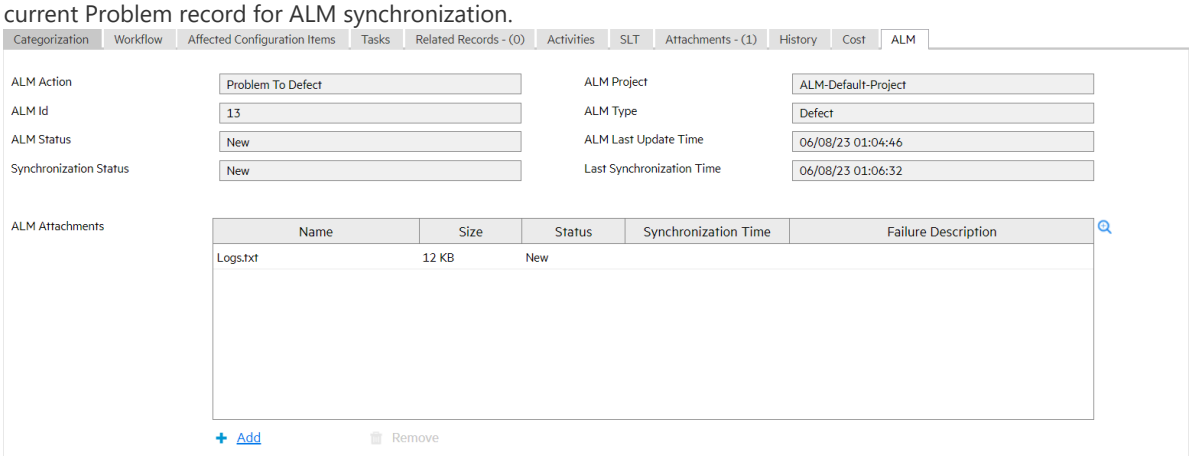

   Meanwhile, the ALM Id, ALM Type, ALM Status, ALM Last Update Time, Synchronization Status, and Last Synchronization Time fields on the **ALM** tab of the Problem in Service Manager have read-only values that are populated automatically.




Service Manager also adds an activity to the **Activity Logs** for defect creation.

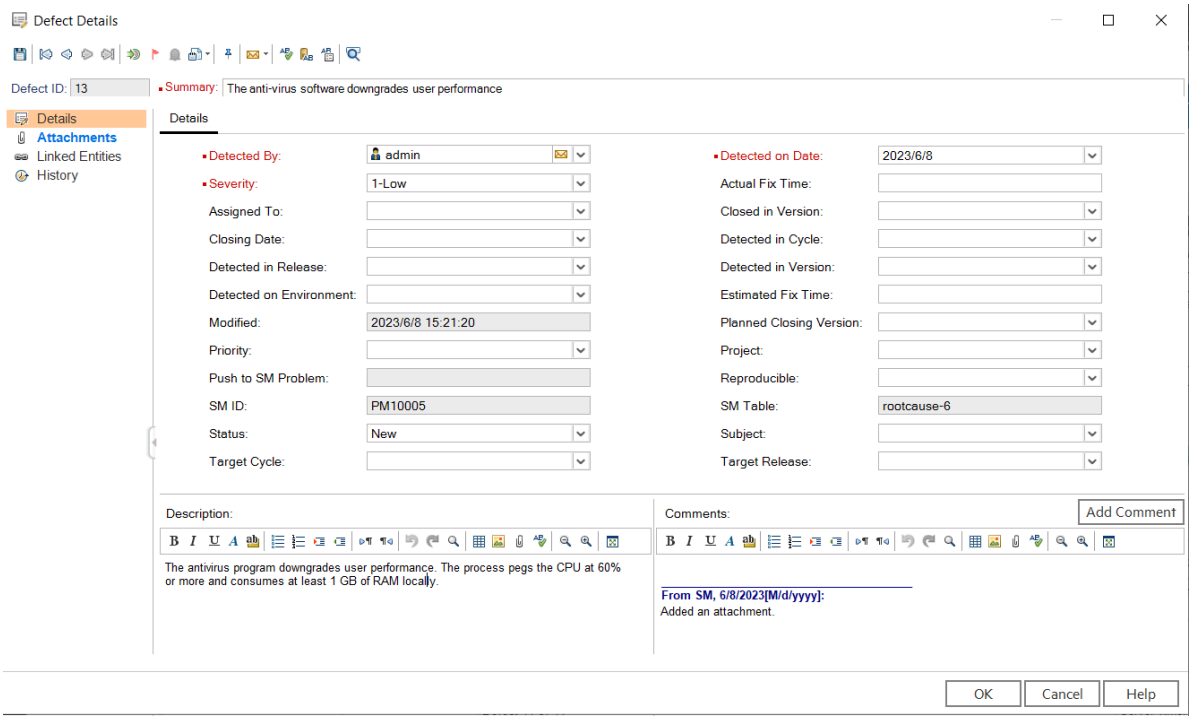


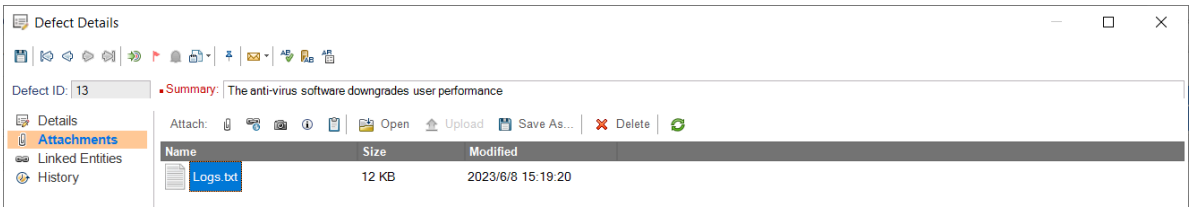

4. The Problem Manager adds an attachment to the Problem and then clicks **Add** to select the attachments from the current Problem record for ALM synchronization.



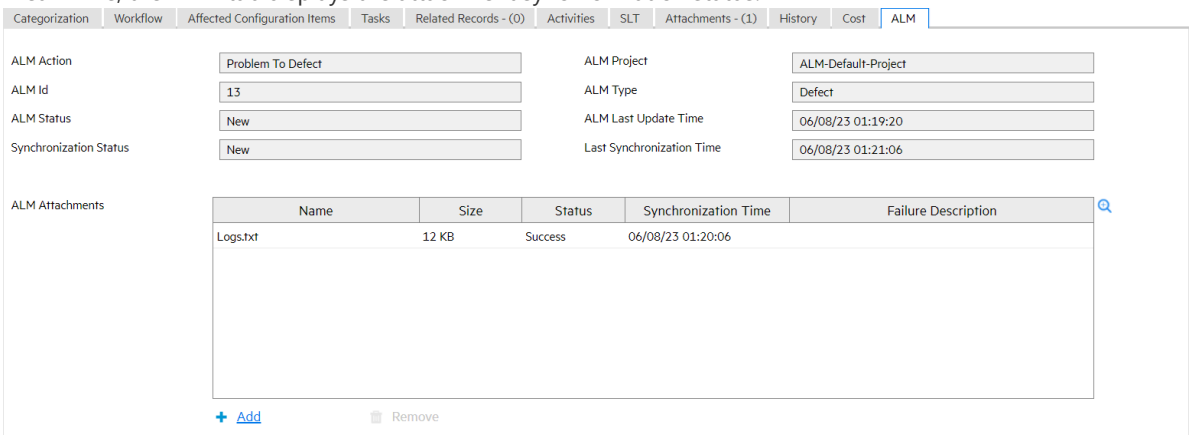

The Problem Manager saves this update. Next, the Problem Manager selects **Update to external service** in the **New Update Type** drop-down list, and then enters some notes in the **New Update** field. The Problem Manager saves this update as well.

5. After synchronization, the Problem Manager's notes in the **New Update** field now appear as a comment in the defect record. The selected attachment in the Problem record now appears as the ALM defect's attachment as well.

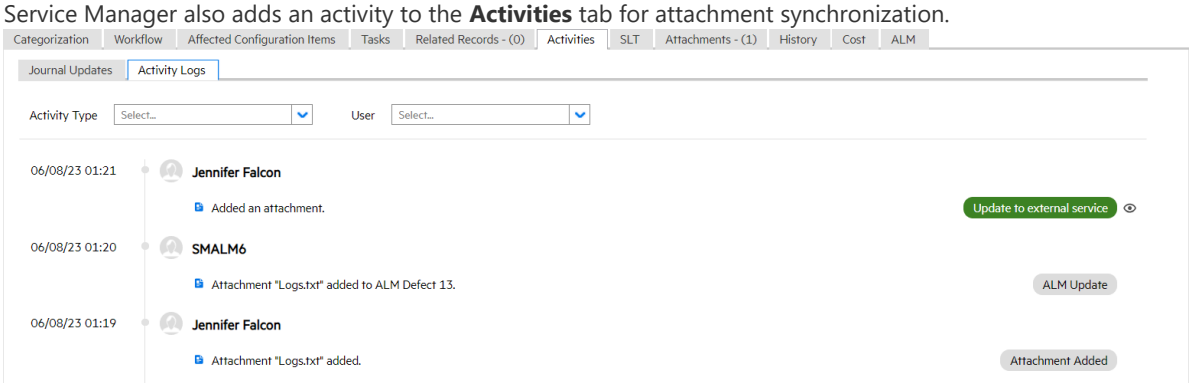


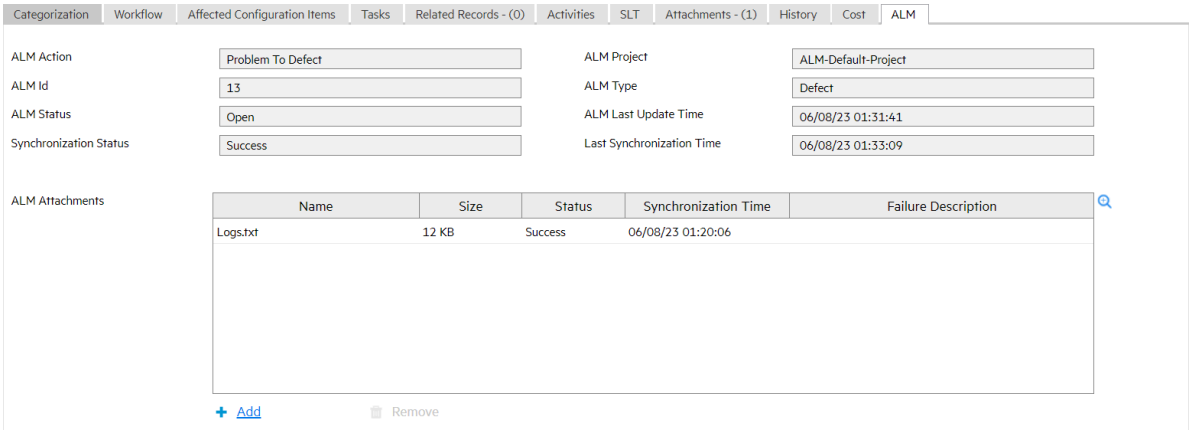

Meanwhile, the **ALM** tab displays the attachment synchronization status.



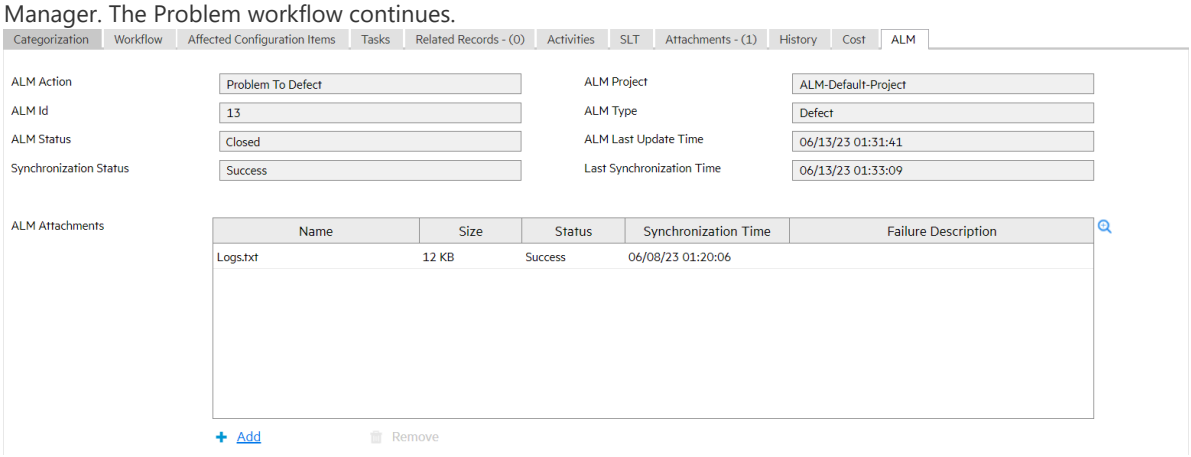

Service Manager also adds an activity to the **Activities** tab for attachment synchronization.




6. As the defect follows the defect management workflows in ALM, the phase updates are synchronized back to Service Manager.




7. Later, when the defect is closed in ALM, the integration synchronizes this status update information to Service Manager. The Problem workflow continues.




### Related topics

ALM

## 10.15. Synchronize ALM defect to Service Manager Problem

This use case describes the following user scenario:

Recently, users reported a lot of not being able to print. After investigation, this issue occurs because the European Printer service is unstable and is always down. This can be a potential problem and it's necessary to investigate its root cause. Since Service Manager has full lifecycle management of problems, the ALM Project Administrator synchronizes the defect to Service Manager as a Problem for further investigation.

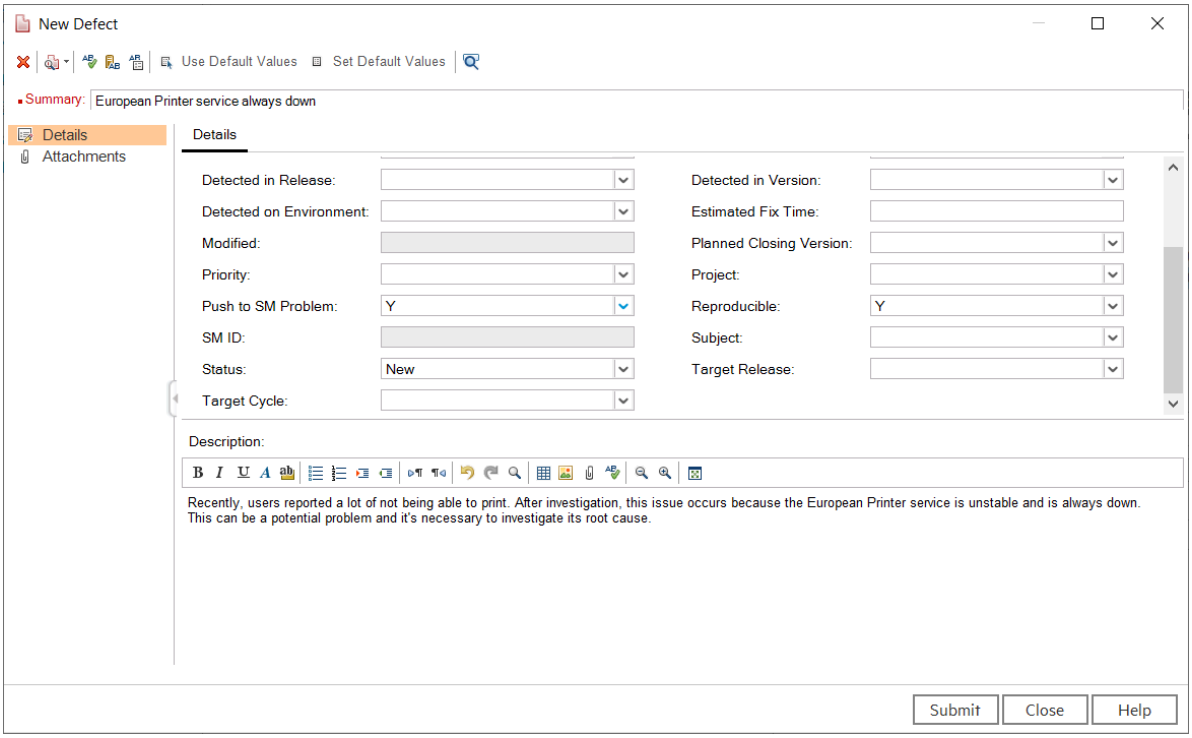
### Prerequisites

As a prerequisite of this use case, the Service Manager system administrator must navigate to **Tailoring** > **Integration Manager** to start the ALM integration instance.

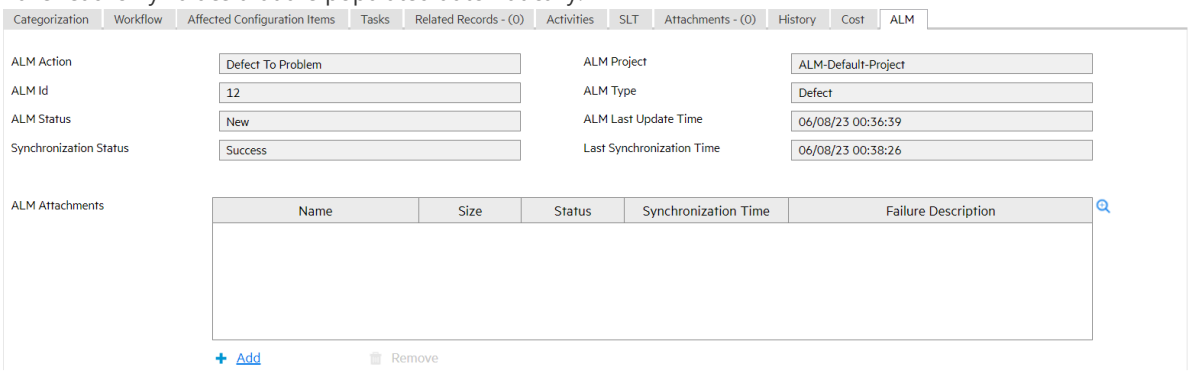
### Synchronize ALM defect to Service Manager Problem

The following steps walk you through this use case:

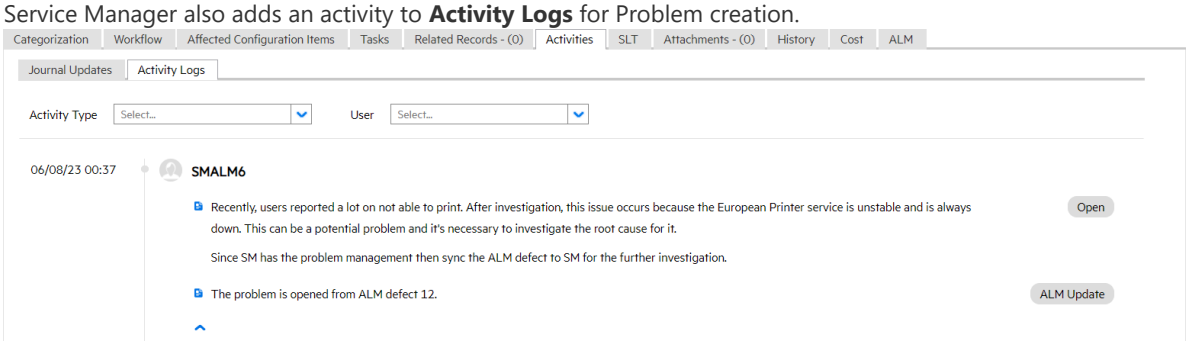
1. The ALM Project Administrator creates a new defect record, fills in the required fields, and then selects **Y** in the **Push to SM Problem** drop-down list.



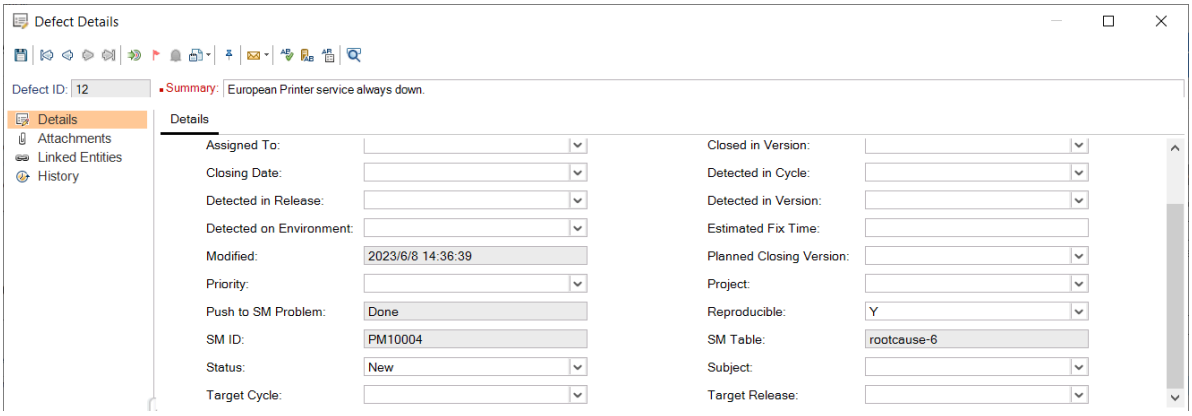

2. The ALM Project Administrator submits the defect and then waits for the background schedule to synchronize the defect record to Service Manager. After synchronization, Service Manager creates a new Problem record.
3. In Service Manager, the Problem Manager clicks the **ALM** tab and then checks that the ALM Action, ALM Project, ALM Id, ALM Type, ALM Status, ALM Last Update Time, Synchronization Status, and Last Synchronization Time fields have read-only values that are populated automatically.



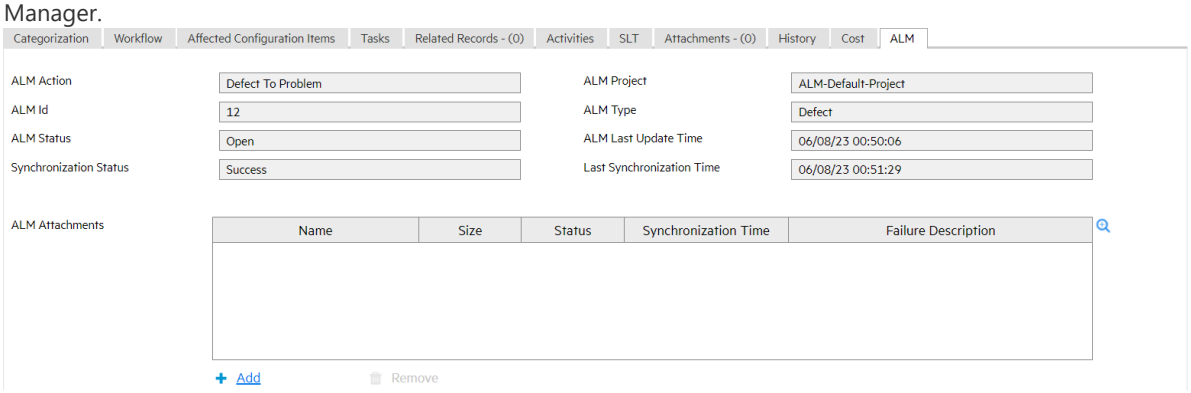

Service Manager also adds an activity to **Activity Logs** for Problem creation.




4. In ALM, the **Push to SM Problem** drop-down list in the defect record becomes read-only and displays **Done**. The SM ID field has a read-only value that's populated automatically.




5. As the defect follows the defect management workflows in ALM, the phase updates are synchronized to Service Manager.




6. Later, after the Problem gets closed in Service Manager, the integration no longer synchronizes the defect's update to Service Manager as the defect management workflow continues.

## 10.16. Synchronize Service Manager Incident to Jira bug

This use case describes the following user scenario:

In Service Manager, an Incident Analyst reviews an Incident where a user reports "Email receiving an error when using a mobile phone as email client". After analysis, the Incident Analyst confirms this issue is a mobility defect, which needs to be fixed by the Mobility Team. Since Jira has a complete bug tracking process in an Agile manner, the Incident Analyst decides to synchronize the Incident to Jira as a bug and hence to fix, test and track it thoroughly.

### Prerequisites

As a prerequisite of this use case, the Service Manager system administrator must complete the following tasks in advance:

1. Navigate to **Tailoring** > **Integration Manager** to start the Jira integration instance.
2. The background synchronization tasks start after you select the Jira synchronization action and the Jira project. There are two ways: manually or automatically.
   To automatically select the Jira synchronization action and the Jira project, check that the **SMJiraCustomerConfig** JavaScript exists in Script Library. Its content may resemble the following.

```
var jiraUtil = lib.SMJiraUtil;
/**
* Update the conditions for automatic synchronization of Change
* vars.$jiraProject: The integration instance ID
* vars.$jiraAction: The type of the Jira issue
* Example: If the assignmentGroup of the Change is "Application",
 *          The current Change record will be synchronized to Jira's SC project as a New Feature
*/
function changeAutoSync(file) {
    var assignmentGroup = file["assign.dept"];
    var projects = jiraUtil.getAllEnabledInstance(funcs.filename(file));
    var isEnable = isEnstanceEable(projects);
    if (isEnable && assignmentGroup == "Application") {
        vars.$jiraProject = projects[0];
        vars.$jiraAction = "New Feature";
    }
}

/**
* Update the conditions for automatic synchronization of Incident
* vars.$jiraProject: The integration instance ID
* vars.$jiraAction: The type of the Jira issue
* Example: If the assignment of the Incident is "Application",
 *          The current Incident record will be synchronized to Jira's SMBT project as a Bug.
*/
function incidentAutoSync(file) {
    var assignment = file["assignment"];
    var projects = jiraUtil.getAllEnabledInstance(funcs.filename(file));
    var isEnable = isEnstanceEable(projects);
    if (isEnable && assignment == "Application") {
        vars.$jiraProject = projects[0];
        vars.$jiraAction = "Bug";
    }
}

/**
* Update the Jira issue types that the Change/Incident module can support.
* vars.$jiraActions: The types of the jira issue, including Bug, New Feature, and Improvement
* vars.$jiraActionsDisPlay: The value displayed in the Jira Action field in SM
*/
function supportJiraTypes(module) {
    if (module == "cm3r") {
        vars.$jiraActions = ["New Feature"];
        vars.$jiraActionsDisplay = ["Change To New Feature"];
    } else if (module == "probsummary") {
        vars.$jiraActions = ["Bug"];
        vars.$jiraActionsDisPlay = ["Incident To Bug"];
    }
}


function isEnstanceEable(projects){
    return projects && projects.length > 0;
}
```

In this use case, `Application` is the Assignment Group of the Incident.
Otherwise, refer to *step 2* below to manually select the Jira synchronization action and the Jira project.

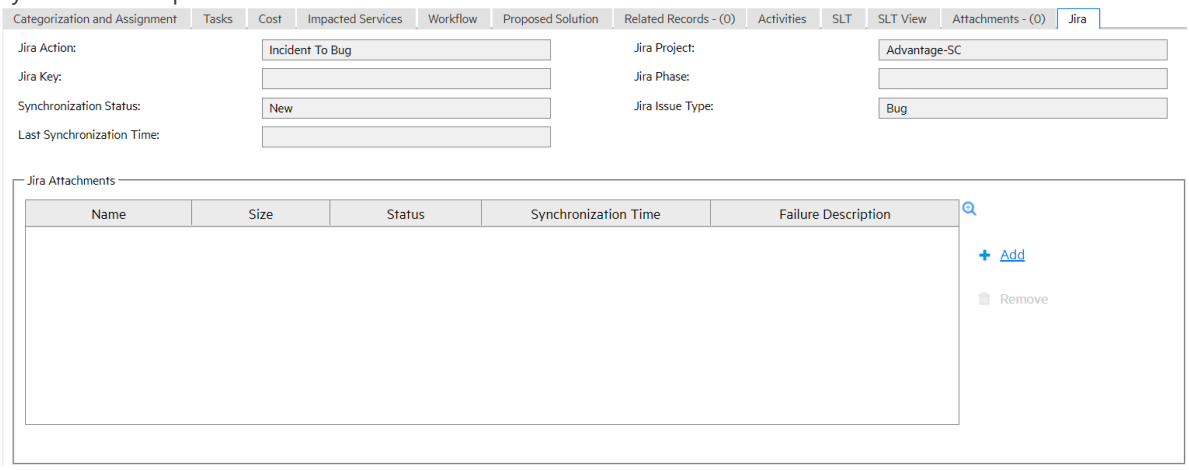
3. Open the workflow of an Incident and click **Edit workflow properties**. Make sure the following rule sets exist on the **Workflow Based Rule Sets** tab.

| Action | Rule set |
|---|---|
| On add | im.jira.assignproject |
| On update | im.jira.assignproject |
| After successful add | im.jira.autoSync |
| After successful update | im.jira.autoSync |

### Synchronize Service Manager Incident to Jira bug

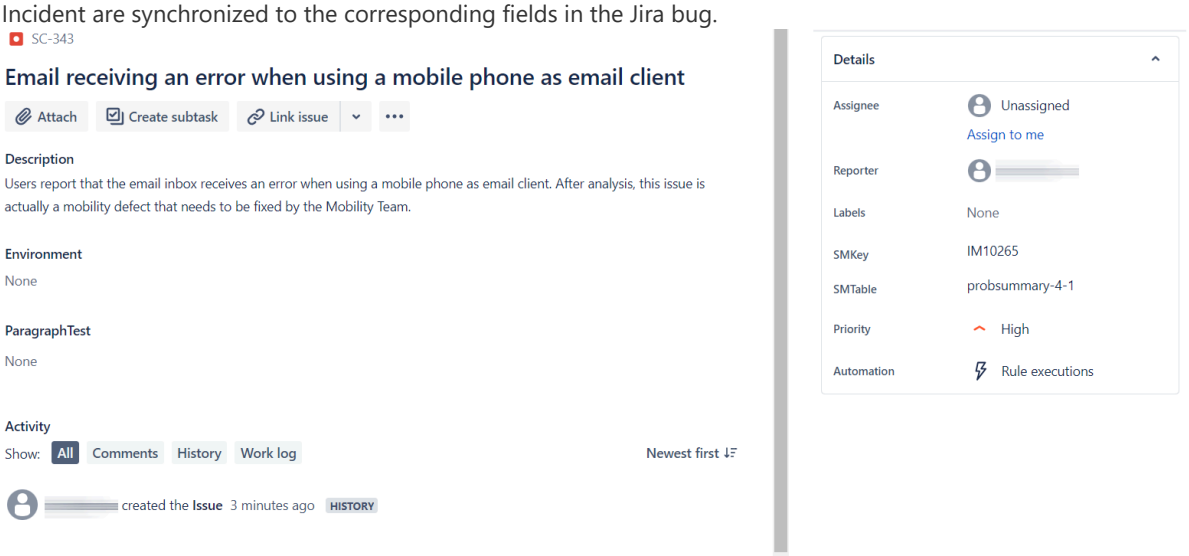
The following steps walk you through this use case:

1. The Incident Analyst opens an Incident record and then sets the **Assignment Group** to Application.
2. The Incident Analyst saves this Incident. The **Jira Action** and **Jira Project** fields are filled automatically and the synchronization process starts.

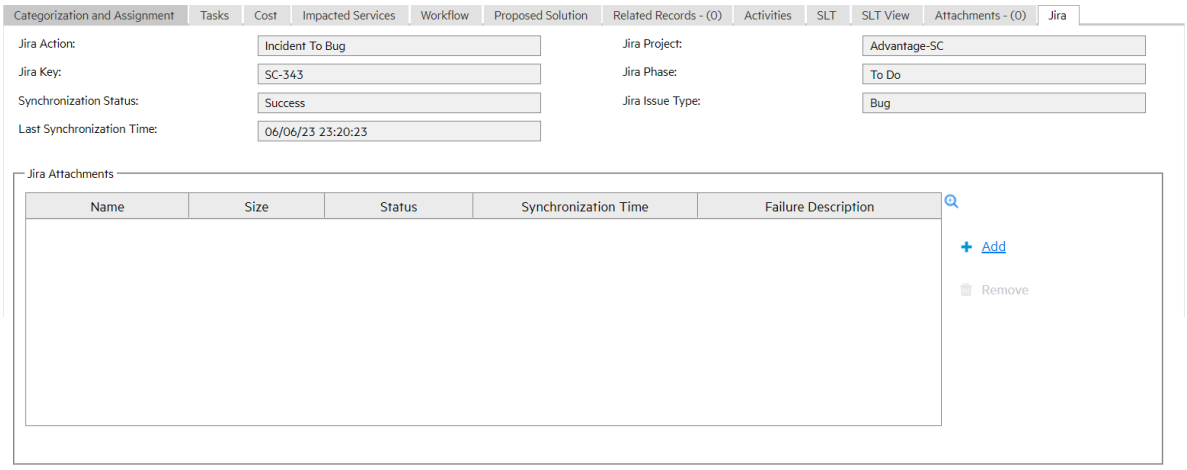



   Otherwise, the Incident Analyst selects **Incident To Bug** in the **Jira Action** drop-down list, selects the Jira project (project alias) in the **Jira Project** drop-down list, and then saves this Incident to commence the synchronization process.

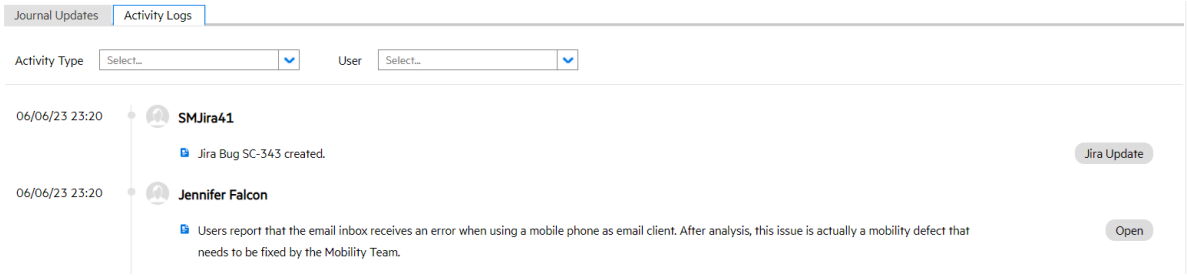
3. After Service Manager runs the integration background process, Jira creates a bug. The title and description of the Incident are synchronized to the corresponding fields in the Jira bug.



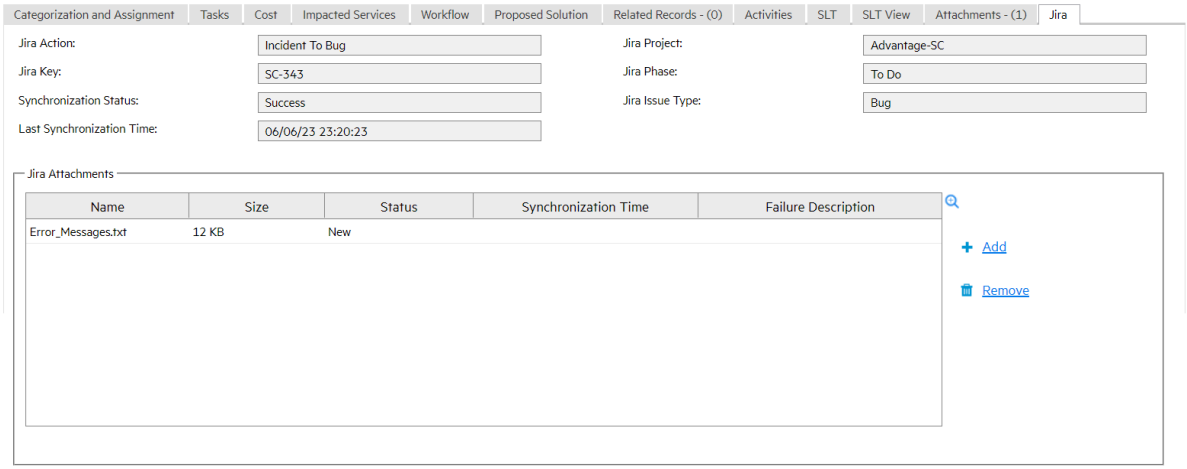

   Meanwhile, the Jira Key, Jira phase, Synchronization Status, Jira Issue Type, and Last Synchronization Time fields on the **Jira** tab of the Incident in Service Manager have read-only values that are populated automatically.



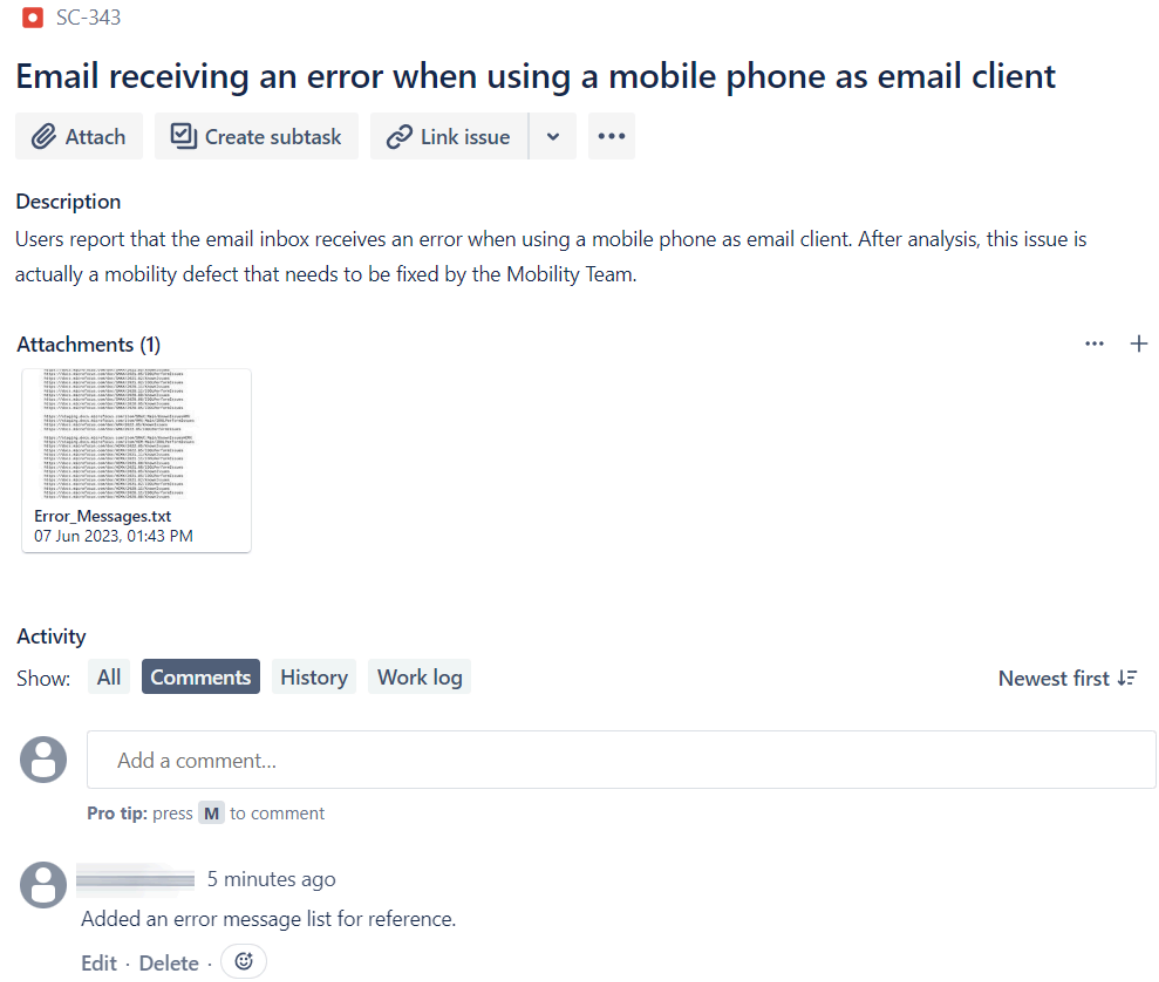

Service Manager also adds an activity to **Activity Logs** for bug creation.



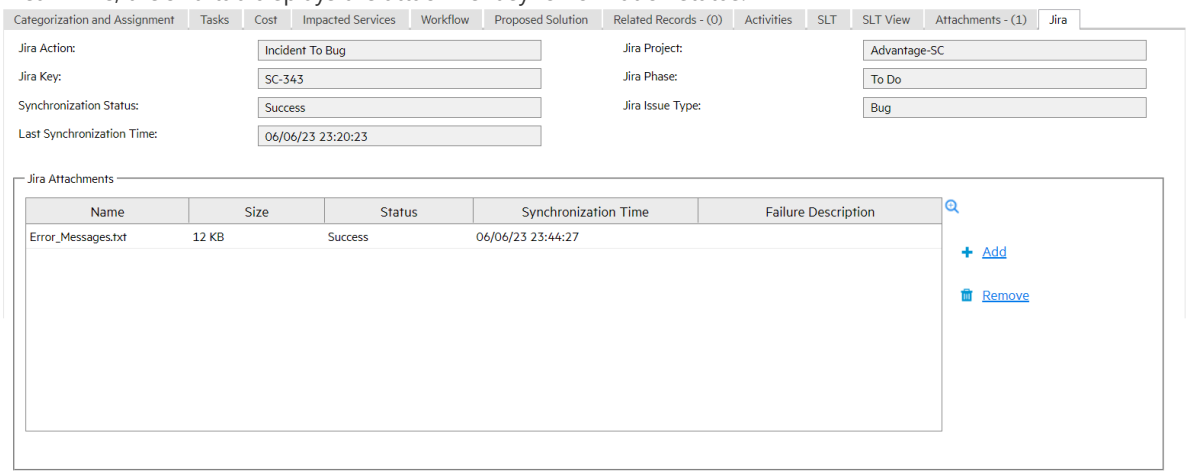

4. The Incident Analyst browses to the **Activities** tab, selects **Update to external service** in the **New Update Type** drop-down list, and then enters some notes in the **Update** field.
   After saving this update, the Incident Analyst adds an attachment to the Incident and clicks **Save**. On the **Jira** tab, the Incident Analyst clicks **Add** to select the attachments from the current Incident record for Jira synchronization.



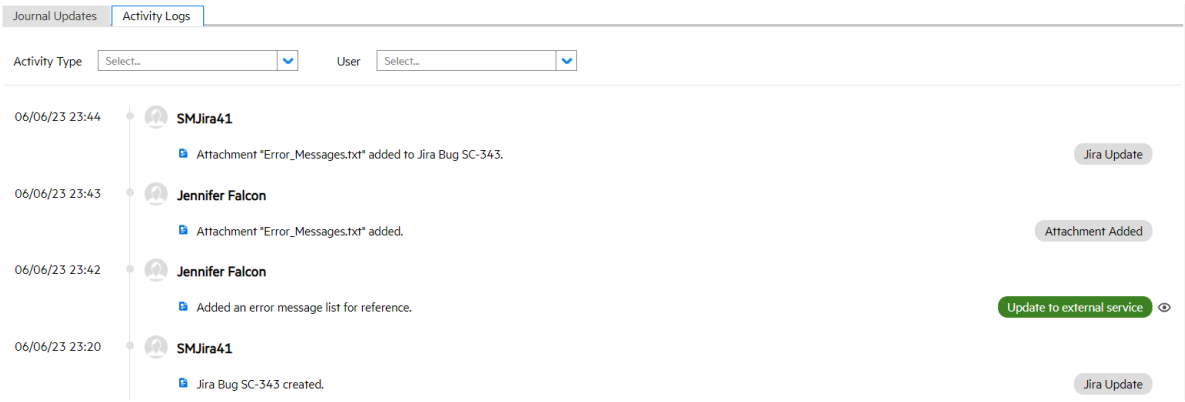

5. After synchronization, the Incident Analyst's notes in the **Update** field now appear as a comment in the bug record. The selected attachment in the Incident record now appears as the Jira bug's attachment as well.



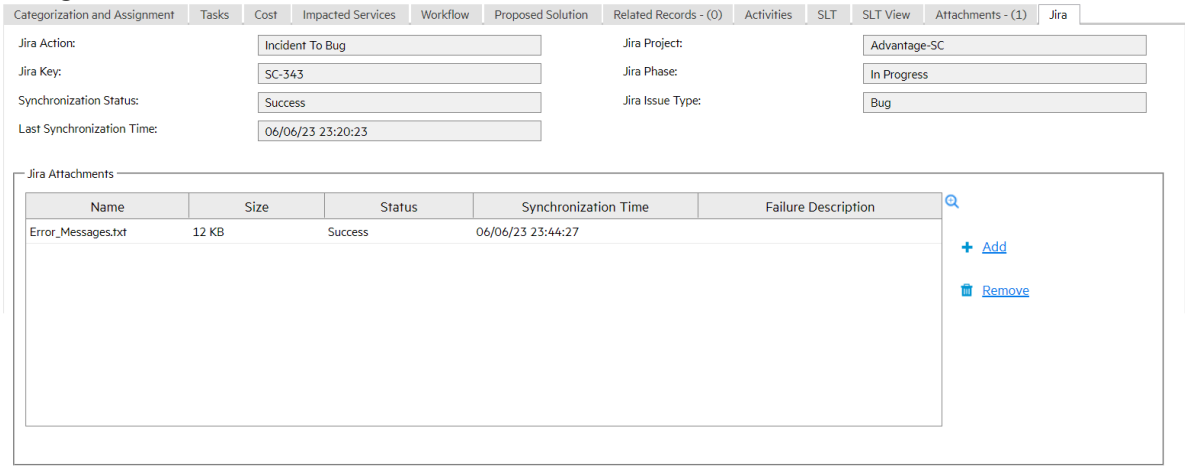

Meanwhile, the **Jira** tab displays the attachment synchronization status.



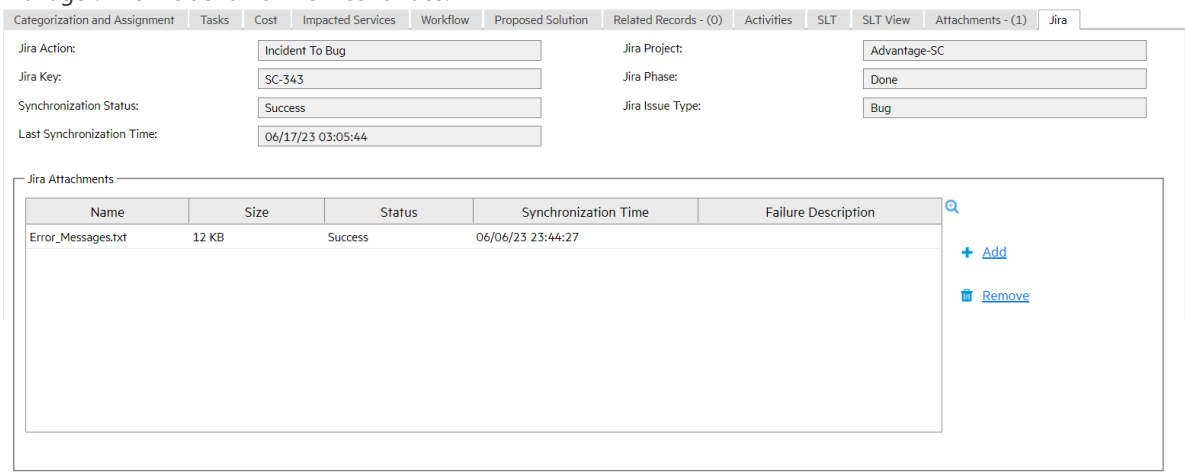

Service Manager also adds an activity to **Activity Logs** for attachment synchronization.




6. As the bug follows the bug management workflows in Jira, the status updates are synchronized back to Service Manager.




7. Later, when a Jira Team Member closes the bug, the integration synchronizes this phase information to Service Manager. The Incident workflow continues.




### Related topics

Atlassian Jira

## 10.17. Synchronize Service Manager Change to Jira new feature

This use case describes the following user scenario:

An online shopping website is exploring foreign markets. To increase its competitiveness, the decision-making department decided to support Paypal as a new payment method. The Product Manager responsible for developing the online platform made a technical evaluation of the decision and proposed to update the payment interface of the platform to be compatible with the new payment method. After assessing the risks and impacts of this change, this proposal is finally

approved. Since Jira has full lifecycle management of features, the Change Manager synchronizes the Change to Jira as a new feature for development, test, and delivery jobs.

**Prerequisites**

As a prerequisite of this use case, the Service Manager system administrator must complete the following tasks in advance:

1. Navigate to **Tailoring** > **Integration Manager** to start the Jira integration instance.
2. The background synchronization tasks start after you select the Jira synchronization action and the Jira project. There are two ways: manually or automatically.
   To automatically select the Jira synchronization action and the Jira project, check that the **SMJiraCustomerConfig** JavaScript exists in Script Library. Its content may resemble the following.

```
var jiraUtil = lib.SMJiraUtil;
/**
* Update the conditions for automatic synchronization of Change
* vars.$jiraProject: The integration instance ID
* vars.$jiraAction: The type of the Jira issue
* Example: If the assignmentGroup of the Change is "Application",
 *          The current Change record will be synchronized to Jira's SC project as a New Feature
*/
function changeAutoSync(file) {
    var assignmentGroup = file["assign.dept"];
    var projects = jiraUtil.getAllEnabledInstance(funcs.filename(file));
    var isEnable = isEnstanceEable(projects);
    if (isEnable && assignmentGroup == "Application") {
        vars.$jiraProject = projects[0];
        vars.$jiraAction = "New Feature";
    }
}

/**
* Update the conditions for automatic synchronization of Incident
* vars.$jiraProject: The integration instance ID
* vars.$jiraAction: The type of the Jira issue
* Example: If the assignment of the Incident is "Application",
 *          The current Incident record will be synchronized to Jira's SMBT project as a Bug.
*/
function incidentAutoSync(file) {
    var assignment = file["assignment"];
    var projects = jiraUtil.getAllEnabledInstance(funcs.filename(file));
    var isEnable = isEnstanceEable(projects);
    if (isEnable && assignment == "Application") {
        vars.$jiraProject = projects[0];
        vars.$jiraAction = "Bug";
    }
}

/**
* Update the Jira issue types that the Change/Incident module can support.
* vars.$jiraActions: The types of the jira issue, including Bug, New Feature, and Improvement
* vars.$jiraActionsDisPlay: The value displayed in the Jira Action field in SM
*/
function supportJiraTypes(module) {
    if (module == "cm3r") {
        vars.$jiraActions = ["New Feature"];
        vars.$jiraActionsDisplay = ["Change To New Feature"];
    } else if (module == "probsummary") {
        vars.$jiraActions = ["Bug"];
        vars.$jiraActionsDisPlay = ["Incident To Bug"];
    }
}


function isEnstanceEable(projects){
    return projects && projects.length > 0;
}
```

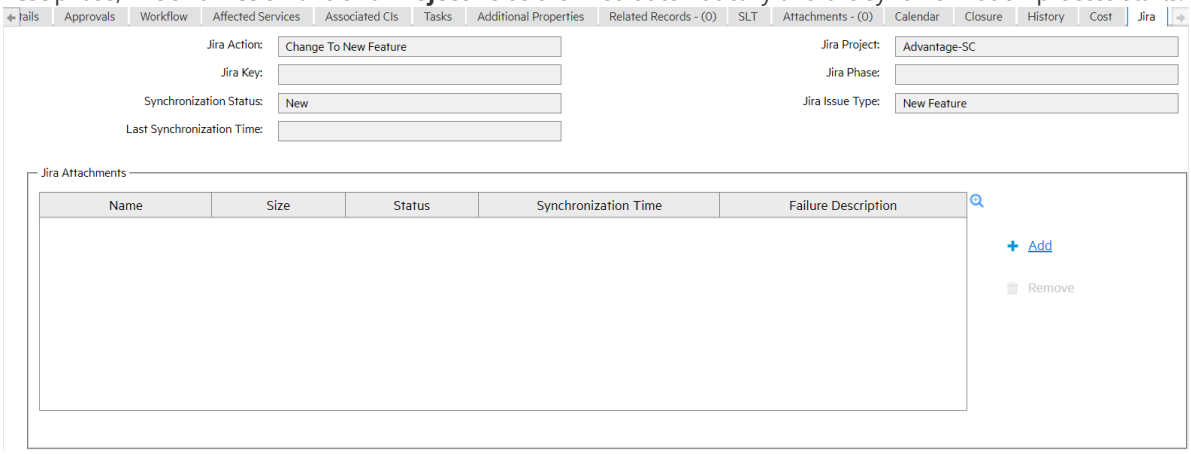
In this use case, `Application` is the Assignment Group of the Change.
Otherwise, refer to *step 2* below to manually select the Jira synchronization action and the Jira project.

3. Open the workflow of a Normal Change and click the **Build And Test** phase. Make sure that the **chm.jira.enter** rule set exists in **On enter** and the **chm.jira.assignproject** rule set exists in **On update**.

### Synchronize Service Manager Change to Jira new feature

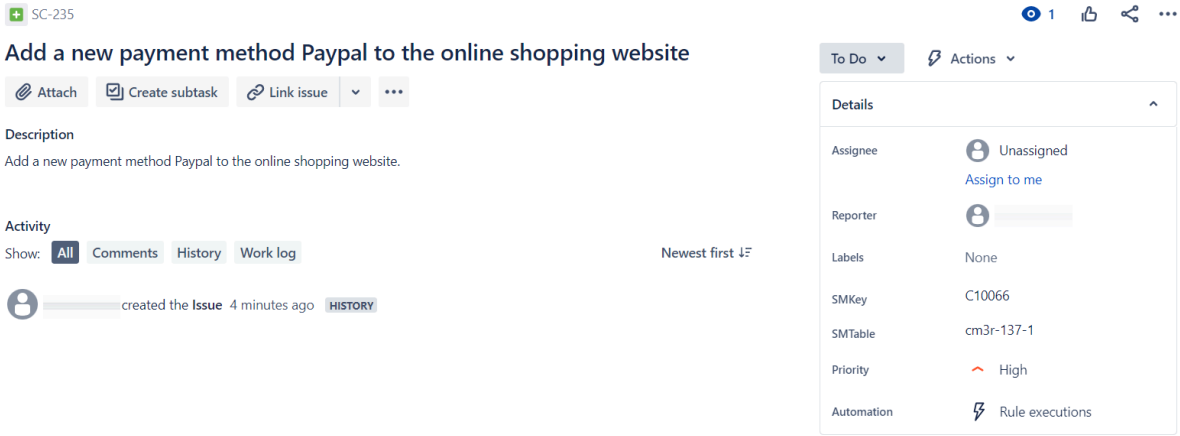
The following steps walk you through this use case:

1. The Change Manager opens a Normal Change and then sets the **Assignment Group** to Application.
2. Suppose the **SMJiraCustomerConfig** JavaScript exists in Script Library. When the Change moves to the **Build and Test** phase, The **Jira Action** and **Jira Project** fields are filled automatically and the synchronization process starts.

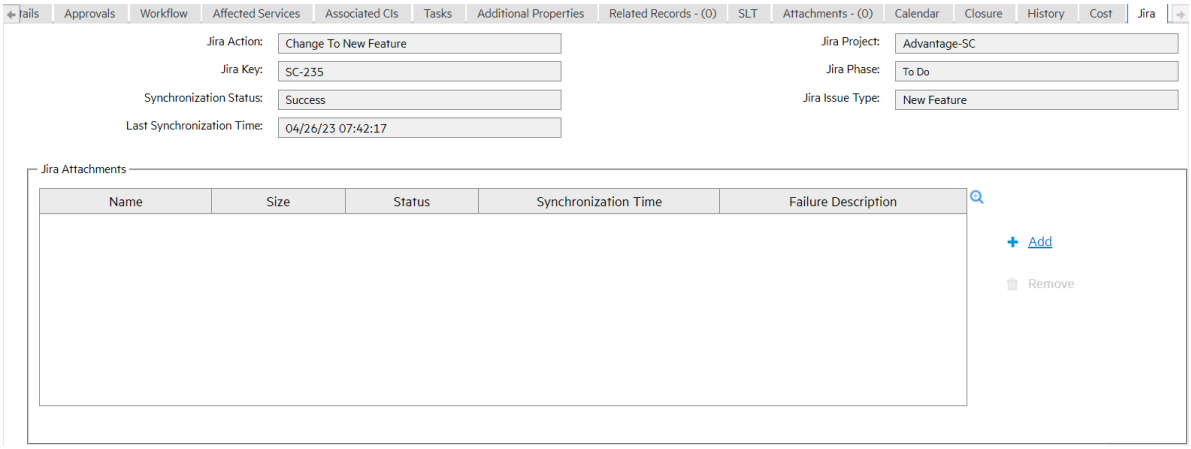



Otherwise, the Change Manager selects **Change To New Feature** in the **Jira Action** drop-down list, selects the Jira project (project alias) in the **Jira Project** drop-down list, and then saves this Change to commence the synchronization process.

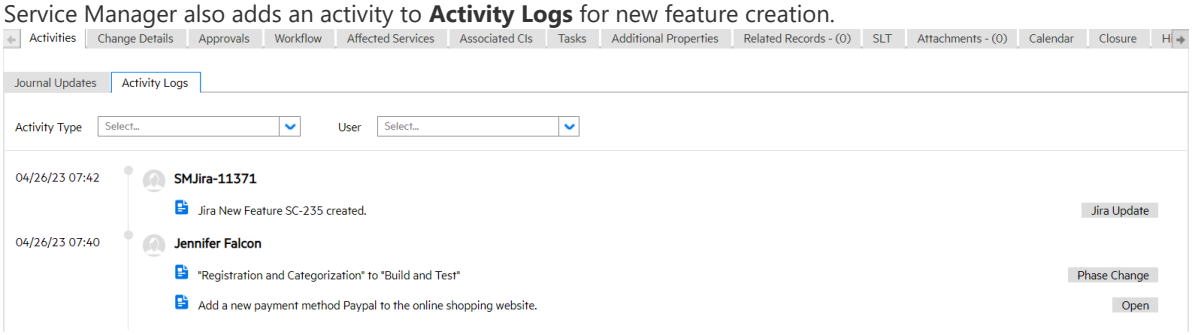
3. After Service Manager runs the integration background process, Jira creates a new feature. The title, description, and urgency of the Change are synchronized to the corresponding fields in the Jira new feature.



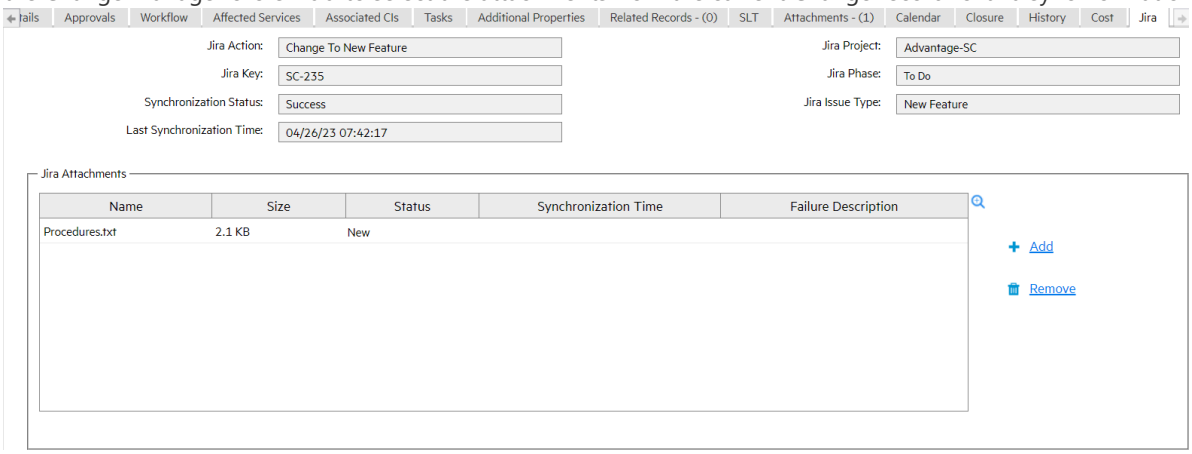

Meanwhile, the Jira Key, Jira phase, Synchronization Status, Jira Issue Type, and Last Synchronization Time fields on the **Jira** tab of the Change in Service Manager have read-only values that are populated automatically.



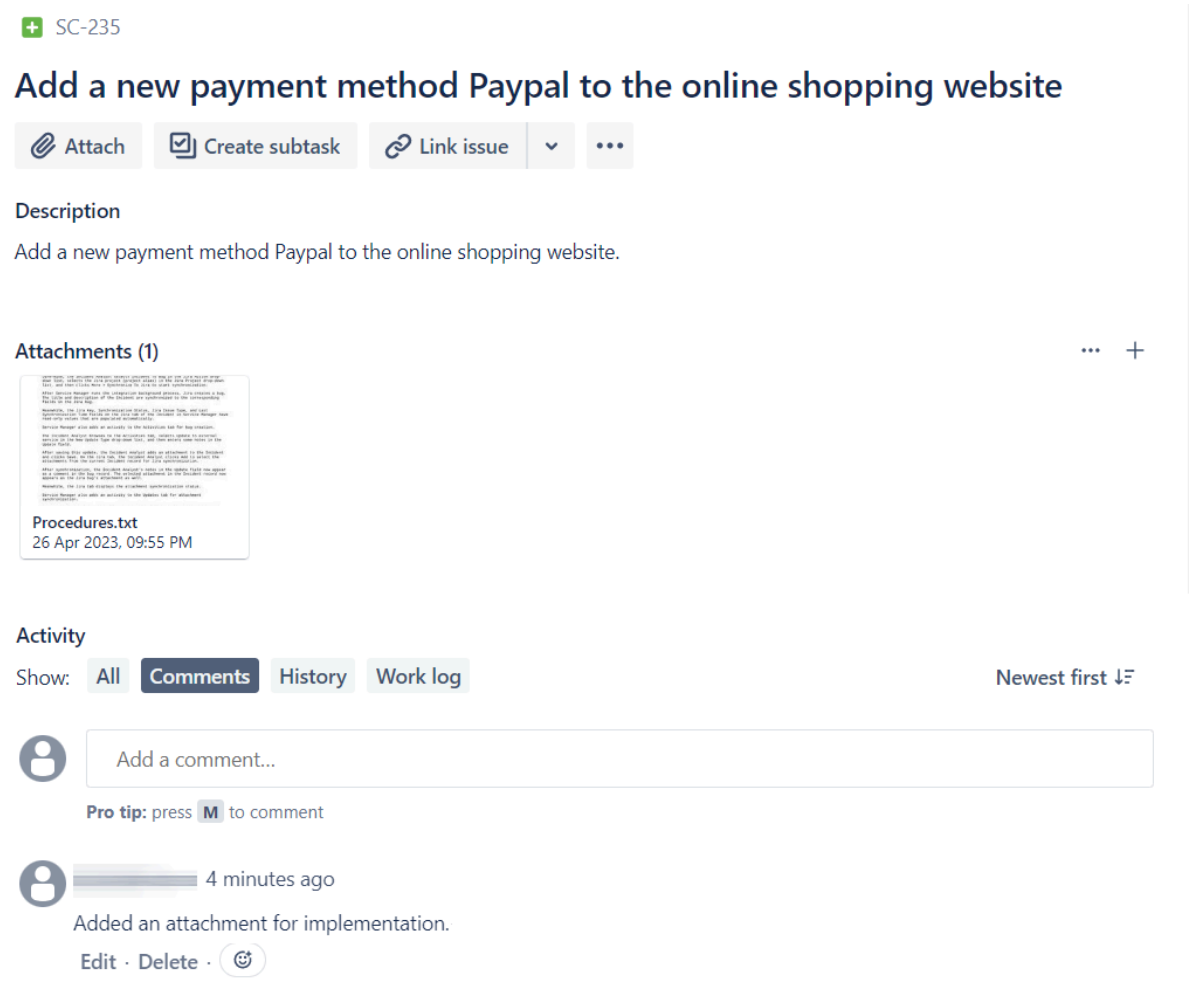

Service Manager also adds an activity to **Activity Logs** for new feature creation.



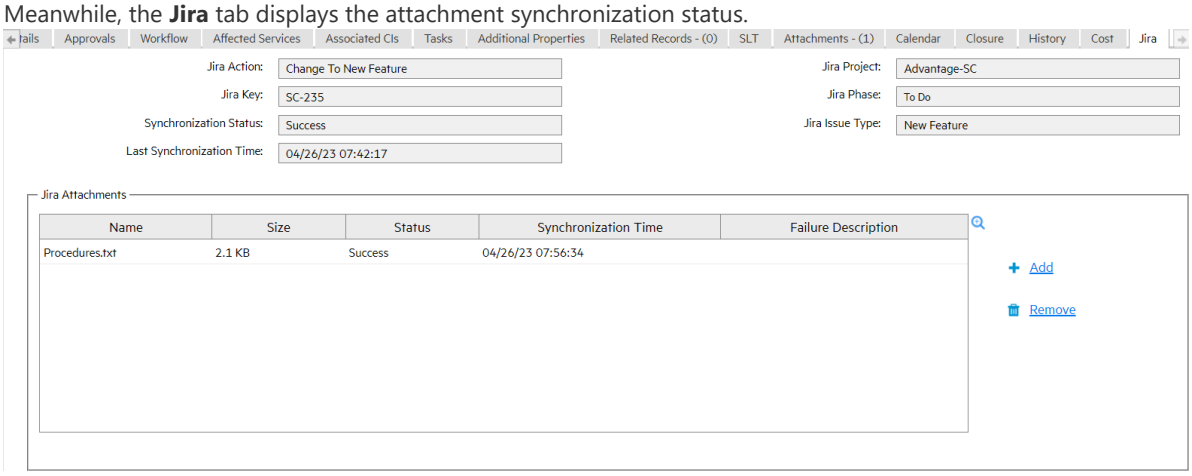

4. The Change Manager browses to the **Journal Updates** tab, selects **Update to external service** in the **New Update Type** drop-down list, and then enters some notes in the **New Update** field.
   After saving this update, the Change Manager adds an attachment to the Change and clicks **Save**. On the **Jira** tab, the Change Manager clicks **Add** to select the attachments from the current Change record for Jira synchronization.



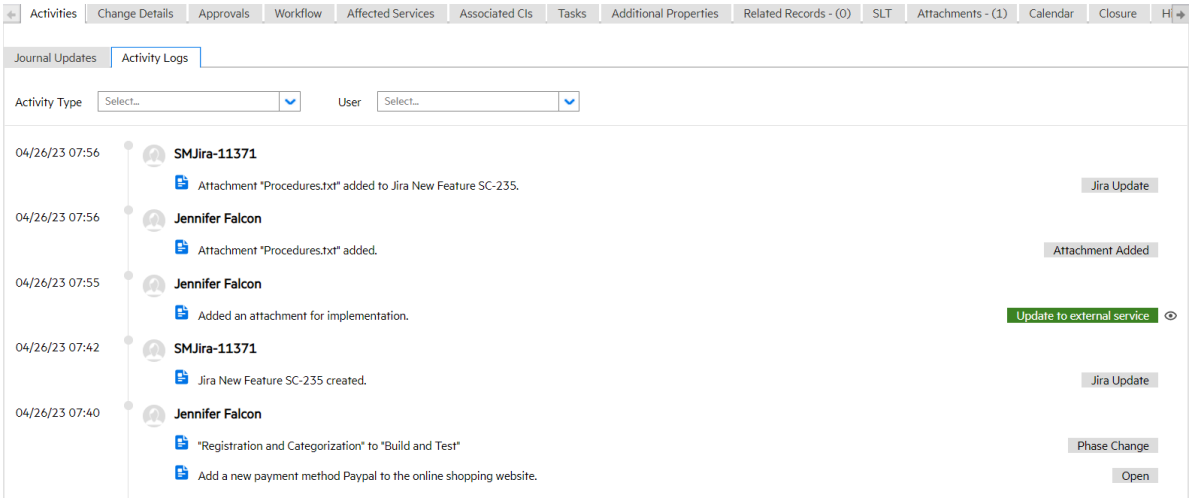

5. After synchronization, the Change Manager's notes in the **New Update** field now appear as a comment in the new feature record. The selected attachment in the Change record now appears as the Jira new feature's attachment as well.



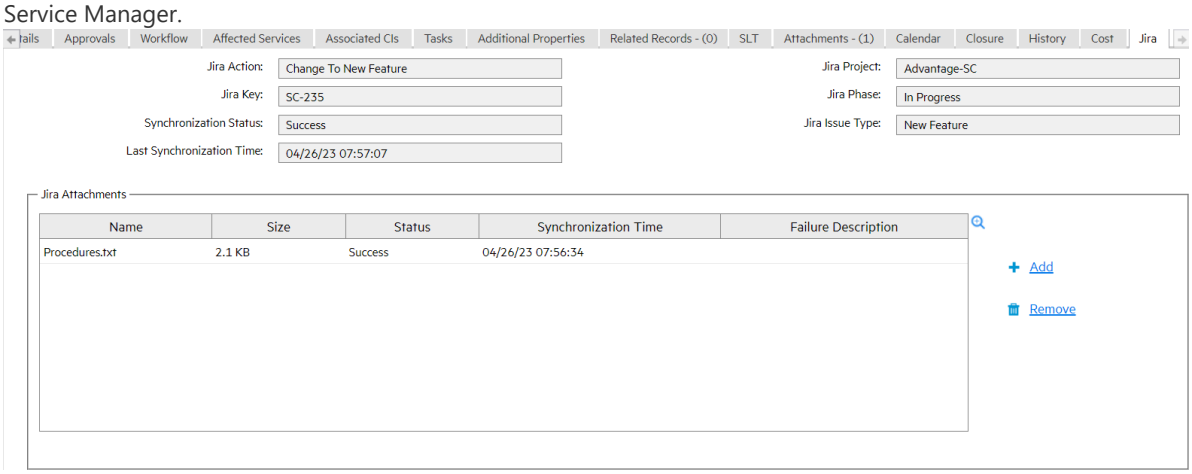

Meanwhile, the **Jira** tab displays the attachment synchronization status.



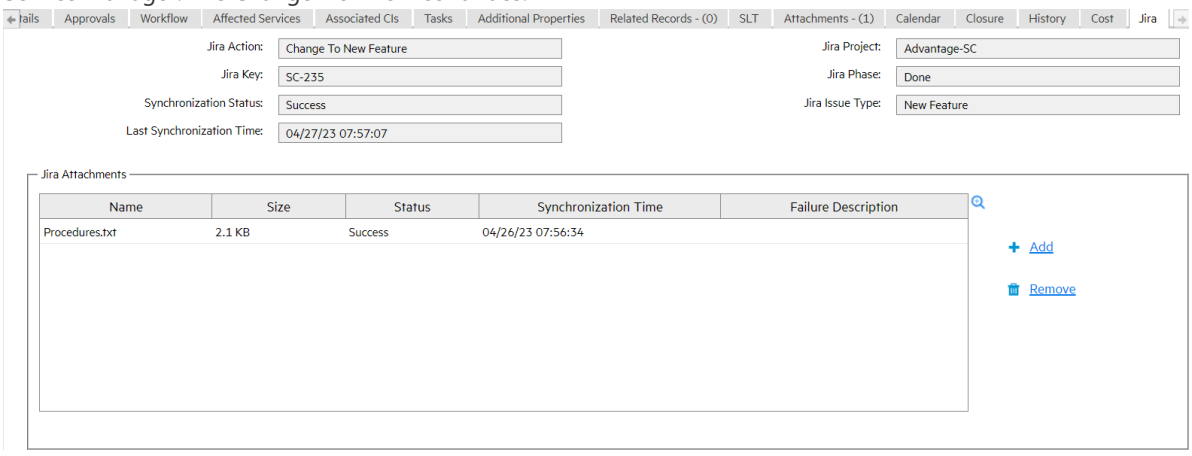

Service Manager also adds an activity to **Activity Logs** for attachment synchronization.




6. As the feature follows the feature management workflows in Jira, the phase updates are synchronized back to Service Manager.




7. Later, when the feature moves to the Done phase in Jira, the integration synchronizes this phase information to Service Manager. The Change workflow continues.




### Related topics

Atlassian Jira

## 11. Glossary

## A

### Activity

An activity is a set of actions defined in the plans or processes that's designed to achieve a particular result. Sets of activities and Procedure ID, such as ST3.2.1.

### Actual CI state

By default, Service Manager only stores and displays the authorized or managed state of configuration items (CIs). However,the actual state of a CI may differ from the authorized state. You can view the actual state of a CI by creating an integration to an Universal CMDB server. The server utilizes a discovery tool that discovers the actual state of a CI, and then Service Manager displays that discovered state in the Configuration Management database format.

### Adaptive learning

Adaptive learning is a feature of Knowledge Management where the system collects words or specific phrases used to search a knowledgebase for documents marked as useful and documents used as solutions for incidents, problems, or interactions. Knowledge Management uses this data to weight the search results hit list so that items with a higher score are given a higher ranking in the list.

### Agreement

An agreement is a document describing a formal understanding between two or more parties. An agreement is not legally binding unless it forms part of a Contract.

### Alert

1. In Change Management, users can define any number of standard or customized alerts, as follows:
    - In which phase the alert will be triggered.
    - Who is notified for each alert.
    - The naming convention used for the alert itself.
2. Alerts support other functions within the system.
3. The Service Manager table that stores alert records.

### Application

Applications are the Service Manager services and their related configuration files. These include:

- Change Management
- Configuration Management
- Knowledge Management
- Incident Management
- Problem Management
- Request Management
- Scheduled Maintenance
- Service Level Management
- Service Desk

### Approval

Official agreement by an approver, or an approval group, to accept the risk, cost, and responsibility to implement a change or complete a task. The System Administrator defines membership in an approval group in each user’s operator record according to their User Role. Change requests and tasks have approval requirements. Approvals give Change Management the ability to stop work and to begin certain work activities.

### Approval delegation

An optional feature that provides the ability to users with approval rights to temporarily delegate their approval authority to another qualified user. Users with the "can delegate" option enabled in their application profiles can delegate some or

all of their approvals by using the Approval Delegation wizard.

**Approval group**

An approval group consists of a list of reviewers and approvers. If you are a member of an approval group, your task is to be a change reviewer or to accept or deny the change. Approval groups set security and approval functions and control movement of change requests from one phase to another.

**Note:** The change sponsor must be a member of one of the groups required to approve a change request.

**Asset**

An asset is a generic term referring to any resource used in service delivery. An asset can be any potential resource or capability that contributes to the delivery of a service. Assets can be any one of the following types: Applications, Financial Capital, Information, Infrastructure, Knowledge Management, Organization, People, Hardware, Software, or Process. Information Technology (IT) assets that need to be individually managed are also Configuration Items (CIs).

**Asset Management Database (AMDB)**

An Asset Management Database (AMDB) is a database containing all data needed to support Availability Management and may be part of the Configuration Management Database. The AMDB is used by Availability Management to gather information needed to support report generation, statistical analysis, and availability forecasting. Implementing an AMDB satisfies control objectives for Information and related Technology (COBIT) governance as well as saves time, staffing requirements, and costs in monitoring areas of COBIT.

**Asset Register**

An Asset Register is a list of Assets or Configuration Items (CIs) that includes their ownership and value. The Asset Register is maintained by Asset Management, which is a subledger of the general ledger. Assets managed in the general ledger should link to CIs managed in the Configuration Management Database (CMDB) to have the ability to track and trace the true status of an item using the fixed asset ID.

**AssetCenter**

Enables IT organizations to manage the physical, financial, and contractual aspects of all IT assets, from request and procurement to retirement and disposal. The ITIL-based processes and best practices integrate AssetCenter with the Universal CMDB.

**Attribute**

In Configuration Management, an attribute is a characteristic or feature of a type of Configuration Item (CI). When setting up Configuration Management, you define general CI attributes in the devtype table and specific CI attributes in device attribute tables.

**Audit tracking**

Once set up, audit tracking audits every field in every record.

**Authorized CI state**

Configuration Management tracks and manages authorized configuration item (CI) states under the Change Management workflows.

**Availability**

The amount of time a Configuration Item (CI) is accessible to perform its role or function in the infrastructure. The usual metric for reporting availability is a percentage of time that the CI is in service during defined service hours. Service Level Management gathers availability data from changes, tasks, and incidents.

**Availability Data**

Availability Data is data measured over a specific period of time based upon the availability of an asset or Configuration Item (CI).

**Availability Plan**

An Availability Plan is a plan to ensure that existing and future Availability Requirements for Information Technology (IT) Services can be provided using effective cost measures.

**Availability Report**

An Availability Report is a document containing availability details of one or more Key Performance Indicators (KPIs) or other important targets that have exceeded their defined thresholds.

## B

**Background agent**

A process that runs in the background on the application server and performs assigned tasks.

**Baseline**

A baseline is the recorded state of an item at any specific point in time, such as a Configuration, a Process, or any other set of data. Examples:

- A baseline can be defined in Continuous Service Improvement to establish a starting point for planning improvements.
- In Capacity Management, you can document performance characteristics during normal operations.
- In Configuration Management, you can enable the Information Technology (IT) infrastructure to automatically restore to a known configuration if a change fails.
- You can also specify a standard configuration for data capture, release, or audit purposes.

**Baseline configuration lifecycle**

Baselines are an optional feature of Configuration Management that allow you to define a set of attributes a configuration item (CI) should have. Typically a baseline only describes the attributes that you expect CIs to share in common, and doesn't include attributes that you expect to vary. A baseline configuration lifecycle indicates at what point the CI is within its lifecycle. A baseline can have the status of Analysis, Authorized, Design, Develop, Retired, or Retiring. Only baselines in the Authorized state can be newly-assigned to a CI.

**Baseline exception reporting**

Baseline records are separate from the configuration item (CI) records they manage. If any baseline attribute differs from the attribute in the authorized state of a CI using this baseline, an exception occurs. However, you may authorize an exception to the baseline, rather than create a new baseline. For example, you might have a baseline requiring that all application servers have 8 GB of RAM, but you have one application server, the Web server, that needs to have 16 GB of RAM. The baseline exception reporting will report this instance within your baseline.

**BLOB/Image**

BLOB is a data type for binary large objects in a relational database management system (RDBMS). Certain RDBMS vendors such as Oracle, call this binary data type BLOB. Other RDBMS vendors like Microsoft SQL Server, call this binary data type image.

**Breach**

A failure to meet the objectives specified in a Service Level Agreement.

**Bundle**

A bundle is a group of items available within a Service Catalog offering.

**Business services**

An Information Technology (IT) Service that directly supports a specific business process, as opposed to an Infrastructure Service which is used internally by the IT Service Provider and is not visible to the Business. Services can be grouped to Business Service groups that include Configuration Items (CIs), documents, resources, or any other components that

comprise a defined business service. The services can then be delivered to Business Customers by Business Units, such as the delivery of financial services to Customers of a bank, or products to Customers in a retail store. Successful delivery of Business Services often depends on one or more IT Services.

## C

**Calendar**

The Service Manager utility supporting the definitions of working hours for different work groups.

- The duty hours (caldutyhours) table defines the normal daily working hours for a particular work or assignment group.
- The holidays (calholiday) table defines the starting and ending dates for valid company holidays.

**Note:** Be aware of time zone differences when executing events within Service Manager. The calendar feature is based on a single time zone and will not compensate for time zone differences.

**Call**

A call is a telephone call to the Service Desk from a User. As a result of a call, an interaction is opened by the Service Desk and tracked through resolution.

**Capability word**

A value in the Execute Capabilities section of an operator record that grants access to specific applications and utilities in Service Manager.

**Capacity Management Database (CDB)**

A Capacity Management Database (CDB) consists of one or more databases that serve as a repository for holding multiple types of data used by its subprocesses. The core infrastructure record, the Configuration Management Database (CMDB), provides information about the state of Configuration Items (CIs) and the relationships between CIs. By providing performance and resource information about the (CIs), a CDB works with the CMDB to provide a complete and reliable view of the IT infrastructure.

**Capital item**

A capital item is an asset or fixed asset, also referred to as a Configuration Item (CI). The CI is registered and managed in the general ledger for depreciation and investment. The item is managed by the organization from the perspective of total value. A capital item has attributes such as:

- Book value depreciation schema
- Depreciation start date
- Accumulated depreciation
- Residual value

**Case-insensitive system**

Case-insensitive systems refer to RDBMS systems that are based on a case-insensitive naming convention, which identifies object names regardless of their cases (capitalizations).

**Catalog categories**

The catalog categories are a hierarchical grouping of requests. Categories can contain the following:

- Subcategories
- Catalog items
- Bundles, such as IT bundles
- HR services
- Facilities services
- Telecom services

**Catalog item**

A catalog item is an entry in the Service Catalog that's associated with a specific product or fulfillment service that needs to be delivered.

**Category**

In Change Management, the category classifies the Configuration Item (CI), change documents, service desk interactions, incidents, requests, and tasks. It determines the data to be collected for a particular change or task, the forms presented to the user for data entry, the approvals needed, and the intervals at which alerts are sent. Change Management includes a series of default categories. Administrators can also create new categories. Each change record must have a category. Each category has at least one phase. Examples of considerations for a change record are as follows:

1. The record contains the details of the change.
2. Each change record documents the lifecycle of a single change.
3. A change record is created for every Request for Change (RFC) received, even those that are subsequently rejected.
4. Each change record references the CIs affected by the Change.
5. The change records are stored in the Configuration Management System (CMS).

In a Service Catalog context, a category describes similar Information Technology (IT) products or services that users can select from an online catalog. The Service Manager Service Catalog feature includes out-of-box categories and subcategories. Administrators can optionally create new categories to meet their operational needs. In Knowledge Management, knowledge documents are assigned to a category or subcategory when the document is created or approved. Knowledge Management administrators can optionally create new categories to meet their operational needs.

**Cause code**

An optional code that links an incident record to a probable cause record. Cause codes allow incident records to be more easily categorized and assigned. By using a standard series of cause codes, users can simplify reporting and track incidents that have a common cause. The cause code field links a record to a related probable cause record.

**Change**

A change is the addition, modification or elimination of an authorized, planned, or supporting service or component and the related documentation. Changes are controlled by the Change Management application and, unlike requests, they affect multiple users or business units. Examples include changes to the network environment, facilities, or telephony. A change has a life cycle that contains approvals, alerts, tasks, phases, and closure. When you submit a change, the Change Management application assigns a unique change record number.

**Change Advisory Board (CAB)**

A group of people that assists the Change Manager with assessing, prioritizing, and scheduling changes. This board is usually made up of representatives from the Information Technology Service Provider (ISP); the business; and third parties, such as Suppliers.

**Change category**

A change category or subcategory is assigned to a change to define or classify the change type. The change category determines the phases of the change and the overall workflow.

**Change initiator**

A Change Management User Role. The person who starts the process of a change.

**Change Management**

1. The ITIL term for the process of controlling changes to an organization's infrastructure to minimize the disruption of services.
2. The Service Manager application used to request, manage, approve, and control changes to an organization's infrastructure.

A change generally affects multiple users or business units. Examples include changes to the network environment, facilities, or telephony. For user requests for products and services, refer to Request Management.

**Change model**

A change model is a method of controlling the category of a change. A change model defines the steps to follow for a change in a specific category. Change models can be simple and require no approvals, such as a password reset, or can be complex and include multiple steps that require approval, such as a major software release or upgrade.

**Change owner**

The person who plans and monitors the activities of a change and gives approval to proceed.

**Change phase**

A change phase is a phase in the life cycle of a change. The life cycle of a change is divided into phases determined by the change category assigned to that phase. Phases are defined with the additional constraint of sequential order, for example, each interaction is executed with a series of milestone dates spread over time that assume minimal or no overlap between activities.

**Change priority**

The urgency of the change, based on business need.

**Change sponsor**

A Change Management User Role. The person who authorizes the change from the business perspective. If the change sponsor doesn't have access to Service Manager, the Change Administrator (CA) must ensure that authorization is obtained from the change sponsor. The CA then approves the change on behalf of the change sponsor on the Service Manager system.

**Change task**

A change task is a unit of work assigned to a person or a group. For example, one or more change tasks are created to assign test and deployment tasks to Information Technology (IT) staff, such as development, Quality Assurance, and Documentation.

**CI group**

A Configuration Item (CI) group is a logical group of related configuration items. A CI group is used to manage a configuration as one item in a group, such as a software release or upgrade. Individual CIs can be grouped into one or more logical groups or clusters with the same characteristics. The group is then used to identify a set of CIs associated with a release and identify any items with problems or items with supported contracts, such as a maintenance contract that covers all printers in one or more locations or a software license for multiple workstations.

**CI type**

A CI type is a category used to classify Configuration Items (CIs). The CI type identifies the required attributes and relationships that comprise a configuration record. Common CI Types include: hardware, software, documents, and users.

**Closure**

The point in time when a service desk interaction, incident, or problem has been resolved.

- Service desk interactions are closed when service is no longer required because the issue is resolved or the interaction escalates to an incident.
- Incidents are closed when the reported issue is corrected, typically when service is restored.
- Problems are closed at the point where the underlying issues have been corrected.

**Collection**

In Knowledge Management, this is a term sometimes used to mean the same thing as knowledgebase, particularly in relation to the search engine used by Knowledge Management.

### Command box

The Service Manager command box refers to the command-line box, which provides quick access to RAD applications. To display the command box, click the command icon on the top-right corner of the Service Manager client.

> **Note**
>
> **Note:** Clicking the command icon only expands and hides the command box. To run a command, type the command in the command box, and then press Enter.

### Configuration Item (CI)

A Configuration Item (CI) is an ITIL term used for a component in the infrastructure or item associated with the infrastructure of a company. CIs are controlled by the Configuration Management application. A CI can be any piece of equipment or component that's tracked through a device record in Configuration Management. CIs vary widely in complexity, size, and type. For example, any object or process within the Service Desk application is recorded as a CI. This can include an entire system with all hardware, software, and documentation; a single software application; a software license; or a hardware device. Service Manager stores CI records in the device table.

### Configuration Management

1. The ITIL term for the process of identifying and defining Configuration Items (CIs), recording and reporting the status of CIs, and verifying their completeness and correctness.
2. The Service Manager application used to manage CIs and the relationships between them.

### Configuration Management Database (CMDB)

The acronym for Configuration Management Database (CMDB). Service Manager refers to CMDB in the Information Technology Infrastructure Library (ITIL) context. In this context, a CMDB represents the authorized configuration of the significant components of the IT environment. A key goal of a CMDB is to help an organization understand the relationships between these components and track their configuration. The CMDB is a fundamental component of the ITIL framework's Configuration Management process.

### Configuration Management System (CMS)

A Configuration Management System (CMS) is a software tool that provides support for Configuration, Change and Release Management. Information Technology Infrastructure Library (ITIL) v3 emphasizes how organizations manage IT services within a company with a focus on service lifecycle rather than in process silos. As a result, the resulting work service-centric instead of process-centric. Effective implementation requires strong integration of Configuration, Change, and Release Management.

### Conflict resolution

Conflicts may occur when you are applying the new Service Manager changes to your existing Service Manager installation, which is likely to include customizations, tailoring, and patches. The conflict resolution phase of the upgrade is to reconcile the differences between the customized objects and the objects provided by the upgrade package.

### Connector

A connector defines the integration between a Service Catalog item and the back end product or fulfillment Service Catalog. For example, a connector to Change Management allows the Service Catalog to include items that are fulfilled by the creation of a change request.

### Control Objectives for Information and related Technology (COBIT)

The acronym for Control Objectives for Information and related Technology (COBIT). COBIT is a set of best practices for IT management created by the Information Systems Audit and Control Association (ISACA) and the IT Governance Institute (ITGI). The framework provides managers, auditors, and IT users with a set of generally accepted measures, indicators, processes, and best practices to assist them in maximizing the benefits and developing IT governance and control in a company.

### Custom upgrade

The custom upgrade is the upgrade build that's created on the development system after applying the upgrade files and resolving conflicts. The custom upgrade is eventually exported from the development system and applied on the production system. A custom upgrade consists of new Service Manager application files that replaced old application files, customized application files that you retained, and merged files that combine prior customization with new application functionality.

### Customer SLA

An agreement with an individual customer group that covers all the services they use. For example, agreements may be reached with an organization's Finance Department covering the Finance System, the Accounting System, the Payroll System, the Billing System, the Procurement System and any other IT systems that they use. In Service Manager, it refers to the one SLA that's associated with an application record based on the contract.

## D

### Data policy

Data Policy enables System Administrators to apply default values, mandatory fields, and lookup validations to a specific table. These policies, once set, are enforced across the entire system, regardless of what form is being used to display the data.

### Database definition language (DDL)

In restricted-access RDBMS environments, Service Manager can create database definition language (DDL) describing the changes proposed by your database dictionary records. The RDBMS administrator can then create the necessary tables and columns for Service Manager manually. After the RDBMS has the necessary tables and columns, the Service Manager administrator can then update the database dictionary records to map to the actual RDBMS objects.

### Database dictionary (Dbdict)

1. Service Manager maintains a logical view of your RDBMS tables and columns in the database dictionary. The database dictionary describes each table and column in your system and how they are mapped to logical entities within Service Manager.
2. The Service Manager utility used to define the pools of a Service Manager file. The Service Manager command to open the Database Dictionary (Dbdict) utility is dbdict.

**Tip:** You can also use the System Definition utility to define fields and keys for a file.

### Database Manager

The Service Manager utility for adding, updating, deleting, viewing, printing, and searching for records. You can search by file or form. The Service Manager command to open Database Manager is `db`.

### Dedicated subscription

A subscription provided with a configuration item (CI) specifically assigned to the subscriber. A dedicated subscription that's not shared by any other subscribers. For example, a request for a laptop results in a subscription between a user and a specific laptop (CI).

### Definite Media Library (DML)

Service Manager refers to a Definitive Media Library (DML) in the Information Technology Infrastructure Library (ITIL) context. The DML was previously called the Definitive Software Library (DSL) in ITIL versions earlier than Version 3. The DML is the secure, logical library in which the definitive authorized versions of all media Configuration Items (CIs) are stored and protected. The DML covers not only software, but also other types of media that require control, such as databases and files. The DML is a single logical storage area that can be linked to multiple locations. All software in the DML is under the control of Change and Release Management and recorded in the Configuration Management System (CMS).

**Department subscription**

A subscription managed by a department subscriber on behalf of the entire department. The departmental subscriber can request or cancel the subscription at any time. Upon fulfillment, all members of the department are considered subscribed to the service.

**Detail log**

The detail.log file includes specific information about the upgrade, such as which files are being signed at any time.

**Development system**

The development system refers to the Service Manager system that mirrors your current production environment. Use the development system to run the Upgrade Utility and build a custom upgrade. This system should not be on the same machine as the production server.

**Display screen**

Display screens are individual records identified by a unique screen ID. Each display screen record defines the attributes of a screen and provides access to the individual records for options and events. A display screen is different from a form.

**Document Engine**

A Service Manager utility that enables System Administrators to define comprehensive processing rules for Service Manager tables.

**Draft documents**

In Knowledge Management, these are documents that are being created, but have not yet been submitted for approval or approved. Users can save draft documents and return to them later.

## E

**Error control**

An ITIL term for the process of correcting Known Errors. Making permanent changes to the infrastructure to reduce the number of current and prospective incidents is a function of Problem Management.

**Event**

The occurrence of a specific, detectable action or condition. For example, the opening of a change or task, an approval, or an update.

**Except log**

The except.log file includes information about any exceptions reported by the upgrade. The except.log file may have important messages about data type mismatches that you should resolve, or database dictionaries that it can't upgrade.

## F

**Federated CI views**

In Configuration Management, Configuration Item (CI) Visualization provides views of relationships between CI components and the current state of each item in the configuration. An extensive library of graphics displays the relationships in a variety of views.

**Field**

A single data element within a database dictionary record. For example, name is a field in the operator database dictionary record, and its value for one user might be Joe. A field is equivalent to a column in a Relational Database Management System (RDBMS).

**Field chooser widget**

A widget is a visual element of a form, such as a button, entry field, or drop-down list. In Service Manager, the field chooser widget allows users to easily add and modify fields on a form, such as customizing views of a View or Contacts list, or adding fields to the field list for forms that you use often.

**File**

A collection of related records. For example, all device records are contained in the device file. A file is equivalent to a Relational Database Management System (RDBMS) table that can be retrieved for manipulation using keyed search methods. In most of the Service Manager documentation, the term file has been replaced with the term table where appropriate.

**Fill**

The Fill function enables you to quickly project related data into a record, based on a source field and its relation rules defined in a link record. For example, you can click Fill to enter a caller's name in the Reported by field in a service desk interaction record. The caller's full name, phone number, location, and email address are automatically filled into the appropriate fields in the service desk interaction record.

**Find**

The Find function enables you to look up information from a source field. Like the Fill function, Find is controlled by relation rules defined in a link record. With the cursor in a source field, the Find function enables you to directly access the related file. For example, you can click Find when the cursor is in the Reported By field in a service desk interaction record to access the related contacts record and view complete details about the caller.

**Floating users**

Operators authorized to log onto Service Manager, based on the number of floating seats available in the license pool. A floating user ID may log onto Service Manager more than once; however, each time an ID is logged onto the system, that user counts as a floating user for that Service Manager system. When the maximum number of floating users licensed to a given system are concurrently logged on, no more floating users can connect to that system.

**Folder entitlement**

In Service Manager, an administrator can enable or disable folder entitlement for Operator Records. With folder entitlement enabled, operators are able to view specific folders, as specified in the Folder Entitlement tab for the Operator Record. When folder entitlement is disabled, the Folder Entitlement tab is no longer visible to operators.

**Form**

A GUI screen layout used to access, add, change, and view records in a specific Service Manager table, or used as part of the user interaction in a Service Manager application.

A form is the data entry mask Service Manager uses to display data. It serves as the interface between the user and the table. A form can be associated with only one Service Manager table; however, a table may have many associated forms. Not all forms refer to specific tables. For example, the login screen is displayed using a form that doesn't refer to any table and is only used for authentication. Typically, the primary form associated with a table carries the same name as the table. For example, if you use Database Manager to open the location file while running a GUI client, you access the location table using the form location.g. Service Manager automatically determines that you are running a GUI client and provides the .g version of the location form, if one is available. However, if you search for forms named location, Service Manager presents a list of several associated forms for your further selection. All these forms work with the location table, but only one is named location. You can create and modify forms with the Service Manager Forms Designer (FD) utility, and add or edit rules for processing data automatically with the Service Manager Format Control (FC) utility. Service Manager stores form records in the format table.

**Format Control (FC)**

1. Background processing rules that Service Manager applies automatically when opening, updating, saving or closing a record.

2. A Service Manager utility that enables administrators to build rules to activate behind-the-scenes processing under controlled conditions. The Service Manager command to open Format Control is `fc`.

**Forms Designer (FD)**

A Service Manager utility used to build, modify, copy, and rename forms, and then validate the forms against associated files. The Service Manager command to open Forms Designer is `fd`.

**Fulfillment services**

Fulfillment services are back-end processes that contribute toward the delivery of a requested Service Catalog item. A variety of processes may be involved including change, procurement, and configuration.

## G

**Glossary**

All user types can read the topics in this information plug-in to learn the definitions of Service Manager terms. The Search feature supports a keyword search. You can query the entire documentation set at once, or you can limit the search to a specific information plug-in. To search, type a value in the **Search** field and then click **Go**. To specify a search, click **Search Scope** and select only the information plug-ins to include in your list of returns. To select a term from a list of terms, click a letter and then click on the corresponding term.

**Group**

1. One or more operators assigned to a common area of responsibility. Typically, each group reflects a business or technical area (or department). Certain operators are designated as having approval authority for these groups. For example, Incident Management assignment groups are responsible for incidents and Change Management change groups are responsible for changes.
2. The Service Manager table that stores group records.

## H

**Historic CI state**

Past configuration states for a configuration item (CI). This information is typically captured by historic changes.

**Hit list**

In Knowledge Management, the list of search results. Typically, the hit list displays the relevancy, document title and summary, document status, and knowledgebase.

## I

**Impact**

The term impact refers to the measure of the effect of an Incident, Problem, or Change on a business process. Impact and Urgency are based on how Service Levels are affected and used to assign Priority. The impact code is a Category used to represent the impact, such as Major, Minor, or Catastrophic.

**Import wizard**

A Service Manager utility that simplifies importing batch data into Service Manager from other applications.

**Inbox**

1. A predefined search displayed in the queue for an operator, now called Views. Views enable a user to save queries that conduct predefined searches with specific parameters that enable you to quickly and easily repeat searches.
2. The name of a Service Manager table that stores views, which were previously called inboxes.

**Incident**

The ITIL term for an event that causes an interruption to or a reduction in service. Typically, when the service desk personnel can't immediately resolve a service desk interaction, they use Incident Management to create an incident record. The difference between an interaction and an incident is the level of support required for the resolution. This may be based upon the complexity of the issue, number of individuals involved, or amount of work required. Any number of interactions may relate to one incident. The resolution of an incident results in the resolution of any related interactions. Service Manager stores incident records in the problem tables.

**Incident Management**

1. The ITIL term for the process of restoring normal operations as quickly as possible after an incident, to minimize the adverse affect on business operation, and ensure the best possible levels of service quality.
2. The Service Manager application used to manage and track the resolution of incidents.

**Incident model**

An incident model, also known as an interaction model, is a method of predefining steps to handle the Incident Management (IM) process. The model includes steps to take, chronological order of the steps, who is responsible for each step, a time scale, and escalation procedures. In addition, the model ideally includes necessary activities required to preserve evidence, such as security and capacity-related incidents. Some incidents occur frequently and require a quick and consistent manner of registration. For this reason, you can create predefined incident models or interaction models that include values for attributes, such as description, priority, impact, and service. You can then apply the predefined models to incidents as they occur.

**Incident record**

The record of an incident, opened in Incident Management. Incident records include the following information: the issue the user has reported, affected item, work history, escalation status, and the resolution actions taken to resolve the issue.

**Index regeneration**

Index regeneration occurs when a database administrator recompiles indexes following changes that have been applied to Service Manager keys.

**Individual subscription**

A subscription assigned to an individual. Some individual subscriptions are also a "dedicated service." For example, an individual may have a subscription that provides a laptop. Other individual subscriptions are assigned as a "shared service." For example, an employee may be individually subscribed to a backup service.

**Information Technology Infrastructure Library (ITIL)**

Information Technology Infrastructure Library (ITIL) is a set of best practices and processes for managing the support and delivery of Information Technology (IT) services. Originally developed for the United Kingdom government in the 1980s, ITIL has become a world-wide de facto standard in the support and delivery of IT Service.

**Information Technology Service Management (ITSM)**

Information Technology Service Management (ITSM) combines proven methods, such as process management and industry best practices, in the area of IT Service Management. This combination helps ensure the quality delivery of IT services that satisfy business needs and achieve performance targets specified within Service Level Agreements (SLAs).

**Initialization file**

A file, such as the sm.ini file or the smstart script, which stores program settings that provide overall configuration and control information for an application. Initialization files must be located in the same directory as the application. You can edit the sm.ini file or the smstart script with any text editor.

**Initiator**

A Service Manager user role. The operator who starts a process.

**Interaction**

An interaction is the contact between a user and the Service Desk when a user requests assistance or requests a service. An interaction typically results in logging an Incident or Service Request.

**Interaction model**

An interaction model, also known as an incident model, is a method of predefining steps to handle the Incident Management (IM) process. The model includes steps to take, chronological order of the steps, who is responsible for each step, a time scale, and escalation procedures. In addition, the model ideally includes necessary activities required to preserve evidence, such as security and capacity-related incidents. Some interactions occur frequently and require a quick and consistent manner of registration. For this reason, you can create predefined incident models or interaction models that include values for attributes such as description, priority, impact, and service. You can then apply the predefined models to interactions as they occur.

**IR Expert**

An intelligent, concept-based information retrieval engine that searches records for similar or related information, based on a simple, natural language query. IR Expert creates its queries based on the IR keys in the dbdict and the terms you specify, after it performs *lexical analysis, stemming, stop words,* and *spelling correction* on them. For example, instead of relying on the exact-match keywords that QBE queries use to select incidents from the Incident Management database, you can search the description of an incident for specific terms using an IR query.

**IR key**

IR (information retrieval) is a key type where the fields in the key are indexed by IR Expert. Only one IR key can be used per dbdict record or IR searches on that file don't work. You can concatenate several fields in an IR key.

**IR query**

A search based on terms you enter in the text box on the IR Expert tab. If the search returns multiple records, you can select the record you want to view from a record list.

**IT Asset Management (ITAM)**

IT Asset Management (ITAM) establishes a lifecycle management process for all Information Technology (IT) assets and manages IT hardware, software, and license assets. Asset Management provides the basis for important Information Technology Infrastructure Library (ITIL) processes, such as Service Level Management (SLM) and Configuration Management (CM), and is required for Service Catalog and Configuration Management Database (CMDB) development.

**IT Service Desk**

The ITIL term for the process of delivering and supporting the Information Technology (IT) services appropriate to the business requirements of an organization.

## K

**Key types**

Fields that define the key fields in the Database Dictionary (Dbdict) of a file. These key fields create the indexes used in queries to make searching for records fast and efficient. Key types allow or disallow certain values across all records within the file for the combined fields that comprise the key, and assist in maintaining data integrity by controlling the field values in a key. The key types include *unique, no nulls, no duplicates,* and *nulls & duplicates*. IR Expert uses separate keys.

**Keys**

1. Keys are abstract entities that provide a logical view of the indexes in your RDBMS. When you create Service Manager logical keys, the server creates corresponding indexes in the back-end RDBMS.
2. The Service Manager table that stores keys.

**Known Error**

The ITIL term for a problem that has been successfully diagnosed and has a temporary work around. A Known Error exists until it is permanently fixed by a change in Change Management.

## L

**Lexical analysis**

The process IR Expert uses to convert the characters in an IR query into a stream of "or" words or tokens.

**Load**

The process of reading saved records from an external unload file into the Service Manager database. You can load files to restore records or apply customizations to an existing Service Manager installation. When loading, you can view the contents of the unload file, specify the level of messaging you want to see, or load the unload file either as a foreground or a background process.

## M

**Managed CI state**

The authorized configuration for a configuration item (CI). This is typically defined by the CI's baseline.

**Messages**

The Change Management application sends specified messages in response to an event. The messages are directed to specific operators listed in the event record and contain values from specified fields in quote, order, and line item records that trigger the initial event.

## N

**Named users**

Specific operators always authorized to log onto Service Manager based on a reserved license seat. Named users may log onto Service Manager at any time, based on their authorization code and license agreement. Their seats are reserved in Service Manager and are always available for login. If a named user logs on multiple times at the same work station, Service Manager only uses one license seat. If a named user logs on to multiple work stations, the named user license is used at the first login, then each subsequent login takes a floating license seat.

**No duplicates**

A type of key where the value of the complete key must be unique in the index or the values of all fields must be null.

**No nulls**

A type of key. At least one field in the key must not be null (empty or missing).

**Notification**

1. A Service Manager message delivered in an on-screen active note, the form window's status bar, or the Service Manager mail file.
2. A message sent externally, for example, using email.

**Nulls & duplicates**

A type of key where all fields can be null (empty or missing) and the complete key value can be in the index more than once.

## O

**OO flow**

An OO flow refers to a flow from Operations Orchestration. An OO flow is a set of linked actions that automate health checks, troubleshooting or any other repetitive IT support tasks. For more information about OO flows, refer to the Operations Orchestration documentation.

**Operational Level Agreement (OLA)**

An Operational Level Agreement (OLA) is an agreement covering the delivery of services supporting an IT organization in its delivery of services. An OLA may also be between two parts of the same IT Service Provider. For example between the Service Desk and a Support Group. An OLA can be created with the facilities department to provide air conditioning, or with the procurement department to obtain hardware at agreed times.

**Operator**

1. A person who logs on to Service Manager to work. This is a smaller group than the contacts who use the devices, software, and services that Service Manager supports.
2. The Service Manager table that stores operator records.

**Order**

A record that defines basic ordering information such as assignee, vendor, required dates, coordinator and description. An order doesn't contain detailed parts information. Service Manager generates an order when a Request Management quote has been approved and is ready for ordering. One quote can generate one order or multiple orders. Each order can have no line items, one line item, or many line items.

## P

**Page**

An individual history record in a set of common records. For example, each time a change is updated, a new page can be written if this option is turned on for the change phase. The new page then becomes part of the change's history.

**Peregrine Four (P4)**

The legacy Database Management System (DBMS) that stored ServiceCenter data in a proprietary flat-file format. All Service Manager data must be converted to a supported Relational Database Management System (RDBMS).

**Phase**

An administrative step within the change, order, task, or quote that's needed to complete the work. A phase determines how forms are viewed by the users, approval requirements, and the intervals at which alerts are sent. Phases are sequential, repeatable steps characteristic of a category. Each category must have at least one phase defined in the category specification and may have any number of phases defined (except line item categories, which have only one defined phase). Each phase can include none, one, or many tasks.

**Planned (pending) CI state**

The future (potential) state of a configuration item (CI), captured by a pending change record.

**Post Implementation Review (PIR)**

A Post Implementation Review (PIR) is a formal project review used to answer the question: "Did we achieve what we set out to do and, if not, what should be done?" For major changes, there can be several PIRs scheduled over time. A PIR checks whether benefits have been achieved and identifies opportunities for further improvement. Without a PIR, you can't effectively demonstrate that your investment was worthwhile.

**Primary file**

The main file associated with a form. You can define the primary file associated with a form using the Format Control. The primary file is $file. Any additional files are $file1, $file2, etc. in the order they appear in the secondary files section of the

Additional File Queries screen of the Format Control record.

**Priority**

The priority is an attribute in Incident Management and Problem Management that describes the degree of urgency with which a corrective action should be taken. The priority is set after considering the severity, frequency, reproducibility, and relevant business conditions that are impacted, such as Service Level Agreements (SLAs).

**Probable cause**

A predefined description of the cause of a service desk interaction or incident. These records contain an optional Cause Code field that links the interaction or incident record to a probable cause record. When a cause code is added to a record using the Fill function, accompanying information, such as the default category and description, also fills the record from the corresponding probable cause record. The out-of-box probcause file contains sample probable cause records.

**Problem**

The ITIL term for the unknown underlying cause of one or more incidents. Use the Problem Management application to resolve Problems. **Note:** Prior to version 6.1, problems were called root causes. Service Manager stores problem records in the rootcause table.

**Problem control**

The ITIL term for the process of identifying the root cause of an incident to prevent future recurrences. Identifying the underlying cause of an incident is a function of Problem Management.

**Problem Management**

1. The ITIL term for the process of identifying and resolving Problems and Known Errors before they become incidents, and the process of resolving existing Problems.
2. The Service Manager application used to control Problems and Errors.

**Note:** Prior to version 6.1, Problem Management was called Root Cause Analysis.

**Process**

1. A connected series of actions, activities, changes, etc. performed with the intent of satisfying a purpose or achieving a goal.
2. An instance of a program running on a computer. For Service Manager, this is either a user client session or a background agent.

**Product catalog**

The product catalog manages a list of products and configurations that may include items such as hardware, software, and contracts. This catalog is managed by an IT configuration analyst. Its contents are aggregated and published in a consumable form within the Service Catalog.

**Production environment**

The production environment is the Service Manager system that runs your approved applications and where you apply a custom upgrade.

**Profile**

The security record that defines the options and authorities available to an operator or group.

**Project Portfolio Management (PPM)**

Project Portfolio Management (PPM) is a term used by project managers and project management organizations to describe methods for analyzing and collectively managing a group of current or proposed projects. Each project is based on several key characteristics. The primary objective of the PPM process is to determine the optimal mix and sequencing of proposed projects to best achieve the overall goal. This is typically expressed in terms of economic measures, business strategy goals, or technical strategy goals while honoring constraints imposed by management or external real-world factors. Typical attributes of projects analyzed in a PPM process include each of the following: Total expected cost,

consumption of resources (human or otherwise), expected timeline and schedule of investment, expected nature, magnitude, timing of benefits to be realized, and relationship or interdependencies with other projects in the portfolio.

**Projected data**

Data copied from fields in an existing (source) record to identically named fields in a newly opened (target) record.

## Q

**Query–by–Example (QBE)**

A search based on values you enter in the appropriate fields on a search form. If the search returns multiple records, you can select the record you want to view or update from a record list.

**Query group**

An optional security feature that limits access to views. A System Administrator can define a view as available only to members of particular query groups. Only users with the necessary query groups defined in their operator record can see the restricted view. Service Manager stores query group records in the querygroups table.

**Queue**

For any Service Manager application, the default list of records of interest to a particular operator. The application displays the list based on the default view for the operator. Typically, the list includes records assigned to the operator or the assignment groups that this operator belongs to.

## R

**RAD comparison utility**

A tool that compares one version of a Service Manager RAD application to a different version of the same application.

**Rapid Application Development (RAD)**

The native system language that Service Manager uses to communicate with its routines and processes.

**Record**

A collection of data that describes a single entity. A record is a row in a Relational Database Management System (RDBMS) table.

**Record list**

1. A list of records returned by a search, also known as a QBE list or a search list.
2. A Service Manager view that has a list of records in one form and a record from the list in another form, both displayed in one window.

**Relational database management system (RDBMS)**

The Relational Database Management System (RDBMS) refers to the database system that Service Manager uses for its storage, such as Oracle and Microsoft SQL Server.

**Request**

A request for products and services. Unlike changes, requests affect only the person making the request, or a subordinate group of employees. Examples include password resets, individual pc upgrades, and new employee setup.

**Request for Change (RFC)**

A type of change. A request for a change to any Configuration Item (CI) within an infrastructure or to procedures and items associated with the infrastructure. An RFC includes details of the proposed Change, and may be recorded on paper or electronically. The term RFC is sometimes mistaken to mean a Change Record, or the Change itself. An RFC is only the detailed request for the change, not the actual change.

**Request Fulfillment**

1. The process of managing user requests for products and services.
2. The Service Manager application used to manage user requests for products and services.
   Requests affect only the person making the request, or a subordinate group of employees. Examples include password resets, individual pc upgrades, and new employee setup.
   For changes to the infrastructure, refer to Change Management.

**Reserved words**

Terms or words that have been set aside for exclusive use within Rapid Application Development (RAD) programming, SQL queries, or tailoring activities.

**Response time**

The amount of time, specified in a Service Level Agreement (SLA), for a service desk interaction, change, or incident to advance to another status. For example, how soon work must be assigned after the initial incident report, or how soon an incident must be resolved. If the specified amount of time elapses without a change in status, the SLA is breached.

**Revert**

In Knowledge Management, a user can choose to delete all changes made to a working copy of document while the document is in a workflow by selecting the Revert button. In this case, the document reverts to its original version and all changes made to the document are deleted.

**Reviewers**

Personnel who can examine the tasks and phases of a Change Management change, but who don't have approval authority.

**Role**

A set of responsibilities, activities, and authorizations.

**Root cause**

The underlying issue that's causing a problem. Problem Management investigates root causes. Service Manager stores problem records in the rootcause table. **Note:** Prior to version 6.1, Problem Management was called Root Cause Analysis.

**Row**

A collection of data that describes a single entity in a Relational Database Management System (RDBMS) table.

**Run–Time Environment (RTE)**

The binary, load library, or executable layers of the Service Manager server.

## S

**Scheduled Maintenance**

1. The process of scheduling, completing, and tracking routinely recurring activities on different components and device types in different locations. To reduce unplanned outages and system failures by servicing systems before they fail.
2. The Service Manager application used to manage recurring activities. Problem Management can generate Scheduled Maintenance activities.

**Self-service**

Enables users to connect to Service Manager using a supported browser to open, update, and track incidents and service requests.

**Note**

**Note:** Self-service doesn't consume floating user licenses.

**Service Catalog**

A Service Catalog contains a comprehensive list of enterprise products and services available to internal and external customers, depending on their business role. Users submit service requests that follow the usual planning and approval workflows. Service Catalogs are created and maintained by the IT Service Provider and are used by all IT Service Management Processes.

**Service Catalog request**

The service request tracks a user's request for Service Catalog items. The request undergoes an approval workflow and links to the product and fulfillment catalog processes that provide the goods and services to the requester.

**Service Contract Management**

The process of managing service contracts, including the type of service available to customers. It usually describes the hours, available services, and costs for special services.

**Service Delivery Unit (SDU)**

A group responsible for resolving a particular type of incident.

**Service desk**

1. The ITIL term for the single point of contact within the IT organization for users of IT services.
2. The Service Manager application used to document and track user interactions received by a service desk, including creating new incidents and associating those incidents to existing problems.

**Service Desk interaction**

A service desk interaction is the record of a telephone call or other communication from a user. If a service request to the Service Desk is not quickly resolved, the Service Desk team can create an incident record to resolve the issue. Service Manager stores service desk interaction records in the incident table.

**Service level**

A description of the services provided as part of a Service Level Agreement (SLA).

**Service Level Agreement (SLA)**

A Service Level Ageement (SLA) is a contract with a company or department for maintaining and repairing equipment or providing other services. This is typically an agreement between an IT Service Provider and a Customer. The SLA describes the service, establishes service level targets, and specifies the responsibilities of the IT Service Provider (ISP) and the customer. A single SLA can cover multiple IT Services or multiple customers. The agreement records the common understanding of services, priorities, responsibilities, guarantees, and the level of service. For example, the agreement can specify the level of availability, serviceability, performance, operation, or other attributes of the service, such as billing or penalties in the case of an SLA violation. In Service Manager, the SLA records are stored in the sla table and accessed using the sla tab.

**Service Level Management**

1. The ITIL term for the process of defining, documenting and managing levels of IT support.
2. The Service Manager application used to manage levels of IT support.

**Service Level Objective (SLO)**

A Service Level Objective (SLO) is a negotiated document that defines the service to be delivered to a customer in qualitative terms. A small number of Key Performance Indicators (KPIs) may also be defined within an SLO. An SLO is

commonly preferred to Service Level Agreements (SLA) because it provides a clearer understanding of the exact nature of the service offered. SLOs focus on the total contribution of the service to the entire business value chain.

**Service SLA**

Where an SLA covers one service, for all the customers of that service. For example, an SLA may be established for an organization's e-mail service that covers all of the customers of that service. In Service Manager it refers to the SLA for the contact's subscription to a Configuration Item (CI). This is only true if the application record references a Business Service, the contact has a subscription to the service, and the subscription references an SLA.

**Severity**

An attribute in Incident Management and Problem Management that describes the degree of impact on the user when the issue occurs. Severity can also indicate the number of users affected. For example, a server crash that caused loss of data would be of critical severity.

- **Critical: Severe business disruption.** The business unit or subunit is unable to operate, or a critical system component has failed or is severely impaired.
- **Major: Major business disruption.** A critical user or user group is unable to operate, or the business unit is experiencing a significant reduction in system performance.
- **Medium: Minor business disruption.** A single user is unable to operate with no available workaround.
- **Low: Minor disruption.** A single user or user group is experiencing incidents, but a workaround is available.
- **Very Low: Inquiry.** A single user or user group requiring assistance, but with no direct impact on business such as a request for information.

**Signature**

A numerical value that Service Manager uses to verify the authenticity of each file in an upgrade. Any change to the contents of the record causes the signature of that record to change, depending on the definitions in the signaturemake file.

**SMOO integration**

An acronym for the Service Manager to Operations Orchestration integration. This integration links Knowledge Management in Service Manager (SM) to flows in Operations Orchestration (OO). System administrators can configure and enable an instance of this integration in Integration Manager. When enabled, this integration synchronizes specified flows from Operations Orchestration to Service Manager, and users can then access, manage and launch these OO flows from Knowledge Management in Service Manager through the web client.

**Software Asset Management (SAM)**

Software Asset Management (SAM) is the practice of integrating people, processes, and technology to allow software licenses and usage to be tracked, evaluated, and managed. The goal of SAM is to reduce IT expenses, human resource overhead, and risks inherent to owning and managing software assets. This includes, but is not limited to, maintaining software license compliance, tracking inventory and usage of software assets, and maintaining policies and procedures surrounding the definition, deployment, configuration, use, and retirement of software assets. SAM represents the software component of IT Asset Management (ITAM), including hardware asset management. SAM is linked to ITAM due to the concept that, without effective inventory hardware controls, efforts to control the software are significantly inhibited.

**SQL query**

A query written using Structured Query Language (SQL) and run in the Expert Search utility. To use the Expert Search utility from a Search form, open the More Actions menu, and select Expert Search. **Note:** This option is only visible if your user profile allows expert searches.

**Stemming**

1. The process IR Expert uses to combine related words in an IR query based on their common roots, stems, and forms.
2. The Knowledge Management system also employs stemming when searching. For a more complete explanation of how Knowledge Management uses stemming, see the online help for Knowledge Management.

**Stop words**

1. A list of words you specify for IR Expert to ignore if they appear in an IR query.
2. The Knowledge Management system also employs stop words when searching. For a more complete explanation of how Knowledge Management uses stop words, see the online help for Knowledge Management.

**System bulletin**

A daily notice posted by a Service Manager administrator listing any system wide service desk issues and the current list of hot records.

**System Definition**

The Service Manager utility used to define the fields, keys, and data policy files. To define pools for a file, use the Database Dictionary (Dbdict) utility.

## T

**Table**

1. A collection of rows in a Relational Database Management System (RDBMS).
2. A Forms Designer (FD) object with multiple rows and columns.

**Tailoring**

Changes made to Service Manager by creating and modifying control records using Service Manager utilities. Tailoring is the normal method of adapting Service Manager to each installation's requirements. Tailoring involves no Rapid Application Development (RAD) programming or coding changes.

**Task**

The work processes necessary to complete a change. For example, a change to replace a hard drive with a larger model might include these tasks: ordering the new drive, backing up the old drive, and installing the new drive. Tasks must belong to a change. Task start and end dates, if specified, must fall within the start and end dates of the parent change. Tasks are classified by categories. Tasks can be broken into phases, if that level of discrimination is necessary. Work can't proceed to the next phase until all tasks beneath it are complete.

**Test environment**

The system that mirrors your current production environment for testing purposes. You use this environment to run and verify the custom upgrade prior to pushing it to the production environment.

**Total Cost of Ownership (TCO)**

Total Cost of Ownership (TCO) a form of full cost accounting. It is a financial estimate designed to help consumers and enterprise managers assess direct and indirect costs commonly related to software or hardware. The total cost includes all financial consequences of owning a Configuration Item (CI) or asset. For example, in addition to the initial purchase price, the total cost includes maintenance, accommodation charges, potential opportunity costs, training costs, consumables required, internal and external support, and interest on capital.

**Trace**

A debugging function that enables you to see the panels encountered by the Rapid Application Development (RAD) flow.

**Trigger**

An automated database activity defined by a rule in the trigger table and activated by a related activity in another Service Manager table. For example, when an operator updates an incident record, the related probsummary record is automatically updated in accordance with rules defined in a trigger record.

## U

### Underpinning Contract (UC)

Underpinning Contract (UC) is a contract between an Information Technology (IT) service provider and a third party. The third party provides goods or services that support the delivery of an IT service to a customer. The UC defines targets and responsibilities required to meet Service Level Targets defined in the Service Level Agreement (SLA).

### Unique

A type of key in a Relational Database Management System (RDBMS) where at least one field in the key must not be null (empty or missing) and the value of the complete key must be unique in the index.

### Unload

1. The native Service Manager export format that stores the database dictionary definition of Service Manager tables in addition to its records.
2. The process of exporting Service Manager records to an external unload file. You can use unload files to archive data and to move data from one Service Manager installation to another. You can unload an individual record or a list of records (using mass unload). You can also create an unload script to specify a list of files and queries to select the records unloaded.

### Unload file

An external file that stores records. You can create unload files by:

- Manually unloading a record or list of records.
- Creating an unload script to automatically unload a collection of files and filter records based on a predefined query.

You can save unload files as binary files, as text, and as formatted text. All three file formats are unique to Service Manager and aren't a means of exporting records to other file formats.

### Upgrade log

The upgrade.log file includes information about the upgrade status. It indicates where the upgrade is at a specific point. This file contains only the main steps of the upgrade.

### Upgrade utility

A set of utilities that are shipped with the new software release for upgrading to the new Service Manager applications. It makes the upgrade easier by automating certain steps.

### Urgency

A measure of the time elapsed until an Incident, Problem, or Change has a significant impact on business. For example, a high impact Incident may have low urgency if the impact doesn't affect the business until the end of the Financial Year. Impact and Urgency are often used to determine the Priority.

### User

- A Service Manager operator.
- An end user or contact.

### User authentication

A set of requirements to prove the identity of a caller before completing changes to data security access.

## V

### Variable

A value that resides in memory only, rather than in a database record. Service Manager variable names always start with $. To ensure that a variable contains the proper value, initialize it in the Initializations section of the Format Control record.

**View**

A predefined search displayed in the queue for an operator, previously called Inbox. Views enable a user to save queries that conduct predefined searches with specific parameters that enable you to quickly and easily repeat searches. **Note:** With the introduction of Service Manager 7, the term inboxes is changed to views. Previous versions of ServiceCenter used the term inboxes for views. The underlying table for views is named the inbox table.

## W

**Weblib**

Weblib refers to a type of knowledgebase created when the web crawler in Knowledge Management is used to crawl a Web site and create a knowledgebase. Web crawling is the process of creating an index so that the contents of the Web site are searchable.

**Workflow**

The term workflow is used in computer programming to capture and develop human to machine interaction. Workflow software provides end users with an easier way to orchestrate or describe complex processing of data in a visual form, such as easily read flow charts that describe the flow of work without the need to understand programming required.

**Working copy**

Knowledge Management uses the term working copy to describe a document that's in the process of being edited and is a copy of a published document.

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